
py3plex Documentation

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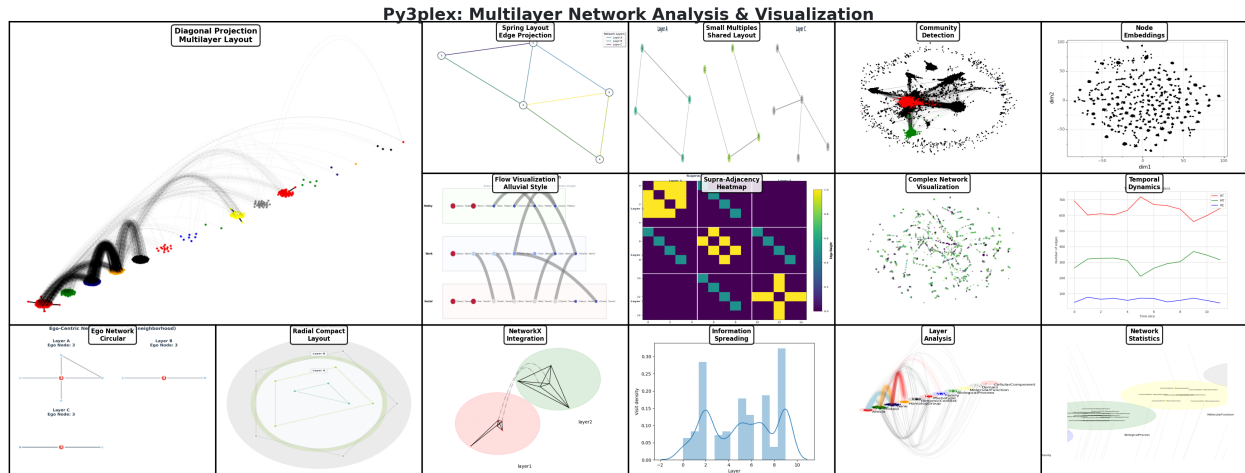
Blaž Škrlj

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py3plex enables scalable analysis and visualization of multilayer and multiplex networks in Python, supporting complex network modeling across diverse scientific and applied domains.

OVERVIEW

py3plex is a lightweight Python library designed specifically for analyzing and visualizing heterogeneous and multilayer networks. Unlike traditional network analysis tools that focus on homogeneous networks (single node and edge type), py3plex provides specialized capabilities for networks with:

- Multiple node types
- Multiple edge types
- Multiple layers of interaction
- Temporal dynamics
- Heterogeneous attributes

Target Users: Researchers in network science, computational biology, complex systems, social network analysis, and applied network modeling.

Key Features:

- Native support for multiplex and multilayer network structures
- Diagonal projection visualization for large multilayer networks
- Comprehensive multilayer centrality measures
- Community detection across network layers
- Network decomposition and feature extraction
- Semantic enrichment with external knowledge bases
- Integration with NetworkX, igraph, and other graph libraries

INSTALLATION

2.1 Install from GitHub

```
pip install git+https://github.com/SkBlaz/py3plex.git
```

2.2 Docker Installation (Alternative)

py3plex is also available as a Docker container with all dependencies pre-installed:

```
# Clone and build
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
docker build -t py3plex:latest .

# Run commands
docker run --rm py3plex:latest --version
docker run --rm py3plex:latest selftest
```

See *Docker Usage Guide* for complete Docker documentation.

2.3 Install from source for development

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
pip install -e .
```

2.4 Optional Dependencies

```
# Advanced community detection with Infomap
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[infomap]

# Additional algorithms (Louvain, cdlib)
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[algos]

# Advanced visualization (plotly, igraph)
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[viz]
```

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```
# For development (includes testing and linting tools)
pip install -e "[dev]"
```

2.5 Requirements

- Python 3.8 or higher
- NetworkX, NumPy, SciPy (automatically installed)
- Optional: Matplotlib, Plotly for visualization

QUICK START

Create a simple multilayer network:

```
from py3plex.core import multinet

# Create a new multilayer network
network = multinet.multi_layer_network()

# Add edges within layers
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1],
    ['B', 'layer1', 'C', 'layer1', 1],
    ['A', 'layer2', 'B', 'layer2', 1],
], input_type="list")

# Display basic statistics
network.basic_stats()

# Visualize
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default([network], display=True)
```

Load from file and analyze:

```
# Load from edge list
network = multinet.multi_layer_network().load_network(
    "data.edgelist", input_type="edgelist", directed=False)

# Compute multilayer statistics
from py3plex.algorithms.statistics import multilayer_statistics as mls
density = mls.layer_density(network, 'layer1')
versatility = mls.versatility centrality(network, centrality_type='degree')

# Community detection
from py3plex.algorithms.community_detection import community_wrapper
communities = community_wrapper.best_partition(network.core_network)
```


WHY PY3PLEX?

4.1 vs. NetworkX

NetworkX excels at single-layer homogeneous networks but lacks native multilayer support. py3plex builds on NetworkX while adding:

- Native multilayer data structures
- Layer-aware algorithms
- Specialized multilayer visualizations
- Heterogeneous network decomposition

4.2 vs. Other Multilayer Tools

py3plex provides:

- Lightweight and easy to integrate
- Comprehensive statistical measures (17+ multilayer metrics)
- Publication-ready visualizations
- Active development and research backing

USE CASES

py3plex is particularly well-suited for:

- **Biological networks:** Protein-protein interactions with multiple evidence types, gene regulatory networks
- **Social networks:** Multi-platform analysis (Twitter + Facebook), relationship type analysis
- **Citation networks:** Author-paper-venue multilayer structures, knowledge graphs
- **Transportation:** Multi-modal networks (bus, train, air), urban mobility
- **Temporal networks:** Evolving social networks, dynamic community evolution

WHAT'S IN THIS DOCUMENTATION?

This documentation is organized to help different types of users find what they need quickly:

New to py3plex? Start with *5-Minute Quickstart* for a 5-minute introduction, then explore *10-Minute Tutorial: Getting Started with py3plex* for a comprehensive walkthrough.

Power users? Jump to the *Working with Networks* and other user guide sections for detailed how-tos, or browse the *Examples* for working code.

Want to understand the concepts? Check out *Multilayer Networks 101* and *py3plex Core Model* for deep conceptual explanations.

Deploying in production? See the *Docker Usage Guide* and *Performance and Scalability Best Practices* guides.

Contributing? Read the *Development Guide* and *Contributing to py3plex* pages.

DOCUMENTATION CONTENTS

7.1 5-Minute Quickstart

Get started with py3plex in just 5 minutes with this minimal example.

7.1.1 Installation

Install py3plex from GitHub:

```
pip install git+https://github.com/SkBlaz/py3plex.git
```

If you prefer Docker, see *Installation and Setup* for container-based setup.

7.1.2 Hello World: Your First Multilayer Network

Create a simple multilayer network with just a few lines of code:

```
from py3plex.core import multinet

# Create a new multilayer network
network = multinet.multi_layer_network()

# Add edges (nodes are created automatically)
# Format: [source_node, source_layer, target_node, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Bob', 'friends', 'Carol', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1]
], input_type="list")

# Display basic statistics
network.basic_stats()
```

Output:

```
Number of nodes: 6
Number of edges: 4
Number of unique nodes (as node-layer tuples): 6
Number of unique node IDs (across all layers): 4
Nodes per layer:
```

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```
Layer 'friends': 3 nodes  
Layer 'colleagues': 3 nodes
```

7.1.3 Visualize Your Network

Create a simple visualization:

```
from py3plex.visualization.multilayer import draw_multilayer_default  
  
# Simple visualization  
draw_multilayer_default([network], display=True)
```

This creates a visual plot showing your multilayer network with nodes colored by layer.

7.1.4 Basic Analysis

Compute some basic network statistics:

```
from py3plex.algorithms.statistics import multilayer_statistics as mls  
  
# How dense is each layer?  
friends_density = mls.layer_density(network, 'friends')  
colleagues_density = mls.layer_density(network, 'colleagues')  
  
print(f"Friends layer density: {friends_density:.3f}")  
print(f"Colleagues layer density: {colleagues_density:.3f}")  
  
# How active is Bob across layers?  
bob_activity = mls.node_activity(network, 'Bob')  
print(f"Bob's activity: {bob_activity:.3f}")
```

Output:

```
Friends layer density: 0.667  
Colleagues layer density: 0.667  
Bob's activity: 1.000
```

Bob appears in both layers (100% activity) and both layers are well-connected.

7.1.5 What's Next?

Congratulations! You've created your first multilayer network. To dive deeper:

- *10-Minute Tutorial: Getting Started with py3plex* - Comprehensive 10-minute tutorial with more features
- *Installation and Setup* - Complete installation guide with optional dependencies
- *Common Issues and Troubleshooting* - Troubleshooting common problems
- *Working with Networks* - Learn more about creating and loading networks
- *Visualization Guide* - Advanced visualization techniques

7.1.6 Need Help?

- **Documentation:** <https://skblaz.github.io/py3plex/>
- **GitHub Issues:** <https://github.com/SkBlaz/py3plex/issues>
- **Examples:** [examples/](#) directory

7.2 10-Minute Tutorial: Getting Started with py3plex

This tutorial provides a quick introduction to py3plex, covering the most common tasks for working with multilayer networks.

7.2.1 What You Will Learn

In 10 minutes, you will learn how to:

1. Create and load multilayer networks
2. Perform basic network analysis
3. Compute multilayer statistics
4. Detect communities
5. Perform random walks for embeddings
6. Visualize networks

7.2.2 Prerequisites

Make sure py3plex is installed:

```
pip install git+https://github.com/SkBlaz/py3plex.git
```

7.2.3 1. Creating Your First Multilayer Network (2 minutes)

Start by creating a simple multilayer network from scratch:

```
from py3plex.core import multinet

# Create a new multilayer network
network = multinet.multi_layer_network()

# Add edges within layers (this automatically creates nodes)
# Format: [source_node, source_layer, target_node, target_layer, weight]
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1],
    ['B', 'layer1', 'C', 'layer1', 1],
    ['A', 'layer2', 'B', 'layer2', 1],
    ['B', 'layer2', 'D', 'layer2', 1]
], input_type="list")

# Display basic statistics
network.basic_stats()
```

Output:

```
Number of nodes: 5
Number of edges: 4
Number of unique nodes (as node-layer tuples): 5
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'layer1': 3 nodes
  Layer 'layer2': 3 nodes
```

7.2.4 2. Loading Networks from Files (1 minute)

py3plex supports multiple input formats. Here is how to load from an edge list:

```
from py3plex.core import multinet

# Load from a multiedgelist file
# Format: source target layer
network = multinet.multi_layer_network().load_network(
    "datasets/multiedgelist.txt",
    input_type="multiedgelist",
    directed=False
)

# Check what we loaded
network.basic_stats()
```

Output:

```
Number of nodes: 5
Number of edges: 4
Number of unique nodes (as node-layer tuples): 5
Number of unique node IDs (across all layers): 5
Nodes per layer:
  Layer '1': 3 nodes
  Layer '2': 2 nodes
```

Supported formats:

- multiedgelist - source, target, layer
- edgelist - simple source, target pairs
- gpickle - NetworkX pickle format
- gml, graphml - standard graph formats

7.2.5 3. Exploring Network Structure (2 minutes)

Iterate Through Nodes and Edges

```
# Loop through all nodes
print("Nodes:")
for node in network.get_nodes(data=True):
    print(node)

# Loop through all edges
```

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```

print("\nEdges:")
for edge in network.get_edges(data=True):
    print(edge)

# Get neighbors of a specific node in a layer
node_of_interest = "1"
layer_id = "1"
neighbors = list(network.get_neighbors(node_of_interest, layer_id=layer_id))
print(f"\nNeighbors of {node_of_interest} in layer {layer_id}:", neighbors)

```

Output (first few items):

```

Nodes (first 5):
(('1', '1'), {'type': '1'})
(('2', '1'), {'type': '1'})
(('6', '2'), {'type': '2'})
(('3', '1'), {'type': '1'})
(('5', '2'), {'type': '2'})

Edges (first 5):
(('1', '1'), ('2', '1'), {'weight': '1'})
(('1', '1'), ('3', '1'), {'weight': '1'})
(('2', '1'), ('6', '2'), {'weight': '1'})
(('3', '1'), ('5', '2'), {'weight': '1'})

Neighbors of 1 in layer 1: [('2', '1'), ('3', '1')]

```

Extract Subnetworks

```

# Extract a single layer
layer_1 = network.subnetwork(['1'], subset_by="layers")
print("Layer 1 nodes:", list(layer_1.get_nodes()))

# Extract specific nodes
node_subset = network.subnetwork(['1', '2'], subset_by="node_names")
print("Node subset:", list(node_subset.get_nodes()))

# Extract specific node-layer pairs
specific_pairs = network.subnetwork(
    [('1', '1'), ('2', '1')],
    subset_by="node_layer_names"
)
print("Specific pairs:", list(specific_pairs.get_nodes()))

```

Output:

```

Layer 1 nodes: 3 nodes
Node subset: 2 node-layer pairs
Specific pairs: [('2', '1'), ('1', '1')]

```

7.2.6 4. Computing Network Metrics (2 minutes)

Basic Centrality Measures

```
# Get a single layer as NetworkX graph
layer_1 = network.subnetwork(['1'], subset_by="layers")

# Compute degree centrality using NetworkX wrapper
degree_centrality = layer_1.monoplex_nx_wrapper("degree_centrality")
print("Degree centrality:", degree_centrality)

# Compute betweenness centrality
betweenness = layer_1.monoplex_nx_wrapper("betweenness_centrality")
print("Betweenness centrality:", betweenness)
```

Output (top nodes shown):

```
Degree centrality (first 5):
('1', '1'): 1.0000
('2', '1'): 0.5000
('3', '1'): 0.5000

Betweenness centrality (first 5):
('1', '1'): 1.0000
('2', '1'): 0.0000
('3', '1'): 0.0000
```

Multilayer Centrality

For multilayer-specific centrality measures:

```
from py3plex.algorithms.multilayer_algorithms.centrality import MultilayerCentrality

# Initialize centrality calculator
calc = MultilayerCentrality(network)

# Compute multilayer degree centrality (considers node participation across layers)
ml_degree = calc.overlapping_degree_centrality(weighted=False)
print("Multilayer degree centrality:", ml_degree)

# Compute multilayer betweenness centrality
ml_betweenness = calc.multilayer_betweenness_centrality()
print("Multilayer betweenness centrality:", ml_betweenness)
```

Output (top nodes shown):

```
Multilayer degree centrality (first 5):
1: 2.0000
2: 1.0000
3: 1.0000
5: 0.0000
6: 0.0000

Multilayer betweenness centrality (first 5):
```

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```
( '1', '1'): 0.6667
( '2', '1'): 0.5000
( '3', '1'): 0.5000
( '6', '2'): 0.0000
( '5', '2'): 0.0000
```

7.2.7 5. Multilayer Network Statistics (2 minutes)

py3plex provides many specialized statistics for analyzing multilayer networks. Here are the most commonly used:

Basic Layer Statistics

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Measure how densely connected each layer is
density_layer1 = mls.layer_density(network, 'layer1')
density_layer2 = mls.layer_density(network, 'layer2')
print(f"Layer 1 density: {density_layer1:.3f}")
print(f"Layer 2 density: {density_layer2:.3f}")

# Measure diversity across layers
entropy = mls.entropy_of_multiplexity(network)
print(f"Layer diversity (entropy): {entropy:.3f} bits")
```

Output:

```
Layer 1 density: 0.667
Layer 2 density: 0.000
Layer diversity (entropy): 0.000 bits
```

Node-Level Statistics

```
# Compute node activity (how many layers a node participates in)
activity = mls.node_activity(network, node='1')
print(f"Node activity for node 1: {activity:.3f}")
```

Output:

```
Node activity for node 1: 0.500
```

Network-Level Statistics

```
# Measure how much layers differ in structure (edge overlap)
edge_overlap_value = mls.edge_overlap(network, 'layer1', 'layer2')
print(f"Edge overlap between layers: {edge_overlap_value:.3f}")

# Measure layer similarity using Jaccard index
jaccard = mls.layer_similarity(network, 'layer1', 'layer2', method='jaccard')
print(f"Jaccard similarity between layers: {jaccard:.3f}")
```

Output:

```
Edge overlap between layers: 0.000
Jaccard similarity between layers: 0.000
```

Available statistics:

See the `py3plex.algorithms.statistics.multilayer_statistics` module for all available statistics including:

- `layer_density` - Density of edges within a layer
- `entropy_of_multiplexity` - Diversity across layers
- `node_activity` - Number of layers a node participates in
- `edge_overlap` - Edge overlap between layers
- `layer_similarity` - Similarity between layers (Jaccard, Pearson, etc.)
- `algebraic_connectivity` - Connectivity measure from Laplacian
- `community_participation_coefficient` - How communities span layers
- `interdependence` - Layer interdependence measure
- `multilayer_clustering_coefficient` - Clustering in multilayer context
- `multiplex_betweenness_centrality` - Betweenness in multiplex networks
- `multiplex_closeness_centrality` - Closeness in multiplex networks
- And more...

7.2.8 6. Community Detection (2 minutes)

py3plex provides Louvain-based community detection that works across multiple layers:

```
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)

# Detect communities using multilayer Louvain
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Resolution parameter
    omega=1.0,      # Inter-layer coupling
    random_state=42 # For reproducibility
)

# Examine results
print("Number of communities:", len(set(partition.values())))
print("\nNode assignments (first 10):")
for node, community in list(partition.items())[:10]:
    print(f" Node {node}: Community {community}")

# Count community sizes
from collections import Counter
community_sizes = Counter(partition.values())
print("\nCommunity sizes:", dict(community_sizes))
```

Output:

```
Number of communities: 3
```

```
Node assignments (first 10):
```

```
Node ('1', '1'): Community 0
Node ('2', '1'): Community 0
Node ('6', '2'): Community 1
Node ('3', '1'): Community 0
Node ('5', '2'): Community 2
```

```
Community sizes (top 5): {0: 3, 1: 1, 2: 1}
```

Note: The `louvain_multilayer` function performs community detection across all layers simultaneously, taking into account both intra-layer and inter-layer connections. Parameters:

- `gamma`: Resolution parameter (higher values = more communities)
- `omega`: Inter-layer coupling strength (higher values = more consistency across layers)
- `random_state`: Seed for reproducibility

7.2.9 7. Random Walks (1 minute)

Random walks are fundamental for graph embeddings (Node2Vec, DeepWalk) and analyzing network structure:

Basic Random Walk

```
from py3plex.algorithms.general.walkers import basic_random_walk

# Load a network
network = multinet.multi_layer_network().load_network(
    "datasets/test.edgelist",
    input_type="edgelist",
    directed=False
)
G = network.core_network

# Perform a random walk
start_node = list(G.nodes())[0]
walk = basic_random_walk(
    G,
    start_node=start_node,
    walk_length=10,
    weighted=True,
    seed=42
)
print(f"Random walk: {walk[:5]}... (length: {len(walk)})")
```

Output:

```
Random walk from ('0', 'null'): [('0', 'null'), ('1', 'null'), ('4872', 'null'), ('786',
↪ 'null'), ('5382', 'null')]... (length: 11)
```

Note: When loading a simple edgelist file (without explicit layer information), py3plex assigns nodes to a default layer named 'null'. Nodes are represented as tuples (node_id, layer_name).

Node2Vec Biased Walks

```
from py3plex.algorithms.general.walkers import node2vec_walk

# Biased walk with Node2Vec parameters
# p: return parameter (higher = less likely to return)
# q: in-out parameter (higher = stay local, lower = explore)

walk_bfs = node2vec_walk(
    G, start_node, walk_length=20,
    p=1.0, q=2.0, # BFS-like (local)
    seed=42
)

walk_dfs = node2vec_walk(
    G, start_node, walk_length=20,
    p=1.0, q=0.5, # DFS-like (explore)
    seed=42
)
```

Output (first 10 nodes of each walk):

```
Node2Vec BFS-like walk: [('0', 'null'), ('1', 'null'), ('5829', 'null'), ('1', 'null'), (
    ↳ '3842', 'null'), ('1', 'null'), ('6364', 'null'), ('1', 'null'), ('3422', 'null'), ('1
    ↳ ', 'null')]
Node2Vec DFS-like walk: [('0', 'null'), ('1', 'null'), ('4872', 'null'), ('786', 'null'),
    ↳ ('5382', 'null'), ('260', 'null'), ('2861', 'null'), ('260', 'null'), ('5128', 'null
    ↳ '), ('260', 'null')]
```

Generating Multiple Walks

```
from py3plex.algorithms.general.walkers import generate_walks

# Generate walks from all nodes for embeddings
walks = generate_walks(
    G,
    num_walks=10,      # Walks per node
    walk_length=10,    # Steps per walk
    p=1.0, q=1.0,      # Node2Vec parameters
    seed=42
)
print(f"Generated {len(walks)} walks")

# Use with Word2Vec for node embeddings
# walks_str = [[str(node) for node in walk] for walk in walks]
# model = Word2Vec(walks_str, vector_size=128, window=10)
```

Output:

```
Generated 63870 walks
Example walk: [('4528', 'null'), ('2611', 'null'), ('4267', 'null'), ('2611', 'null'), (
    ↳ '2894', 'null'), ('2611', 'null'), ('6234', 'null'), ('2611', 'null'), ('3073', 'null
    ↳ '), ('479', 'null'), ('3125', 'null')]
```

Key parameters:

- `walk_length`: Number of steps in the walk
- `p`: Return parameter (controls backtracking)
- `q`: In-out parameter (controls exploration vs. local search)
- `weighted`: Use edge weights in sampling
- `seed`: Random seed for reproducibility

7.2.10 8. Basic Visualization (1 minute)

Visualize your multilayer network:

```
from py3plex.visualization.multilayer import hairball_plot
import matplotlib.pyplot as plt

# Get network for visualization
network_colors, graph = network.get_layers(style="hairball")

# Create a simple hairball plot
plt.figure(figsize=(10, 10))
hairball_plot(
    graph,
    network_colors,
    layout_algorithm="force",
    layout_parameters={"iterations": 50}
)
plt.title("Multilayer Network Visualization")
plt.savefig("my_network.png", dpi=150, bbox_inches='tight')
plt.close()
print("Visualization saved to my_network.png")
```

Output:

```
Visualization saved to my_network.png
```

For more advanced visualizations with community colors:

```
from py3plex.visualization.colors import colors_default

# Get communities
partition = louvain_multilayer(network)

# Select top N communities
top_n = 5
community_counts = Counter(partition.values())
top_communities = [c for c, _ in community_counts.most_common(top_n)]

# Assign colors
color_map = dict(zip(
    top_communities,
    colors_default[:top_n]
))
```

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```

network_colors = [
    color_map.get(partition.get(node), "gray")
    for node in network.get_nodes()
]

# Plot with community colors
plt.figure(figsize=(10, 10))
hairball_plot(graph, network_colors, layout_algorithm="force")
plt.title("Multilayer Network with Communities")
plt.savefig("my_network_communities.png", dpi=150, bbox_inches='tight')
plt.close()
print("Community visualization saved to my_network_communities.png")

```

Output:

```
Community visualization saved to my_network_communities.png
```

7.2.11 Complete Example: Putting It All Together

Here's a complete workflow:

```

from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    ↳ multilayer
from py3plex.visualization.multilayer import hairball_plot
from py3plex.visualization.colors import colors_default
from collections import Counter
import matplotlib.pyplot as plt

# Load network
network = multinet.multi_layer_network().load_network(
    "datasets/multiedgelist2.txt",
    input_type="multiedgelist",
    directed=False
)

# Analyze structure
print("=== Network Statistics ===")
network.basic_stats()

# Compute centrality for one layer
layer_1 = network.subnetwork(['1'], subset_by="layers")
degree_cent = layer_1.monoplex_nx_wrapper("degree_centrality")
print("\n=== Top 5 Nodes by Degree (Layer 1) ===")
for node, score in sorted(degree_cent.items(), key=lambda x: x[1], reverse=True)[:5]:
    print(f"{node}: {score:.3f}")

# Detect communities using multilayer method
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
print(f"\n=== Communities ===")
print(f"Number of communities: {len(set(partition.values()))}")

```

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```
# Visualize with communities
network_colors, graph = network.get_layers(style="hairball")
top_n = 3
community_counts = Counter(partition.values())
top_communities = [c for c, _ in community_counts.most_common(top_n)]
color_map = dict(zip(top_communities, colors_default[:top_n]))
network_colors = [
    color_map.get(partition.get(node), "lightgray")
    for node in network.get_nodes()
]

plt.figure(figsize=(12, 12))
hairball_plot(graph, network_colors, layout_algorithm="force")
plt.title("Multilayer Network Analysis")
plt.savefig("complete_analysis.png", dpi=150, bbox_inches='tight')
plt.close()
print("\nComplete analysis saved to complete_analysis.png")
```

Output:

```
=== Network Statistics ===
Number of nodes: 184
Number of edges: 1691
Number of unique nodes (as node-layer tuples): 184
Number of unique node IDs (across all layers): 46
Nodes per layer:
  Layer '1': 46 nodes
  Layer '2': 46 nodes
  Layer '3': 46 nodes
  Layer '4': 46 nodes

=== Top 5 Nodes by Degree (Layer 1) ===
('15', '1'): 0.244
('7', '1'): 0.244
('27', '1'): 0.244
('28', '1'): 0.244
('1', '1'): 0.244

=== Communities ===
Number of communities: 46

Complete analysis saved to complete_analysis.png
```

Note: In this example, the high number of communities (equal to the number of nodes) indicates that the network structure or parameters may need adjustment for meaningful community detection. In practice, adjust γ (resolution) and ω (inter-layer coupling) parameters to achieve desired community granularity.

7.2.12 Next Steps

Now that you've completed this tutorial, explore more advanced features:

- **Multilayer Statistics:** Explore all available statistics in `py3plex.algorithms.statistics.multilayer_statistics`
- **Multilayer Centrality:** See examples in `examples/network_analysis/`
- **More Examples:** Check the `examples/` directory for 80+ examples organized by topic
- **Full Documentation:** Visit <https://skblaz.github.io/py3plex/>

7.2.13 Common Issues

File Not Found

Make sure you're running from the repository root or use absolute paths:

```
import os
base_path = "/path/to/py3plex"
network = multinet.multi_layer_network().load_network(
    os.path.join(base_path, "datasets/multiedgelist.txt"),
    input_type="multiedgelist",
    directed=False
)
```

Visualization Not Showing

If using Jupyter, add `%matplotlib inline` at the top of your notebook. For scripts, use `plt.show()` instead of `plt.close()`.

Missing Dependencies

Some features require optional dependencies:

```
# For advanced visualization
pip install plotly

# For community detection with Infomap
pip install infomap

# For all extras
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[viz,algos]
```

7.2.14 Tips for Success

1. **Start Simple:** Begin with small test networks before scaling to large datasets
2. **Check Stats:** Always run `basic_stats()` after loading to verify your network
3. **Use Subnetworks:** Extract layers or subsets for faster prototyping
4. **Seed Your Random:** Use seed parameters in algorithms for reproducible results
5. **Visualize Early:** Quick plots help catch data loading issues early

Happy network analysis!

7.3 Installation and Setup

This guide covers all aspects of installing py3plex and setting up your environment.

7.3.1 Basic Installation

Install from GitHub (Recommended)

The latest version is always available on GitHub:

```
pip install git+https://github.com/SkBlaz/py3plex.git
```

This installs the core py3plex library with all required dependencies.

Install from Source

For development or to access the latest features:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex
pip install -e .
```

The `-e` flag installs in editable mode, so changes to the source code are immediately available.

Docker Installation (Alternative)

For a containerized environment with all dependencies pre-installed, use Docker:

```
# Clone the repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build the Docker image
docker build -t py3plex:latest .

# Run py3plex commands
docker run --rm py3plex:latest --version
docker run --rm py3plex:latest selftest
```

Benefits of Docker installation:

- No Python environment setup required
- All dependencies pre-installed and tested
- Consistent environment across different systems
- Isolated from system Python installation

See [Docker Usage Guide](#) for complete Docker documentation including:

- Building and running the container
- Working with data files via volume mounts
- Docker Compose usage
- Helper scripts (`py3plex-docker.sh`)
- Batch processing and CI/CD integration

7.3.2 Optional Dependencies

py3plex offers several optional feature sets that can be installed separately:

Advanced Community Detection

Install Infomap for advanced overlapping community detection:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[infomap]
```

Additional Algorithms

Install extra community detection algorithms (Louvain, cdlib):

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[algos]
```

Advanced Visualization

Install Plotly and igraph for interactive and advanced visualizations:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[viz]
```

Apache Arrow Support

Install pyarrow for high-performance I/O with Arrow/Parquet formats:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[arrow]
```

Arrow format provides 2-3x faster read/write operations and better compression compared to JSON. See *I/O and Serialization* for details on using Arrow format.

Multiple Extras

Install multiple feature sets at once:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[infomap,viz,algos,  
↪arrow]
```

Development Installation

For contributors, install with development tools (testing, linting):

```
git clone https://github.com/SkBlaz/py3plex.git  
cd py3plex  
pip install -e ".[dev]"
```

This installs:

- pytest and coverage tools
- black, ruff, isort for code formatting
- mypy for type checking
- sphinx for documentation building

7.3.3 System Requirements

Python Version

py3plex requires Python 3.8 or higher. We test on:

- Python 3.8
- Python 3.9
- Python 3.10
- Python 3.11
- Python 3.12

Platform Support

py3plex is cross-platform and tested on:

- **Linux** (Ubuntu 20.04, 22.04)
- **macOS** (macOS 11, 12, 13)
- **Windows** (Windows Server 2019, 2022)

Core Dependencies

These are automatically installed with py3plex:

- `networkx>=2.5` - Graph data structures and algorithms
- `numpy>=0.8` - Numerical computing
- `scipy>=1.1.0` - Scientific computing and sparse matrices
- `pandas` - Data manipulation
- `matplotlib` - Static visualization
- `scikit-learn` - Machine learning utilities
- `tqdm` - Progress bars
- `rdflib>=0.1` - Semantic web support
- `bitarray>=2.0.0` - Efficient boolean arrays

7.3.4 External Binaries (Optional)

Note: As of October 2025, py3plex no longer bundles external binaries to reduce repository size and improve licensing clarity.

Infomap Community Detection

For advanced community detection using Infomap:

Option 1: Use the bundled Python implementation (recommended):

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[infomap]
```

Option 2: Download the C++ binary from the official source:

- Website: <https://www.mapequation.org/infomap/>
- Follow platform-specific installation instructions

Alternative: Use the built-in Louvain algorithm (no binary needed):

```
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(network.core_network)
```

Node2Vec Embeddings

For Node2Vec node embeddings, use pure Python alternatives:

Option 1: Use pecanpy (fast parallel implementation):

```
pip install pecanpy
```

Option 2: Use node2vec package:

```
pip install node2vec
```

Option 3: Use py3plex's built-in random walk implementation:

```
from py3plex.algorithms.general.walkers import node2vec_walk, generate_walks
walks = generate_walks(network.core_network, num_walks=10, walk_length=80)
```

7.3.5 Virtual Environment Setup

We strongly recommend using a virtual environment:

Using venv

```
# Create virtual environment
python3 -m venv py3plex-env

# Activate (Linux/macOS)
source py3plex-env/bin/activate

# Activate (Windows)
py3plex-env\Scripts\activate

# Install py3plex
pip install git+https://github.com/SkBlaz/py3plex.git
```

Using conda

```
# Create conda environment
conda create -n py3plex python=3.10

# Activate
conda activate py3plex

# Install py3plex
pip install git+https://github.com/SkBlaz/py3plex.git
```

7.3.6 Common Installation Issues

Issue: “No module named ‘numpy’”

Solution: Install numpy first:

```
pip install numpy
pip install git+https://github.com/SkBlaz/py3plex.git
```

Issue: Compilation errors on Windows

Solution: Install Visual C++ Build Tools:

1. Download from: <https://visualstudio.microsoft.com/visual-cpp-build-tools/>
2. Install “Desktop development with C++”
3. Retry installation

Issue: “Permission denied” on Linux/macOS

Solution: Use pip install with `--user` flag:

```
pip install --user git+https://github.com/SkBlaz/py3plex.git
```

Or use a virtual environment (recommended).

Issue: Old version installed

Solution: Upgrade to latest version:

```
pip install --upgrade git+https://github.com/SkBlaz/py3plex.git
```

7.3.7 Verifying Installation

Test that py3plex is correctly installed:

```
import py3plex
print(py3plex.__version__)

from py3plex.core import multinet
network = multinet.multi_layer_network()
print("py3plex installed successfully!")
```

Run the test suite:

```
# Clone repository (if not already)
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Run tests
python run_tests.py
```

7.3.8 License Considerations

Main Library License

py3plex core is distributed under the MIT License (permissive, commercial-friendly).

Bundled Code Licenses

Important: The repository contains AGPLv3-licensed code in `py3plex/algorithms/community_detection/infomap/`.

If you use Infomap-based community detection functions, your application may be subject to AGPLv3 requirements (copyleft).

License Matrix

Feature Category	License	Commercial Use	Notes
Core multilayer functionality	MIT	[OK] Yes	Safe for proprietary use
Network visualization	MIT	[OK] Yes	Safe for proprietary use
I/O operations	MIT	[OK] Yes	Safe for proprietary use
Louvain community detection	BSD-3-Clause	[OK] Yes	Safe for proprietary use
Label propagation	MIT	[OK] Yes	Safe for proprietary use
Infomap community detection	AGPLv3	WARNING Restricted	Viral license - requires open-sourcing derived works

Recommendations

- For commercial/proprietary projects: Avoid Infomap functions or use the pure Python `infomap` package separately
- For open-source projects: All features are safe to use
- When in doubt: Use alternative algorithms (Louvain, label propagation) which are BSD/MIT licensed

7.3.9 Next Steps

After installation, proceed to:

- *5-Minute Quickstart* - Quick 5-minute introduction
- *10-Minute Tutorial: Getting Started with py3plex* - Comprehensive 10-minute tutorial
- *Working with Networks* - Detailed usage guide

7.3.10 Getting Help

If you encounter installation issues:

1. Check [GitHub Issues](#)
2. Search for similar problems
3. Open a new issue with:
 - Your operating system and version
 - Python version (`python --version`)
 - Error message and full traceback

- Installation command used

For any errors, please open an issue at <https://github.com/SkBlaz/py3plex/issues>

7.4 Common Issues and Troubleshooting

This guide helps you troubleshoot common problems when getting started with py3plex.

7.4.1 Installation Issues

“No module named ‘numpy’”

Problem: Import error when trying to use py3plex.

Solution: Install numpy first:

```
pip install numpy
pip install git+https://github.com/SkBlaz/py3plex.git
```

Compilation Errors on Windows

Problem: C++ compilation errors during installation on Windows.

Solution: Install Visual C++ Build Tools:

1. Download from: <https://visualstudio.microsoft.com/visual-cpp-build-tools/>
2. Install “Desktop development with C++”
3. Retry installation

```
pip install git+https://github.com/SkBlaz/py3plex.git
```

“Permission denied” on Linux/macOS

Problem: Permission errors during pip install.

Solution 1: Use pip install with `-user` flag:

```
pip install --user git+https://github.com/SkBlaz/py3plex.git
```

Solution 2: Use a virtual environment (recommended):

```
python3 -m venv py3plex-env
source py3plex-env/bin/activate # On Windows: py3plex-env\Scripts\activate
pip install git+https://github.com/SkBlaz/py3plex.git
```

Old Version Installed

Problem: Changes not reflecting or documentation mentions newer features.

Solution: Upgrade to the latest version:

```
pip install --upgrade git+https://github.com/SkBlaz/py3plex.git
```

7.4.2 Data Loading Issues

File Not Found Errors

Problem: `FileNotFoundError` when loading networks.

Solution 1: Use absolute paths:

```
import os

network_path = "/absolute/path/to/data.edgelist"
network = multinet.multi_layer_network().load_network(
    network_path, input_type="edgelist", directed=False
)
```

Solution 2: Ensure you're running from the correct directory:

```
import os

# Check current directory
print("Current directory:", os.getcwd())

# List files
print("Files:", os.listdir('.'))
```

Solution 3: Use paths relative to script location:

```
import os

script_dir = os.path.dirname(os.path.abspath(__file__))
data_path = os.path.join(script_dir, "data", "network.edgelist")
```

Empty or Malformed Networks

Problem: Network loads but has no nodes or edges.

Solution: Check file format matches `input_type`:

```
# For simple edge lists (source target)
network.load_network("data.txt", input_type="edgelist")

# For multilayer edge lists (source target layer)
network.load_network("data.txt", input_type="multiedgelist")

# Always verify after loading
network.basic_stats()
```

7.4.3 Visualization Issues

Visualization Not Showing

Problem: No window appears when calling visualization functions.

Solution 1: For Jupyter notebooks, add at the top:

```
%matplotlib inline
```


Solution 2: For scripts, explicitly show the plot:

```
import matplotlib.pyplot as plt

# Your visualization code
draw_multilayer_default([network], display=False)

# Explicitly show
plt.show()
```

Solution 3: Save to file instead of displaying:

```
from py3plex.visualization.multilayer import hairball_plot
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 10))
hairball_plot(network.core_network, layout_algorithm="force")
plt.savefig("network.png", dpi=150, bbox_inches='tight')
plt.close()
```

“No Display Found” Error

Problem: Error on headless servers or SSH sessions without X11.

Solution: Use Agg backend for matplotlib:

```
import matplotlib
matplotlib.use('Agg') # Must be before importing pyplot
import matplotlib.pyplot as plt

# Now visualization will save to file instead of displaying
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default([network], display=False)
plt.savefig("output.png")
```

Blurry or Low-Quality Plots

Problem: Plots look pixelated or blurry.

Solution: Increase DPI when saving:

```
plt.savefig("network.png", dpi=300, bbox_inches='tight')
```

7.4.4 Missing Dependencies

“No module named ‘plotly’”

Problem: Interactive visualization features not working.

Solution: Install visualization extras:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[viz]
```

“No module named ‘infomap’”

Problem: Infomap community detection not working.

Solution: Install Infomap support:

```
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[infomap]
```

Or use alternative community detection methods:

```
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(network.core_network)
```

7.4.5 Performance Issues

Very Slow Loading

Problem: Loading large networks takes too long.

Solution 1: Use Arrow/Parquet format for large networks:

```
# Install Arrow support
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[arrow]
```

```
from py3plex.io import read, write

# Save in efficient format
write(network, "network.arrow")

# Load much faster
network = read("network.arrow")
```

Solution 2: Process in chunks or use subnetworks:

```
# Extract a single layer for testing
layer_1 = network.subnetwork(['layer1'], subset_by="layers")

# Work with the smaller network
layer_1.basic_stats()
```

Out of Memory Errors

Problem: Python crashes or raises MemoryError with large networks.

Solution: See *Performance and Scalability Best Practices* for:

- Memory-efficient loading strategies
- Batch processing techniques
- Sparse matrix representations
- Subnetwork extraction

7.4.6 Algorithm Issues

Community Detection Returns Trivial Results

Problem: Each node in its own community or all nodes in one community.

Solution: Adjust algorithm parameters:

```
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)

# Try different resolution parameters
partition = louvain_multilayer(
    network,
    gamma=0.5,      # Lower = fewer, larger communities
    omega=1.5,      # Higher = more layer consistency
    random_state=42
)
```

Random Walk or Embedding Errors

Problem: Random walk or Node2Vec functions fail.

Solution: Ensure network is connected:

```
import networkx as nx

# Check connectivity
if not nx.is_connected(network.core_network.to_undirected()):
    print("Warning: Network is not connected")

# Get largest component
largest = max(nx.connected_components(
    network.core_network.to_undirected()
), key=len)

# Create subgraph
subgraph = network.core_network.subgraph(largest)
```

7.4.7 Docker Issues

Docker Build Fails

Problem: Docker build fails or takes very long.

Solution: Clear Docker cache and rebuild:

```
docker system prune -a
docker build --no-cache -t py3plex:latest .
```

Cannot Access Files in Docker

Problem: Docker container can't see your data files.

Solution: Mount volumes correctly:

```
# Mount current directory
docker run -v $(pwd):/data py3plex:latest /data/network.edgelist

# On Windows (PowerShell)
docker run -v ${PWD}:/data py3plex:latest /data/network.edgelist
```

See *Docker Usage Guide* for complete Docker documentation.

7.4.8 Getting Help

If you encounter an issue not covered here:

1. **Search GitHub Issues:** <https://github.com/SkBlaz/py3plex/issues>
2. **Check Documentation:** <https://skblaz.github.io/py3plex/>
3. **Browse Examples:** <https://github.com/SkBlaz/py3plex/tree/master/examples>

When reporting issues, include: * Python version (`python --version`) * py3plex version (`pip show py3plex`) * Operating system * Full error traceback * Minimal code to reproduce the problem

7.4.9 Related Pages

- *Installation and Setup* - Complete installation guide
- *5-Minute Quickstart* - Quick introduction
- *10-Minute Tutorial: Getting Started with py3plex* - Comprehensive tutorial
- *Performance and Scalability Best Practices* - Performance optimization

7.5 Multilayer Networks 101

This guide explains what multilayer networks are and why they matter for modern network analysis.

7.5.1 What are Multilayer Networks?

A **multilayer network** is a complex network structure that goes beyond traditional single-layer graphs by incorporating multiple types of relationships, node types, or interaction contexts. They model the reality that most real-world systems involve multiple, interconnected types of relationships.

Traditional vs. Multilayer

Traditional (Single-Layer) Networks:

- One type of node (e.g., only people)
- One type of edge (e.g., only friendship)
- Homogeneous structure
- Limited ability to model complex systems

Multilayer Networks:

- Multiple node types (e.g., people, organizations, documents)
- Multiple edge types (e.g., friendship, collaboration, citation)
- Multiple layers of interaction

- Inter-layer and intra-layer connections
- Rich, heterogeneous structure

7.5.2 Types of Multilayer Networks

py3plex supports several common multilayer network paradigms:

Multiplex Networks

Definition: Multiple layers with the **same set of nodes** but different types of edges.

Characteristics:

- Node set is identical across layers
- Each layer represents a different relationship type
- Inter-layer edges typically connect the same node across layers

Examples:

- **Social networks:** The same people connected via friendship, colleague, and family relationships
- **Transportation:** Cities connected via air, rail, and road networks
- **Communication:** Users interacting via email, phone, and instant messaging

Visual representation:

Layer 1 (Friends):	Layer 2 (Colleagues):
A --- B --- C	A --- B
\	
\	D
D	

Code example:

```
from py3plex.core import multinet

network = multinet.multi_layer_network(network_type="multiplex")

# Same nodes, different relationship types
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1],
], input_type="list")
```

Heterogeneous Networks (HINs)

Definition: Networks with **different node types** and type-specific relationships.

Characteristics:

- Multiple node types (e.g., authors, papers, venues)
- Edges connect nodes of specific types
- Meta-paths describe relationship sequences

Examples:

- **Academic networks:** Authors write papers published in venues

- **E-commerce:** Users purchase products from sellers
- **Biomedical:** Drugs treat diseases via molecular targets

Visual representation:

```
Authors      Papers      Venues
[Alice] ----> [P1] -----> [ICML]
  |             ^             ^
  |             |             |
+-----> [P2] -----+
[Bob]
```

Code example:

```
network = multinet.multi_layer_network()

# Different node types on different layers
network.add_edges([
    ['Alice', 'authors', 'Paper1', 'papers', 1],
    ['Paper1', 'papers', 'ICML', 'venues', 1],
], input_type="list")
```

Temporal Networks

Definition: Networks that **evolve over time**, with time-sliced layers.

Characteristics:

- Each layer represents a time period
- Nodes may appear/disappear over time
- Edges show relationships at specific times
- Inter-layer edges connect temporal states

Examples:

- **Communication networks:** Who-contacts-whom over different time periods
- **Social dynamics:** Friendship evolution over years
- **Disease spread:** Contact networks during epidemic progression

Visual representation:

```
t=1: A --- B --- C
      |
      D

t=2: A --- B --- C --- E
      |       |
      D -----+

t=3: A --- B --- C --- E
              |
              D
```

Code example:

```

network = multinet.multi_layer_network()

# Different time periods as layers
network.add_edges([
    ['A', 't1', 'B', 't1', 1],
    ['A', 't2', 'B', 't2', 1],
    ['B', 't2', 'D', 't2', 1],
], input_type="list")

```

Interdependent Networks

Definition: Multiple networks where **nodes in one network depend on nodes in another**.

Characteristics:

- Networks with distinct functions
- Dependencies between networks
- Cascading failures possible
- Critical infrastructure modeling

Examples:

- **Infrastructure:** Power grid depends on communication network
- **Supply chain:** Manufacturing depends on logistics network
- **Cyber-physical systems:** Software layer depends on hardware layer

Visual representation:

Power Grid:	Communication Net:
P1 --- P2 --- P3	C1 --- C2 --- C3
Depends on	C4
v	v
C1 --- C2	P1 --- P2

7.5.3 When to Use Multilayer Networks

Use multilayer networks when:

1. Multiple Relationship Types Matter

If your system has multiple types of connections that interact, a multilayer model is essential.

Example: Studying information diffusion in social networks where both online and offline relationships matter.

2. Node Roles Vary by Context

When nodes play different roles in different contexts or layers.

Example: A person might be central in their work network but peripheral in their hobby network.

3. Layer Interactions Are Important

When understanding how layers interact is crucial to your analysis.

Example: How transportation failures in one mode (air) affect alternatives (rail, road).

4. Temporal Evolution Matters

When the timing and sequence of relationships are important.

Example: Understanding how community structure evolves over time in a social network.

5. System-Level Properties Emerge

When whole-system properties can't be understood by analyzing layers independently.

Example: Resilience of infrastructure systems depends on cross-layer dependencies.

7.5.4 Comparison with Single-Layer Approaches

Why Not Just Aggregate?

You might ask: "Why not just combine all layers into one network?"

Problems with aggregation:

1. **Loss of Information:** Different relationship types get conflated
2. **Misleading Metrics:** Centrality computed on aggregated network can be wrong
3. **Hidden Patterns:** Layer-specific patterns are lost
4. **Incorrect Analysis:** Communities and pathways depend on layer structure

Example:

```
# DON'T: Naive aggregation loses information
all_edges = []
all_edges.extend(friends_edges)
all_edges.extend(colleague_edges)
single_layer = nx.Graph(all_edges) # Information about edge types lost!

# DO: Use multilayer representation
network = multinet.multi_layer_network()
network.add_edges(friends_edges + colleague_edges, input_type="list")
```

Why Not Analyze Each Layer Separately?

You might ask: "Why not just analyze each layer independently?"

Problems with separate analysis:

1. **Ignores Cross-Layer Effects:** Nodes may be important because of multi-layer presence
2. **Misses Inter-Layer Patterns:** Communities often span multiple layers
3. **Incomplete View:** Node importance depends on cross-layer activity
4. **Lost Synergies:** Complementary information across layers is ignored

Example: A person who is moderately connected in multiple social contexts may be more influential than someone highly connected in just one context.

7.5.5 Key Concepts

Intra-Layer Edges

Edges **within** a single layer connecting nodes in that layer.

Example: Friendships within the "friends" layer.

Inter-Layer Edges

Edges **between** layers, typically connecting the same node across layers or different nodes in different layers.

Example: Connecting Alice in the “friends” layer to Alice in the “colleagues” layer.

Node-Layer Pairs

In py3plex, nodes are represented as (node_id, layer_id) tuples.

Example: ('Alice', 'friends') and ('Alice', 'colleagues') are different node-layer pairs.

Supra-Adjacency Matrix

A matrix representation that stacks layer adjacency matrices into a block structure, encoding both intra-layer and inter-layer connections.

See *py3plex Core Model* for implementation details.

7.5.6 Real-World Applications

py3plex has been used for analyzing:

Biological Networks:

- Protein-protein interactions with multiple evidence types
- Gene regulatory networks with transcription, translation, and metabolic layers
- Brain networks with structural and functional connectivity

Social Networks:

- Multi-platform social media analysis (Twitter + Facebook + Instagram)
- Relationship type analysis (friends, family, colleagues)
- Online-offline integration studies

Citation Networks:

- Author-paper-venue multilayer structures
- Knowledge graphs with entities, relationships, and contexts
- Research collaboration networks

Transportation:

- Multi-modal networks (bus, train, metro, air)
- Urban mobility analysis
- Resilience and failure analysis

Infrastructure:

- Critical infrastructure interdependencies
- Smart city systems
- Cyber-physical systems

7.5.7 Further Reading

- *py3plex Core Model* - How py3plex represents multilayer networks internally
- *Design Principles* - Design philosophy and API principles
- *Algorithm Landscape* - Overview of multilayer algorithms
- *Working with Networks* - Creating and loading multilayer networks in code

Academic References:

- Kivelä et al. (2014). “Multilayer networks.” *Journal of Complex Networks* 2(3): 203-271.
- Boccaletti et al. (2014). “The structure and dynamics of multilayer networks.” *Physics Reports* 544(1): 1-122.
- De Domenico et al. (2013). “Mathematical formulation of multilayer networks.” *Physical Review X* 3(4): 041022.

7.6 py3plex Core Model

This guide explains how py3plex represents multilayer networks internally and how to work with the core data structures.

7.6.1 The `multi_layer_network` Class

The central data structure in py3plex is the `multi_layer_network` class, which provides a high-level API for working with multilayer networks while maintaining full compatibility with NetworkX.

Basic Structure

```
from py3plex.core import multinet

# Create a multilayer network
network = multinet.multi_layer_network()

# The underlying NetworkX graph
nx_graph = network.core_network # MultiDiGraph or MultiGraph

# Layer management
layers = network.get_layers() # List of layer names
layer_map = network.layer_name_map # Layer name to ID mapping
```

Key Components

1. Core NetworkX Graph

At its heart, py3plex uses a NetworkX `MultiDiGraph` (for directed networks) or `MultiGraph` (for undirected networks) to store all nodes and edges.

```
# Access the underlying graph
G = network.core_network

# Use any NetworkX function
import networkx as nx
betweenness = nx.betweenness centrality(G)
```

2. Layer Encoding

Nodes are internally encoded as (node_id, layer_id) tuples to distinguish the same logical node across different layers.

```
# Node 'Alice' in layer 'friends' is stored as ('Alice', 'friends')
network.add_nodes([('Alice', 'friends')])

# Access all node-layer pairs
for node_layer_pair in network.get_nodes():
    print(node_layer_pair)  # ('Alice', 'friends'), ('Bob', 'friends'), ...
```

3. Layer Mapping

py3plex maintains bidirectional mappings between layer names and internal layer IDs for efficient lookups.

```
# Get layer mapping
layer_map = network.layer_name_map
# {'friends': 0, 'colleagues': 1, ...}
```

4. Attributes

Node and edge attributes are stored directly in the underlying NetworkX graph and can be accessed and modified using standard NetworkX patterns.

```
# Add node with attributes
network.core_network.nodes[('Alice', 'friends')]['age'] = 30

# Add edge with attributes
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), {'weight': 0.8, 'type': 'close'}]
])
```

7.6.2 Internal Representation

Node-Layer Pairs

The key concept in py3plex's internal model is the **node-layer pair**: a tuple (node_id, layer_id) that uniquely identifies a node in a specific layer.

```
# Same logical node, different layers
alice_friends = ('Alice', 'friends')
alice_colleagues = ('Alice', 'colleagues')

# These are treated as different nodes internally
network.add_nodes([alice_friends, alice_colleagues])
```

This representation allows:

- The same logical entity to appear in multiple layers
- Different attributes per node-layer pair
- Efficient layer-specific operations

Edge Types

Intra-Layer Edges

Edges connecting nodes within the same layer:

```
# Edge within 'friends' layer
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), 1.0]
])
```

Inter-Layer Edges

Edges connecting nodes across different layers:

```
# Connect Alice across layers (identity edge)
network.add_edges([
    [('Alice', 'friends'), ('Alice', 'colleagues'), 1.0]
])

# Connect different nodes across layers
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'colleagues'), 0.5]
])
```

7.6.3 Supra-Adjacency Matrix

The supra-adjacency matrix is a block matrix representation that encodes the entire multilayer network structure.

Mathematical Definition

For a multilayer network with L layers, the supra-adjacency matrix $\mathbf{S}$ is:

$$\mathbf{S} = \begin{bmatrix} A_1 & C_{12} & \cdots & C_{1L} \\ C_{21} & A_2 & \cdots & C_{2L} \\ \vdots & \vdots & \ddots & \vdots \\ C_{L1} & C_{L2} & \cdots & A_L \end{bmatrix}$$

Where:

- A_{α} is the adjacency matrix of layer α (intra-layer connections)
- $C_{\alpha\beta}$ is the coupling matrix between layers α and β (inter-layer connections)

Construction

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
# ... add nodes and edges ...

# Get supra-adjacency matrix (sparse by default for efficiency)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Dense version (for small networks only)
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)

# Shape: (total_nodes, total_nodes)
# where total_nodes = sum of nodes across all layers
print(f"Shape: {supra_adj.shape}")
```

Visualization

Visualize the supra-adjacency matrix structure:

```
from py3plex.core import multinet, random_generators
import matplotlib.pyplot as plt

# Generate a small multilayer network
network = random_generators.random_multilayer_ER(
    num_nodes=50, num_layers=3, probability=0.1, directed=False
)

# Visualize the supra-adjacency matrix
network.visualize_matrix({"display": True})
```

The visualization shows the block structure with:

- Diagonal blocks: intra-layer connections
- Off-diagonal blocks: inter-layer connections

7.6.4 Network Construction

Creating Networks from Scratch

Method 1: Add edges directly

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add edges in list format: [source_node, source_layer, target_node, target_layer,
#                             ↪weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1.0],
], input_type="list")
```

Method 2: Add nodes and edges separately

```
network = multinet.multi_layer_network()

# Add nodes explicitly
network.add_nodes([
    ('Alice', 'friends'),
    ('Bob', 'friends'),
    ('Alice', 'colleagues'),
    ('Bob', 'colleagues')
])

# Add edges between existing nodes
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), 1.0],
    [('Alice', 'colleagues'), ('Bob', 'colleagues'), 1.0]
])
```

Method 3: Define layers first

```

network = multinet.multi_layer_network()

# Define layers explicitly
network.add_layer('friends')
network.add_layer('colleagues')

# Add content (nodes are created automatically with edges)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0]
], input_type="list")

```

Loading from Files

py3plex supports multiple input formats. See *I/O and Serialization* for complete documentation.

```

from py3plex.core import multinet

# From simple edge list (source target)
network = multinet.multi_layer_network().load_network(
    "data.edgelist",
    input_type="edgelist",
    directed=False
)

# From multilayer edge list (source target layer)
network = multinet.multi_layer_network().load_network(
    "data.multiedgelist",
    input_type="multiedgelist",
    directed=False
)

# From GraphML
network = multinet.multi_layer_network().load_network(
    "data.graphml",
    input_type="graphml"
)

# From NetworkX graph
import networkx as nx
G = nx.karate_club_graph()
network = multinet.multi_layer_network()
network.load_network_from_networkx(G)

```

7.6.5 Network Operations

Querying Network Elements

```

# Get all node-layer pairs
nodes = network.get_nodes(data=True) # With attributes
nodes = network.get_nodes(data=False) # Without attributes

```

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```
# Get nodes in a specific layer
layer_nodes = network.get_nodes(layer='friends')

# Get all edges
edges = network.get_edges(data=True)

# Get neighbors of a node in a layer
neighbors = list(network.get_neighbors('Alice', layer_id='friends'))

# Get all layers
layers = network.get_layers()
print(f"Layers: {layers}")
```

Basic Statistics

```
# Get basic network statistics
network.basic_stats()

# Output:
# Number of nodes: X
# Number of edges: Y
# Number of unique nodes (as node-layer tuples): Z
# Number of unique node IDs (across all layers): W
# Nodes per layer:
#   Layer 'friends': N1 nodes
#   Layer 'colleagues': N2 nodes
```

Subnetwork Extraction

```
# Extract a single layer
layer_1 = network.subnetwork(['friends'], subset_by="layers")

# Extract specific nodes (all their appearances)
node_subset = network.subnetwork(['Alice', 'Bob'], subset_by="node_names")

# Extract specific node-layer pairs
specific_pairs = network.subnetwork(
    [('Alice', 'friends'), ('Bob', 'friends')],
    subset_by="node_layer_names"
)
```

Matrix Representations

```
# Get supra-adjacency matrix (all layers stacked)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Get adjacency matrix for a single layer
layer_subgraph = network.subnetwork(['friends'], subset_by="layers")
layer_adj = nx.adjacency_matrix(layer_subgraph.core_network)
```


7.6.6 Integration with NetworkX

Full NetworkX Compatibility

Since py3plex builds on NetworkX, you can use any NetworkX algorithm:

```
import networkx as nx
from py3plex.core import multinet

# Create py3plex network
mlnet = multinet.multi_layer_network()
mlnet.add_edges([
    ['A', 'L1', 'B', 'L1', 1],
    ['B', 'L1', 'C', 'L1', 1],
], input_type="list")

# Access underlying NetworkX graph
G = mlnet.core_network

# Use any NetworkX function
betweenness = nx.betweenness centrality(G)
print(f"Betweenness centrality: {betweenness}")

# Shortest paths
try:
    path = nx.shortest_path(G, ('A', 'L1'), ('C', 'L1'))
    print(f"Shortest path: {path}")
except nx.NetworkXNoPath:
    print("No path exists")

# Connected components
if not G.is_directed():
    components = list(nx.connected_components(G))
    print(f"Number of components: {len(components)}")
```

NetworkX Wrappers

py3plex provides convenience wrappers for common NetworkX functions:

```
# Extract a layer
layer_1 = network.subnetwork(['layer1'], subset_by="layers")

# Apply NetworkX algorithms via wrapper
degree centrality = layer_1.monoplex_nx_wrapper("degree centrality")
betweenness = layer_1.monoplex_nx_wrapper("betweenness centrality")

# Results are dictionaries mapping node-layer pairs to values
print(f"Degree centrality: {degree centrality}")
```

Converting Between Formats

```
# From NetworkX to py3plex
import networkx as nx
from py3plex.core import multinet
```

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```

G = nx.karate_club_graph()
mlnet = multinet.multi_layer_network()
# NetworkX graphs are imported as single-layer networks
mlnet.load_network_from_networkx(G)

# From py3plex to NetworkX (already a NetworkX graph!)
G_export = mlnet.core_network

# Save as standard NetworkX formats
nx.write_graphml(G_export, "output.graphml")
nx.write_gml(G_export, "output.gml")

```

7.6.7 Performance Considerations

Sparse vs. Dense Matrices

For large networks, always use sparse matrix representations:

```

# Good: Sparse matrix (efficient for large networks)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Bad: Dense matrix (memory-intensive)
# Only use for small networks (< 1000 nodes)
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)

```

Memory Usage

Node-layer pairs mean that a node appearing in L layers occupies L times the memory of a single-layer network. For networks with many layers:

- Consider layer-specific analysis when possible
- Use subnetwork extraction to work with smaller portions
- See *Performance and Scalability Best Practices* for optimization strategies

Tensor-Like Indexing

You can access nodes and edges using tensor-like notation:

```

from py3plex.core import multinet, random_generators

# Create a network
network = random_generators.random_multilayer_ER(100, 3, 0.05, directed=False)

# Get some nodes and edges
some_nodes = list(network.get_nodes())[0:5]
some_edges = list(network.get_edges())[0:5]

# Access node directly (returns node attributes)
node_data = network[some_nodes[0]]
print(f"Node {some_nodes[0]} data: {node_data}")

```

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```
# Access edge (returns edge attributes)
edge_data = network[some_edges[0][0]][some_edges[0][1]]
print(f"Edge data: {edge_data}")
```

7.6.8 Design Principles

py3plex follows several key design principles:

1. NetworkX Compatibility

All operations maintain compatibility with NetworkX, allowing seamless use of the rich NetworkX ecosystem.

2. Lazy Evaluation

Expensive operations like supra-adjacency matrix construction are computed on demand and cached when appropriate.

3. Graceful Degradation

Optional features (like Infomap community detection or advanced visualizations) fail gracefully with informative error messages if dependencies are missing.

4. Extensibility

The modular design makes it easy to extend py3plex with custom algorithms and visualizations.

5. Flexibility

Multiple ways to construct and manipulate networks to suit different workflows and use cases.

7.6.9 Comparison with Other Representations

Why Node-Layer Pairs?

Alternative representations exist, but py3plex's node-layer pair approach offers key advantages:

Alternative: Separate graphs per layer

Drawback: Loses inter-layer information, requires manual bookkeeping for cross-layer operations.

Alternative: Single graph with edge type attributes

Drawback: Cannot distinguish node context across layers, loses layer-specific node attributes.

py3plex approach: Node-layer pairs

Advantages:

- Preserves both intra-layer and inter-layer structure
- Allows layer-specific node attributes
- Compatible with NetworkX algorithms
- Efficient for multilayer-specific operations

7.6.10 Next Steps

- *Working with Networks* - Creating and manipulating networks in practice
- *Network Statistics* - Computing multilayer network statistics
- *Design Principles* - High-level design philosophy
- *Algorithm Landscape* - Overview of available algorithms

- *I/O and Serialization* - Complete I/O documentation
- *API Documentation* - Full API reference

Academic References:

- De Domenico et al. (2013). “Mathematical formulation of multilayer networks.” *Physical Review X* 3(4): 041022.
- Kivelä et al. (2014). “Multilayer networks.” *Journal of Complex Networks* 2(3): 203-271.

7.7 Design Principles

py3plex is built on a foundation of key design principles that guide its development and usage. Understanding these principles helps you use the library effectively and extend it for your needs.

7.7.1 Core Philosophy

Simple, Off-the-Shelf Functionality

Principle: Provide ready-to-use tools that work out of the box with minimal configuration.

In practice:

```
from py3plex.core import multinet

# Simple, intuitive API
network = multinet.multi_layer_network()
network.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type="list")
network.basic_stats() # Works immediately
```

Rationale: Researchers and practitioners need tools that are easy to adopt without extensive setup or configuration. py3plex aims to minimize the barrier to entry for multilayer network analysis.

NetworkX Compatibility

Principle: Build on NetworkX rather than reinventing the wheel.

In practice:

```
import networkx as nx

# Direct access to NetworkX graph
G = network.core_network

# Use any NetworkX function
betweenness = nx.betweenness_centrality(G)
communities = nx.community.louvain_communities(G)
```

Rationale: NetworkX is the de facto standard for network analysis in Python. By building on it, py3plex:

- Leverages a mature, well-tested codebase
- Provides access to hundreds of NetworkX algorithms
- Ensures interoperability with the broader Python ecosystem
- Reduces learning curve for users familiar with NetworkX

Modular Architecture

Principle: Separate concerns into independent, composable modules.

Structure:

```
py3plex/
├── core/                # Data structures
├── algorithms/          # Analysis methods
│   ├── community_detection/
│   ├── statistics/
│   └── multilayer_algorithms/
├── visualization/      # Plotting
└── wrappers/           # High-level interfaces
```

In practice:

```
# Import only what you need
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.visualization.multilayer import hairball_plot
```

Rationale: Modular design allows users to:

- Import only the functionality they need
- Understand and maintain code more easily
- Extend the library with custom modules
- Mix and match components flexibly

7.7.2 Implementation Principles

Flexibility

Principle: Support multiple ways to accomplish tasks, accommodating different workflows.

Examples:

```
# Multiple ways to create networks

# Method 1: Direct edge addition
network.add_edges([['A', 'L1', 'B', 'L1', 1]], input_type="list")

# Method 2: Load from file
network.load_network("data.edgelist", input_type="edgelist")

# Method 3: From NetworkX
network.load_network_from_networkx(nx_graph)

# Multiple ways to access nodes
nodes1 = network.get_nodes()
nodes2 = list(network.core_network.nodes())
nodes3 = network.get_nodes(layer='L1')
```

Rationale: Different users have different preferences and workflows. Flexibility ensures py3plex adapts to the user's needs, not vice versa.

Lazy Evaluation

Principle: Compute expensive operations only when needed.

In practice:

```
# Supra-adjacency matrix is not computed on network creation
network = multinet.multi_layer_network()
network.add_edges([...]) # Fast

# Matrix is computed only when requested
supra_adj = network.get_supra_adjacency_matrix() # Computed here

# Subsequent calls may use cached result (where appropriate)
```

Rationale: Many multilayer operations (e.g., matrix construction) are computationally expensive. Lazy evaluation:

- Improves performance for common operations
- Reduces memory usage
- Allows working with large networks when full matrices aren't needed

Graceful Degradation

Principle: Optional features should fail gracefully, not crash the entire application.

In practice:

```
# If Infomap is not installed
try:
    communities = infomap_communities(network)
except ImportError as e:
    print("Infomap not available. Using Louvain instead.")
    communities = louvain_partition(network)
```

Rationale: Not all users need all features. By degrading gracefully, py3plex:

- Allows users to install only what they need
- Reduces dependency bloat
- Provides helpful error messages
- Suggests alternatives when dependencies are missing

Explicit Over Implicit

Principle: Make behavior explicit and predictable, avoiding surprising defaults.

Good examples:

```
# Explicit parameter names
network.load_network(
    "data.txt",
    input_type="edgelist",      # Explicit format
    directed=False             # Explicit directionality
)
```

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```
# Explicit layer specification
neighbors = network.get_neighbors('Alice', layer_id='friends')
```

Rationale: Explicit code is:

- Easier to understand and debug
- Less prone to errors
- Self-documenting
- More maintainable

7.7.3 Performance Considerations

Sparse Representations

Principle: Use sparse data structures for large networks.

In practice:

```
# Sparse matrix by default (efficient for large networks)
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Dense only for small networks or specific algorithms
supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)
```

Rationale: Real-world networks are typically sparse (few edges relative to potential edges). Sparse representations:

- Reduce memory usage by orders of magnitude
- Enable analysis of networks with millions of nodes
- Maintain computational efficiency

Efficient Data Structures

Principle: Choose appropriate data structures for common operations.

Examples:

- NetworkX graphs for topology (edge lookups, traversals)
- Dictionaries for layer mappings (O(1) lookups)
- Sparse matrices for linear algebra
- Sets for membership testing

Rationale: The right data structure makes operations fast and memory-efficient.

Algorithmic Defaults

Principle: Provide sensible default parameters based on empirical research.

In practice:

```
# Default parameters work well for most cases
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Standard resolution
```

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```
    omega=1.0,      # Balanced inter-layer coupling
    random_state=42 # Reproducibility
)
```

Rationale: Good defaults:

- Make the library easy to use for beginners
- Reduce parameter tuning overhead
- Are based on published research and empirical validation

7.7.4 Extensibility

Plugin-Friendly Architecture

Principle: Make it easy to add custom functionality.

In practice:

```
# Easy to add custom algorithms
def my_centrality(ml_network):
    """Custom centrality measure."""
    G = ml_network.core_network
    # Implement your algorithm
    return centrality_scores

# Easy to add custom visualizations
def my_plot(ml_network, **kwargs):
    """Custom visualization."""
    # Use py3plex drawing machinery
    from py3plex.visualization import drawing_machinery as dm
    # Implement your plot
    pass
```

Rationale: Research needs evolve. Extensibility ensures py3plex can grow with new methods without requiring core library changes.

Clear Interfaces

Principle: Define clear, stable APIs for modules.

In practice:

```
# All statistics follow same interface
from py3plex.algorithms.statistics import multilayer_statistics as mls

density = mls.layer_density(network, layer)
activity = mls.node_activity(network, node)
overlap = mls.edge_overlap(network, layer1, layer2)

# Consistent parameter names and return types
```

Rationale: Clear interfaces make py3plex:

- Easier to learn (patterns repeat)

- More predictable
- More maintainable
- Easier to extend

7.7.5 Documentation and Testing

Comprehensive Documentation

Principle: Every feature should be documented with examples.

In practice:

- Detailed docstrings for all public functions
- Code examples in documentation
- Real-world use cases
- API reference with parameter descriptions

Rationale: Good documentation:

- Reduces barrier to entry
- Serves as a reference for experienced users
- Helps maintainers understand code
- Acts as specification

Extensive Testing

Principle: Test all core functionality and edge cases.

In practice:

```
# Comprehensive test suite
make test          # Unit tests
make benchmark     # Performance tests
make fuzz-property # Property-based tests
```

Rationale: Tests ensure:

- Code works as intended
- Changes don't break existing functionality
- Edge cases are handled correctly
- Performance regressions are caught

7.7.6 Interoperability

Standard Formats

Principle: Support widely-used data formats.

Formats supported:

- EdgeList (simple, human-readable)
- GraphML (XML-based, preserves attributes)
- GML (Graph Modeling Language)

- Pickle (NetworkX native)
- JSON/Arrow/Parquet (modern, efficient)

Rationale: Standard formats enable:

- Data sharing between tools
- Integration with other software
- Long-term data preservation

Cross-Platform Compatibility

Principle: Work consistently across operating systems.

Tested on:

- Linux (Ubuntu 20.04, 22.04)
- macOS (11, 12, 13)
- Windows (Server 2019, 2022)

Rationale: Users work on different platforms. Cross-platform support ensures py3plex works everywhere.

7.7.7 Research-Oriented Design

Citation and Attribution

Principle: Make it easy to cite algorithms and give proper attribution.

In practice:

```
# Algorithms include citations in docstrings
help(louvain_multilayer)
# Docstring includes:
# References:
# Mucha et al. (2010). "Community structure in time-dependent..."
```

See: *Citation and References*

Rationale: Proper attribution:

- Gives credit to algorithm developers
- Helps users find original papers
- Supports reproducibility

Reproducibility

Principle: Support reproducible research.

In practice:

```
# Random seed parameters for reproducibility
partition = louvain_multilayer(
    network,
    random_state=42 # Reproducible results
)

walks = generate_walks(
```

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```
network,
seed=42 # Reproducible walks
)
```

Rationale: Reproducibility is fundamental to science. Seed parameters ensure:

- Results can be verified
- Comparisons are fair
- Bugs can be diagnosed

Validation and Benchmarking

Principle: Validate algorithms against published results.

In practice:

- Benchmark suite comparing to reference implementations
- Validation against known network properties
- Performance metrics for large networks

Rationale: Validation ensures:

- Implementations are correct
- Performance is acceptable
- Results match published literature

7.7.8 Practical Implications

For Users

These design principles mean you can:

1. **Start quickly** with minimal setup
2. **Use NetworkX knowledge** and tools directly
3. **Choose your workflow** with multiple approaches
4. **Work efficiently** with large networks
5. **Extend easily** with custom algorithms
6. **Trust results** through validation and testing

For Contributors

When contributing to py3plex, follow these principles:

1. **Maintain NetworkX compatibility** in new features
2. **Document thoroughly** with examples
3. **Test extensively** including edge cases
4. **Provide sensible defaults** based on research
5. **Fail gracefully** with helpful error messages
6. **Follow existing patterns** for consistency

See *Contributing to py3plex* for detailed guidelines.

7.7.9 Comparison with Other Libraries

py3plex vs. Pure NetworkX

NetworkX:

- Excellent for single-layer networks
- Lacks native multilayer support
- General-purpose graph library

py3plex:

- Native multilayer data structures
- Layer-aware algorithms
- Specialized multilayer visualizations
- Built on NetworkX (inherits all features)

py3plex vs. Other Multilayer Tools

Other tools (e.g., muxViz, pymnet):

- Often specialized or platform-specific
- May have steep learning curves
- Limited algorithm libraries

py3plex:

- Pure Python (easy installation)
- Comprehensive algorithm library
- NetworkX integration
- Active development
- Research-backed

7.7.10 Related Documentation

- *Multilayer Networks 101* - Conceptual introduction to multilayer networks
- *py3plex Core Model* - Internal representation details
- *Algorithm Landscape* - Overview of available algorithms
- *Development Guide* - Contributing to py3plex
- *Architecture and Design* - Internal architecture details

Academic References:

- Škrlj et al. (2019). “Py3plex toolkit for visualization and analysis of multilayer networks.” *Applied Network Science* 4(1): 94.

7.8 Algorithm Landscape

This guide provides a narrative overview of the algorithm families available in py3plex, helping you understand when to use each type of algorithm for multilayer network analysis.

7.8.1 Overview

py3plex provides algorithms in seven main categories:

1. **Community Detection** - Finding groups of densely connected nodes
2. **Centrality & Importance** - Measuring node and edge importance
3. **Network Statistics** - Computing network-level and layer-specific metrics
4. **Random Walks & Embeddings** - Node representations and sampling
5. **Node Ranking & Classification** - Ordering and classifying nodes
6. **Visualization & Layout** - Rendering and spatial arrangement
7. **Benchmarking & Generation** - Testing and synthetic network creation

For detailed parameter lists and API reference, see [Algorithm Roadmap](#).

7.8.2 Community Detection

Goal: Identify groups of nodes that are more densely connected to each other than to the rest of the network.

When to Use

Use community detection when you want to:

- Discover functional modules in biological networks
- Find social groups in multilayer social networks
- Identify topical clusters in citation networks
- Understand organizational structure
- Detect anomalies (nodes that don't fit any community)

Algorithm Families

Modularity-Based Methods

Best for: Fast community detection on large networks

Examples:

- **Louvain** - Fast, greedy modularity optimization
- **Leiden** - Improved version of Louvain with better stability
- **Multilayer Louvain** - Extends Louvain to multiple layers with inter-layer coupling

Trade-offs:

- ✓ Fast (scales to millions of nodes)
- ✓ Easy to use (minimal parameters)
- Resolution limit (may miss small communities)
- Stochastic (different runs give different results)

```
from py3plex.algorithms.community_detection import community_louvain

# Single-layer Louvain
partition = community_louvain.best_partition(network.core_network)

# Multilayer Louvain
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)
partition = louvain_multilayer(network, gamma=1.0, omega=1.0)
```

Flow-Based Methods

Best for: Finding communities based on information flow

Examples:

- **Infomap** - Minimizes description length of random walks
- **Walktrap** - Based on random walk distances

Trade-offs:

- ✓ Theoretically grounded (information theory)
- ✓ Finds hierarchical structure
- Slower than modularity methods
- May require external binaries

```
from py3plex.algorithms.community_detection import community_wrapper

# Infomap (requires binary)
partition = community_wrapper.infomap_communities(
    network.core_network,
    binary_path="/path/to/infomap"
)
```

Overlapping Community Detection

Best for: Networks where nodes belong to multiple communities

Examples:

- **NoRC** - Network of Ranked Communities
- **Clique Percolation** - Based on overlapping cliques

Trade-offs:

- ✓ More realistic for many real networks
- ✓ Captures multi-role nodes
- More complex to interpret
- Computationally expensive

```
from py3plex.algorithms.community_detection import community_wrapper
```

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```
# NoRC overlapping communities
partition = community_wrapper.NoRC_communities(network, alpha=0.5)
```

Choosing a Community Detection Method

Use this decision tree:

```
Do you need overlapping communities?
├ Yes → NoRC or Clique Percolation
└ No
    └ Is your network multilayer?
        ├── Yes → Multilayer Louvain or Leiden
        └ No
            └ What's most important?
                ├── Speed → Louvain
                ├── Stability → Leiden
                └ Theory → Infomap
```

7.8.3 Centrality & Importance

Goal: Quantify the importance of nodes or edges in the network.

When to Use

Use centrality measures when you want to:

- Identify influential nodes (e.g., key proteins, opinion leaders)
- Find bottlenecks in flow networks
- Prioritize nodes for intervention or study
- Understand network vulnerability
- Analyze node roles across layers

Algorithm Families

Degree-Based Centrality

Measures: Local connectivity

Examples:

- **Degree Centrality** - Number of connections
- **Multilayer Degree** - Degree summed across layers
- **Versatility** - Participation across layers

When to use:

- Quick assessment of connectivity
- Identifying hubs
- Local importance measures

```
import networkx as nx

# Single-layer degree centrality
degree_cent = nx.degree_centrality(network.core_network)

# Multilayer versatility
from py3plex.algorithms.statistics import multilayer_statistics as mls
versatility = mls.versatility_centrality(network, centrality_type='degree')
```

Distance-Based Centrality

Measures: Global position in network

Examples:

- **Closeness Centrality** - Average distance to all other nodes
- **Betweenness Centrality** - Fraction of shortest paths through node
- **Harmonic Centrality** - Harmonic mean of distances

When to use:

- Identifying central nodes
- Finding information brokers
- Understanding information flow

```
# Betweenness centrality
betweenness = nx.betweenness_centrality(network.core_network)

# Closeness centrality
closeness = nx.closeness_centrality(network.core_network)
```

Eigenvector-Based Centrality

Measures: Importance based on connections to important nodes

Examples:

- **Eigenvector Centrality** - First eigenvector of adjacency matrix
- **PageRank** - Google's ranking algorithm
- **HITS** - Hubs and authorities

When to use:

- Ranking web pages or citations
- Finding influential nodes in social networks
- Authority/hub analysis

```
# PageRank
pagerank = nx.pagerank(network.core_network)

# Eigenvector centrality
eigenvector = nx.eigenvector_centrality(network.core_network)
```


Choosing a Centrality Measure

Consider these questions:

1. **Local vs. Global?** - Local → Degree centrality - Global → Closeness or betweenness
2. **Direct connections or influence?** - Direct → Degree - Influence → PageRank or eigenvector
3. **Network type?** - Directed → PageRank or HITS - Undirected → Any method - Weighted → Use weight-aware versions
4. **Computational cost?** - Large network → Degree or PageRank (fast) - Small network → Betweenness (expensive but informative)

7.8.4 Network Statistics

Goal: Characterize network structure at the global, layer, or node level.

When to Use

Use network statistics when you want to:

- Compare networks quantitatively
- Understand structural properties
- Validate network models
- Monitor network evolution
- Detect anomalies

Statistic Families

Global Statistics

Examples:

- Number of nodes, edges, layers
- Density, clustering coefficient
- Average path length
- Degree distribution

When to use: Basic network characterization

```
# Basic stats
network.basic_stats()

# Clustering
import networkx as nx
clustering = nx.average_clustering(network.core_network)

# Density
from py3plex.algorithms.statistics import multilayer_statistics as mls
density = mls.layer_density(network, 'layer1')
```

Layer-Specific Statistics

Examples:

- Layer density

- Layer similarity (Jaccard, correlation)
- Edge overlap between layers
- Inter-layer degree correlation

When to use: Comparing layers, understanding layer relationships

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Layer density
density_l1 = mls.layer_density(network, 'layer1')

# Layer similarity
similarity = mls.layer_similarity(network, 'layer1', 'layer2', method='jaccard')

# Edge overlap
overlap = mls.edge_overlap(network, 'layer1', 'layer2')
```

Node-Level Statistics

Examples:

- Node activity (presence across layers)
- Participation coefficient
- Cross-layer degree correlation

When to use: Characterizing node behavior across layers

```
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Node activity
activity = mls.node_activity(network, 'Alice')

# Participation coefficient
participation = mls.community_participation_coefficient(network, partition)
```

7.8.5 Random Walks & Embeddings

Goal: Generate node representations or sample the network through traversal.

When to Use

Use random walks and embeddings when you want to:

- Create low-dimensional node representations for ML
- Sample large networks efficiently
- Measure node similarity
- Generate features for classification/clustering
- Understand structural equivalence

Algorithm Families

Random Walk Algorithms

Examples:

- **Basic Random Walk** - Uniform random traversal
- **Node2Vec** - Biased walks (BFS/DFS-like)
- **DeepWalk** - Uniform random walks for embeddings

Parameters:

- `walk_length` - Steps per walk
- `num_walks` - Walks per node
- `p` (Node2Vec) - Return parameter
- `q` (Node2Vec) - In-out parameter

```
from py3plex.algorithms.general.walkers import generate_walks, node2vec_walk

# Basic walks
walks = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80
)

# Node2Vec walks (BFS-like: stay local)
walks_bfs = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80,
    p=1.0, q=2.0 # BFS-like
)

# Node2Vec walks (DFS-like: explore)
walks_dfs = generate_walks(
    network.core_network,
    num_walks=10,
    walk_length=80,
    p=1.0, q=0.5 # DFS-like
)
```

Embedding Methods

Examples:

- **Node2Vec** - Skip-gram on random walks
- **DeepWalk** - Word2Vec on uniform walks
- **LINE** - First/second-order proximity preservation

When to use:

- Node classification
- Link prediction

- Visualization in 2D/3D
- Clustering
- Similarity computation

```
from py3plex.wrappers import node2vec_embedding

# Generate embeddings
embeddings = node2vec_embedding.generate_embeddings(
    network.core_network,
    dimensions=128,
    walk_length=80,
    num_walks=10,
    p=1.0,
    q=1.0
)

# Use embeddings for ML
# embeddings is a numpy array: (num_nodes, dimensions)
```

Choosing Walk Parameters

walk_length:

- Short (10-20): Fast, local neighborhood
- Medium (40-80): Balanced, standard choice
- Long (100+): Global structure, slower

p (return parameter):

- $p < 1$: More likely to return to previous node
- $p = 1$: Balanced
- $p > 1$: Less likely to return

q (in-out parameter):

- $q < 1$: DFS-like (explore distant nodes)
- $q = 1$: Balanced
- $q > 1$: BFS-like (stay in local neighborhood)

7.8.6 Visualization & Layout

Goal: Create visual representations of multilayer networks.

When to Use

Use visualization when you want to:

- Explore network structure
- Present findings
- Communicate patterns
- Identify visual anomalies

- Create publication-ready figures

Layout Families

Force-Directed Layouts

Examples:

- Spring layout
- Fruchterman-Reingold
- Force Atlas

Best for: General-purpose visualization, showing community structure

```
from py3plex.visualization.multilayer import hairball_plot

hairball_plot(
    network.core_network,
    layout_algorithm="force",
    layout_parameters={"iterations": 50}
)
```

Specialized Multilayer Layouts

Examples:

- Diagonal projection
- Layered stacking
- Circular layer arrangement

Best for: Emphasizing multilayer structure

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Diagonal layout for multilayer networks
draw_multilayer_default(
    [network],
    display=True,
    layout_style='diagonal'
)
```

Hierarchical Layouts

Examples:

- Reingold-Tilford tree
- Radial tree
- Sugiyama layered

Best for: Trees and DAGs

For complete visualization documentation, see [Visualization Guide](#).

7.8.7 Benchmarking & Generation

Goal: Create synthetic networks for testing and validation.

When to Use

Use synthetic network generation when you want to:

- Test algorithm performance
- Validate implementations
- Create controlled experiments
- Generate training data
- Benchmark scalability

Generator Families

Random Network Models

Examples:

- **Erdős-Rényi** - Random edges with fixed probability
- **Barabási-Albert** - Preferential attachment (scale-free)
- **Watts-Strogatz** - Small-world networks

```
from py3plex.core import random_generators

# Random multilayer ER network
network = random_generators.random_multilayer_ER(
    num_nodes=100,
    num_layers=3,
    probability=0.05,
    directed=False
)
```

Benchmark Networks

Examples:

- LFR benchmark (with ground-truth communities)
- Configuration model (preserving degree distribution)
- Block model (with planted partition)

7.8.8 Algorithm Combinations

Common Workflows

Community Detection + Visualization:

```
# Detect communities
partition = louvain_multilayer(network)

# Visualize with community colors
from py3plex.visualization.multilayer import hairball_plot
from py3plex.visualization.colors import colors_default
```

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```

from collections import Counter

top_communities = [c for c, _ in Counter(partition.values()).most_common(5)]
color_map = dict(zip(top_communities, colors_default[:5]))
node_colors = [color_map.get(partition.get(node), 'gray')
                for node in network.get_nodes()]

hairball_plot(network.core_network, node_colors, layout_algorithm="force")

```

Centrality + Ranking:

```

# Compute centrality
centrality = nx.betweenness_centrality(network.core_network)

# Rank nodes
ranked = sorted(centrality.items(), key=lambda x: x[1], reverse=True)

# Top 10
print("Top 10 nodes:", ranked[:10])

```

Embeddings + Classification:

```

# Generate embeddings
embeddings = node2vec_embedding.generate_embeddings(network.core_network)

# Use for classification (with sklearn)
from sklearn.ensemble import RandomForestClassifier
clf = RandomForestClassifier()
# clf.fit(embeddings, labels)
# predictions = clf.predict(embeddings_test)

```

7.8.9 Performance Considerations

Algorithm Complexity

Algorithm Family	Typical Complexity	Notes
Degree Centrality	$O(m)$	Fast, scales well
Betweenness	$O(nm)$ or $O(nm + n^2 \log n)$	Expensive for large networks
Louvain	$O(n \log n)$	Fast community detection
Node2Vec	$O(k \cdot l \cdot n)$	k =walks, l =length, n =nodes
Force Layout	$O(n^2)$	Slow for >1000 nodes

When working with large networks:

- Use approximate algorithms (e.g., sampling for betweenness)
- Consider layer-by-layer analysis
- Use sparse representations
- See *Performance and Scalability Best Practices*

7.8.10 Next Steps

- *Community Detection Tutorial* - Detailed community detection guide
- *Network Statistics* - Complete statistics reference
- *Random Walk Algorithms* - Embeddings and walks
- *Visualization Guide* - Visualization techniques
- *Algorithm Roadmap* - Detailed algorithm parameters and API

Academic References:

- Fortunato & Hric (2016). “Community detection in networks: A user guide.” *Physics Reports* 659: 1-44.
- Grover & Leskovec (2016). “node2vec: Scalable feature learning for networks.” *KDD*.
- Newman (2018). “Networks” (2nd ed.). Oxford University Press.

7.9 Working with Networks

This guide covers everything you need to know about creating, loading, and manipulating multilayer networks in py3plex.

7.9.1 Creating Networks from Scratch

Method 1: Add Edges Directly

The simplest way to create a network is to add edges directly. Nodes are created automatically.

```
from py3plex.core import multinet

# Create empty network
network = multinet.multi_layer_network()

# Add edges in list format
# Format: [source_node, source_layer, target_node, target_layer, weight]
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Bob', 'friends', 'Carol', 'friends', 1.0],
    ['Alice', 'colleagues', 'Bob', 'colleagues', 1.0],
    ['Bob', 'colleagues', 'Dave', 'colleagues', 1.0]
], input_type="list")

# Verify
network.basic_stats()
```

Output:

```
Number of nodes: 6
Number of edges: 4
Number of unique nodes (as node-layer tuples): 6
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```


Method 2: Add Nodes and Edges Separately

For more control, add nodes explicitly before adding edges:

```
from py3plex.core import multinet

network = multinet.multi_layer_network()

# Add nodes explicitly (as node-layer tuples)
network.add_nodes([
    ('Alice', 'friends'),
    ('Bob', 'friends'),
    ('Carol', 'friends'),
    ('Alice', 'colleagues'),
    ('Bob', 'colleagues'),
    ('Dave', 'colleagues')
])

# Add edges between existing nodes
network.add_edges([
    [('Alice', 'friends'), ('Bob', 'friends'), 1.0],
    [('Bob', 'friends'), ('Carol', 'friends'), 1.0],
    [('Alice', 'colleagues'), ('Bob', 'colleagues'), 1.0],
    [('Bob', 'colleagues'), ('Dave', 'colleagues'), 1.0]
])

network.basic_stats()
```

Method 3: Define Layers First

You can define layers explicitly before adding content:

```
network = multinet.multi_layer_network()

# Define layers
network.add_layer('friends')
network.add_layer('colleagues')
network.add_layer('family')

# Check layers
print("Layers:", network.get_layers())

# Add edges (nodes created automatically)
network.add_edges([
    ['Alice', 'friends', 'Bob', 'friends', 1.0],
    ['Alice', 'family', 'Carol', 'family', 1.0]
], input_type="list")
```

Output:

```
Layers: ['friends', 'colleagues', 'family']
```

Method 4: Build from NetworkX Graphs

Start with existing NetworkX graphs for each layer:

```
import networkx as nx
from py3plex.core import multinet

# Create layer graphs
friends = nx.Graph()
friends.add_edges_from([('Alice', 'Bob'), ('Bob', 'Carol')])

colleagues = nx.Graph()
colleagues.add_edges_from([('Alice', 'Bob'), ('Bob', 'Dave')])

# Combine into multilayer network
network = multinet.multi_layer_network()

# Add edges from each graph with layer label
for u, v in friends.edges():
    network.add_edges([[u, 'friends', v, 'friends', 1.0]], input_type="list")

for u, v in colleagues.edges():
    network.add_edges([[u, 'colleagues', v, 'colleagues', 1.0]], input_type="list")
```

7.9.2 Loading Networks from Files

py3plex supports multiple file formats for loading networks. See *I/O and Serialization* for complete I/O documentation.

From Simple Edge Lists

Format: Simple source target pairs (one per line)

File example (data.edgelist):

```
Alice Bob
Bob Carol
Carol Dave
```

Code:

```
from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data.edgelist",
    input_type="edgelist",
    directed=False
)

network.basic_stats()
```

Output:

```
Number of nodes: 4
Number of edges: 3
Number of unique nodes (as node-layer tuples): 4
```

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```
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'null': 4 nodes
```

Note: Simple edge lists create a single layer named 'null' by default.

From Multilayer Edge Lists

Format: source target layer (space or tab-separated)

File example (data.multiedgelist):

```
Alice Bob friends
Bob Carol friends
Alice Bob colleagues
Bob Dave colleagues
```

Code:

```
network = multinet.multi_layer_network().load_network(
    "data.multiedgelist",
    input_type="multiedgelist",
    directed=False
)

network.basic_stats()
```

Output:

```
Number of nodes: 6
Number of edges: 4
Number of unique nodes (as node-layer tuples): 6
Number of unique node IDs (across all layers): 4
Nodes per layer:
  Layer 'friends': 3 nodes
  Layer 'colleagues': 3 nodes
```

From GraphML

GraphML is an XML-based format that preserves node and edge attributes:

```
network = multinet.multi_layer_network().load_network(
    "data.graphml",
    input_type="graphml"
)
```

From GML

Graph Modeling Language is another standard format:

```
network = multinet.multi_layer_network().load_network(
    "data.gml",
    input_type="gml"
)
```

From NetworkX Pickle

NetworkX's native pickle format:

```
network = multinet.multi_layer_network().load_network(
    "data.gpickle",
    input_type="gpickle"
)
```

From NetworkX Graphs

Load directly from a NetworkX graph object:

```
import networkx as nx
from py3plex.core import multinet

# Create or load a NetworkX graph
G = nx.karate_club_graph()

# Convert to py3plex
network = multinet.multi_layer_network()
network.load_network_from_networkx(G)

# Result is a single-layer network
network.basic_stats()
```

Modern I/O with Arrow/Parquet

For high-performance I/O, use the modern schema-based API. See *I/O and Serialization* for details.

```
from py3plex.io import read, write

# Read (auto-detects format from extension)
network = read('network.arrow') # or .parquet, .json

# Write
write(network, 'output.arrow')
```

7.9.3 Network Operations

Querying Network Elements

Get all nodes:

```
# Without attributes
nodes = network.get_nodes(data=False)
print("Nodes:", list(nodes)[:5])

# With attributes
nodes_with_data = network.get_nodes(data=True)
for node, attrs in list(nodes_with_data)[:3]:
    print(f"Node: {node}, Attributes: {attrs}")
```

Get nodes in a specific layer:

```
# Filter by layer
friends_nodes = network.get_nodes(layer='friends')
print("Friends layer nodes:", list(friends_nodes))
```

Get all edges:

```
# Without attributes
edges = network.get_edges(data=False)
print("Edges:", list(edges)[:5])

# With attributes
edges_with_data = network.get_edges(data=True)
for u, v, attrs in list(edges_with_data)[:3]:
    print(f"Edge: {u} -> {v}, Attributes: {attrs}")
```

Get neighbors:

```
# Get neighbors of a node in a specific layer
neighbors = list(network.get_neighbors('Alice', layer_id='friends'))
print(f"Alice's friends: {neighbors}")
```

Get layers:

```
layers = network.get_layers()
print(f"Layers: {layers}")
print(f"Number of layers: {len(layers)}")
```

Extracting Subnetworks

Extract a single layer:

```
# Get all nodes and edges in one layer
friends_subnet = network.subnetwork(['friends'], subset_by="layers")
friends_subnet.basic_stats()
```

Extract multiple layers:

```
# Combine multiple layers
social_subnet = network.subnetwork(['friends', 'family'], subset_by="layers")
```

Extract specific nodes (all layers):

```
# Get all appearances of specific nodes
alice_bob_subnet = network.subnetwork(['Alice', 'Bob'], subset_by="node_names")
alice_bob_subnet.basic_stats()
```

Extract specific node-layer pairs:

```
# Get specific node-layer combinations
specific_subnet = network.subnetwork(
    [('Alice', 'friends'), ('Bob', 'friends')],
    subset_by="node_layer_names"
)
specific_subnet.basic_stats()
```

Iterating Over Network Elements

Iterate over nodes:

```
# Simple iteration
for node in network.get_nodes():
    print(node) # Prints (node_id, layer_id) tuples

# With attributes
for node, attrs in network.get_nodes(data=True):
    print(f"{node}: {attrs}")
```

Iterate over edges:

```
# Simple iteration
for u, v in network.get_edges():
    print(f"{u} -> {v}")

# With attributes
for u, v, attrs in network.get_edges(data=True):
    print(f"{u} -> {v}: weight={attrs.get('weight', 1.0)}")
```

Iterate over layers:

```
for layer_name in network.get_layers():
    layer_subnet = network.subnetwork([layer_name], subset_by="layers")
    num_nodes = len(list(layer_subnet.get_nodes()))
    num_edges = len(list(layer_subnet.get_edges()))
    print(f"Layer {layer_name}: {num_nodes} nodes, {num_edges} edges")
```

7.9.4 Modifying Networks

Adding Elements

Add new nodes:

```
# Add single node
network.add_nodes([('Eve', 'friends')])

# Add multiple nodes
network.add_nodes([
    ('Eve', 'colleagues'),
    ('Frank', 'colleagues')
])
```

Add new edges:

```
# Add single edge
network.add_edges([
    ['Alice', 'friends', 'Eve', 'friends', 1.0]
], input_type="list")

# Add multiple edges
network.add_edges([
    ['Eve', 'colleagues', 'Frank', 'colleagues', 1.0],
```

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```
['Bob', 'colleagues', 'Frank', 'colleagues', 1.0]
], input_type="list")
```

Add new layer:

```
network.add_layer('family')
```

Removing Elements

To remove nodes or edges, work with the underlying NetworkX graph:

```
# Access NetworkX graph
G = network.core_network

# Remove node (removes all edges)
G.remove_node(('Alice', 'friends'))

# Remove edge
G.remove_edge(('Bob', 'friends'), ('Carol', 'friends'))

# Remove multiple nodes
G.remove_nodes_from([('Alice', 'colleagues'), ('Bob', 'colleagues')])
```

Modifying Attributes

Node attributes:

```
# Access NetworkX graph
G = network.core_network

# Set node attribute
G.nodes[('Alice', 'friends')]['age'] = 30
G.nodes[('Alice', 'friends')]['role'] = 'admin'

# Get node attribute
age = G.nodes[('Alice', 'friends')].get('age')
print(f"Alice's age: {age}")
```

Edge attributes:

```
# Set edge attribute
G[('Alice', 'friends')][('Bob', 'friends')]['weight'] = 0.8
G[('Alice', 'friends')][('Bob', 'friends')]['type'] = 'close'

# Get edge attribute
weight = G[('Alice', 'friends')][('Bob', 'friends')].get('weight')
print(f"Edge weight: {weight}")
```

7.9.5 Network Properties

Basic Statistics

```
# Get comprehensive stats
network.basic_stats()

# Output shows:
# - Total nodes
# - Total edges
# - Unique node-layer pairs
# - Unique node IDs
# - Nodes per layer
```

Manual counting:

```
# Count elements manually
num_nodes = len(list(network.get_nodes()))
num_edges = len(list(network.get_edges()))
num_layers = len(network.get_layers())

print(f"Nodes: {num_nodes}, Edges: {num_edges}, Layers: {num_layers}")
```

Network Type

```
# Check if directed
G = network.core_network
is_directed = G.is_directed()
print(f"Directed: {is_directed}")

# Check if multigraph
is_multi = G.is_multigraph()
print(f"Multigraph: {is_multi}")
```

Connectivity

```
import networkx as nx

G = network.core_network

# For undirected networks
if not G.is_directed():
    is_connected = nx.is_connected(G)
    num_components = nx.number_connected_components(G)
    print(f"Connected: {is_connected}")
    print(f"Number of components: {num_components}")

# For directed networks
else:
    is_weakly_connected = nx.is_weakly_connected(G)
    is_strongly_connected = nx.is_strongly_connected(G)
    print(f"Weakly connected: {is_weakly_connected}")
    print(f"Strongly connected: {is_strongly_connected}")
```


Matrix Representations

```
# Get supra-adjacency matrix
supra_adj = network.get_supra_adjacency_matrix(sparse=True)
print(f"Supra-adjacency shape: {supra_adj.shape}")

# For small networks, get dense version
if num_nodes < 1000:
    supra_adj_dense = network.get_supra_adjacency_matrix(sparse=False)
```

See *py3plex Core Model* for details on supra-adjacency matrices.

7.9.6 NetworkX Integration

Direct Access

The `core_network` attribute gives you direct access to the NetworkX graph:

```
import networkx as nx

# Get NetworkX graph
G = network.core_network

# Use any NetworkX function
degree = dict(G.degree())
betweenness = nx.betweenness_centrality(G)
clustering = nx.clustering(G)

print(f"Average clustering: {nx.average_clustering(G):.3f}")
```

NetworkX Wrappers

py3plex provides convenience wrappers for common NetworkX functions:

```
# Extract a layer first
friends_layer = network.subnetwork(['friends'], subset_by="layers")

# Apply NetworkX algorithms via wrapper
degree_centrality = friends_layer.monoplex_nx_wrapper("degree_centrality")
betweenness = friends_layer.monoplex_nx_wrapper("betweenness_centrality")
pagerank = friends_layer.monoplex_nx_wrapper("pagerank")

# Results are dictionaries
top_degree = sorted(degree_centrality.items(), key=lambda x: x[1], reverse=True)[:5]
print("Top 5 by degree:", top_degree)
```

Converting to NetworkX

Since py3plex networks ARE NetworkX graphs, conversion is trivial:

```
# Export to standard NetworkX formats
import networkx as nx

G = network.core_network
```

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```
# Save as GraphML
nx.write_graphml(G, "output.graphml")

# Save as GML
nx.write_gml(G, "output.gml")

# Save as pickle
nx.write_gpickle(G, "output.gpickle")
```

See *I/O and Serialization* for more export options.

7.9.7 Best Practices

File Organization

```
# Use absolute paths or path relative to script
import os

# Get directory of current script
script_dir = os.path.dirname(os.path.abspath(__file__))
data_dir = os.path.join(script_dir, "data")
network_path = os.path.join(data_dir, "network.edgelist")

# Load
network = multinet.multi_layer_network().load_network(
    network_path,
    input_type="edgelist"
)
```

Always Verify After Loading

```
# Load network
network = multinet.multi_layer_network().load_network(...)

# ALWAYS check stats
network.basic_stats()

# Verify it matches expectations
assert len(network.get_layers()) == 3, "Expected 3 layers"
assert len(list(network.get_nodes())) > 0, "Network is empty!"
```

Use Appropriate Data Structures

```
# For large networks, use sparse matrices
supra_adj = network.get_supra_adjacency_matrix(sparse=True) # Good

# Avoid dense matrices for large networks
# supra_adj = network.get_supra_adjacency_matrix(sparse=False) # Bad for n > 1000
```

7.9.8 Common Patterns

Pattern 1: Load, Analyze, Visualize

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import draw_multilayer_default

# Load
network = multinet.multi_layer_network().load_network(
    "data.multiedgelist", input_type="multiedgelist"
)

# Analyze
network.basic_stats()

# Visualize
draw_multilayer_default([network], display=True)
```

Pattern 2: Create from Multiple Sources

```
# Combine data from multiple files
network = multinet.multi_layer_network()

# Load friends from file
import pandas as pd
friends_df = pd.read_csv("friends.csv")
for _, row in friends_df.iterrows():
    network.add_edges([[row['source'], 'friends', row['target'], 'friends', 1]],
                      input_type="list")

# Load colleagues from database (pseudocode)
# colleagues = query_database("SELECT * FROM colleagues")
# for source, target in colleagues:
#     network.add_edges([[source, 'colleagues', target, 'colleagues', 1]],
#                       input_type="list")
```

Pattern 3: Layer-by-Layer Analysis

```
# Analyze each layer separately
for layer_name in network.get_layers():
    print(f"\n=== Layer: {layer_name} ===")

    # Extract layer
    layer_subnet = network.subnetwork([layer_name], subset_by="layers")

    # Compute metrics
    layer_subnet.basic_stats()

    # Optional: visualize
    # draw_multilayer_default([layer_subnet], display=True)
```

7.9.9 Next Steps

- *Network Statistics* - Computing network metrics
- *Community Detection Tutorial* - Finding communities
- *I/O and Serialization* - Complete I/O documentation
- *Visualization Guide* - Visualization techniques
- *py3plex Core Model* - Understanding the internal representation

Related Examples:

- `example_multilayer_functionality.py` - Core operations
- `example_IO.py` - Loading and saving
- `example_manipulation.py` - Network manipulation
- `example_networkx_wrapper.py` - NetworkX integration

Repository: <https://github.com/SkBlaz/py3plex/tree/master/examples>

7.10 Network Statistics

This guide covers the statistical measures available in py3plex for analyzing multilayer networks.

7.10.1 Overview

py3plex provides three levels of network statistics:

1. **Global Statistics** - Whole network properties (density, clustering, etc.)
2. **Layer-Specific Statistics** - Per-layer measures and comparisons
3. **Node-Level Statistics** - Node activity and participation across layers

For centrality measures (degree, betweenness, PageRank, etc.), see *Algorithm Landscape*.

7.10.2 Basic Network Statistics

Quick Stats

The fastest way to get basic information:

```
from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data.multiedgelist", input_type="multiedgelist"
)

# Display comprehensive stats
network.basic_stats()
```

Output:

```
Number of nodes: 184
Number of edges: 1691
Number of unique nodes (as node-layer tuples): 184
Number of unique node IDs (across all layers): 46
```

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Nodes per layer:

```

Layer '1': 46 nodes
Layer '2': 46 nodes
Layer '3': 46 nodes
Layer '4': 46 nodes

```

Manual Counting

```

# Count elements
num_nodes = len(list(network.get_nodes()))
num_edges = len(list(network.get_edges()))
num_layers = len(network.get_layers())

# Unique node IDs (across all layers)
unique_nodes = set()
for node, layer in network.get_nodes():
    unique_nodes.add(node)
num_unique_nodes = len(unique_nodes)

print(f"Nodes (node-layer pairs): {num_nodes}")
print(f"Edges: {num_edges}")
print(f"Layers: {num_layers}")
print(f"Unique node IDs: {num_unique_nodes}")

```

7.10.3 Layer-Specific Statistics

The `multilayer_statistics` module provides comprehensive statistics:

```
from py3plex.algorithms.statistics import multilayer_statistics as mls
```

Layer Density

Definition: Fraction of possible edges that exist in a layer.

Formula: $\text{density} = \frac{\text{edges}}{\text{nodes} \times (\text{nodes} - 1)}$ for undirected graphs

Use case: Measure how connected a layer is.

```

# Density of individual layers
density_layer1 = mls.layer_density(network, 'layer1')
density_layer2 = mls.layer_density(network, 'layer2')

print(f"Layer 1 density: {density_layer1:.4f}")
print(f"Layer 2 density: {density_layer2:.4f}")

```

Interpretation:

- 0.0 = No edges (empty layer)
- 1.0 = Complete graph (all possible edges exist)
- Typical real-world networks: 0.001 - 0.1

Layer Similarity

Definition: How similar two layers are in structure.

Methods: Jaccard index, Pearson correlation, cosine similarity

Use case: Identify redundant or complementary layers.

```
# Jaccard similarity (based on edges)
jaccard = mls.layer_similarity(
    network, 'layer1', 'layer2', method='jaccard'
)

# Pearson correlation
pearson = mls.layer_similarity(
    network, 'layer1', 'layer2', method='pearson'
)

print(f"Jaccard similarity: {jaccard:.4f}")
print(f"Pearson correlation: {pearson:.4f}")
```

Interpretation:

- 1.0 = Identical layers
- 0.0 = Completely different
- Negative values (Pearson) = Anti-correlated

Edge Overlap

Definition: Fraction of edges that appear in both layers.

Use case: Measure redundancy between layers.

```
overlap = mls.edge_overlap(network, 'layer1', 'layer2')
print(f"Edge overlap: {overlap:.4f}")
```

Interpretation:

- 1.0 = All edges in common
- 0.0 = No edges in common

Inter-Layer Degree Correlation

Definition: Correlation between node degrees in different layers.

Use case: Determine if “hubs” in one layer are hubs in another.

```
correlation = mls.inter_layer_degree_correlation(
    network, 'layer1', 'layer2'
)
print(f"Degree correlation: {correlation:.4f}")
```

Interpretation:

- 1.0 = Perfect positive correlation (hubs in layer1 are hubs in layer2)
- 0.0 = No correlation
- -1.0 = Perfect negative correlation

7.10.4 Node-Level Statistics

Node Activity

Definition: Fraction of layers in which a node appears.

Use case: Identify nodes that participate in multiple contexts.

```
# Activity for a single node
activity_alice = mls.node_activity(network, 'Alice')
print(f"Alice's activity: {activity_alice:.4f}")

# Compute for all nodes
all_activities = {}
unique_nodes = set(node for node, layer in network.get_nodes())
for node in unique_nodes:
    all_activities[node] = mls.node_activity(network, node)

# Top 5 most active nodes
top_active = sorted(all_activities.items(), key=lambda x: x[1], reverse=True)[:5]
print("Most active nodes:", top_active)
```

Interpretation:

- 1.0 = Node appears in all layers
- 0.5 = Node appears in half of layers
- Close to 0.0 = Node appears in few layers

Versatility Centrality

Definition: Node importance considering activity across layers.

Use case: Find nodes that are important across multiple layers.

```
# Versatility based on degree
versatility_degree = mls.versatility_centrality(
    network, centrality_type='degree'
)

# Versatility based on betweenness
versatility_betweenness = mls.versatility_centrality(
    network, centrality_type='betweenness'
)

# Top versatile nodes
top_versatile = sorted(
    versatility_degree.items(),
    key=lambda x: x[1],
    reverse=True
)[:10]
print("Top 10 versatile nodes:", top_versatile)
```

Interpretation:

- Higher values = More important across multiple layers
- Combines centrality within layers with cross-layer participation

Participation Coefficient

Definition: Measures how evenly a node's connections are distributed across layers.

Use case: Identify nodes that bridge different layers.

```
from py3plex.algorithms.community_detection.multilayer_modularity import (
    louvain_multilayer
)

# Detect communities first
partition = louvain_multilayer(network)

# Compute participation coefficient
participation = mls.community_participation_coefficient(network, partition)

# Top bridging nodes
top_bridging = sorted(
    participation.items(),
    key=lambda x: x[1],
    reverse=True
)[:10]
print("Top bridging nodes:", top_bridging)
```

Interpretation:

- 1.0 = Connections evenly distributed across layers
- 0.0 = All connections in one layer

7.10.5 Network-Level Statistics

Entropy of Multiplexity

Definition: Measures layer diversity (how evenly nodes/edges are distributed across layers).

Use case: Quantify structural diversity of the multilayer network.

```
entropy = mls.entropy_of_multiplexity(network)
print(f"Entropy of multiplexity: {entropy:.4f} bits")
```

Interpretation:

- 0.0 = All activity in one layer (no diversity)
- Higher values = More evenly distributed across layers
- Maximum = $\log_2(\text{number of layers})$

Algebraic Connectivity

Definition: Second smallest eigenvalue of the Laplacian matrix.

Use case: Measure network robustness (higher = more robust).

```
algebraic_conn = mls.algebraic_connectivity(network, 'layer1')
print(f"Algebraic connectivity: {algebraic_conn:.4f}")
```

Interpretation:

- 0.0 = Disconnected network
- Higher values = Better connectivity and robustness

Multilayer Clustering Coefficient

Definition: Extension of clustering coefficient to multilayer networks.

Use case: Measure local cohesion across layers.

```
clustering = mls.multilayer_clustering_coefficient(network)
print(f"Multilayer clustering: {clustering:.4f}")
```

Interpretation:

- 1.0 = Every node's neighbors form a clique
- 0.0 = No clustering (tree-like structure)

7.10.6 Using NetworkX Statistics

Since py3plex networks are NetworkX graphs, you can use any NetworkX statistic:

Basic NetworkX Metrics

```
import networkx as nx

G = network.core_network

# Clustering coefficient
clustering = nx.average_clustering(G)
print(f"Average clustering: {clustering:.4f}")

# Transitivity
transitivity = nx.transitivity(G)
print(f"Transitivity: {transitivity:.4f}")

# Density
density = nx.density(G)
print(f"Density: {density:.4f}")
```

Degree Distribution

```
import networkx as nx
from collections import Counter

G = network.core_network

# Get degree sequence
degrees = [d for n, d in G.degree()]

# Degree distribution
degree_dist = Counter(degrees)
print("Degree distribution:", dict(sorted(degree_dist.items())))
```

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```
# Average degree
avg_degree = sum(degrees) / len(degrees)
print(f"Average degree: {avg_degree:.2f}")
```

Path-Based Statistics

```
import networkx as nx

G = network.core_network

# Check if connected first
if nx.is_connected(G.to_undirected()):
    # Average shortest path length
    avg_path = nx.average_shortest_path_length(G)
    print(f"Average shortest path: {avg_path:.2f}")

    # Diameter
    diameter = nx.diameter(G)
    print(f"Diameter: {diameter}")
else:
    print("Network is not connected")

    # Largest component
    largest_cc = max(nx.connected_components(G.to_undirected()), key=len)
    largest_size = len(largest_cc)
    print(f"Largest component: {largest_size} nodes")
```

7.10.7 Statistical Comparison

Comparing Networks

Compare two multilayer networks:

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Load two networks
network1 = multinet.multi_layer_network().load_network("data1.multiedgelist")
network2 = multinet.multi_layer_network().load_network("data2.multiedgelist")

# Compare basic stats
print("Network 1:")
network1.basic_stats()

print("\nNetwork 2:")
network2.basic_stats()

# Compare layer densities (if same layers)
for layer in network1.get_layers():
    if layer in network2.get_layers():
        density1 = mls.layer_density(network1, layer)
```

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```
density2 = mls.layer_density(network2, layer)
print(f"{layer}: {density1:.4f} vs {density2:.4f}")
```

Layer-by-Layer Analysis

Systematic comparison of all layers:

```
import pandas as pd
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Collect stats for each layer
layer_stats = []
for layer in network.get_layers():
    # Extract layer
    layer_subnet = network.subnetwork([layer], subset_by="layers")
    G_layer = layer_subnet.core_network

    # Compute stats
    stats = {
        'layer': layer,
        'nodes': G_layer.number_of_nodes(),
        'edges': G_layer.number_of_edges(),
        'density': mls.layer_density(network, layer),
        'clustering': nx.average_clustering(G_layer)
    }
    layer_stats.append(stats)

# Display as table
df = pd.DataFrame(layer_stats)
print(df.to_string(index=False))
```

Output:

layer	nodes	edges	density	clustering
1	46	143	0.1384	0.4521
2	46	139	0.1346	0.4123
3	46	136	0.1317	0.3892
4	46	134	0.1298	0.3756

7.10.8 Exporting Statistics

Save to CSV

```
import pandas as pd

# Collect node statistics
node_stats = []
unique_nodes = set(node for node, layer in network.get_nodes())
for node in unique_nodes:
    stats = {
        'node': node,
        'activity': mls.node_activity(network, node),
```

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```

        # Add more stats as needed
    }
    node_stats.append(stats)

# Save
df = pd.DataFrame(node_stats)
df.to_csv("node_statistics.csv", index=False)

```

Save to JSON

```

import json

# Collect statistics
stats = {
    'num_nodes': len(list(network.get_nodes())),
    'num_edges': len(list(network.get_edges())),
    'num_layers': len(network.get_layers()),
    'layers': {}
}

# Layer stats
for layer in network.get_layers():
    stats['layers'][layer] = {
        'density': float(mls.layer_density(network, layer)),
        # Add more stats
    }

# Save
with open("network_stats.json", 'w') as f:
    json.dump(stats, f, indent=2)

```

7.10.9 Best Practices

1. Always check basic stats first

```
network.basic_stats() # Before doing any analysis
```

2. Extract layers for layer-specific analysis

```
layer1 = network.subnetwork(['layer1'], subset_by="layers")
# Now apply NetworkX functions
```

3. Cache expensive computations

```

# Compute once
versatility = mls.versatility_centrality(network, centrality_type='degree')

# Reuse
top_nodes = sorted(versatility.items(), key=lambda x: x[1], reverse=True)

```

4. Handle edge cases

```
# Check for empty layers
layer_subnet = network.subnetwork(['layer1'], subset_by="layers")
if len(list(layer_subnet.get_edges())) == 0:
    print("Layer is empty, skipping...")
else:
    density = mls.layer_density(network, 'layer1')
```

7.10.10 Complete Example

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
import networkx as nx

# Load network
network = multinet.multi_layer_network().load_network(
    "data.multiedgelist", input_type="multiedgelist"
)

print("=== Basic Statistics ===")
network.basic_stats()

print("\n=== Layer Statistics ===")
for layer in network.get_layers():
    density = mls.layer_density(network, layer)
    print(f"{layer}: density = {density:.4f}")

print("\n=== Node Activity ===")
unique_nodes = set(node for node, layer in network.get_nodes())
activities = {node: mls.node_activity(network, node) for node in unique_nodes}
top_active = sorted(activities.items(), key=lambda x: x[1], reverse=True)[:5]
for node, activity in top_active:
    print(f"{node}: {activity:.4f}")

print("\n=== Layer Similarity ===")
layers = network.get_layers()
if len(layers) >= 2:
    similarity = mls.layer_similarity(
        network, layers[0], layers[1], method='jaccard'
    )
    print(f"{layers[0]} vs {layers[1]}: {similarity:.4f}")

print("\n=== Global Metrics ===")
entropy = mls.entropy_of_multiplexity(network)
print(f"Entropy of multiplexity: {entropy:.4f} bits")
```

7.10.11 Next Steps

- *Community Detection Tutorial* - Finding communities
- *Working with Networks* - Creating and loading networks
- *Visualization Guide* - Visualizing statistics
- *Algorithm Landscape* - Overview of all algorithms

- *Algorithm Roadmap* - Complete API reference

Related Examples:

- `example_multilayer_statistics.py` - Statistical analysis examples
- `example_layer_comparison.py` - Comparing layers
- `example_node_metrics.py` - Node-level metrics

Repository: <https://github.com/SkBlaz/py3plex/tree/master/examples>

7.11 Community Detection Tutorial

This tutorial demonstrates **community detection algorithms** for **multilayer networks** using py3plex.

7.11.1 Overview

Community detection identifies **groups of nodes** that are more **densely connected** to each other than to the rest of the network. For **multilayer networks**, communities can **span multiple layers**, accounting for both **intra-layer** and **inter-layer** structure.

7.11.2 Supported Algorithms

py3plex provides **several community detection algorithms**:

- **Louvain - Fast modularity optimization** (recommended for most use cases)
- **Infomap - Flow-based** community detection (requires external binary)
- **Label Propagation - Semi-supervised** approach with known seed communities
- **Multilayer Modularity - True multilayer** community detection (Mucha et al. 2010)

7.11.3 Louvain Algorithm

Basic Usage

Fastest algorithm for large networks:

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection import community_louvain

# Create or load network
network = multinet.multi_layer_network()
network.load_network("data.graphml", input_type="graphml")

# Detect communities using Louvain
communities = community_louvain.best_partition(network.core_network)

# Print results
for node, community_id in communities.items():
    print(f"Node {node} -> Community {community_id}")
```

Parameters

```
# With custom resolution parameter
communities = community_louvain.best_partition(
    network.core_network,
    resolution=1.0 # Higher = more communities, Lower = fewer communities
)
```

Resolution parameter:

- resolution=1.0 - **Standard modularity**
- resolution>1.0 - **More, smaller communities**
- resolution<1.0 - **Fewer, larger communities**

Advantages

- **Very fast:** $O(n \log n)$
- **Scales to millions** of nodes
- **BSD license** (commercial-friendly)
- **Well-established** and widely used

Disadvantages

- **Non-deterministic** (random initialization)
- **Cannot find overlapping** communities
- **Resolution limit** issues

7.11.4 Infomap Algorithm

Basic Usage

Flow-based approach for detecting communities:

```
from py3plex.algorithms.community_detection import community_wrapper

# Detect communities using Infomap
communities = community_wrapper.infomap_communities(
    network.core_network,
    binary_path="/path/to/infomap" # Path to Infomap binary
)
```

With Hierarchical Structure

```
# Get hierarchical community structure
hierarchical_communities = community_wrapper.infomap_communities(
    network.core_network,
    binary_path="/path/to/infomap",
    hierarchical=True
)
```

Advantages

- Can detect overlapping communities
- Flow-based (natural for many applications)
- Hierarchical structure
- Information-theoretic foundation

Disadvantages

- Requires external binary
- AGPLv3 license (viral copyleft - problematic for commercial use)
- Slower than Louvain

7.11.5 Label Propagation

Semi-Supervised Detection

Use when you have some known community memberships:

```
from py3plex.algorithms.community_detection import label_propagation

# Provide seed labels for some nodes
seed_labels = {
    'node1': 0,
    'node2': 0,
    'node3': 1,
    'node4': 1
}

# Propagate labels to unlabeled nodes
communities = label_propagation.propagate(
    network.core_network,
    seed_labels=seed_labels,
    max_iter=100
)
```

Fully Unsupervised

```
# Without seed labels (random initialization)
communities = label_propagation.propagate(
    network.core_network,
    max_iter=100
)
```

Advantages

- Very fast: $O(m)$ linear in edges
- Can incorporate prior knowledge
- MIT license
- Simple and interpretable

Disadvantages

- Non-deterministic
- Sensitive to initialization
- May not converge
- Lower quality than Louvain/Infomap

7.11.6 Multilayer Modularity

True Multilayer Detection

Accounts for multilayer structure following Mucha et al. (2010):

```
from py3plex.algorithms.community_detection import multilayer_modularity as mlm

# Get supra-adjacency matrix
supra_adj = network.get_supra_adjacency_matrix(sparse=True)

# Detect communities with multilayer modularity
communities = mlm.multilayer_louvain(
    supra_adj,
    gamma=1.0,      # Resolution parameter
    omega=1.0       # Inter-layer coupling strength
)
```

Parameter Tuning

```
# Emphasize layer-specific structure
communities = mlm.multilayer_louvain(
    supra_adj,
    gamma=1.0,
    omega=0.1 # Low coupling = layer-specific communities
)

# Emphasize cross-layer structure
communities = mlm.multilayer_louvain(
    supra_adj,
    gamma=1.0,
    omega=10.0 # High coupling = cross-layer communities
)
```

Mathematical Formulation

Multilayer modularity is defined as:

$$Q^{ML} = \frac{1}{2\mu} \sum_{ij\alpha\beta} \left[(A_{ij}^{[\alpha]} - \gamma_{\alpha} P_{ij}^{[\alpha]}) \delta_{\alpha\beta} + \omega_{\alpha\beta} \delta_{ij} \right] \delta(g_i^{[\alpha]}, g_j^{[\beta]})$$

Where:

- $A_{ij}^{[\alpha]}$ is the adjacency matrix of layer α
- $P_{ij}^{[\alpha]}$ is the null model (e.g., configuration model)
- γ_{α} is the resolution parameter for layer α

- $\omega_{\alpha\beta}$ is the coupling strength between layers
- $\delta(g_i^{[\alpha]}, g_j^{[\beta]})$ is 1 if nodes are in the same community, 0 otherwise

Advantages

- Accounts for multilayer structure
- Implements state-of-the-art algorithm
- Configurable inter-layer coupling
- Published in Science (Mucha et al. 2010)

Disadvantages

- More computationally expensive
- Requires parameter tuning
- May not scale to very large networks (>100k nodes)

7.11.7 Evaluating Community Quality

Modularity Score

```
import networkx as nx

# Compute modularity
modularity = nx.community.modularity(network.core_network, communities)
print(f"Modularity: {modularity:.3f}")
```

Interpretation:

- $Q > 0.3$: Good community structure
- $Q > 0.5$: Strong community structure
- $Q < 0.3$: Weak or no community structure

Coverage and Performance

```
# Coverage: fraction of edges within communities
coverage = nx.community.coverage(network.core_network, communities)

# Performance: fraction of correctly classified node pairs
performance = nx.community.performance(network.core_network, communities)

print(f"Coverage: {coverage:.3f}")
print(f"Performance: {performance:.3f}")
```

7.11.8 Visualizing Communities

Color by Community

```
from py3plex.visualization.multilayer import hairball_plot
import matplotlib.pyplot as plt
```

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```
# Map communities to colors
node_colors = [communities.get(node, 0) for node in network.core_network.nodes()]

# Visualize with community colors
hairball_plot(
    network.core_network,
    node_color=node_colors,
    layout_algorithm='force',
    cmap='tab10'
)
plt.show()
```

Community Size Distribution

```
from collections import Counter
import matplotlib.pyplot as plt

# Count community sizes
community_sizes = Counter(communities.values())
sizes = list(community_sizes.values())

# Plot distribution
plt.hist(sizes, bins=20)
plt.xlabel('Community Size')
plt.ylabel('Frequency')
plt.title('Community Size Distribution')
plt.show()
```

7.11.9 Comparing Algorithms

Run Multiple Algorithms

```
# Run different algorithms
louvain_comms = community_louvain.best_partition(network.core_network)
label_prop_comms = label_propagation.propagate(network.core_network)

# Compare number of communities
print(f"Louvain: {len(set(louvain_comms.values()))} communities")
print(f"Label Prop: {len(set(label_prop_comms.values()))} communities")

# Compare modularity
louvain_mod = nx.community.modularity(network.core_network,
                                     [set(n for n, c in louvain_comms.items() if c == i)
                                      for i in set(louvain_comms.values())])
label_mod = nx.community.modularity(network.core_network,
                                    [set(n for n, c in label_prop_comms.items() if c ==
↪ i)
                                     for i in set(label_prop_comms.values())])

print(f"Louvain modularity: {louvain_mod:.3f}")
print(f"Label Prop modularity: {label_mod:.3f}")
```

Normalized Mutual Information

Compare similarity between community structures:

```
from sklearn.metrics import normalized_mutual_info_score

# Convert to lists
louvain_list = [louvain_comms[node] for node in network.core_network.nodes()]
label_list = [label_prop_comms[node] for node in network.core_network.nodes()]

# Compute NMI
nmi = normalized_mutual_info_score(louvain_list, label_list)
print(f"NMI between Louvain and Label Prop: {nmi:.3f}")
```

Interpretation:

- NMI = 1.0: Identical community structures
- NMI = 0.0: Completely different structures
- NMI > 0.5: Similar structures

7.11.10 Best Practices

Algorithm Selection

Network Size	Speed Priority	Quality Priority	Recommendation
Small (<1K)	Any	Any	Try all algorithms
Medium (1K-10K)	Louvain	Louvain/Infomap	Louvain (good balance)
Large (10K-100K)	Louvain/Label Prop	Louvain	Louvain
Very Large (>100K)	Label Prop	Louvain	Label Prop or sample

Parameter Guidelines

Louvain resolution:

- Start with resolution=1.0
- Increase if communities are too large
- Decrease if communities are too fragmented

Multilayer coupling (omega):

- omega=1.0 - Default, balanced
- omega<1.0 - Emphasize layer-specific structure
- omega>1.0 - Emphasize cross-layer structure

Validation

Always validate community detection results:

1. **Visual inspection** - Plot and examine communities
2. **Modularity** - Check modularity score (>0.3 is good)
3. **Size distribution** - Check for giant communities or singletons
4. **Domain knowledge** - Do communities make sense for your application?

7.11.11 References

Louvain:

Blondel, V. D., et al. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics*.

Infomap:

Rosvall, M., & Bergstrom, C. T. (2008). Maps of random walks on complex networks reveal community structure. *PNAS*, 105(4), 1118-1123.

Multilayer Modularity:

Mucha, P. J., et al. (2010). Community structure in time-dependent, multiscale, and multiplex networks. *Science*, 328(5980), 876-878.

See `../citation` for complete citations with DOIs.

7.11.12 Next Steps

- `./multilayer_centrality` - Centrality measures tutorial
- `./multilayer_modularity` - Multilayer modularity details
- `../algorithm_guide` - Algorithm selection guide
- Examples in `examples/communities/example_community_detection.py`

7.12 Random Walk Algorithms

Py3plex provides comprehensive random walk primitives for graph-based algorithms, including Node2Vec, DeepWalk, and diffusion processes. These implementations support weighted, directed, and multilayer networks with deterministic reproducibility.

7.12.1 Overview

Random walks are fundamental building blocks for many graph algorithms:

- **Node2Vec**: Learn node embeddings using biased random walks
- **DeepWalk**: Generate node embeddings through uniform random walks
- **Personalized PageRank**: Compute node importance via random walks with restart
- **Diffusion processes**: Model information or disease propagation
- **Community detection**: Identify network structure through walk patterns

Key Features

- [OK] Basic random walks with proper edge weight handling
- [OK] Second-order (biased) random walks following Node2Vec logic
- [OK] Multiple simultaneous walks with deterministic seeding
- [OK] Support for directed, weighted, and multigraphs
- [OK] Multilayer network-aware walks with layer constraints
- [OK] Efficient sparse adjacency matrix handling
- [OK] Comprehensive test suite validating correctness properties

7.12.2 Basic Random Walk

The basic random walk samples the next node proportionally to normalized edge weights.

Simple Example

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx

# Create a simple graph
G = nx.karate_club_graph()

# Perform a random walk
walk = basic_random_walk(
    G,
    start_node=0,
    walk_length=10,
    seed=42 # for reproducibility
)

print(f"Walk: {walk}")
print(f"Length: {len(walk)}") # 11 nodes (includes start)
```

Weighted Graphs

Edge weights are automatically used when `weighted=True` (default):

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx

# Create weighted graph
G = nx.Graph()
G.add_weighted_edges_from([
    (0, 1, 10.0), # high weight -> more likely
    (0, 2, 1.0), # low weight -> less likely
    (1, 2, 5.0),
])

# Weighted walk (respects edge weights)
walk = basic_random_walk(G, 0, walk_length=20, weighted=True, seed=42)

# Unweighted walk (uniform transitions)
walk_uniform = basic_random_walk(G, 0, walk_length=20, weighted=False, seed=42)
```

Directed Graphs

Directed edges are handled correctly:

```
import networkx as nx
from py3plex.algorithms.general.walkers import basic_random_walk

# Create directed graph
G = nx.DiGraph()
G.add_edges_from([(0, 1), (1, 2), (2, 0), (1, 3)])
```

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```
# Walk respects edge direction
walk = basic_random_walk(G, 0, walk_length=10, seed=42)

# Verify all transitions follow directed edges
for i in range(len(walk) - 1):
    assert G.has_edge(walk[i], walk[i+1])
```

7.12.3 Node2Vec Biased Random Walk

Second-order random walks with return parameter p and in-out parameter q following the Node2Vec algorithm (Grover & Leskovec, 2016).

Parameters

When transitioning from node $t \rightarrow v \rightarrow x$, the probability of choosing x is biased:

- If $x == t$ (return to previous): weight / p
- If x is neighbor of t (stay close): $\text{weight} / 1$
- If x is not neighbor of t (explore): weight / q

Parameter Effects:

- $p > 1$: Less likely to return to previous node (explore forward)
- $p < 1$: More likely to return (backtrack)
- $q > 1$: Less likely to explore distant nodes (stay local, BFS-like)
- $q < 1$: More likely to explore distant nodes (venture far, DFS-like)
- $p = q = 1$: Equivalent to basic random walk

Basic Usage

```
from py3plex.algorithms.general.walkers import node2vec_walk
import networkx as nx

G = nx.karate_club_graph()

# Balanced walk (p=1, q=1)
walk_balanced = node2vec_walk(G, 0, walk_length=20, p=1.0, q=1.0, seed=42)

# BFS-like walk (stay local)
walk_bfs = node2vec_walk(G, 0, walk_length=20, p=1.0, q=2.0, seed=42)

# DFS-like walk (explore outward)
walk_dfs = node2vec_walk(G, 0, walk_length=20, p=1.0, q=0.5, seed=42)
```

Exploring vs Backtracking

```
from py3plex.algorithms.general.walkers import node2vec_walk
import networkx as nx
```

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```

# Triangle graph
G = nx.Graph()
G.add_edges_from([(0, 1), (1, 2), (2, 0)])

# Low p, high q: tends to backtrack
walk_backtrack = node2vec_walk(G, 0, walk_length=20, p=0.1, q=10.0, seed=42)

# High p, low q: tends to explore outward
walk_explore = node2vec_walk(G, 0, walk_length=20, p=10.0, q=0.1, seed=42)

# Count backtracking patterns
backtracks = sum(1 for i in range(2, len(walk_backtrack))
                  if walk_backtrack[i] == walk_backtrack[i-2])
print(f"Backtracks: {backtracks}")

```

7.12.4 Multiple Walk Generation

Generate multiple walks efficiently with deterministic seeding.

Generate Walks from All Nodes

```

from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate 10 walks from each node
walks = generate_walks(
    G,
    num_walks=10,
    walk_length=10,
    seed=42
)

print(f"Total walks: {len(walks)}") # 340 (34 nodes × 10 walks)
print(f"First walk: {walks[0]}")

```

Generate from Specific Nodes

```

from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate walks only from nodes 0, 1, 2
walks = generate_walks(
    G,
    num_walks=5,
    walk_length=15,
    start_nodes=[0, 1, 2],
    seed=42
)

```

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```
)

print(f"Total walks: {len(walks)}") # 15 (3 nodes × 5 walks)
```

Node2Vec-Style Walks

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate biased walks with p=0.5, q=2.0
walks = generate_walks(
    G,
    num_walks=10,
    walk_length=20,
    p=0.5,
    q=2.0,
    seed=42
)
```

Edge Sequences

Return walks as edge sequences instead of node sequences:

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

G = nx.karate_club_graph()

# Generate edge sequences
edge_walks = generate_walks(
    G,
    num_walks=5,
    walk_length=10,
    return_edges=True,
    seed=42
)

# First walk is a list of edges
print(edge_walks[0]) # [(0, 3), (3, 2), (2, 1), ...]
```

7.12.5 Multilayer Network Walks

Perform random walks on multilayer networks with layer constraints.

Layer-Constrained Walks

```
from py3plex.algorithms.general.walkers import layer_specific_random_walk
from py3plex.core import multinet
```

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```

# Create multilayer network
network = multinet.multi_layer_network()
network.add_layer("social")
network.add_layer("biological")

network.add_nodes([
    ("A", "social"), ("B", "social"), ("C", "social"),
    ("A", "biological"), ("B", "biological")
])

network.add_edges([
    (("A", "social"), ("B", "social")),
    (("B", "social"), ("C", "social")),
    (("A", "biological"), ("B", "biological")),
])

# Walk constrained to social layer
walk = layer_specific_random_walk(
    network.core_network,
    start_node="A---social",
    walk_length=10,
    layer="social",
    cross_layer_prob=0.0, # never cross layers
    seed=42
)

# All nodes in walk are in social layer
print(walk)

```

Cross-Layer Transitions

Allow occasional transitions between layers:

```

from py3plex.algorithms.general.walkers import layer_specific_random_walk

# Same network as above

# Walk with 10% probability of crossing layers
walk = layer_specific_random_walk(
    network.core_network,
    start_node="A---social",
    walk_length=20,
    layer="social",
    cross_layer_prob=0.1, # 10% chance to cross
    seed=42
)

# May contain nodes from both layers
print(walk)

```

7.12.6 Reproducibility

All walk functions support deterministic seeding for reproducibility.

Fixed Seed

```
from py3plex.algorithms.general.walkers import basic_random_walk
import networkx as nx

G = nx.karate_club_graph()

# Same seed produces identical walks
walk1 = basic_random_walk(G, 0, 50, seed=42)
walk2 = basic_random_walk(G, 0, 50, seed=42)

assert walk1 == walk2 # Identical
```

Different Seeds

```
# Different seeds produce different walks
walk1 = basic_random_walk(G, 0, 50, seed=42)
walk2 = basic_random_walk(G, 0, 50, seed=123)

assert walk1 != walk2 # Different
```

Batch Reproducibility

```
from py3plex.algorithms.general.walkers import generate_walks

G = nx.karate_club_graph()

# Same seed produces identical walk sets
walks1 = generate_walks(G, num_walks=100, walk_length=10, seed=42)
walks2 = generate_walks(G, num_walks=100, walk_length=10, seed=42)

assert walks1 == walks2
```

7.12.7 Performance Tips

Large Graphs

For large sparse graphs (>100k nodes), use basic walks for better performance:

```
from py3plex.algorithms.general.walkers import generate_walks
import networkx as nx

# Large sparse graph
G = nx.erdos_renyi_graph(100000, 0.0001, seed=42)

# Use basic walk (p=1, q=1) for speed
walks = generate_walks(
    G,
    num_walks=10,
```

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```
walk_length=100,  
p=1.0, # No bias = faster  
q=1.0,  
seed=42  
)
```

Parallel Processing

For generating many walks, consider parallel processing:

```
from py3plex.algorithms.general.walkers import generate_walks  
import networkx as nx  
from multiprocessing import Pool  
  
G = nx.karate_club_graph()  
  
def generate_batch(seed):  
    return generate_walks(G, num_walks=10, walk_length=20, seed=seed)  
  
# Generate walks in parallel  
with Pool(4) as pool:  
    batch_walks = pool.map(generate_batch, range(10))  
  
# Flatten results  
all_walks = [walk for batch in batch_walks for walk in batch]
```

7.12.8 Correctness Validation

The implementation has been validated against the following properties:

Uniformity Test

On unweighted regular graphs, transition probabilities are uniform:

```
import networkx as nx  
from py3plex.algorithms.general.walkers import basic_random_walk  
from collections import Counter  
  
# Complete graph (regular)  
G = nx.complete_graph(10)  
  
# Count visits to neighbors from node 0  
visits = Counter()  
for i in range(10000):  
    walk = basic_random_walk(G, 0, 1, weighted=False, seed=i)  
    if len(walk) > 1:  
        visits[walk[1]] += 1  
  
# All neighbors should be visited roughly equally  
print(visits)  
# Expected: ~1111 visits to each of 9 neighbors
```

Conservation Test

Sum of transition probabilities equals 1.0 (within machine epsilon):

```
import networkx as nx
import numpy as np

G = nx.karate_club_graph()

# For each node, verify probability conservation
for node in G.nodes():
    neighbors = list(G.neighbors(node))
    if len(neighbors) > 0:
        weights = np.array([G[node][n].get('weight', 1.0) for n in neighbors])
        probs = weights / weights.sum()
        assert abs(probs.sum() - 1.0) < 1e-10 # Machine epsilon
```

Bias Consistency Test

Node2Vec parameters p and q produce expected biases:

```
import networkx as nx
from py3plex.algorithms.general.walkers import node2vec_walk

# Triangle graph
G = nx.Graph()
G.add_edges_from([(0, 1), (1, 2), (2, 0)])

# Low p should encourage backtracking
backtracks_low_p = 0
for i in range(1000):
    walk = node2vec_walk(G, 0, 10, p=0.1, q=1.0, seed=i)
    for j in range(2, len(walk)):
        if walk[j] == walk[j-2]:
            backtracks_low_p += 1

# High p should discourage backtracking
backtracks_high_p = 0
for i in range(1000):
    walk = node2vec_walk(G, 0, 10, p=10.0, q=1.0, seed=i)
    for j in range(2, len(walk)):
        if walk[j] == walk[j-2]:
            backtracks_high_p += 1

assert backtracks_low_p > backtracks_high_p * 2 # Significantly more
```

Edge Weight Test

Visit frequency matches edge weight ratios:

```
import networkx as nx
from py3plex.algorithms.general.walkers import basic_random_walk
from collections import Counter
```

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```

G = nx.Graph()
G.add_weighted_edges_from([
    (0, 1, 1.0),
    (0, 2, 2.0),
    (0, 3, 3.0),
])

visits = Counter()
for i in range(10000):
    walk = basic_random_walk(G, 0, 1, weighted=True, seed=i)
    if len(walk) > 1:
        visits[walk[1]] += 1

# Expected ratio: 1:2:3
total = sum(visits.values())
ratios = [visits[1]/total, visits[2]/total, visits[3]/total]
expected = [1/6, 2/6, 3/6]

# Within 5% tolerance
for obs, exp in zip(ratios, expected):
    assert abs(obs - exp) < 0.05

```

7.12.9 References

Node2Vec:

Grover, A., & Leskovec, J. (2016). node2vec: Scalable feature learning for networks. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 855-864. <https://doi.org/10.1145/2939672.2939754>

DeepWalk:

Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). DeepWalk: Online learning of social representations. *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 701-710. <https://doi.org/10.1145/2623330.2623732>

Random Walks on Graphs:

Lovász, L. (1993). Random walks on graphs: A survey. *Combinatorics, Paul Erdős is Eighty*, 2, 1-46.

7.12.10 API Reference

See `py3plex.algorithms.general.walkers` for complete API documentation.

7.12.11 Examples

Complete examples can be found in:

- `examples/advanced/example_random_walks.py` - Basic usage examples
- `tests/test_random_walks.py` - Comprehensive test suite

Repository: <https://github.com/SkBlaz/py3plex>

7.13 I/O and Serialization

py3plex provides a comprehensive I/O system for reading and writing multilayer graphs in various formats. The system is designed to be extensible, efficient, and easy to use.

7.13.1 Supported Formats

The I/O system supports multiple file formats, each with different trade-offs:

- **JSON** - Human-readable, widely compatible, good for small to medium networks
- **JSONL** - Streaming JSON format, efficient for large networks
- **CSV** - Spreadsheet-compatible, easy to edit manually
- **Arrow/Feather** - High-performance columnar format (requires pyarrow)
- **Parquet** - Compressed columnar format, best for storage (requires pyarrow)

7.13.2 Basic Usage

The I/O system provides two main functions: `read()` and `write()`.

Reading Graphs

```
from py3plex.io import read

# Auto-detect format from extension
graph = read('network.json')
graph = read('network.csv')
graph = read('network.arrow')

# Or specify format explicitly
graph = read('myfile.dat', format='json')
```

Writing Graphs

```
from py3plex.io import write

# Auto-detect format from extension
write(graph, 'network.json')
write(graph, 'network.arrow')
write(graph, 'network.parquet')

# Or specify format explicitly
write(graph, 'myfile.dat', format='json')
```

7.13.3 Creating Graphs with the Schema API

The modern I/O system uses a schema-based API for creating graphs:

```
from py3plex.io import MultiLayerGraph, Node, Layer, Edge

# Create graph
graph = MultiLayerGraph(
```

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```

    directed=True,
    attributes={'name': 'Social Network'}
)

# Add layers
graph.add_layer(Layer(id='facebook', attributes={'type': 'social'}))
graph.add_layer(Layer(id='twitter', attributes={'type': 'social'}))

# Add nodes
graph.add_node(Node(id='alice', attributes={'age': 30}))
graph.add_node(Node(id='bob', attributes={'age': 25}))

# Add edges
graph.add_edge(Edge(
    src='alice',
    dst='bob',
    src_layer='facebook',
    dst_layer='facebook',
    attributes={'weight': 0.8}
))

```

7.13.4 Apache Arrow Format

Apache Arrow is a high-performance columnar format designed for efficient data interchange. py3plex supports Arrow through two sub-formats:

- **Feather** - Fast, uncompressed format ideal for temporary storage
- **Parquet** - Compressed format ideal for long-term storage

Installing Arrow Support

Arrow support requires the pyarrow package:

```

pip install 'py3plex[arrow]'
# or directly
pip install pyarrow

```

Using Arrow Format

```

from py3plex.io import read, write

# Feather format (fast, uncompressed)
write(graph, 'network.arrow')
graph = read('network.arrow')

# Parquet format (compressed)
write(graph, 'network.parquet', format='parquet')
graph = read('network.parquet', format='parquet')

```


Benefits of Arrow Format

1. **Performance:** Columnar storage enables fast read/write operations
2. **Compression:** Parquet format provides excellent compression ratios
3. **Interoperability:** Arrow is an industry-standard format supported by:
 - pandas, polars (Python data analysis)
 - Apache Spark (big data processing)
 - R, Julia (statistical computing)
 - DuckDB (analytical database)
4. **Type Safety:** Schema preservation with strong typing
5. **Zero-Copy:** Efficient in-memory representation

Performance Comparison

For a typical multilayer network with 1000 nodes and ~5000 edges:

Format	Write Time	Read Time	File Size
Arrow	0.016s	0.008s	0.46 MB
Parquet	0.020s	0.010s	0.35 MB
JSON	0.046s	0.030s	1.09 MB

Arrow format is **2-3x faster** for writes and provides **2-3x better compression** compared to JSON.

When to Use Each Format

Use Arrow/Feather when:

- You need maximum read/write performance
- Working with large networks (>10k nodes)
- Interoperating with data science tools (pandas, polars)
- Building data pipelines

Use Parquet when:

- Long-term storage is important
- Minimizing storage costs
- Sharing data across platforms
- Archiving networks

Use JSON when:

- Human readability is important
- Working with small networks
- Debugging or manual editing
- Maximum compatibility needed

Use CSV when:

- Working with spreadsheet tools (Excel)

- Simple edge lists
- Manual data entry/editing

7.13.5 CSV Format with Sidecars

CSV format supports optional sidecar files for node and layer attributes:

```
from py3plex.io import read, write

# Write with sidecars
write(graph, 'edges.csv', format='csv', write_sidecars=True)
# Creates: edges.csv, nodes.csv, layers.csv

# Read with sidecars
graph = read('edges.csv', format='csv',
            nodes_file='nodes.csv',
            layers_file='layers.csv')
```

7.13.6 Integration with NetworkX

Convert between py3plex I/O format and NetworkX:

```
from py3plex.io import read, to_networkx, from_networkx

# Load graph
graph = read('network.json')

# Convert to NetworkX
G = to_networkx(graph, mode='union') # Merge all layers
# or
G = to_networkx(graph, mode='multiplex') # Preserve layers as (node, layer)

# Convert back from NetworkX
graph = from_networkx(G, mode='multiplex')
```

7.13.7 Example: Complete Workflow

Here's a complete example demonstrating the I/O system:

```
from py3plex.io import (
    MultiLayerGraph, Node, Layer, Edge,
    read, write, to_networkx
)

# Create a multilayer network
graph = MultiLayerGraph(directed=True)

# Add layers
for layer_id in ['social', 'work', 'family']:
    graph.add_layer(Layer(id=layer_id))

# Add nodes
for name in ['alice', 'bob', 'charlie']:
```

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```

graph.add_node(Node(id=name))

# Add edges
edges = [
    ('alice', 'bob', 'social', 'social', 0.8),
    ('bob', 'charlie', 'work', 'work', 0.6),
    ('alice', 'charlie', 'family', 'family', 0.9),
]

for src, dst, src_layer, dst_layer, weight in edges:
    graph.add_edge(Edge(
        src=src, dst=dst,
        src_layer=src_layer, dst_layer=dst_layer,
        attributes={'weight': weight}
    ))

# Save in multiple formats
write(graph, 'network.json')
write(graph, 'network.arrow')
write(graph, 'network.parquet')

# Load back
loaded = read('network.arrow')

# Convert to NetworkX for analysis
G = to_networkx(loaded, mode='union')

# Use NetworkX algorithms
import networkx as nx
centrality = nx.degree_centrality(G)
print(f"Most central node: {max(centrality, key=centrality.get)}")

```

7.13.8 Checking Supported Formats

You can query which formats are available at runtime:

```

from py3plex.io import supported_formats

formats = supported_formats()
print(f"Read formats: {formats['read']}")
print(f"Write formats: {formats['write']}")

```

This is useful for checking if optional dependencies (like pyarrow) are installed.

7.13.9 Schema Validation

The I/O system includes automatic validation:

```

from py3plex.io import (
    MultiLayerGraph, Node, Edge,
    ReferentialIntegrityError
)

```

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```
graph = MultiLayerGraph()
graph.add_node(Node(id='alice'))

try:
    # This will fail - bob doesn't exist
    graph.add_edge(Edge(
        src='alice', dst='bob',
        src_layer='l1', dst_layer='l1'
    ))
except ReferentialIntegrityError as e:
    print(f"Validation error: {e}")
```

Validation ensures:

1. All edge endpoints reference existing nodes
2. All edge layers reference existing layers
3. All attributes are JSON-serializable
4. No duplicate edges (by src, dst, src_layer, dst_layer, key)

7.13.10 Advanced: Custom Formats

The I/O system is extensible. You can register custom format readers/writers:

```
from py3plex.io import register_reader, register_writer

def my_reader(filepath, **kwargs):
    # Custom reading logic
    graph = MultiLayerGraph()
    # ... populate graph ...
    return graph

def my_writer(graph, filepath, **kwargs):
    # Custom writing logic
    with open(filepath, 'w') as f:
        # ... write graph ...
    pass

# Register
register_reader('myformat', my_reader)
register_writer('myformat', my_writer)

# Now you can use it
write(graph, 'network.myformat')
graph = read('network.myformat')
```

7.13.11 Examples

Complete examples are available in `examples/io_and_data/`:

- `example_new_io.py` - Comprehensive I/O demonstration
- `example_save_to_arrow.py` - Apache Arrow format usage

- `example_save_to_gpickle.py` - NetworkX pickle format
- `example_save_to_edgelist.py` - Edge list format
- `example_schema_validation.py` - Schema validation examples

7.13.12 See Also

- *5-Minute Quickstart* - Getting started guide
- *Working with Networks* - Basic network operations
- *py3plex Core Model* - NetworkX integration details
- *Performance and Scalability Best Practices* - Performance optimization tips

7.14 Visualization Guide



This guide covers multilayer network visualization in Py3plex, including preset modes, customization options, and best practices for different network scales.

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7.14.1 Quick Start

Basic Multilayer Visualization

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import draw_multilayer_default
import matplotlib.pyplot as plt

# Load network
network = multinet.multi_layer_network()
network.load_network("network.csv", input_type="multiedgelist")

# Visualize with defaults
draw_multilayer_default(
    network.get_layers(),
    display=True,
    labels=True
)
```

7.14.2 Preset Visualization Modes

Py3plex provides three preset modes optimized for different network scales and use cases.

Minimal Mode (Large Networks >1000 nodes)

Optimized for networks with many nodes where detail isn't critical.

```
from py3plex.visualization.multilayer import draw_multilayer_default

# Minimal preset for large networks
draw_multilayer_default(
    network.get_layers(),
    # Node settings
    node_size=5,           # Small nodes
    labels=False,         # No node labels (too cluttered)
    node_labels=False,    # No node IDs
```

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```

# Edge settings
edge_size=0.5,          # Thin edges
alphalevel=0.3,         # Transparent edges (reduce clutter)

# Layout
background_shape="circle", # Circular layout

# Display
display=True,
remove_isolated_nodes=True # Clean up isolated nodes
)

```

Use cases: - Large social networks (>1000 nodes) - Overview visualizations - Pattern detection at macro level - Network topology understanding

Advantages: - Fast rendering - Reduced visual clutter - Shows overall structure - Works with many nodes

Balanced Mode (Medium Networks 100-1000 nodes)

Default settings that work well for most networks. Good balance between detail and clarity.

```

# Balanced preset (this is the default)
draw_multilayer_default(
    network.get_layers(),
    # Node settings
    node_size=10,          # Medium nodes
    labels=True,           # Show layer labels
    node_labels=False,     # Node IDs hidden by default

    # Edge settings
    edge_size=1.0,         # Normal edge width
    alphalevel=0.13,       # Semi-transparent edges

    # Layout
    background_shape="circle", # Circular layout
    networks_color="rainbow",  # Auto-assign colors

    # Display
    display=True
)

```

Use cases: - Most research networks - Exploratory data analysis - Publication figures (moderate detail) - Interactive exploration

Advantages: - Good readability - Reasonable performance - Balanced aesthetics - Suitable for most publications

Dense Mode (Small Networks <100 nodes)

Maximum detail for small networks where every node and edge matters.

```

# Dense preset for small, detailed networks
draw_multilayer_default(
    network.get_layers(),

```

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```

# Node settings
node_size=20,           # Large nodes
labels=True,           # Show all labels
node_labels=True,       # Show node IDs
node_font_size=10,      # Readable font
scale_by_size=True,     # Scale by degree

# Edge settings
edge_size=2.0,          # Thick edges
alphalevel=0.8,         # Mostly opaque
arrowsize=0.5,          # Visible arrows (if directed)

# Layout
background_shape="rectangle", # Rectangular layout
networks_color="rainbow",

# Display
display=True
)

```

Use cases: - Small case studies - Detailed analysis - Node-level examination - High-quality publication figures

Advantages: - Maximum detail - Clear node identification - Edge weights visible - Professional appearance

7.14.3 Layout Options

Py3plex supports multiple layout algorithms for different network structures.

Circular Layout (Default)

Arranges layers in a circle. Good for showing inter-layer connections.

```

draw_multilayer_default(
    network.get_layers(),
    background_shape="circle",
    rectanglex=1.0, # Circle radius x
    rectangley=1.0 # Circle radius y
)

```

Best for: - 2-10 layers - Symmetric layer interactions - Publication figures

Rectangular Layout

Arranges layers in a grid. Good for many layers or hierarchical structures.

```

draw_multilayer_default(
    network.get_layers(),
    background_shape="rectangle",
    rectanglex=2.0, # Width of layout
    rectangley=1.0 # Height of layout
)

```

Best for: - Many layers (>10) - Hierarchical networks - Time-series data - Wide figures

7.14.4 Auto-Scaling Features

Py3plex automatically adjusts visualization parameters based on network size.

Automatic Node Sizing

Scale node sizes by degree (number of connections):

```
draw_multilayer_default(
    network.get_layers(),
    scale_by_size=True,      # Enable auto-scaling
    node_size=10            # Base size (scaled up/down by degree)
)
```

How it works:

- Nodes with more connections → larger size
- Isolated nodes → smaller size
- Hub nodes are easily identifiable

Automatic Color Assignment

Py3plex uses colorblind-safe palettes by default:

```
# Rainbow colors (automatic assignment)
draw_multilayer_default(
    network.get_layers(),
    networks_color="rainbow" # Auto-assign distinct colors
)

# Custom palette
draw_multilayer_default(
    network.get_layers(),
    networks_color=["#FF6B6B", "#4ECDC4", "#45B7D1", "#FFA07A"]
)
```

Available palettes (from `py3plex.config`):

- `colorblind_safe`: 8 colors safe for colorblind viewers
- `wong`: 7-color palette (scientifically validated)
- `tol_bright`: 7 bright colors
- `rainbow`: Automatic generation

Automatic Layout Adjustment

Layout parameters auto-adjust to network size:

```
# For small networks, layout is compact
small_network = multinet.multi_layer_network()
# ... load 50-node network ...
draw_multilayer_default(small_network.get_layers()) # Compact layout

# For large networks, layout expands
large_network = multinet.multi_layer_network()
```

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```
# ... load 1000-node network ...
draw_multilayer_default(large_network.get_layers()) # Expanded layout
```

7.14.5 Customization Options

Complete Parameter Reference

```
draw_multilayer_default(
    network_list,                # List of layer subgraphs

    # Display control
    display=True,                # Show plot immediately
    axis=None,                   # Matplotlib axis (None = create new)

    # Node appearance
    node_size=10,                # Base node size
    scale_by_size=False,         # Scale by degree?
    node_labels=False,           # Show node IDs?
    node_font_size=5,            # Font size for node labels

    # Edge appearance
    edge_size=1.0,               # Edge thickness
    alphalevel=0.13,             # Edge transparency (0=invisible, 1=opaque)
    arrowsize=0.5,               # Arrow size (for directed graphs)

    # Layout
    background_shape="circle",   # "circle" or "rectangle"
    rectanglex=1.0,              # Layout width/radius
    rectangley=1.0,              # Layout height/radius

    # Colors
    networks_color="rainbow",    # "rainbow" or list of colors
    background_color="rainbow",  # Background color scheme

    # Labels
    labels=True,                 # Show layer labels?
    label_position=1.0,          # Label distance from center

    # Cleanup
    remove_isolated_nodes=False, # Remove disconnected nodes?

    # Debugging
    verbose=False                # Print debug info?
)
```

Layer-Specific Customization

Customize individual layers:

```
import matplotlib.pyplot as plt
from py3plex.visualization.multilayer import draw_multilayer_default
```

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```
# Get layers
layers = network.get_layers()

# Modify specific layer (e.g., highlight layer 0)
for node in layers[0].nodes():
    layers[0].nodes[node]['color'] = 'red'
    layers[0].nodes[node]['size'] = 20

# Visualize with custom layer
draw_multilayer_default(layers, display=True)
```

Color-Coding by Community

Color nodes by community membership:

```
from py3plex.algorithms.community_detection import community_louvain
import matplotlib.pyplot as plt
import matplotlib.cm as cm

# Detect communities
communities = community_louvain.best_partition(network.core_network)

# Assign colors
num_communities = len(set(communities.values()))
colors = cm.rainbow([i/num_communities for i in range(num_communities)])

# Apply to network
for node, comm_id in communities.items():
    network.core_network.nodes[node]['color'] = colors[comm_id]

# Visualize
draw_multilayer_default(network.get_layers(), display=True)
```

7.14.6 Exporting Visualizations

Save to File

```
import matplotlib.pyplot as plt

# Create figure
fig, ax = plt.subplots(1, 1, figsize=(12, 8))

# Draw network
draw_multilayer_default(
    network.get_layers(),
    display=False,
    axis=ax
)

# Customize and save
plt.title("Multilayer Network Visualization")
plt.savefig('network.png', dpi=300, bbox_inches='tight')
```

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```
plt.savefig('network.pdf', bbox_inches='tight') # Vector format
print("Visualization saved to network.png and network.pdf")
```

High-Quality Publications

```
import matplotlib.pyplot as plt

# Use publication settings
plt.rcParams['font.family'] = 'serif'
plt.rcParams['font.size'] = 10
plt.rcParams['figure.dpi'] = 300

# Large figure for detail
fig, ax = plt.subplots(1, 1, figsize=(10, 8))

# Dense mode for publications
draw_multilayer_default(
    network.get_layers(),
    display=False,
    axis=ax,
    node_size=15,
    labels=True,
    alphalevel=0.5,
    background_shape="circle"
)

plt.title("Figure 1: Multilayer Network Structure", fontsize=12)
plt.savefig('publication_figure.pdf', bbox_inches='tight')
plt.savefig('publication_figure.png', dpi=300, bbox_inches='tight')
```

7.14.7 Interactive Visualizations

Using Plotly (Optional)

Py3plex provides interactive visualization capabilities through Plotly, allowing you to explore networks dynamically in your web browser.

Installation:

```
# Install plotly separately
pip install plotly

# Or install py3plex with visualization extras
pip install git+https://github.com/SkBlaz/py3plex.git#egg=py3plex[viz]
```

Basic Interactive Hairball Plot

The `interactive_hairball_plot` function creates an interactive 2D network visualization:

```
import networkx as nx
from py3plex.core import random_generators
from py3plex.visualization.multilayer import interactive_hairball_plot
```

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```

# Generate a network
multilayer_net = random_generators.random_multilayer_ER(50, 3, 0.15, directed=False)

# Convert to NetworkX graph
network_colors, G = multilayer_net.get_layers(style="hairball")

# Compute layout
pos = nx.spring_layout(G, iterations=50, seed=42)

# Compute node sizes based on degree
degrees = dict(G.degree())
max_degree = max(degrees.values())
node_sizes = [10 + 40 * (degrees[node] / max_degree) for node in G.nodes()]

# Create color mapping
color_mapping = {node: node_sizes[i] for i, node in enumerate(G.nodes())}

# Generate interactive visualization
fig = interactive_hairball_plot(
    G,
    nsizes=node_sizes,
    final_color_mapping=color_mapping,
    pos=pos,
    colorscale="Viridis" # Options: Viridis, Rainbow, Blues, Reds, etc.
)

# Save to HTML file
if fig:
    fig.write_html("network.html")
    print("Interactive visualization saved to network.html")

```

Interactive Features:

- **Hover:** Move your mouse over nodes to see details
- **Zoom:** Use mouse wheel to zoom in/out
- **Pan:** Click and drag to move the view
- **Select:** Click nodes to highlight connections

Color Scales Available:

- **Viridis:** Perceptually uniform, colorblind-friendly
- **Rainbow:** Full color spectrum
- **Blues, Reds, Greens:** Single-hue gradients
- **RdBu, YlOrRd:** Diverging color schemes
- **Jet, Hot, Electric:** High-contrast schemes

Advanced Interactive Multilayer Example

For more complex multilayer networks with custom styling:

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import interactive_hairball_plot
import networkx as nx

# Create multilayer network
network = multinet.multi_layer_network()

# Add nodes to different layers
layers = ['social', 'professional', 'family']
nodes_per_layer = {
    'social': ['Alice', 'Bob', 'Charlie', 'David', 'Eve'],
    'professional': ['Alice', 'Bob', 'Frank', 'Grace', 'Henry'],
    'family': ['Alice', 'Charlie', 'David', 'Ivan', 'Jane']
}

for layer in layers:
    for node in nodes_per_layer[layer]:
        network.add_nodes([{'source': node, 'type': layer}])

# Add edges (example for one layer)
network.add_edges([
    {'source': 'Alice', 'target': 'Bob', 'source_type': 'social', 'target_type': 'social'
    ↪},
    {'source': 'Bob', 'target': 'Charlie', 'source_type': 'social', 'target_type':
    ↪'social'},
])

# Convert to aggregate graph
G = nx.Graph()
for node in set(n for layer_nodes in nodes_per_layer.values() for n in layer_nodes):
    G.add_node(node)

# Compute layout and visualize
pos = nx.spring_layout(G, seed=42)
degrees = dict(G.degree())
node_sizes = [20 + 80 * (degrees[node] / max(degrees.values())) for node in G.nodes()]

# Assign colors by layer membership
layer_colors = {'social': 0, 'professional': 50, 'family': 100}
color_mapping = {}
for node in G.nodes():
    # Find primary layer for this node
    for layer, nodes in nodes_per_layer.items():
        if node in nodes:
            color_mapping[node] = layer_colors[layer]
            break

# Create interactive visualization
fig = interactive_hairball_plot(G, node_sizes, color_mapping, pos, colorscale="Viridis")
```

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```
# Customize the figure
if fig:
    fig.update_layout(
        title="Interactive Multilayer Network",
        hovermode='closest',
        plot_bgcolor='rgba(240, 240, 240, 0.9)'
    )
    fig.write_html("multilayer_network.html")
```

Complete Working Examples:

See the following example scripts in the repository:

- `examples/visualization/example_interactive_hairball.py` - Basic interactive visualization
- `examples/visualization/example_interactive_multilayer.py` - Advanced multilayer interactive plot

These examples are standalone and can be run directly:

```
cd examples/visualization
python example_interactive_hairball.py
# Opens browser with interactive visualization
```

Jupyter Notebook Integration

Interactive visualizations work seamlessly in Jupyter notebooks:

```
# In Jupyter notebook
from py3plex.visualization.multilayer import interactive_hairball_plot

# ... prepare your graph, positions, etc. ...

fig = interactive_hairball_plot(G, node_sizes, color_mapping, pos)

# The figure will be displayed inline in the notebook
# No need to call fig.show() - it's called automatically
```

Notebook Tips:

1. The interactive plot appears directly in the notebook
2. All interactive features work in the notebook interface
3. Use `fig.write_html()` to save for sharing
4. Interactive plots work in JupyterLab, Jupyter Notebook, and VS Code notebooks

Saving Interactive Visualizations

Save your interactive visualizations for sharing or publishing:

```
# Create the figure
fig = interactive_hairball_plot(G, node_sizes, color_mapping, pos)

# Save as standalone HTML
fig.write_html("network.html")
```

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```
# Save as HTML with custom configuration
fig.write_html(
    "network.html",
    config={
        'displayModeBar': True, # Show toolbar
        'displaylogo': False,   # Hide Plotly logo
        'modeBarButtonsToRemove': ['pan2d', 'lasso2d'] # Hide specific tools
    }
)

# Save as static image (requires kaleido)
# pip install kaleido
fig.write_image("network.png", width=1200, height=800)
fig.write_image("network.pdf") # Vector format
```

HTML File Features:

- Standalone - works without internet connection
- No dependencies needed in browser
- Full interactivity preserved
- Can be embedded in web pages
- Works on mobile devices

Embedding in Web Applications

Embed interactive visualizations in your web applications:

```
<!DOCTYPE html>
<html>
<head>
    <title>Network Visualization</title>
</head>
<body>
    <h1>My Network Analysis</h1>

    <!-- Embed the interactive plot -->
    <iframe src="network.html" width="100%" height="600px" frameborder="0"></iframe>

    <p>Description of your network...</p>
</body>
</html>
```

Or include the plot div directly:

```
# Generate only the div (not full HTML page)
fig_html = fig.to_html(
    include_plotlyjs='cdn', # Use CDN for Plotly.js
    div_id='my-network-plot'
)

# Use fig_html in your web framework (Flask, Django, etc.)
```

Troubleshooting Interactive Visualizations

Issue: Plotly not available

```
# Solution: Install plotly
pip install plotly
```

Issue: Figure doesn't show in Jupyter

```
# Solution: Call fig.show() explicitly
fig = interactive_hairball_plot(G, node_sizes, color_mapping, pos)
fig.show()
```

Issue: Slow performance with large networks

```
# Solution 1: Sample nodes for interactive view
import random
sample_nodes = random.sample(list(G.nodes()), min(500, G.number_of_nodes()))
G_sample = G.subgraph(sample_nodes)

# Solution 2: Use simpler layout
pos = nx.random_layout(G) # Faster than spring_layout

# Solution 3: Reduce node sizes and simplify colors
node_sizes = [5] * G.number_of_nodes() # Constant size
color_mapping = {node: 0 for node in G.nodes()} # Single color
```

Issue: HTML file is too large

```
# Solution: Reduce data in the plot
# 1. Use fewer nodes (sample or filter)
# 2. Use constant node sizes instead of varying
# 3. Simplify color scheme
# 4. Remove edge traces if not needed
```

Performance Guidelines

Network Size Recommendations:

- **< 100 nodes:** All features work smoothly
- **100-500 nodes:** Good performance, all features
- **500-1000 nodes:** Usable, may have slight lag
- **1000-5000 nodes:** Sample nodes or use simplified styling
- **> 5000 nodes:** Use static visualizations instead

Optimization Tips:

1. Use constant node sizes for large networks
2. Simplify color schemes (single color or discrete categories)
3. Use `nx.random_layout()` instead of `nx.spring_layout()` for speed
4. Sample nodes for interactive exploration
5. Consider static visualization for very large networks

Jupyter Notebook Integration

```
# Enable inline plotting
%matplotlib inline

# Or for interactive plots
%matplotlib notebook

# Visualize
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default(network.get_layers(), display=True)
```

7.14.8 Performance Tips

For Large Networks

1. Use **minimal mode** (small nodes, no labels)
2. **Sample nodes** if network is too large:

```
import random

# Sample 1000 random nodes
all_nodes = list(network.get_nodes())
sample_nodes = random.sample(all_nodes, min(1000, len(all_nodes)))

# Create subnetwork
subnetwork = network.get_subnetwork(sample_nodes)

# Visualize sample
draw_multilayer_default(subnetwork.get_layers(), display=True)
```

3. **Remove isolated nodes:**

```
draw_multilayer_default(
    network.get_layers(),
    remove_isolated_nodes=True # Faster rendering
)
```

4. Use **lower resolution** for interactive exploration:

```
plt.figure(figsize=(8, 6), dpi=100) # Lower DPI for speed
```

7.14.9 Troubleshooting

Visualization Not Showing

Issue: Plot doesn't appear

Solutions:

```
# Jupyter notebook: Enable inline plots
%matplotlib inline

# Python script: Add plt.show()
```

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```
draw_multilayer_default(network.get_layers(), display=False)
plt.show()

# Headless server: Save to file
import matplotlib
matplotlib.use('Agg') # Must be before importing pyplot
```

Nodes Overlapping

Issue: Nodes overlap and are hard to distinguish

Solutions:

```
# 1. Use smaller node size
draw_multilayer_default(network.get_layers(), node_size=5)

# 2. Increase layout area
draw_multilayer_default(
    network.get_layers(),
    rectanglex=2.0, # Wider layout
    rectangley=2.0 # Taller layout
)

# 3. Sample nodes
# (See "Performance Tips" above)
```

Labels Too Crowded

Issue: Layer labels overlap or are too dense

Solutions:

```
# 1. Disable labels for large networks
draw_multilayer_default(network.get_layers(), labels=False)

# 2. Adjust label position
draw_multilayer_default(
    network.get_layers(),
    labels=True,
    label_position=1.2 # Move labels outward
)

# 3. Use smaller font
draw_multilayer_default(
    network.get_layers(),
    node_font_size=3 # Smaller font
)
```

7.14.10 Advanced Visualization Modes

Py3plex provides multiple specialized visualization modes for multilayer networks, each optimized for different analysis goals. These modes can be accessed through the unified `visualize_multilayer_network` API or by calling individual plot functions.

Overview of Visualization Modes

The following visualization modes are available:

1. **diagonal** (default): Layer-centric diagonal layout with inter-layer edges
2. **small_multiples**: Side-by-side comparison of layers in a grid
3. **edge_colored_projection**: Aggregated view with color-coded edges by layer
4. **supra_adjacency_heatmap**: Matrix representation of the multilayer structure
5. **radial_layers**: Concentric circles with layers as rings
6. **ego_multilayer**: Focus on a single node's neighborhood across layers
7. **flow/alluvial**: Layered flow visualization with horizontal bands
8. **sankey**: Sankey diagram showing inter-layer flow strength

Unified API

All visualization modes can be accessed through the `visualize_multilayer_network` function:

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import visualize_multilayer_network

# Load network
network = multinet.multi_layer_network()
network.load_network("network.txt", input_type="multiedgelist")

# Use any visualization mode
fig = visualize_multilayer_network(
    network,
    visualization_type="small_multiples" # or any other mode
)
```

Small Multiples View

Displays each layer as a separate subplot in a grid layout, making it easy to compare layer structures side-by-side.



example_images/multilayer_small_multiples_shared.png

```
from py3plex.visualization.multilayer import visualize_multilayer_network

# Shared layout (nodes appear at same positions across layers)
fig = visualize_multilayer_network(
    network,
    visualization_type="small_multiples",
    shared_layout=True,          # Same node positions in all layers
    layout="spring",            # Layout algorithm
    node_size=300,
    max_cols=3,                 # Maximum columns in grid
    show_layer_titles=True,
    with_labels=True
)

# Independent layouts (optimized per layer)
```

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```
fig = visualize_multilayer_network(  
    network,  
    visualization_type="small_multiples",  
    shared_layout=False,      # Different layouts per layer  
    layout="circular"  
)
```

Best for:

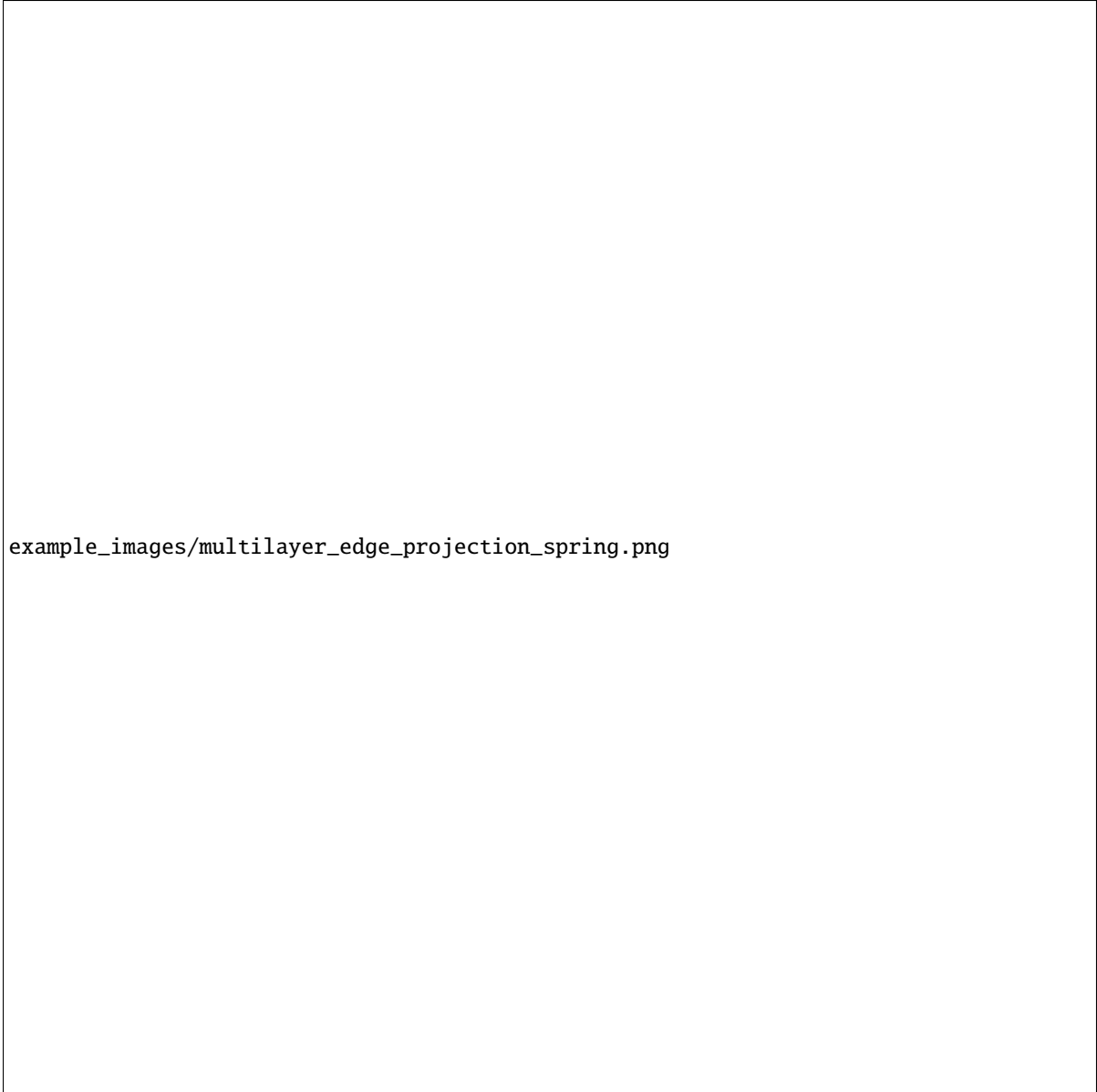
- Comparing layer structures
- Identifying layer-specific patterns
- Understanding node presence across layers
- Networks with 2-10 layers

Parameters:

- `shared_layout`: Use same node positions across layers
- `layout`: "spring", "circular", "random", "kamada_kawai"
- `max_cols`: Maximum number of columns in grid
- `show_layer_titles`: Display layer names

Edge-Colored Projection

Projects all layers onto a single 2D graph, using edge colors to indicate layer membership. Useful for seeing the overall structure while maintaining layer information.



example_images/multilayer_edge_projection_spring.png

```
from py3plex.visualization.multilayer import visualize_multilayer_network

# Basic projection
fig = visualize_multilayer_network(
    network,
    visualization_type="edge_colored_projection",
    layout="spring",
    node_size=500,
    edge_alpha=0.7,           # Edge transparency
    with_labels=True
)

# Custom layer colors
custom_colors = {
    'layer1': 'red',
```

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```
'layer2': 'blue',
'layer3': 'green'
}

fig = visualize_multilayer_network(
    network,
    visualization_type="edge_colored_projection",
    layer_colors=custom_colors
)
```

Best for:

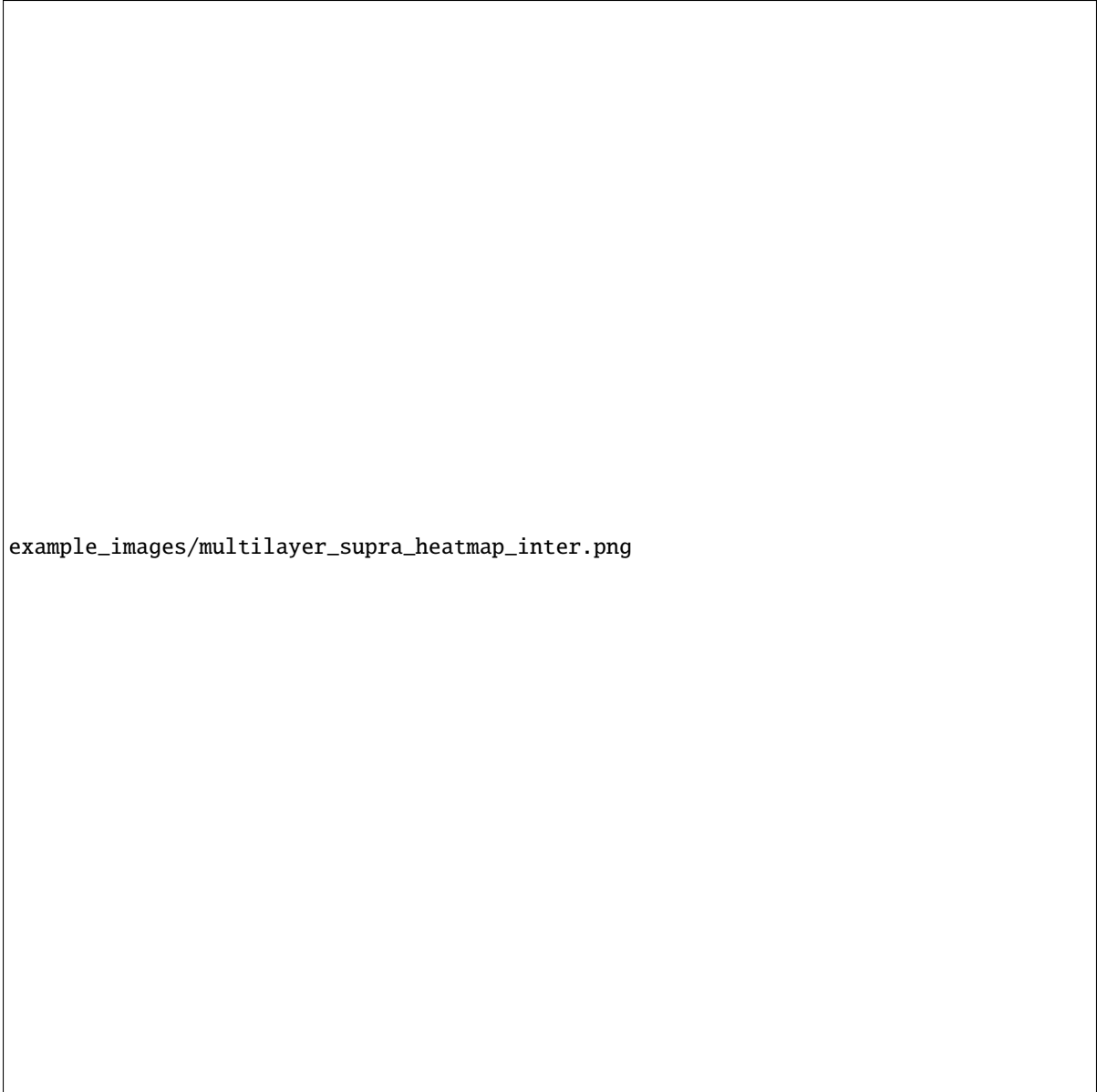
- Aggregate network structure
- Comparing edge distributions across layers
- Identifying layer-dominant connections
- Networks where edges don't heavily overlap

Parameters:

- layout: Layout algorithm for the aggregated graph
- layer_colors: Dict mapping layer names to colors
- edge_alpha: Transparency level for edges (0-1)
- figsize: Figure dimensions

Supra-Adjacency Heatmap

Shows the multilayer network as a block matrix where each block represents the adjacency matrix of one layer. Can optionally include inter-layer connections.



example_images/multilayer_supra_heatmap_inter.png

```
from py3plex.visualization.multilayer import visualize_multilayer_network

# Intra-layer only (block-diagonal structure)
fig = visualize_multilayer_network(
    network,
    visualization_type="supra_adjacency_heatmap",
    include_inter_layer=False,
    cmap="Blues"
)

# With inter-layer coupling
fig = visualize_multilayer_network(
    network,
    visualization_type="supra_adjacency_heatmap",
    include_inter_layer=True,
```

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```
inter_layer_weight=0.5,    # Weight for inter-layer edges
cmap="viridis"
)
```

Best for:

- Mathematical analysis of structure
- Identifying block patterns
- Understanding coupling between layers
- Spectral analysis preparation

Parameters:

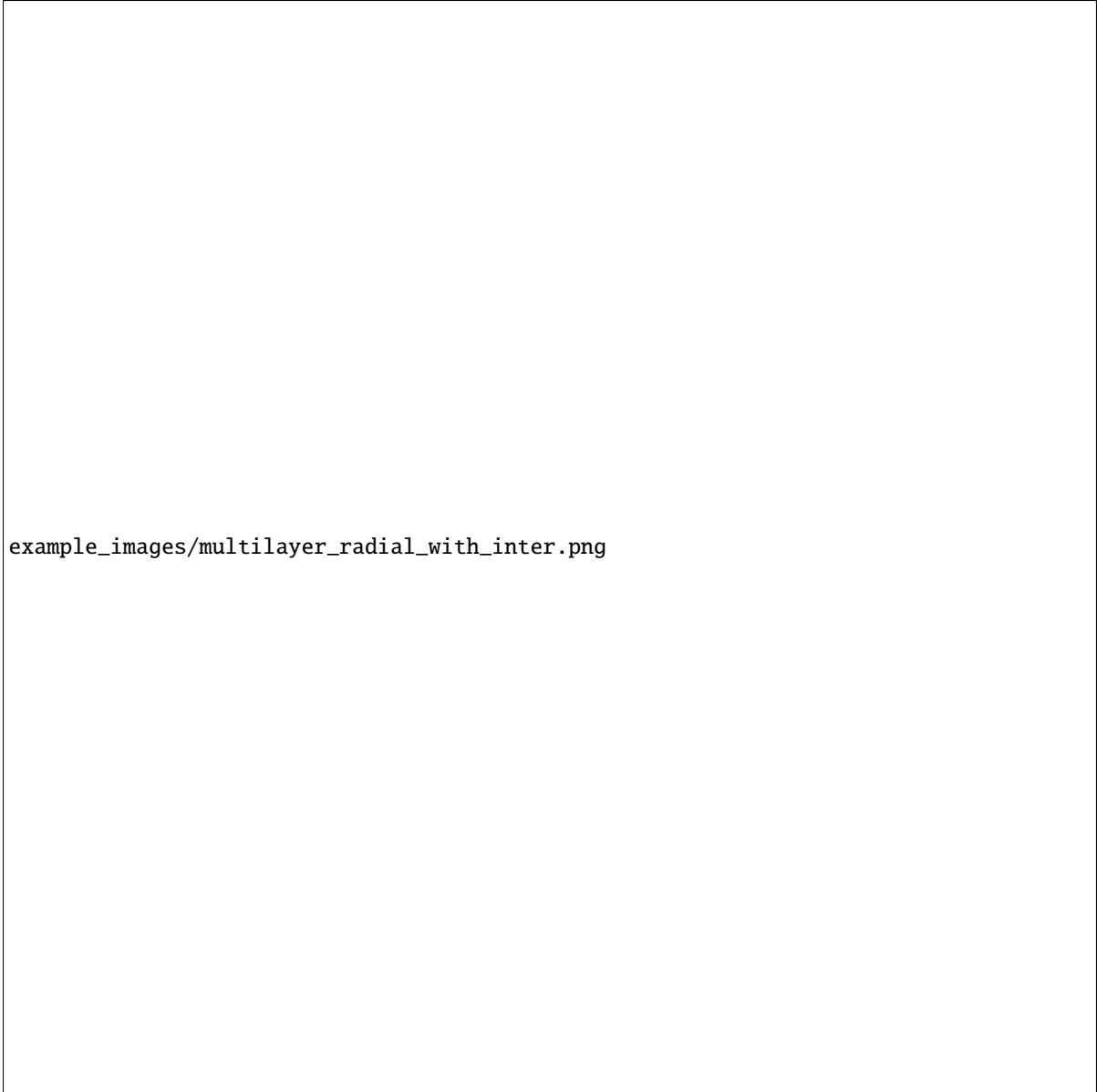
- `include_inter_layer`: Show inter-layer connections
- `inter_layer_weight`: Default weight for inter-layer edges
- `cmap`: Matplotlib colormap name
- `figsize`: Figure dimensions

Interpretation:

- Diagonal blocks = intra-layer adjacency matrices
- Off-diagonal blocks = inter-layer connections
- White grid lines = layer boundaries

Radial/Concentric Layers

Arranges layers as concentric circles, with nodes positioned on rings and inter-layer edges shown as radial connections.



example_images/multilayer_radial_with_inter.png

```
from py3plex.visualization.multilayer import visualize_multilayer_network

# With inter-layer edges
fig = visualize_multilayer_network(
    network,
    visualization_type="radial_layers",
    base_radius=1.0,          # Radius of innermost layer
    radius_step=1.5,         # Distance between layers
    node_size=200,
    draw_inter_layer_edges=True,
    edge_alpha=0.5
)

# Intra-layer only
fig = visualize_multilayer_network(
```

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```
network,  
visualization_type="radial_layers",  
draw_inter_layer_edges=False  
)
```

Best for:

- Temporal networks (layers as time steps)
- Hierarchical multilayer networks
- Visualizing node evolution across layers
- Networks with clear layer ordering

Parameters:

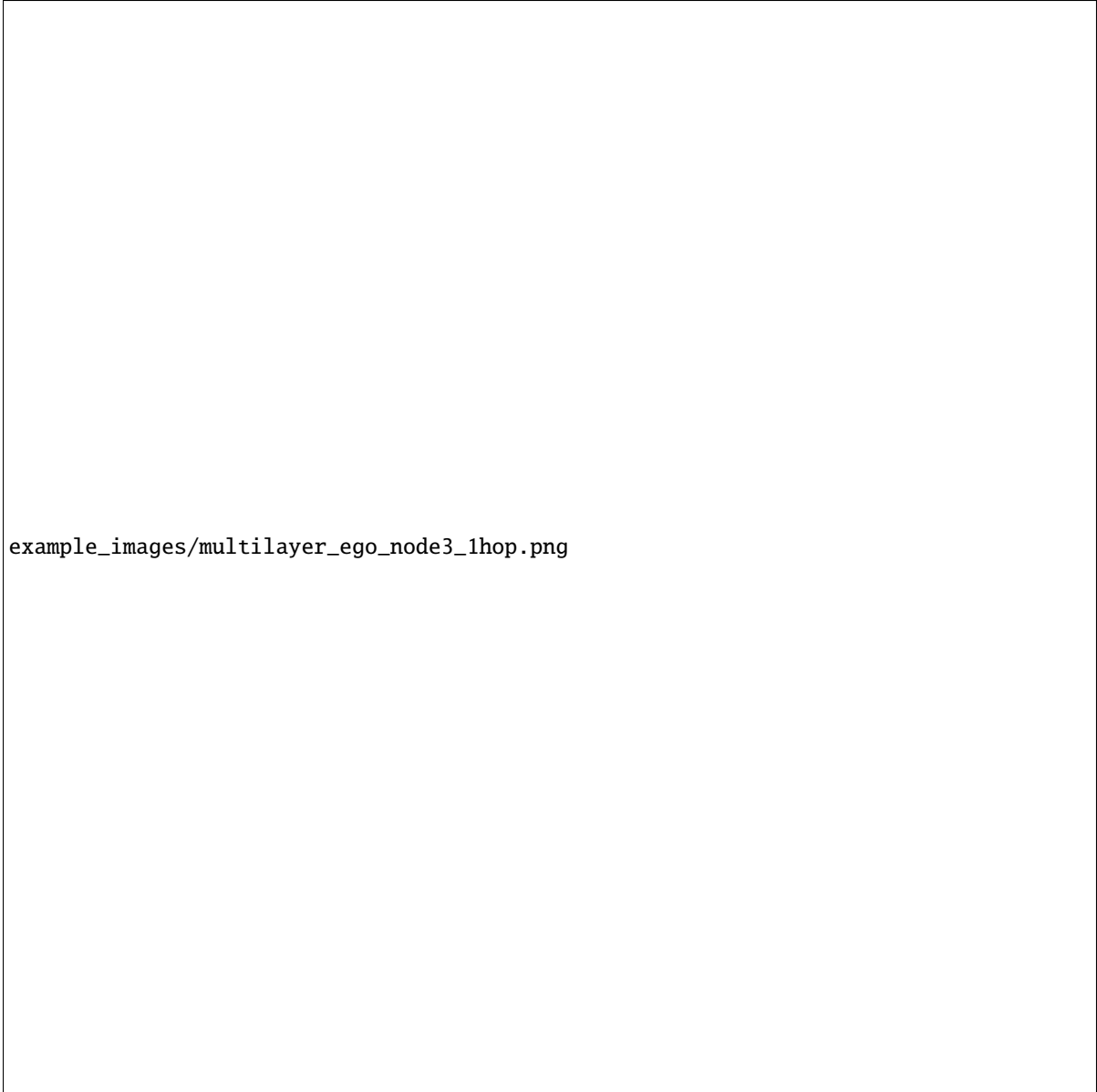
- `base_radius`: Radius of innermost ring
- `radius_step`: Distance between consecutive rings
- `draw_inter_layer_edges`: Show inter-layer connections
- `node_size`: Size of nodes

Interpretation:

- Each ring = one layer
- Same node appears at same angle on all rings
- Dashed lines = inter-layer connections

Ego-Centric Multilayer View

Focuses on a single node (the “ego”) and shows its neighborhood across different layers, highlighting the ego node’s position in each layer context.



example_images/multilayer_ego_node3_1hop.png

```
from py3plex.visualization.multilayer import visualize_multilayer_network

# Basic ego view
fig = visualize_multilayer_network(
    network,
    visualization_type="ego_multilayer",
    ego='node_id',          # Node to focus on
    max_depth=1,           # Neighborhood depth (hops)
    layout="spring",
    node_size=100,
    ego_node_size=400      # Highlight ego node
)

# Specific layers only
fig = visualize_multilayer_network(
```

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```
network,
visualization_type="ego_multilayer",
ego='node_id',
layers=['layer1', 'layer2'], # Only these layers
max_depth=2,                # 2-hop neighborhood
with_labels=True
)
```

Best for:

- Analyzing individual node behavior
- Comparing local structure across layers
- Identifying layer-specific influential neighbors
- Understanding node role variations

Parameters:

- ego: Node ID to focus on
- layers: Specific layers to visualize (or None for all)
- max_depth: Neighborhood depth in hops
- layout: Layout algorithm per ego graph
- ego_node_size: Size of the highlighted ego node

Interpretation:

- Red node = the ego (focal node)
- Blue nodes = neighbors of the ego
- Each subplot = ego's neighborhood in one layer

Flow and Sankey Diagrams

Py3plex provides two complementary visualizations for understanding inter-layer connectivity: flow/alluvial diagrams and Sankey diagrams. Both show how nodes and connections flow between layers, but with different visual approaches.

Flow/Alluvial Visualization

The flow visualization shows each layer as a horizontal band with nodes positioned along the x-axis. Inter-layer connections are drawn as flowing Bezier curves, with node activity encoded by color and size.

```
from py3plex.core import multinet

# Load network
network = multinet.multi_layer_network()
network.load_network("network.txt", input_type="multiedgelist")

# Flow visualization
ax = network.visualize_network(style='flow', show=True)

# Or use 'alluvial' (same as 'flow')
ax = network.visualize_network(style='alluvial', show=True)
```


Best for:

- Visualizing node-level connectivity across layers
- Understanding individual node trajectories
- Showing how specific nodes connect between layers
- Networks with up to 5-6 layers

Sankey Diagram for Inter-Layer Flows

The Sankey-style diagram provides an aggregate view of inter-layer connection strength. The visualization uses text and arrows where width represents the number of connections between layers. This is ideal for understanding overall flow patterns rather than individual node connections.

```
from py3plex.core import multinet

# Load network
network = multinet.multi_layer_network()
network.load_network("network.txt", input_type="multiedgelist")

# Sankey-style diagram showing inter-layer flow strength
ax = network.visualize_network(style='sankey', show=True)
```

You can also use the function directly:

```
from py3plex.visualization.sankey import draw_multilayer_sankey

# Get layers
labels, graphs, multilinks = network.get_layers()

# Create Sankey-style diagram
ax = draw_multilayer_sankey(
    graphs,
    multilinks,
    labels=labels,
    display=True
)
```

Best for:

- Analyzing aggregate inter-layer connection patterns
- Understanding flow strength between layers
- Identifying dominant layer connections
- Networks with 2-5 layers
- Summarizing inter-layer connectivity

Interpretation:

- Text shows layer-to-layer connections with counts
- Arrows indicate connection direction
- Arrow width proportional to connection strength
- Useful for identifying which layer pairs have strongest connections

Note:

This uses a simplified flow diagram approach rather than matplotlib's traditional Sankey class, as it's better suited for multilayer network inter-layer connections.

Example Workflows**Comparing Visualization Modes**

Different modes reveal different aspects of the same network:

```
from py3plex.visualization.multilayer import visualize_multilayer_network
import matplotlib.pyplot as plt

# Load network
network = multinet.multi_layer_network()
network.load_network("network.txt", input_type="multiedgelist")

# Compare multiple views
modes = [
    "small_multiples",
    "edge_colored_projection",
    "supra_adjacency_heatmap",
    "radial_layers"
]

for mode in modes:
    fig = visualize_multilayer_network(network, visualization_type=mode)
    fig.savefig(f"network_{mode}.png", dpi=150, bbox_inches='tight')
    plt.close()
```

Analyzing Specific Nodes

Use ego-centric view to understand individual nodes:

```
# Find high-degree nodes
import networkx as nx

# Get aggregated network
agg_graph = nx.Graph()
for node in network.get_nodes():
    agg_graph.add_node(node[0])
for u, v in network.get_edges():
    agg_graph.add_edge(u[0], v[0])

# Get top nodes by degree
degrees = dict(agg_graph.degree())
top_nodes = sorted(degrees, key=degrees.get, reverse=True)[:5]

# Visualize each top node's ego network
for node in top_nodes:
    fig = visualize_multilayer_network(
        network,
        visualization_type="ego_multilayer",
```

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```

    ego=node,
    max_depth=1
)
fig.savefig(f"ego_{node}.png", bbox_inches='tight')
plt.close()

```

Choosing the Right Visualization

Use this guide to select the appropriate visualization mode:

For structural comparison:

- Use `small_multiples` to compare layer structures side-by-side
- Use `edge_colored_projection` to see aggregated structure with layer info

For mathematical analysis:

- Use `supra_adjacency_heatmap` for matrix-based analysis
- Useful for spectral methods and linear algebra operations

For temporal or hierarchical networks:

- Use `radial_layers` when layers have natural ordering
- Shows progression or hierarchy clearly

For local analysis:

- Use `ego_multilayer` to understand individual nodes
- Compare how a node's role varies across layers

For presentations:

- Use `edge_colored_projection` for clean, simple overview
- Use `small_multiples` when detail matters
- Use `radial_layers` for visually striking displays

7.14.11 Next Steps

- *Working with Networks* - Network operations
- *Community Detection Tutorial* - Detect communities for coloring
- *I/O and Serialization* - Load data from various formats
- *Performance and Scalability Best Practices* - Optimize for large networks

For more examples, see the [visualization examples](#) in the GitHub repository.

7.15 Analysis Recipes & Workflows

This guide provides **practical recipes** for **common analysis workflows** in py3plex. Each recipe is a **complete, ready-to-use solution** for a real-world task.

Recipe Index

- *Data Import & Setup*
 - *Recipe 1: Load Network from Multiple File Formats*
 - *Recipe 2: Create Synthetic Benchmark Networks*
 - *Recipe 3: Convert Between Network Formats*
- *Network Exploration & Analysis*
 - *Recipe 4: Quick Network Summary*
 - *Recipe 5: Layer-by-Layer Analysis*
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- *Centrality & Node Importance*
 - *Recipe 7: Single-Layer Centrality Analysis*
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- *Community Detection*
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 - *Recipe 17: Benchmark and Profile Your Analysis*
- *Tips & Best Practices*
 - *General Tips*
 - *Performance Tips*
 - *Visualization Tips*
 - *Community Detection Tips*
- *Common Issues & Solutions*
 - *Issue: “Network appears to be empty”*
 - *Issue: “No module named ‘community’”*
 - *Issue: “Visualization doesn’t appear”*
 - *Issue: “Memory error with large network”*

- Issue: “Community detection is slow”
- Further Reading

7.15.1 Data Import & Setup

Recipe 1: Load Network from Multiple File Formats

Task: Load **multilayer networks** from various **file formats** (edgelist, GraphML, CSV).

```
from py3plex.core import multinet

# From multiedgelist (node1, layer1, node2, layer2, weight)
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist",
    directed=False
)

# From GraphML
network = multinet.multi_layer_network().load_network(
    "data/network.graphml",
    input_type="graphml"
)

# From edges list programmatically
network = multinet.multi_layer_network()
network.add_edges([
    ['A', 'layer1', 'B', 'layer1', 1.0],
    ['B', 'layer1', 'C', 'layer1', 1.0],
    ['A', 'layer2', 'C', 'layer2', 0.5],
], input_type="list")

# Verify the network
network.basic_stats()
```

Expected Output:

```
Number of nodes: 3
Number of edges: 3
Number of unique nodes (as node-layer tuples): 5
Number of unique node IDs (across all layers): 3
Nodes per layer:
  Layer 'layer1': 3 nodes
  Layer 'layer2': 2 nodes
```

When to use: Starting any **analysis project** with external data.

Recipe 2: Create Synthetic Benchmark Networks

Task: Generate **synthetic multilayer networks** for **testing algorithms**.

```
from py3plex.core.random_generators import random_multilayer_ER
import numpy as np

# Set random seed for reproducibility
np.random.seed(42)

# Create a random Erdős-Rényi multilayer network
# Parameters: n_nodes, n_layers, edge_probability
network = random_multilayer_ER(
    n=100,          # 100 nodes
    L=3,           # 3 layers
    p=0.1,         # 10% edge probability
    directed=False
)

network.basic_stats()
```

When to use: Algorithm validation, testing, benchmarking, or when real data is unavailable.

Recipe 3: Convert Between Network Formats

Task: Export networks to **different formats** for use in **other tools**.

```
from py3plex.core import multinet

# Load network
network = multinet.multi_layer_network().load_network(
    "input.txt",
    input_type="multiedgelist"
)

# Save as GraphML (standard format, compatible with Gephi, Cytoscape)
network.save_network("output.graphml", output_type="graphml")

# Save as edge list with node mapping
inverse_map = network.serialize_to_edgelist("output_edges.txt")

# Save as pickle (preserves all Python objects)
import pickle
with open("network.pkl", "wb") as f:
    pickle.dump(network, f)
```

When to use: Sharing data with collaborators, importing to visualization tools, or archiving results.

7.15.2 Network Exploration & Analysis

Recipe 4: Quick Network Summary

Task: Get comprehensive overview of network structure.

```
from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
```

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```

    input_type="multiedgelist"
)

# Basic statistics
print("=== Basic Statistics ===")
network.basic_stats()

# Layer information
layers = network.get_layers()
print(f"\nLayers: {layers}")
print(f"Number of layers: {len(layers)}")

# Node and edge counts
nodes = list(network.get_nodes())
edges = list(network.get_edges())
print(f"Total nodes: {len(nodes)}")
print(f"Total edges: {len(edges)}")

# Get unique nodes (nodes may appear in multiple layers)
unique_nodes = set([n[0] for n in nodes])
print(f"Unique nodes: {len(unique_nodes)}")

```

Expected Output:

```

=== Basic Statistics ===
Number of nodes: 250
Number of edges: 180
Number of unique nodes (as node-layer tuples): 250
Number of unique node IDs (across all layers): 120
Nodes per layer:
  Layer 'collaboration': 85 nodes
  Layer 'citation': 90 nodes
  Layer 'coauthor': 75 nodes

Layers: ['collaboration', 'citation', 'coauthor']
Number of layers: 3
Total nodes: 250
Total edges: 180
Unique nodes: 120

```

When to use: Initial data exploration, quality checks, reporting.**Recipe 5: Layer-by-Layer Analysis****Task:** Analyze each layer independently and compare properties.

```

from py3plex.core import multinet
import networkx as nx

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

```

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```

# Analyze each layer
layers = network.get_layers()
layer_stats = {}

for layer_name in layers:
    # Extract single layer
    layer = network.subnetwork([layer_name], subset_by="layers")

    # Compute layer statistics
    stats = {}
    stats['nodes'] = len(list(layer.get_nodes()))
    stats['edges'] = len(list(layer.get_edges()))

    # NetworkX metrics on layer
    G = layer.core_network
    if len(G) > 0: # Check if layer has nodes
        stats['density'] = nx.density(G)
        stats['avg_degree'] = sum(dict(G.degree()).values()) / len(G)

    # Check connectivity
    if not layer.directed:
        stats['connected'] = nx.is_connected(G)
        if stats['connected']:
            stats['diameter'] = nx.diameter(G)

    layer_stats[layer_name] = stats

# Display results
print("\n=== Layer-by-Layer Analysis ===")
for layer, stats in layer_stats.items():
    print(f"\nLayer: {layer}")
    for metric, value in stats.items():
        print(f"    {metric}: {value}")

```

When to use: Understanding layer-specific properties, identifying important layers.

Recipe 6: Compute Multilayer Statistics

Task: Calculate multilayer-specific metrics that account for cross-layer structure.

```

from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

layers = network.get_layers()

# === Layer-Level Statistics ===

```

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```

print("=== Layer Statistics ===")
for layer in layers:
    density = mls.layer_density(network, layer)
    print(f"Layer {layer} density: {density:.4f}")

# Layer diversity
entropy = mls.entropy_of_multiplexity(network)
print(f"\nLayer diversity (entropy): {entropy:.4f} bits")

# === Node-Level Statistics ===
print("\n=== Node Activity ===")
sample_nodes = list(network.get_nodes())[:5]
for node_tuple in sample_nodes:
    node = node_tuple[0] # Extract node name
    activity = mls.node_activity(network, node)
    print(f"Node {node} active in {activity*100:.1f}% of layers")

# === Cross-Layer Analysis ===
if len(layers) >= 2:
    print("\n=== Cross-Layer Similarity ===")
    layer1, layer2 = str(layers[0]), str(layers[1])

    # Cosine similarity of adjacency matrices
    similarity = mls.layer_similarity(network, layer1, layer2, method='cosine')
    print(f"Cosine similarity ({layer1} vs {layer2}): {similarity:.4f}")

    # Jaccard similarity of edge sets
    overlap = mls.edge_overlap(network, layer1, layer2)
    print(f"Edge overlap (Jaccard): {overlap:.4f}")

# === Versatility Analysis ===
print("\n=== Node Versatility (Cross-Layer Importance) ===")
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_nodes = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:5]
for node, score in top_nodes:
    print(f"  {node}: {score:.4f}")

```

When to use: Quantifying multilayer structure, comparing layers, identifying versatile nodes.

7.15.3 Centrality & Node Importance

Recipe 7: Single-Layer Centrality Analysis

Task: Identify important nodes within individual layers.

```

from py3plex.core import multinet

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Select a layer

```

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```

layers = network.get_layers()
target_layer = [str(layers[0])]
layer = network.subnetwork(target_layer, subset_by="layers")

# Compute multiple centrality measures
centrality_measures = {
    'degree': layer.monoplex_nx_wrapper("degree_centrality"),
    'betweenness': layer.monoplex_nx_wrapper("betweenness_centrality"),
    'closeness': layer.monoplex_nx_wrapper("closeness_centrality"),
    'eigenvector': layer.monoplex_nx_wrapper("eigenvector_centrality"),
}

# Display top nodes for each measure
print(f"=== Centrality Analysis for Layer {target_layer[0]} ===\n")
for measure_name, scores in centrality_measures.items():
    print(f"{measure_name.capitalize()} Centrality (Top 5):")
    top_nodes = sorted(scores.items(), key=lambda x: x[1], reverse=True)[:5]
    for node, score in top_nodes:
        print(f"    {node}: {score:.4f}")
    print()

```

When to use: Identifying influential nodes, hub detection, network simplification.

Recipe 8: Multiplex Participation Coefficient

Task: Measure how evenly nodes participate across layers in a multiplex network.

```

from py3plex.core import multinet
from py3plex.algorithms.multicentrality import multiplex_participation_coefficient

# Load or create multiplex network (same nodes across all layers)
network = multinet.multi_layer_network().load_network(
    "data/multiplex_network.txt",
    input_type="multiedgelist"
)

# Compute MPC (requires true multiplex: identical node set per layer)
try:
    mpc = multiplex_participation_coefficient(
        network,
        normalized=True,
        check_multiplex=True
    )

    # Display results
    print("=== Multiplex Participation Coefficient ===")
    print("(0 = active in one layer only, 1 = equally active across all layers)\n")

    # Sort by MPC score
    sorted_nodes = sorted(mpc.items(), key=lambda x: x[1], reverse=True)

    print("Top 10 most versatile nodes:")

```

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```

    for node, score in sorted_nodes[:10]:
        print(f" {node}: {score:.4f}")

    print("\nTop 10 most specialized nodes:")
    for node, score in sorted_nodes[-10:]:
        print(f" {node}: {score:.4f}")

except ValueError as e:
    print(f"Error: {e}")
    print("Network is not a true multiplex (layers have different node sets)")

```

When to use: Analyzing multiplex networks, identifying cross-layer hubs, understanding node specialization.

7.15.4 Community Detection

Recipe 9: Multilayer Community Detection

Task: Detect communities that span multiple layers.

```

from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    multilayer
from collections import Counter

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Run multilayer Louvain algorithm
# gamma: resolution parameter (higher = more communities)
# omega: inter-layer coupling strength (higher = communities span layers)
partition = louvain_multilayer(
    network,
    gamma=1.0,      # Standard resolution
    omega=1.0,      # Standard coupling
    random_state=42 # For reproducibility
)

# Analyze results
num_communities = len(set(partition.values()))
print(f"Detected {num_communities} communities")

# Community size distribution
community_sizes = Counter(partition.values())
print("\nTop 10 communities by size:")
for comm_id, size in community_sizes.most_common(10):
    print(f" Community {comm_id}: {size} nodes")

# Check which nodes are in largest community
largest_comm = community_sizes.most_common(1)[0][0]
largest_comm_nodes = [node for node, comm in partition.items()
                      if comm == largest_comm]

```

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```
print(f"\nNodes in largest community: {largest_comm_nodes[:10]}...")
```

When to use: Understanding network organization, identifying functional modules, clustering nodes.

Recipe 10: Compare Community Detection Algorithms

Task: Run multiple community detection methods and compare results.

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection import community_wrapper as cw
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    multilayer
from collections import Counter
import numpy as np

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Method 1: Single-layer Louvain on aggregated network
partition_louvain = cw.louvain_communities(network)

# Method 2: Multilayer Louvain
partition_multilayer = louvain_multilayer(
    network,
    gamma=1.0,
    omega=1.0,
    random_state=42
)

# Compare results
print("=== Community Detection Comparison ===\n")
print(f"Single-layer Louvain: {len(set(partition_louvain.values()))} communities")
print(f"Multilayer Louvain: {len(set(partition_multilayer.values()))} communities")

# Calculate normalized mutual information (if sklearn available)
try:
    from sklearn.metrics import normalized_mutual_info_score

    # Align node sets
    common_nodes = set(partition_louvain.keys()) & set(partition_multilayer.keys())
    labels1 = [partition_louvain[n] for n in common_nodes]
    labels2 = [partition_multilayer[n] for n in common_nodes]

    nmi = normalized_mutual_info_score(labels1, labels2)
    print(f"\nNormalized Mutual Information: {nmi:.4f}")
    print("(1.0 = identical partitions, 0.0 = independent)")
except ImportError:
    print("\nInstall scikit-learn to compute NMI: pip install scikit-learn")
```

When to use: Method validation, algorithm selection, robustness analysis.

7.15.5 Visualization

Recipe 11: Basic Network Visualization

Task: Create publication-ready network visualizations.

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import hairball_plot
import matplotlib
matplotlib.use('Agg') # For non-interactive environments
import matplotlib.pyplot as plt

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Get network in visualization format
network_colors, graph = network.get_layers(style="hairball")

# Create visualization
plt.figure(figsize=(12, 12))
hairball_plot(
    graph,
    network_colors,
    layout_algorithm="force",          # Options: "force", "circular", "spring"
    layout_parameters={"iterations": 100},
    node_size=5,
    edge_width=0.5,
    alpha_channel=0.8,
    scale_by_size=True
)
plt.title("Multilayer Network", fontsize=16, fontweight='bold')
plt.axis('off')
plt.tight_layout()
plt.savefig("network_viz.png", dpi=150, bbox_inches='tight',
           facecolor='white', edgecolor='none')
plt.close()
print("Visualization saved to network_viz.png")
```

When to use: Presentations, papers, reports, exploratory analysis.

Recipe 12: Visualize Communities on Network

Task: Color nodes by community membership in network plot.

```
from py3plex.core import multinet
from py3plex.visualization.multilayer import hairball_plot
from py3plex.visualization.colors import colors_default
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    multilayer
from collections import Counter
import matplotlib.pyplot as plt
import matplotlib
matplotlib.use('Agg')
```

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```

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Detect communities
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)

# Select top N communities to color
top_n = 10
community_counts = Counter(partition.values())
top_communities = [c for c, _ in community_counts.most_common(top_n)]

# Create color mapping
color_map = dict(zip(top_communities, colors_default[:top_n]))

# Assign colors to nodes
network_colors = [
    color_map.get(partition.get(node), "lightgray")
    for node in network.get_nodes()
]

# Get network for visualization
_, graph = network.get_layers(style="hairball")

# Create plot
plt.figure(figsize=(14, 14))
hairball_plot(
    graph,
    network_colors,
    layout_algorithm="force",
    layout_parameters={"iterations": 150},
    node_size=8,
    edge_width=0.5,
    alpha_channel=0.7,
    scale_by_size=True
)
plt.title(f"Network with {top_n} Largest Communities",
          fontsize=16, fontweight='bold')
plt.axis('off')
plt.tight_layout()
plt.savefig("network_communities.png", dpi=150,
            bbox_inches='tight', facecolor='white')
plt.close()
print("Community visualization saved to network_communities.png")

```

When to use: Visualizing analysis results, understanding community structure.

7.15.6 Advanced Workflows

Recipe 13: Complete Analysis Pipeline

Task: Full workflow from data loading to results export.

```
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    ↪ multilayer
from collections import Counter
import json
import numpy as np

# Set random seed
np.random.seed(42)

# === 1. Load Data ===
print("=== Loading Network ===")
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist",
    directed=False
)
network.basic_stats()

# === 2. Compute Statistics ===
print("\n=== Computing Multilayer Statistics ===")
layers = network.get_layers()

stats = {
    'num_layers': len(layers),
    'layer_densities': {},
    'layer_diversity': mls.entropy_of_multiplexity(network),
}

for layer in layers:
    stats['layer_densities'][str(layer)] = mls.layer_density(network, layer)

# Versatility
versatility = mls.versatility_centrality(network, centrality_type='degree')
top_versatile = sorted(versatility.items(), key=lambda x: x[1], reverse=True)[:10]
stats['top_versatile_nodes'] = {str(k): float(v) for k, v in top_versatile}

# === 3. Community Detection ===
print("\n=== Detecting Communities ===")
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
community_sizes = Counter(partition.values())

stats['num_communities'] = len(set(partition.values()))
stats['community_sizes'] = dict(community_sizes.most_common(10))

# === 4. Export Results ===
print("\n=== Exporting Results ===")
```

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```

# Save statistics as JSON
with open("analysis_results.json", "w") as f:
    json.dump(stats, f, indent=2)
print("Statistics saved to analysis_results.json")

# Save community assignments
with open("communities.txt", "w") as f:
    for node, comm in partition.items():
        f.write(f"{node}\t{comm}\n")
print("Communities saved to communities.txt")

# Save processed network
network.save_network("processed_network.graphml", output_type="graphml")
print("Network saved to processed_network.graphml")

print("\n=== Analysis Complete ===")

```

When to use: Reproducible research workflows, batch processing, automated analysis.

Recipe 14: Network Comparison Study

Task: Compare multiple networks systematically.

```

from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls
import pandas as pd
import numpy as np

# List of networks to compare
network_files = [
    ("Network A", "data/network_a.txt"),
    ("Network B", "data/network_b.txt"),
    ("Network C", "data/network_c.txt"),
]

results = []

for name, filepath in network_files:
    print(f"\nAnalyzing {name}...")

    # Load network
    network = multinet.multi_layer_network().load_network(
        filepath,
        input_type="multiedgelist",
        directed=False
    )

    # Compute metrics
    layers = network.get_layers()
    nodes = list(network.get_nodes())
    edges = list(network.get_edges())

```

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```

metrics = {
    'Network': name,
    'Layers': len(layers),
    'Total Nodes': len(nodes),
    'Unique Nodes': len(set([n[0] for n in nodes])),
    'Total Edges': len(edges),
    'Layer Diversity': mls.entropy_of_multiplexity(network),
}

# Average layer density
densities = [mls.layer_density(network, layer) for layer in layers]
metrics['Avg Layer Density'] = np.mean(densities)
metrics['Std Layer Density'] = np.std(densities)

results.append(metrics)

# Create comparison table
df = pd.DataFrame(results)
print("\n=== Network Comparison ===")
print(df.to_string(index=False))

# Save to CSV
df.to_csv("network_comparison.csv", index=False)
print("\nComparison saved to network_comparison.csv")

```

When to use: Comparative studies, evaluating datasets, selecting networks for analysis.

Recipe 15: Random Walks for Node Embeddings

Task: Generate node embeddings using random walks (Node2Vec approach).

```

from py3plex.core import multinet
from py3plex.algorithms.general.walkers import generate_walks, node2vec_walk
import numpy as np

# Set random seed
np.random.seed(42)

network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)

# Get core NetworkX graph
G = network.core_network

# Generate random walks
# p: return parameter (higher = less likely to return to previous node)
# q: in-out parameter (higher = more BFS-like, lower = more DFS-like)
walks = generate_walks(
    G,

```

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```

num_walks=10,      # Number of walks per node
walk_length=80,    # Length of each walk
p=1.0,            # Return parameter
q=1.0,            # In-out parameter
seed=42
)

print(f"Generated {len(walks)} random walks")
print(f"First walk (truncated): {walks[0][:10]}...")

# Use walks for downstream tasks:
# 1. Train Word2Vec model (if gensim available)
try:
    from gensim.models import Word2Vec

    # Convert walks to strings for Word2Vec
    walks_str = [[str(node) for node in walk] for walk in walks]

    # Train embedding model
    model = Word2Vec(
        walks_str,
        vector_size=128,    # Embedding dimension
        window=5,          # Context window
        min_count=1,
        sg=1,              # Skip-gram
        workers=4,
        epochs=5,
        seed=42
    )

    print(f"\nTrained embedding model with {len(model.wv)} nodes")

    # Get embedding for a node
    sample_node = str(list(G.nodes())[0])
    if sample_node in model.wv:
        embedding = model.wv[sample_node]
        print(f"Embedding for node {sample_node}: {embedding[:5]}... (dim=
↪{len(embedding)})")

        # Find similar nodes
        similar = model.wv.most_similar(sample_node, topn=5)
        print(f"\nMost similar nodes to {sample_node}:")
        for node, similarity in similar:
            print(f" {node}: {similarity:.4f}")

    # Save embeddings
    model.wv.save_word2vec_format("node_embeddings.txt")
    print("\nEmbeddings saved to node_embeddings.txt")

except ImportError:
    print("\nInstall gensim for embedding generation: pip install gensim")

```

When to use: Node classification, link prediction, network clustering, feature extraction.

7.15.7 Performance & Optimization

Recipe 16: Handle Large Networks Efficiently

Task: Process large multilayer networks without running out of memory.

```
from py3plex.core import multinet
import gc

# === 1. Load network efficiently ===
# Use generators instead of loading all at once
network = multinet.multi_layer_network()

# For very large files, load in chunks or stream
# (This is a conceptual example - actual implementation may vary)

# === 2. Process layers independently ===
layers = network.get_layers()

layer_results = {}
for layer_name in layers:
    print(f"Processing layer {layer_name}...")

    # Extract and process one layer at a time
    layer = network.subnetwork([layer_name], subset_by="layers")

    # Compute statistics for this layer
    nodes = len(list(layer.get_nodes()))
    edges = len(list(layer.get_edges()))

    layer_results[str(layer_name)] = {'nodes': nodes, 'edges': edges}

    # Clean up to free memory
    del layer
    gc.collect()

print("\n=== Results ===")
for layer, stats in layer_results.items():
    print(f"{layer}: {stats['nodes']} nodes, {stats['edges']} edges")

# === 3. Use sparse representations ===
# When computing supra-adjacency matrix
supra_adj = network.get_supra_adjacency_matrix(sparse=True)
print(f"\nSupra-adjacency matrix: {supra_adj.shape} (sparse)")

# === 4. Sample for exploratory analysis ===
# For visualization or initial exploration, work with a sample
all_nodes = list(network.get_nodes())
import random
random.seed(42)
sample_size = min(1000, len(all_nodes))
sampled_nodes = random.sample(all_nodes, sample_size)

# Create subnetwork with sampled nodes
```

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```
# (Implementation depends on network structure)
print(f"\nSampled {sample_size} nodes for exploratory analysis")
```

When to use: Networks with >10,000 nodes, limited RAM, large-scale studies.

Tips: - Process layers sequentially rather than all at once - Use sparse matrices for large supra-adjacency representations - Sample networks for visualization and initial exploration - Save intermediate results to disk - Use generators and iterators instead of lists

Recipe 17: Benchmark and Profile Your Analysis

Task: Measure performance and identify bottlenecks.

```
from py3plex.core import multinet
from py3plex.algorithms.community_detection.multilayer_modularity import louvain_
    multilayer
import time
import psutil
import os

def get_memory_usage():
    """Get current memory usage in MB"""
    process = psutil.Process(os.getpid())
    return process.memory_info().rss / 1024 / 1024

# Load network
print("=== Performance Benchmarking ===\n")
start_mem = get_memory_usage()

start = time.time()
network = multinet.multi_layer_network().load_network(
    "data/network.txt",
    input_type="multiedgelist"
)
load_time = time.time() - start
load_mem = get_memory_usage() - start_mem

print(f"Network loading:")
print(f"  Time: {load_time:.3f} seconds")
print(f"  Memory: {load_mem:.2f} MB")

# Benchmark statistics computation
start = time.time()
network.basic_stats()
stats_time = time.time() - start
print(f"\nBasic statistics:")
print(f"  Time: {stats_time:.3f} seconds")

# Benchmark community detection
start_mem = get_memory_usage()
start = time.time()
partition = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
comm_time = time.time() - start
```

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```
comm_mem = get_memory_usage() - start_mem

print(f"\nCommunity detection:")
print(f"  Time: {comm_time:.3f} seconds")
print(f"  Memory: {comm_mem:.2f} MB")
print(f"  Communities found: {len(set(partition.values()))}")

# Total resource usage
print(f"\n=== Total Performance ===")
print(f"Total time: {load_time + stats_time + comm_time:.3f} seconds")
print(f"Peak memory: {get_memory_usage():.2f} MB")
```

When to use: Optimizing workflows, comparing algorithms, capacity planning.

Note: Requires psutil package: `pip install psutil`

7.15.8 Tips & Best Practices

General Tips

1. Always set random seeds for reproducibility:

```
import numpy as np
np.random.seed(42)
```

2. Verify network structure after loading:

```
network.basic_stats() # Quick sanity check
```

3. Handle errors gracefully:

```
try:
    result = network.some_operation()
except Exception as e:
    print(f"Operation failed: {e}")
```

4. Save intermediate results for long analyses:

```
import pickle
with open("checkpoint.pkl", "wb") as f:
    pickle.dump(results, f)
```

5. Use appropriate file formats:

- GraphML: Standard, compatible with other tools
- Pickle: Fast, preserves Python objects, not portable
- Edgelist: Simple, human-readable, large file size

Performance Tips

1. For large networks, process layers independently
2. Use sparse matrices when working with supra-adjacency
3. Sample networks for visualization (>1000 nodes gets crowded)

4. **Profile your code** to find bottlenecks
5. **Close plots** after saving to free memory: `plt.close()`

Visualization Tips

1. **Choose layout carefully:**
 - Force-directed: General purpose, slow for >5000 nodes
 - Circular: Fast, good for small networks
 - Spring: Balance between quality and speed
2. **Adjust node size** based on network size:

```
node_size = max(1, 100 / np.sqrt(num_nodes))
```

3. **Use alpha channel** for edge-dense networks:

```
alpha_channel=0.3 # More transparent
```

4. **Save high-resolution** for publications:

```
plt.savefig("figure.png", dpi=300)
```

Community Detection Tips

1. **Tune resolution parameter** (γ):
 - Higher γ → more, smaller communities
 - Lower γ → fewer, larger communities
2. **Tune coupling parameter** (ω) for multilayer:
 - Higher ω → communities span layers
 - Lower ω → layer-specific communities
3. **Run multiple times** with different random seeds to check stability
4. **Compare multiple algorithms** to validate findings

7.15.9 Common Issues & Solutions

Issue: “Network appears to be empty”

Cause: Incorrect file format or delimiter.

Solution::

```
# Check file format
with open("data.txt", "r") as f:
    print(f.readlines()[:5]) # Inspect first lines

# Try different input_type
network = multinet.multi_layer_network().load_network(
    "data.txt",
    input_type="multiedgelist" # or "edgelist", "graphml"
)
```

Issue: “No module named ‘community’”

Cause: Optional dependency not installed.

Solution::

```
pip install python-louvain
```

Issue: “Visualization doesn’t appear”

Cause: Using non-interactive backend or missing display.

Solution::

```
import matplotlib
matplotlib.use('Agg') # For saving files
import matplotlib.pyplot as plt

# ... create plot ...
plt.savefig("output.png") # Save instead of show
```

Issue: “Memory error with large network”

Cause: Loading entire network into RAM at once.

Solution: See Recipe 16 for memory-efficient processing.

Issue: “Community detection is slow”

Cause: Large network or complex structure.

Solution::

```
# Use simpler algorithm for quick results
from py3plex.algorithms.community_detection import community_wrapper as cw
partition = cw.louvain_communities(network) # Faster than multilayer
```

7.15.10 Further Reading

Documentation:

- Installation guide - See main documentation
- Quick start tutorial - See main documentation
- Core concepts - See main documentation
- Algorithm selection guide - See main documentation

Examples:

See `examples/` directory for 50+ complete, working examples covering all major functionality.

Research:

- Kivelä et al. (2014) - Multilayer networks theory
- De Domenico et al. (2013) - Mathematical framework
- See Citations section in main documentation for complete reference list

7.16 Docker Usage Guide

This guide provides comprehensive instructions for using Py3plex via Docker.

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7.16.1 Quick Start

The fastest way to get started with Py3plex using Docker:

```
# Clone the repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Build the Docker image
docker build -t py3plex:latest .

# Run a self-test to verify installation
docker run --rm py3plex:latest selftest

# Display help
docker run --rm py3plex:latest help
```

7.16.2 Building the Image

Using Docker

```
# Build from the repository root
docker build -t py3plex:latest .

# Build with a specific tag
docker build -t py3plex:0.95a .

# Build without cache (if needed)
docker build --no-cache -t py3plex:latest .
```

Using Docker Compose

```
# Build using docker-compose
docker-compose build

# Force rebuild
docker-compose build --no-cache
```

7.16.3 Running Commands

Basic Commands

The Docker container is set up with py3plex as the entrypoint, so you can run any py3plex CLI command directly:

```
# Show version
docker run --rm py3plex:latest --version

# Run self-test
docker run --rm py3plex:latest selftest

# Show help
docker run --rm py3plex:latest help

# Show help for a specific command
docker run --rm py3plex:latest create --help
```

Using Docker Compose

With docker-compose, commands are slightly more verbose but easier to manage:

```
# Show version
docker-compose run --rm py3plex --version

# Run self-test
docker-compose run --rm py3plex selftest -v

# Show help
docker-compose run --rm py3plex help
```

7.16.4 Working with Files

To work with network files, you need to mount a local directory to the container's /data directory.

Creating a Data Directory

```
# Create a local data directory
mkdir -p data
```

Mounting Volumes

With docker run:

```
# Mount current directory's data folder
docker run --rm -v $(pwd)/data:/data py3plex:latest create --nodes 100 --layers 3 --
↳output /data/network.edgelist

# On Windows (PowerShell)
docker run --rm -v ${PWD}/data:/data py3plex:latest create --nodes 100 --layers 3 --
↳output /data/network.edgelist

# On Windows (CMD)
docker run --rm -v %cd%/data:/data py3plex:latest create --nodes 100 --layers 3 --output.
↳/data/network.edgelist
```

With docker-compose:

The docker-compose.yml file already configures volume mounting from ./data to /data, so you can simply:

```
docker-compose run --rm py3plex create --nodes 100 --layers 3 --output /data/network.
↳edgelist
```

Complete Workflow Example

```
# Create data directory
mkdir -p data

# 1. Create a multilayer network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  create --nodes 100 --layers 3 --type random --probability 0.1 --output /data/network.
↳edgelist

# 2. Load and display network information
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  load /data/network.edgelist --info

# 3. Compute statistics
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  stats /data/network.edgelist --measure all --output /data/stats.json

# 4. Detect communities
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain --output /data/communities.json

# 5. Visualize the network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png --layout multilayer

# 6. View the results
ls -lh data/
```

7.16.5 Advanced Usage

Interactive Shell

To explore the container interactively:

```
# Start an interactive shell in the container
docker run --rm -it -v $(pwd)/data:/data --entrypoint /bin/bash py3plex:latest

# Inside the container, you can run:
# py3plex --version
# py3plex selftest
# python -c "import py3plex; print(py3plex.__version__)"
```

Using Python API

You can also use py3plex as a Python library inside the container:

```
# Create a Python script
cat > data/analyze.py << 'EOF'
from py3plex.core import multinet

# Create a network
network = multinet.multi_layer_network()
nodes = [{"source": f"node{i}", "type": "layer1"} for i in range(10)]
network.add_nodes(nodes, input_type="dict")

print(f"Created network with {network.core_network.number_of_nodes()} nodes")
EOF

# Run the script in the container
docker run --rm -v $(pwd)/data:/data --entrypoint python py3plex:latest /data/analyze.py
```

Custom Image Builds

If you need additional packages:

```
# Create a custom Dockerfile
FROM py3plex:latest

# Install additional packages
RUN pip install --no-cache-dir pandas jupyter

# Or system packages
USER root
RUN apt-get update && apt-get install -y vim && rm -rf /var/lib/apt/lists/*
USER py3plex
```

Docker Compose with Multiple Services

You can extend the `docker-compose.yml` to include related services:

```
version: '3.8'

services:
  py3plex:
    build: .
    image: py3plex:latest
    volumes:
```

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```

- ./data:/data

jupyter:
  image: py3plex:latest
  ports:
    - "8888:8888"
  volumes:
    - ./data:/data
    - ./notebooks:/notebooks
  command: jupyter notebook --ip=0.0.0.0 --port=8888 --no-browser --allow-root
  working_dir: /notebooks

```

7.16.6 Helper Scripts

The repository includes convenient helper scripts to simplify Docker usage.

py3plex-docker.sh

A wrapper script that simplifies running py3plex commands via Docker:

```

# The script automatically handles:
# - Docker installation checks
# - Image building (if needed)
# - Data directory creation
# - Volume mounting

# Usage examples:
./py3plex-docker.sh --version
./py3plex-docker.sh selftest
./py3plex-docker.sh create --nodes 100 --layers 3 --output /data/network.edgelist
./py3plex-docker.sh load /data/network.edgelist --info

```

Features:

- Automatically checks if Docker is installed
- Prompts to build the image if it doesn't exist
- Creates the data/ directory if needed
- Mounts the local data/ directory to /data in the container
- Passes all arguments directly to py3plex

Script location: py3plex-docker.sh in the repository root

test-docker-setup.sh

A comprehensive test script to validate your Docker setup:

```

# Run the validation script
./test-docker-setup.sh

```

What it tests:

1. Docker installation verification

2. Dockerfile existence check
3. .dockerignore existence check
4. docker-compose.yml existence check
5. Docker image build process
6. Container version command
7. Container help command
8. Container selftest
9. Volume mounting and file creation

The script provides colored output ([OK] PASS / [X] FAIL) and a summary report.

Script location: `test-docker-setup.sh` in the repository root

7.16.7 Docker Configuration Files

.dockerignore

The `.dockerignore` file controls which files are excluded from the Docker build context:

Key exclusions:

- **Git and version control:** `.git/`, `.github/`, `.gitignore`
- **Python artifacts:** `__pycache__/`, `*.pyc`, `dist/`, `build/`
- **Virtual environments:** `venv/`, `env/`
- **Testing and coverage:** `.pytest_cache/`, `htmlcov/`, `coverage.*`
- **Documentation builds:** `docs/_build/`, `docfiles/`
- **Examples and datasets:** `examples/`, `datasets/`, `multilayer_datasets/`
- **Development files:** `tests/`, `run_tests.py`, `validate_*.py`

Why this matters:

- **Faster builds:** Smaller build context means faster Docker builds
- **Smaller images:** Excludes unnecessary files from the final image
- **Security:** Prevents accidental inclusion of sensitive files
- **Efficiency:** Only includes files needed to run py3plex

The `.dockerignore` file is located in the repository root.

7.16.8 Troubleshooting

Testing the Docker Setup

To verify your Docker setup is working correctly, use the included test script:

```
# Run the test script
./test-docker-setup.sh
```

This script will:

- Check Docker installation
- Verify all Docker-related files exist

- Build the Docker image
- Run basic py3plex commands (`--version`, `help`, `selftest`)
- Test volume mounting and file creation
- Clean up test artifacts

Permission Issues

If you encounter permission issues with mounted volumes:

```
# On Linux, you might need to run with your user ID
docker run --rm -v $(pwd)/data:/data --user $(id -u):$(id -g) py3plex:latest create --
↳output /data/network.edgelist
```

Container Not Found

If the image isn't found:

```
# List available images
docker images | grep py3plex

# Rebuild if necessary
docker build -t py3plex:latest .
```

Network Issues During Build

If you encounter network timeouts during build:

```
# Increase timeout
docker build --build-arg PIP_DEFAULT_TIMEOUT=300 -t py3plex:latest .

# Or use a different PyPI mirror
docker build --build-arg PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/simple -t
↳py3plex:latest .
```

Out of Disk Space

Clean up unused Docker resources:

```
# Remove unused images
docker image prune -a

# Remove all stopped containers, unused networks, and dangling images
docker system prune -a
```

Debugging Build Issues

Build with verbose output:

```
docker build --progress=plain --no-cache -t py3plex:latest .
```

7.16.9 Best Practices

1. **Use Volume Mounts:** Always mount volumes for input/output files
2. **Use `--rm` Flag:** Remove containers after execution with `--rm` to save space
3. **Tag Images:** Use specific tags (e.g., `py3plex:0.95a`) for reproducibility
4. **Keep Data Separate:** Store network files in the mounted data directory
5. **Use Docker Compose:** For repeated operations, docker-compose simplifies commands
6. **Regular Updates:** Rebuild the image periodically to get latest py3plex updates

7.16.10 Practical Examples

Basic Network Analysis

Create and analyze a multilayer network:

```
# Create data directory
mkdir -p data

# Create a network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  create --nodes 100 --layers 3 --type random --probability 0.05 \
  --output /data/network.edgelist

# Analyze the network
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  load /data/network.edgelist --info --stats

# Visualize
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png
```

Community Detection

```
# Detect communities using Louvain
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain \
  --output /data/communities.json

# View community statistics
cat data/communities.json | python -m json.tool | head -20
```

Centrality Analysis

```
# Compute degree centrality
docker run --rm -v $(pwd)/data:/data py3plex:latest \
  centrality /data/network.edgelist --measure degree \
  --top 10 --output /data/centrality.json

# Compute betweenness centrality
docker run --rm -v $(pwd)/data:/data py3plex:latest \
```

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```
centrality /data/network.edgelist --measure betweenness \
--top 10 --output /data/betweenness.json
```

Batch Processing

Process multiple networks (save as batch-process.sh):

```
#!/bin/bash
# Process multiple networks in batch
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

# Create several networks
for i in {1..5}; do
    echo "Creating network $i..."
    docker run --rm -v $DATA_DIR:/data py3plex:latest \
        create --nodes $((50 * i)) --layers 3 \
        --type random --probability 0.1 \
        --output /data/network_$i.edgelist
done

# Analyze each network
for i in {1..5}; do
    echo "Analyzing network $i..."
    docker run --rm -v $DATA_DIR:/data py3plex:latest \
        stats /data/network_$i.edgelist --measure all \
        --output /data/stats_$i.json
done

echo "Done! Results in $DATA_DIR"
```

Custom Analysis Script

Create a Python script (analysis.py) and run it in the container:

Python script:

```
# analysis.py
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

# Load network
network = multinet.multi_layer_network()
network.load_network("/data/network.edgelist", input_type="multiedgelist")

# Compute statistics
layers = ["layer1", "layer2", "layer3"]
for layer in layers:
    density = mls.layer_density(network, layer)
    print(f"{layer} density: {density:.4f}")
```

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```
print("Analysis complete!")
```

Run it:

```
# Copy script to data directory
cp analysis.py data/

# Run in container
docker run --rm -v $(pwd)/data:/data \
  --entrypoint python py3plex:latest /data/analysis.py
```

Complete Analysis Pipeline

A comprehensive analysis pipeline:

```
#!/bin/bash
# complete-pipeline.sh
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

echo "Step 1: Creating network..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  create --nodes 200 --layers 3 --type ba --probability 0.05 \
  --output /data/network.edgelist

echo "Step 2: Computing statistics..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  stats /data/network.edgelist --measure all \
  --output /data/stats.json

echo "Step 3: Detecting communities..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  community /data/network.edgelist --algorithm louvain \
  --output /data/communities.json

echo "Step 4: Computing centrality..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  centrality /data/network.edgelist --measure pagerank \
  --top 20 --output /data/centrality.json

echo "Step 5: Visualizing..."
docker run --rm -v $DATA_DIR:/data py3plex:latest \
  visualize /data/network.edgelist --output /data/network.png \
  --layout multilayer --width 16 --height 12

echo "Pipeline complete! Check $DATA_DIR for results."
```

Comparative Network Analysis

Compare different network types:

```
#!/bin/bash
# compare-networks.sh
set -e

DATA_DIR=$(pwd)/data
mkdir -p $DATA_DIR

TYPES=("random" "er" "ba" "ws")

for type in "${TYPES[@]}; do
  echo "Creating $type network..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    create --nodes 100 --layers 3 --type $type --probability 0.1 \
    --output /data/network_${type}.edgelist

  echo "Analyzing $type network..."
  docker run --rm -v $DATA_DIR:/data py3plex:latest \
    stats /data/network_${type}.edgelist --measure all \
    --output /data/stats_${type}.json
done

echo "Comparison complete!"
```

7.16.11 CI/CD Integration

GitHub Actions

Example workflow for network analysis:

```
# .github/workflows/network-analysis.yml
name: Network Analysis with Docker

on: [push, pull_request]

jobs:
  analyze:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3

      - name: Build Docker image
        run: docker build -t py3plex:latest .

      - name: Create test network
        run: |
          mkdir -p data
          docker run --rm -v $PWD/data:/data py3plex:latest \
            create --nodes 50 --layers 2 --output /data/test.edgelist

      - name: Run analysis
```

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```
run: |
  docker run --rm -v $PWD/data:/data py3plex:latest \
    stats /data/test.edgelist --measure all --output /data/stats.json

- name: Upload results
  uses: actions/upload-artifact@v3
  with:
    name: analysis-results
    path: data/
```

GitLab CI

Example GitLab CI configuration:

```
# .gitlab-ci.yml
image: docker:latest

services:
  - docker:dind

variables:
  DOCKER_DRIVER: overlay2

stages:
  - build
  - analyze

build:
  stage: build
  script:
    - docker build -t py3plex:latest .
    - docker save py3plex:latest > py3plex-image.tar
  artifacts:
    paths:
      - py3plex-image.tar

analyze:
  stage: analyze
  script:
    - docker load < py3plex-image.tar
    - mkdir -p data
    - docker run --rm -v $PWD/data:/data py3plex:latest create --nodes 100 --layers 3 --
      ↪output /data/network.edgelist
    - docker run --rm -v $PWD/data:/data py3plex:latest stats /data/network.edgelist --
      ↪measure all --output /data/stats.json
  artifacts:
    paths:
      - data/
```

7.16.12 Additional Resources

- [Py3plex Documentation](#)
- [CLI Tutorial](#)
- [Docker Documentation](#)
- [Docker Compose Documentation](#)
- [DOCKER.md](#) - Main Docker guide in the repository

7.16.13 Support

For issues related to:

- **Py3plex functionality:** Open an issue at <https://github.com/SkBlaz/py3plex/issues>
- **Docker setup:** Check this guide first, then open an issue if problems persist
- **General questions:** See the main README.md and documentation

7.17 Performance and Scalability Best Practices

This guide provides recommendations for optimizing Py3plex performance and handling large multilayer networks.

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- *Network Scale Guidelines*
- *Sparse Matrix Backend*
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 - *When Py3plex Isn't Enough*
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 - *Before Running on Large Network*
 - *Optimization Order*
- *Next Steps*

7.17.1 Network Scale Guidelines

Py3plex is optimized for research-scale networks. This table shows expected performance characteristics:

Table 1: Network Scale Performance

Network Size	Performance	Visualization	Recommendations
Small (<100 nodes)	Excellent	Fast, detailed	Use dense visualization mode
Medium (100-1k nodes)	Good	Fast, balanced	Default settings work well
Large (1k-10k nodes)	Good	Slower, minimal	Use sparse matrices, sampling
Very Large (>10k nodes)	Variable	Very slow	Sampling required, use NetworkX/igraph

7.17.2 Sparse Matrix Backend

Why Sparse Matrices?

Most real-world networks are **sparse** (few edges compared to possible edges). Sparse matrices:

- **Reduce memory usage** by 10-100x for typical networks
- **Speed up** matrix operations (multiplication, inversion)
- **Enable** analysis of larger networks

Example:

```
import numpy as np
from scipy.sparse import csr_matrix

# Dense representation (10k × 10k network)
# Memory: 10,000^2 × 8 bytes = 800 MB

# Sparse representation (with 1% density)
# Memory: ~8 MB (100x reduction!)
```

Automatic Sparse Matrix Usage

Py3plex automatically uses sparse matrices for:

- Supra-adjacency matrix operations
- Large network storage (>1000 nodes)
- Matrix-based algorithms (PageRank, spectral methods)

Verify sparse usage:

```
from py3plex.core import multinet

network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")

# Get sparse adjacency matrix
adj_sparse = network.get_sparse_adjacency_matrix()

print(f"Matrix size: {adj_sparse.shape}")
print(f"Non-zero entries: {adj_sparse.nnz}")
print(f"Sparsity: {1 - adj_sparse.nnz / (adj_sparse.shape[0]**2):.2%}")
```

Force Sparse Operations

For custom algorithms, explicitly use sparse operations:

```
from scipy.sparse import csr_matrix, lil_matrix
import numpy as np

# Create sparse adjacency matrix
adj = lil_matrix((n_nodes, n_nodes))

for u, v in network.core_network.edges():
    i, j = node_to_idx[u], node_to_idx[v]
    adj[i, j] = 1
    adj[j, i] = 1 # Undirected

# Convert to efficient format for operations
adj_csr = adj.tocsr()

# Sparse matrix operations are MUCH faster
result = adj_csr @ adj_csr # Matrix multiplication
eigvals = scipy.sparse.linalg.eigs(adj_csr, k=10) # Top eigenvalues
```

7.17.3 Network Sampling

When to Sample

Sampling is necessary when:

- Network is too large to visualize (>5k nodes)
- Algorithms take too long (>10 minutes)
- Memory usage is excessive (>8GB RAM)
- You need quick exploratory analysis

Random Node Sampling

```
import random
from py3plex.core import multinet

# Load full network
network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")

# Sample 1000 random nodes
all_nodes = list(network.get_nodes())
sample_nodes = random.sample(all_nodes, min(1000, len(all_nodes)))

# Create subnetwork
subnetwork = network.get_subnetwork(sample_nodes)

print(f"Original: {len(all_nodes)} nodes")
print(f"Sample: {len(sample_nodes)} nodes")
print(f"Sampling ratio: {len(sample_nodes)/len(all_nodes):.1%}")
```

Stratified Sampling (Preserve Layer Distribution)

```
# Sample proportionally from each layer
layers = network.get_layer_names()
sample_nodes_per_layer = {}

for layer in layers:
    layer_nodes = [n for n in network.get_nodes() if n[1] == layer]
    sample_size = len(layer_nodes) // 10 # 10% sample
    sample_nodes_per_layer[layer] = random.sample(layer_nodes, sample_size)

# Combine samples
all_samples = [node for nodes in sample_nodes_per_layer.values() for node in nodes]
subnetwork = network.get_subnetwork(all_samples)
```

Hub-Based Sampling (Keep Important Nodes)

```
import networkx as nx

# Sample high-degree nodes (hubs)
degrees = dict(network.core_network.degree())
```

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```
# Sort by degree and take top 1000
sorted_nodes = sorted(degrees.items(), key=lambda x: x[1], reverse=True)
hub_nodes = [node for node, deg in sorted_nodes[:1000]]

subnetwork = network.get_subnetwork(hub_nodes)
print(f"Sampled top 1000 hubs")
```

7.17.4 Algorithm Optimization

Choose Efficient Algorithms

Some algorithms scale better than others:

Table 2: Algorithm Complexity

Algorithm	Complexity	Recommendations
Degree centrality	$O(n + m)$	Fast, use freely
Betweenness centrality	$O(nm)$	Slow for large networks, sample first
PageRank	$O(\text{iterations} \times m)$	Fast if sparse, limit iterations
Community detection (Louvain)	$O(m \log n)$	Fast, recommended
Shortest paths (all pairs)	$O(n^2m)$	Very slow, use sampling or approximate
Force-directed layout	$O(n^2)$	Slow for >5k nodes, use alternatives

Example - Fast centrality:

```
import networkx as nx

G = network.core_network

# FAST: Degree centrality
degree_cent = nx.degree_centrality(G) #  $O(n+m)$  - instant

# SLOW: Betweenness centrality
# For large networks, sample first or use approximate algorithm
if G.number_of_nodes() < 1000:
    between_cent = nx.betweenness_centrality(G)
else:
    # Use approximate algorithm
    between_cent = nx.betweenness_centrality(G, k=100) # Sample 100 nodes
```

Limit Algorithm Iterations

```
import networkx as nx

# PageRank with iteration limit
pagerank = nx.pagerank(
    network.core_network,
    max_iter=50,          # Limit iterations
    tol=1e-4              # Tolerance for convergence
)
```

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```
# Community detection with resolution limit
from py3plex.algorithms.community_detection import community_louvain
communities = community_louvain.best_partition(
    network.core_network,
    resolution=1.0      # Adjust resolution parameter
)
```

7.17.5 Parallel Processing

Multi-Core Processing

Use joblib for parallel node/edge operations:

```
from joblib import Parallel, delayed
import networkx as nx

def compute_node Centrality(node, graph):
    """Compute centrality for a single node."""
    # Custom centrality computation
    neighbors = list(graph.neighbors(node))
    return node, len(neighbors)

# Parallel processing
nodes = list(network.core_network.nodes())
results = Parallel(n_jobs=-1)( # Use all cores
    delayed(compute_node Centrality)(node, network.core_network)
    for node in nodes
)

centralities = dict(results)
```

GPU Acceleration (Advanced)

For very large networks, use GPU acceleration:

```
# Install CuPy for GPU NumPy operations
pip install cupy-cuda11x # Replace 11x with CUDA version
```

```
# Requires NVIDIA GPU with CUDA
try:
    import cupy as cp

    # Convert to GPU array
    adj_matrix = network.get_sparse_adjacency_matrix().toarray()
    gpu_adj = cp.array(adj_matrix)

    # GPU-accelerated matrix operations
    gpu_result = cp.dot(gpu_adj, gpu_adj)

    # Transfer back to CPU
    result = cp.asnumpy(gpu_result)
```

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```
except ImportError:
    print("CuPy not available, using CPU")
```

7.17.6 Visualization Optimization

Reduce Visual Complexity

```
from py3plex.visualization.multilayer import draw_multilayer_default

# For large networks (>1000 nodes)
draw_multilayer_default(
    network.get_layers(),
    node_size=3,           # Tiny nodes
    labels=False,         # No labels
    edge_size=0.3,        # Thin edges
    alphalevel=0.2,       # Very transparent
    remove_isolated_nodes=True # Remove disconnected
)
```

Save to File Instead of Display

```
import matplotlib.pyplot as plt

# Don't show interactively (faster)
fig, ax = plt.subplots(1, 1, figsize=(10, 8))
draw_multilayer_default(
    network.get_layers(),
    display=False, # Don't show
    axis=ax
)

# Save directly to file
plt.savefig('network.png', dpi=150, bbox_inches='tight')
plt.close() # Free memory
```

Use Lower Resolution

```
# For quick exploration, use low DPI
plt.figure(figsize=(8, 6), dpi=72) # Low resolution

# For publications, use high DPI
plt.figure(figsize=(10, 8), dpi=300) # High resolution
```

7.17.7 Memory Management

Monitor Memory Usage

```
import psutil
import os
```

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```
def get_memory_usage():
    """Get current memory usage in MB."""
    process = psutil.Process(os.getpid())
    return process.memory_info().rss / 1024 / 1024

print(f"Memory before loading: {get_memory_usage():.1f} MB")

network = multinet.multi_layer_network()
network.load_network("large_network.csv", input_type="multiedgelist")

print(f"Memory after loading: {get_memory_usage():.1f} MB")
```

Free Memory When Done

```
# Delete network when no longer needed
del network

# Force garbage collection
import gc
gc.collect()
```

Use Generators Instead of Lists

```
# BAD: Loads all nodes into memory
all_nodes = list(network.get_nodes())
for node in all_nodes:
    process(node)

# GOOD: Processes nodes one at a time
for node in network.get_nodes():
    process(node)
```

7.17.8 Batch Processing

Process Networks in Batches

For multiple networks:

```
import os
import gc

network_files = ["net1.csv", "net2.csv", "net3.csv", ...]

results = []
for i, file in enumerate(network_files):
    print(f"Processing {i+1}/{len(network_files)}: {file}")

    # Load network
    network = multinet.multi_layer_network()
    network.load_network(file, input_type="multiedgelist")
```

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```

# Compute statistics
result = analyze_network(network)
results.append(result)

# Free memory
del network
gc.collect()

# Save intermediate results every 10 networks
if (i + 1) % 10 == 0:
    save_results(results, f"results_batch_{i+1}.json")

```

7.17.9 Benchmark Results

Performance Benchmarks (2025)

Tested on: Intel i7-10700K, 32GB RAM, Python 3.10

Table 3: Operation Performance

Operation	100 nodes	1k nodes	10k nodes	100k nodes
Load from CSV	<1s	<1s	2s	20s
Basic statistics	<1s	<1s	<1s	3s
Degree centrality	<1s	<1s	1s	10s
PageRank	<1s	<1s	2s	25s
Louvain communities	<1s	1s	5s	60s
Visualization (sparse)	<1s	2s	15s	N/A*

* Visualization not recommended for >10k nodes without sampling

Scaling Recommendations

Based on network size:

<1k nodes:

- Use any algorithms
- Full visualization
- No sampling needed

1k-10k nodes:

- Use sparse matrices
- Minimal visualization
- Sample for some algorithms

10k-100k nodes:

- Sparse matrices required
- Sample for visualization
- Use approximate algorithms

- Consider igraph for speed

>100k nodes:

- Use specialized tools (igraph, graph-tool, NetworkKit)
- Sample heavily for Py3plex operations
- Focus on specific analyses

7.17.10 Alternative Tools for Scale

When Py3plex Isn't Enough

For networks >100k nodes, consider:

igraph (C-based, very fast):

```
import igraph as ig

# 10-100x faster for large networks
g = ig.Graph.Read_GraphML("large_network.graphml")
communities = g.community_multilevel()

# Export back to Py3plex if needed
# (via GraphML or edge list)
```

graph-tool (C++, fastest):

```
import graph_tool.all as gt

# Fastest for >1M edges
g = gt.load_graph("large_network.graphml")
communities = gt.community_structure.minimize_blockmodel_dl(g)
```

NetworkKit (C++, parallel):

```
import networkit as nk

# Excellent for parallel algorithms
G = nk.readGraph("large_network.edgelist", nk.Format.EdgeList)
communities = nk.community.detectCommunities(G)
```

7.17.11 Quick Performance Checklist

Before Running on Large Network

```
[ ] Enable sparse matrices (automatic for most ops)
[ ] Sample network if >10k nodes
[ ] Choose efficient algorithms (degree > betweenness)
[ ] Limit visualization detail
[ ] Monitor memory usage
[ ] Use batch processing for multiple networks
[ ] Consider alternative tools if >100k nodes
```

Optimization Order

1. **Use sparse matrices** (biggest impact, usually automatic)
2. **Sample network** (if >10k nodes)
3. **Choose efficient algorithms** (avoid $O(n^3)$ operations)
4. **Parallelize** (if multi-core available)
5. **GPU acceleration** (only if CUDA GPU available)

7.17.12 Next Steps

- *I/O and Serialization* - Efficient data loading
- *Visualization Guide* - Optimize visualizations
- *Docker Usage Guide* - Docker deployment for production

For performance issues, open an issue on [GitHub Issues](#).

7.18 Py3plex GUI

A production-ready web-based GUI for **py3plex** multilayer network analysis, running locally via Docker Compose.

7.18.1 Visual Overview

The Py3plex GUI provides an intuitive web interface for multilayer network analysis. Below are screenshots of the main interface pages:



Figure 1: Load Data page for uploading network files in various formats

7.18.2 Quick Start

```
# Navigate to the gui directory
cd gui

# Copy environment configuration
cp .env.example .env
```

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```
# Start all services (builds automatically)
make up

# Open in browser
# Application: http://localhost:8080
# Data Job Monitor (Flower): http://localhost:5555
```

First time? The build takes ~3-5 minutes. Subsequent starts are instant.

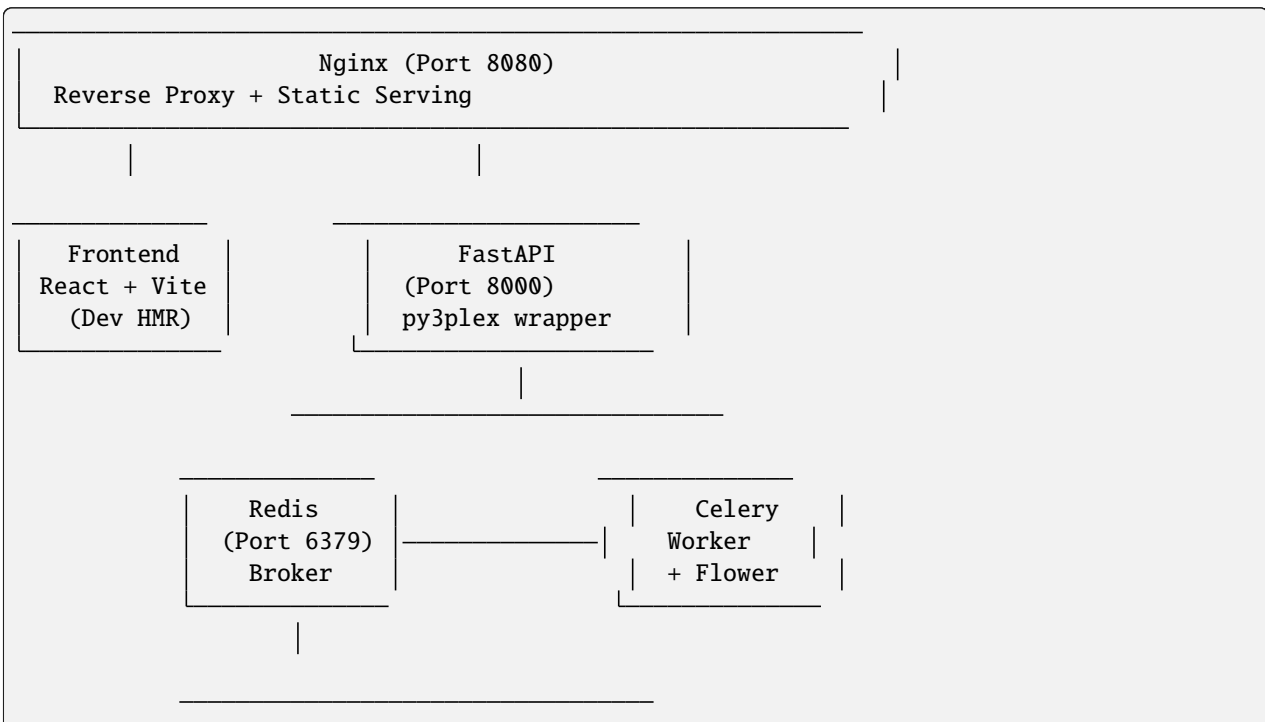
That's it! The application will be running with:

- React + TypeScript frontend with hot reload
- FastAPI backend with py3plex integration
- Celery workers for async jobs
- Redis broker
- Nginx reverse proxy

7.18.3 Prerequisites

- Docker (≥ 20.10)
- Docker Compose (≥ 2.0)
- 4GB RAM minimum
- Ports available: 8080 (GUI), 5555 (Flower), 8000 (API), 6379 (Redis)

7.18.4 Architecture



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Shared Volume: /data - uploads/ - artifacts/ - workspaces/
--

Key Components:

- **Frontend:** React 18 + Vite + TypeScript + Tailwind CSS
- **API:** FastAPI (Python 3.12) with py3plex integration
- **Worker:** Celery for long-running analysis jobs
- **Broker:** Redis for job queue
- **Proxy:** Nginx for unified access
- **Monitoring:** Flower dashboard for job inspection

7.18.5 Features

Data Loading

- Upload network files (.edgelist, .txt, .gml, .gpickle)
- Automatic format detection
- Multilayer network support
- Real-time file preview

Visualization



Figure 2: Network visualization with layer controls and node inspection panel

Features:

- Layer-centric view
- Interactive node/edge inspection
- Configurable layouts

- Position caching

Analysis (Async via Celery)



Figure 3: Analysis page with job controls and real-time progress tracking

Available Analysis Tools:

- **Layouts:** Spring, Kamada-Kawai, Circular, Random
- **Centrality:** Degree, Betweenness, Closeness, Eigenvector, PageRank

- **Community Detection:** Louvain, Label Propagation, Greedy Modularity
- Real-time progress tracking
- Result caching

Export



example_images/gui_export.png

Figure 4: Export page for saving analysis results and workspace bundles

Export Options:

- CSV summaries (centrality, communities)
- JSON position data
- PNG snapshots (planned)
- Workspace bundles (data + state + results)

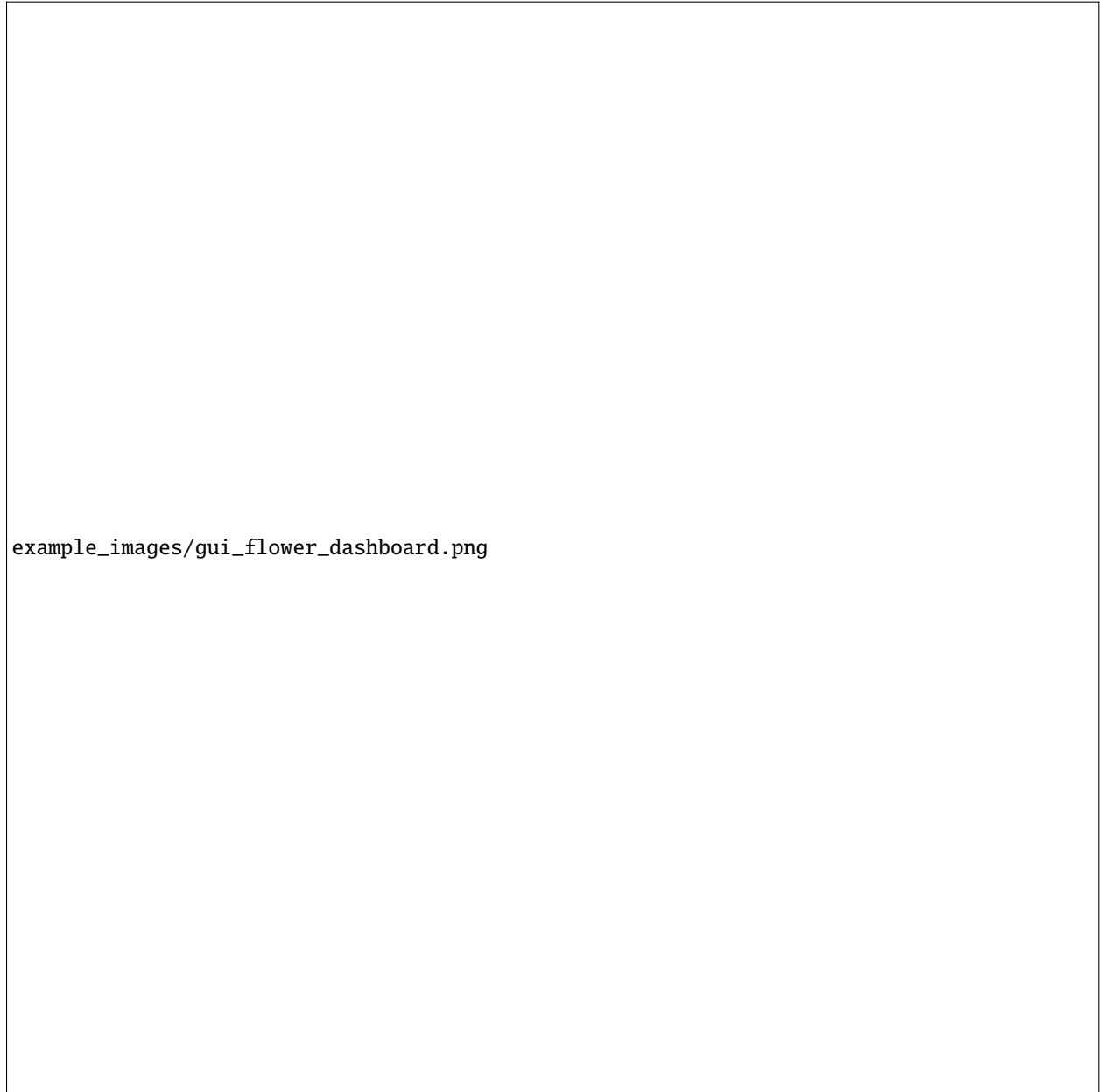
Workspace Bundles

Save and restore complete analysis sessions:

```
# Bundle includes:  
# - Original network file  
# - Computed layouts  
# - Centrality results  
# - Community assignments  
# - UI view state
```

Job Monitoring

The GUI includes Flower dashboard for real-time monitoring of background jobs:



example_images/gui_flower_dashboard.png

Figure 5: Flower dashboard showing task history and worker status

Monitoring Features:

- Track task execution in real-time
- View worker status and performance
- Inspect task history and results
- Monitor task queue and completion rates
- Access at <http://localhost:5555> when running

7.18.6 Development Guide

Project Structure

```

gui/
├── docker-compose.yml      # Main orchestration
├── compose.gpu.yml        # GPU override (optional)
├── Makefile                # Convenience commands
├── .env.example            # Configuration template
├── nginx/
│   └── nginx.conf          # Reverse proxy config
├── api/
│   ├── Dockerfile.api      # API container
│   ├── pyproject.toml      # Python dependencies
│   └── app/
│       ├── main.py         # FastAPI app
│       ├── schemas.py      # Pydantic models
│       ├── deps.py         # DI & config
│       ├── routes/         # API endpoints
│       ├── services/       # Business logic (py3plex wrappers)
│       ├── workers/        # Celery tasks
│       └── utils/          # Helpers
├── worker/
│   └── Dockerfile          # Worker container
├── frontend/
│   ├── Dockerfile.frontend # Frontend container
│   ├── package.json        # Node dependencies
│   ├── vite.config.ts      # Vite config
│   └── src/
│       ├── App.tsx         # Root component
│       ├── lib/api.ts      # API client
│       ├── pages/          # Page components
│       └── components/     # Reusable components
└── data/                   # Shared volume (gitignored)
    ├── uploads/
    ├── artifacts/
    └── workspaces/

```

Makefile Commands

```

make setup      # Copy .env.example to .env
make up         # Start all services (build if needed)
make down       # Stop and remove containers + volumes
make restart    # Restart all services
make build      # Rebuild Docker images
make logs       # Tail logs from all services

# Development helpers
make bash-api   # Shell into API container
make bash-worker # Shell into worker container
make bash-frontend # Shell into frontend container

# Testing

```

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```

make test-api      # Run API tests
make e2e           # Run end-to-end tests (WIP)

# Cleanup
make clean          # Remove containers, volumes, and data

```

Local Development Against py3plex

The py3plex repository root is bind-mounted read-only into containers at `/workspace`:

```

# In Dockerfile.api
ARG PY3PLEX_PATH=/workspace
RUN pip install -e ${PY3PLEX_PATH}

```

Benefits:

- Changes to py3plex core reflect immediately (no rebuild)
- Editable install for development
- Isolated GUI code under `gui/`

Note: The mount is read-only to prevent accidental writes from containers.

7.18.7 Configuration

Environment Variables (.env)

```

# API
API_WORKERS=2           # Uvicorn workers
MAX_UPLOAD_MB=512       # Max file size

# Celery
CELERY_CONCURRENCY=2    # Worker threads
REDIS_URL=redis://redis:6379/0

# Frontend
VITE_API_URL=http://localhost:8080/api

```

GPU Support (Optional)

Enable NVIDIA GPU acceleration:

```
docker compose -f docker-compose.yml -f compose.gpu.yml up
```

Requirements:

- NVIDIA Docker runtime installed
- CUDA-compatible GPU

7.18.8 Data Formats

Accepted Input Formats

1. Edge List (.edgelist, .txt)

```
# Multilayer format: node1 node2 layer weight
1 2 social 1.0
1 3 work 1.0
2 3 hobby 0.5

# Simple format: node1 node2
A B
B C
C D
```

2. GML (.gml)

- Graph Modeling Language
- Preserves attributes and metadata

3. NetworkX Pickle (.gpickle)

- Native NetworkX serialization
- Fastest for large graphs

Example Dataset

Try the included toy network:

```
# From host
curl -F "file=@gui/toy_network.edgelist" http://localhost:8080/api/upload

# Or use the Web UI at http://localhost:8080
```

7.18.9 Testing

Continuous Integration

GUI tests run automatically on GitHub Actions for any changes to the `gui/` directory:

```
# Workflow: .github/workflows/gui-tests.yml
- API unit tests (pytest)
- Integration tests (Docker Compose)
- Frontend build verification
```

Status:

Tests include:

- Health endpoint validation
- File upload with toy network
- Layout job execution and completion
- Centrality computation

- Service health checks

API Tests

```
# Run inside API container
make bash-api
pytest ci/api-tests/ -v

# Or directly
docker compose exec api pytest ci/api-tests/ -v
```

Integration Tests

The CI workflow runs full integration tests:

```
# Upload and analyze a network
cd gui
curl -F "file=@toy_network.edgelist" http://localhost:8080/api/upload
# Run layout, centrality, and community detection jobs
```

Frontend Tests (WIP)

```
# Playwright smoke tests
make e2e
```

Manual Testing Workflow

1. **Upload** `toy_network.edgelist` via Web UI
2. **Visualize** the network (6 nodes, 14 edges, 3 layers)
3. **Analyze**:
 - Run Spring Layout
 - Compute Centrality (degree + betweenness)
 - Detect Communities (Louvain)
4. **Monitor** jobs at <http://localhost:5555> (Flower)
5. **Export** workspace bundle

7.18.10 Production Deployment

Static Build (Recommended)

For production, serve pre-built frontend assets:

```
# Modify Dockerfile.frontend to use multi-stage build
FROM node:20-alpine AS builder
WORKDIR /app
COPY frontend/ .
RUN npm install && npm run build

FROM nginx:alpine
```

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```
COPY --from=builder /app/dist /usr/share/nginx/html
COPY nginx/nginx.conf /etc/nginx/nginx.conf
```

Update docker-compose.yml:

```
frontend:
  build:
    context: .
    dockerfile: Dockerfile.frontend.prod
  # Remove dev server volume mounts
```

Security Considerations

- Set CORS `allow_origins` to specific domains (not *)
- Add authentication/authorization middleware
- Use HTTPS (add TLS termination in Nginx)
- Validate file uploads strictly
- Set resource limits per user/job
- Enable Celery task rate limiting

Current Status: Development mode - suitable for local use only. Do not expose to public internet without proper security hardening.

7.18.11 Troubleshooting

Issue: Port already in use

```
# Find process using port 8080
lsof -ti:8080 | xargs kill -9

# Or change port in docker-compose.yml
ports:
  - "8081:80" # Use 8081 instead
```

Issue: Permission denied on /data

```
# Fix volume permissions
sudo chown -R $(whoami):$(whoami) gui/data/
```

Issue: py3plex import error in containers

```
# Verify mount
docker compose exec api ls -la /workspace

# Reinstall if needed
docker compose exec api pip install -e /workspace
```

Issue: Frontend can't reach API

Check Nginx proxy config:

```
docker compose exec nginx cat /etc/nginx/nginx.conf

# Test API directly
curl http://localhost:8000/api/health
```

7.18.12 API Documentation

Interactive docs available at:

- **Swagger UI:** <http://localhost:8080/api/docs>
- **ReDoc:** <http://localhost:8080/api/redoc>

Key Endpoints

POST	/api/upload	# Upload network file
GET	/api/graphs/{id}/summary	# Graph statistics
POST	/api/graphs/{id}/layout	# Compute layout (async)
POST	/api/graphs/{id}/analysis/centrality	# Compute centrality (async)
POST	/api/graphs/{id}/analysis/community	# Detect communities (async)
GET	/api/jobs/{id}	# Poll job status
POST	/api/workspaces/save	# Save workspace bundle

Example: Run Analysis

```
# 1. Upload
GRAPH_ID=$(curl -F "file=@toy_network.edgelist" \
  http://localhost:8080/api/upload | jq -r .graph_id)

# 2. Start layout job
JOB_ID=$(curl -X POST \
  -H "Content-Type: application/json" \
  -d '{"algorithm": "spring", "seed": 42, "dimensions": 2}' \
  http://localhost:8080/api/graphs/$GRAPH_ID/layout | jq -r .job_id)

# 3. Poll job
curl http://localhost:8080/api/jobs/$JOB_ID

# 4. Get positions
curl http://localhost:8080/api/graphs/$GRAPH_ID/positions
```

7.18.13 Contributing

This GUI is part of the [py3plex](#) project.

Guidelines:

- Keep all GUI code under `gui/`
- Do not modify `py3plex` core
- Follow existing code style (Black, Ruff for Python; ESLint for TypeScript)

- Add tests for new features
- Update documentation with new features

7.18.14 License

This GUI follows the py3plex repository license:

- **GUI code** (under gui/): BSD-3-Clause (same as py3plex)
- **Dependencies**: See individual package licenses

7.18.15 Acknowledgments

Built with:

- **py3plex** - Multilayer network analysis
- **FastAPI** - Modern Python web framework
- **Celery** - Distributed task queue
- **React** - UI library
- **Vite** - Frontend build tool
- **Tailwind CSS** - Utility-first CSS

7.18.16 Support

- **Issues**: <https://github.com/SkBlaz/py3plex/issues>
- **Docs**: <https://skblaz.github.io/py3plex/>
- **py3plex Paper**: [Applied Network Science 2019](#)

7.19 GUI Deployment Guide

This guide covers deploying the py3plex web interface for production use.

7.19.1 Overview

The py3plex GUI can be deployed in several ways:

1. **Local Development** - Run directly on your machine
2. **Docker Container** - Isolated, reproducible environment
3. **Production Server** - Behind reverse proxy with TLS
4. **Cloud Deployment** - AWS, Azure, GCP, etc.

See *Py3plex GUI* for basic usage and *Py3plex GUI Architecture* for technical details.

7.19.2 Local Development Setup

Quick Start

```
# Clone repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Install with GUI dependencies
pip install -e ".[gui]"

# Start development server
python gui/app.py
```

The GUI will be available at `http://localhost:5000`.

Configuration

Create a configuration file `gui/config.py`:

```
# Development configuration
DEBUG = True
HOST = '0.0.0.0'
PORT = 5000
SECRET_KEY = 'dev-secret-key-change-in-production'

# Upload settings
UPLOAD_FOLDER = './uploads'
MAX_CONTENT_LENGTH = 100 * 1024 * 1024 # 100 MB max file size

# Allowed file extensions
ALLOWED_EXTENSIONS = {'txt', 'edgelist', 'graphml', 'gml', 'json', 'arrow'}
```

7.19.3 Docker Deployment

Basic Docker Setup

The repository includes a Dockerfile for the GUI:

```
# Build image
docker build -t py3plex-gui:latest -f gui/Dockerfile .

# Run container
docker run -d \
  -p 5000:5000 \
  -v $(pwd)/data:/app/data \
  --name py3plex-gui \
  py3plex-gui:latest
```

The GUI will be available at `http://localhost:5000`.

Docker Compose

Create `docker-compose.yml`:

```
version: '3.8'

services:
```

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```
gui:
  build:
    context: .
    dockerfile: gui/Dockerfile
  ports:
    - "5000:5000"
  volumes:
    - ./data:/app/data
    - ./uploads:/app/uploads
  environment:
    - FLASK_ENV=production
    - SECRET_KEY=${SECRET_KEY}
  restart: unless-stopped
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:5000/health"]
    interval: 30s
    timeout: 10s
    retries: 3
```

Start with:

```
# Set secret key
export SECRET_KEY=$(python -c 'import secrets; print(secrets.token_hex(32))')

# Start services
docker-compose up -d

# Check logs
docker-compose logs -f gui
```

Environment Variables

Configure via environment variables:

Variable	Description	Default
FLASK_ENV	Environment (development/production)	development
SECRET_KEY	Session secret key (required)	None
HOST	Bind address	0.0.0.0
PORT	Bind port	5000
MAX_WORKERS	Worker processes	4
UPLOAD_FOLDER	Upload directory	./uploads
DATA_FOLDER	Data directory	./data

7.19.4 Production Deployment

Using Gunicorn

For production, use a production WSGI server like Gunicorn:

```
# Install Gunicorn
pip install gunicorn
```

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```
# Run with Gunicorn
gunicorn \
  --bind 0.0.0.0:5000 \
  --workers 4 \
  --timeout 120 \
  --access-logfile - \
  --error-logfile - \
  gui.app:app
```

Supervisor Configuration

Use Supervisor to manage the GUI process:

Create /etc/supervisor/conf.d/py3plex-gui.conf:

```
[program:py3plex-gui]
command=/path/to/venv/bin/gunicorn --bind 127.0.0.1:5000 --workers 4 gui.app:app
directory=/path/to/py3plex
user=www-data
autostart=true
autorestart=true
redirect_stderr=true
stdout_logfile=/var/log/py3plex-gui.log
environment=FLASK_ENV="production",SECRET_KEY="your-secret-key"
```

Start with:

```
sudo supervisorctl reread
sudo supervisorctl update
sudo supervisorctl start py3plex-gui
```

Systemd Service

Alternative: use systemd:

Create /etc/systemd/system/py3plex-gui.service:

```
[Unit]
Description=Py3plex GUI
After=network.target

[Service]
Type=notify
User=www-data
WorkingDirectory=/path/to/py3plex
Environment="FLASK_ENV=production"
Environment="SECRET_KEY=your-secret-key"
ExecStart=/path/to/venv/bin/gunicorn --bind 127.0.0.1:5000 --workers 4 gui.app:app
ExecReload=/bin/kill -s HUP $MAINPID
KillMode=mixed
TimeoutStopSec=5
PrivateTmp=true
```

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```
[Install]
WantedBy=multi-user.target
```

Enable and start:

```
sudo systemctl daemon-reload
sudo systemctl enable py3plex-gui
sudo systemctl start py3plex-gui
sudo systemctl status py3plex-gui
```

7.19.5 Reverse Proxy Setup

Nginx Configuration

Configure Nginx as a reverse proxy:

Create `/etc/nginx/sites-available/py3plex-gui`:

```
server {
    listen 80;
    server_name py3plex.yourdomain.com;

    # Redirect to HTTPS
    return 301 https://$server_name$request_uri;
}

server {
    listen 443 ssl http2;
    server_name py3plex.yourdomain.com;

    # SSL certificates (use Let's Encrypt)
    ssl_certificate /etc/letsencrypt/live/py3plex.yourdomain.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/py3plex.yourdomain.com/privkey.pem;

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN" always;
    add_header X-Content-Type-Options "nosniff" always;
    add_header X-XSS-Protection "1; mode=block" always;

    # Upload size limit
    client_max_body_size 100M;

    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;

        # Timeouts for long-running operations
        proxy_connect_timeout 300s;
        proxy_send_timeout 300s;
```

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```

    proxy_read_timeout 300s;
}

# Static files (if serving separately)
location /static {
    alias /path/to/py3plex/gui/static;
    expires 30d;
    add_header Cache-Control "public, immutable";
}
}

```

Enable the site:

```

sudo ln -s /etc/nginx/sites-available/py3plex-gui /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx

```

Apache Configuration

Alternative: use Apache as reverse proxy:

```

<VirtualHost *:80>
    ServerName py3plex.yourdomain.com
    Redirect permanent / https://py3plex.yourdomain.com/
</VirtualHost>

<VirtualHost *:443>
    ServerName py3plex.yourdomain.com

    SSLEngine on
    SSLCertificateFile /etc/letsencrypt/live/py3plex.yourdomain.com/fullchain.pem
    SSLCertificateKeyFile /etc/letsencrypt/live/py3plex.yourdomain.com/privkey.pem

    ProxyPreserveHost On
    ProxyPass / http://127.0.0.1:5000/
    ProxyPassReverse / http://127.0.0.1:5000/

    # Security headers
    Header always set X-Frame-Options "SAMEORIGIN"
    Header always set X-Content-Type-Options "nosniff"
</VirtualHost>

```

SSL/TLS with Let's Encrypt

```

# Install certbot
sudo apt-get install certbot python3-certbot-nginx

# Obtain certificate
sudo certbot --nginx -d py3plex.yourdomain.com

# Auto-renewal (certbot sets this up automatically)
sudo certbot renew --dry-run

```

7.19.6 Security Considerations

Secret Key Management

Never hardcode secret keys in code or configuration files:

```
# Generate strong secret key
python -c 'import secrets; print(secrets.token_hex(32))' > .secret_key

# Set in environment
export SECRET_KEY=$(cat .secret_key)

# Or use environment file
echo "SECRET_KEY=$(python -c 'import secrets; print(secrets.token_hex(32))')" > .env
```

File Upload Security

Configure upload restrictions:

```
# In config.py
ALLOWED_EXTENSIONS = {'txt', 'edgelist', 'graphml', 'gml', 'json', 'arrow'}
MAX_CONTENT_LENGTH = 100 * 1024 * 1024 # 100 MB

# Validate uploads
def allowed_file(filename):
    return '.' in filename and \
        filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS
```

Rate Limiting

Implement rate limiting for API endpoints:

```
from flask_limiter import Limiter
from flask_limiter.util import get_remote_address

limiter = Limiter(
    app,
    key_func=get_remote_address,
    default_limits=["200 per day", "50 per hour"]
)

@app.route("/api/upload")
@limiter.limit("10 per hour")
def upload():
    pass
```

Authentication

Add user authentication for production:

```
from flask_login import LoginManager, login_required

login_manager = LoginManager()
login_manager.init_app(app)
```

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```
@app.route("/dashboard")
@login_required
def dashboard():
    pass
```

7.19.7 Monitoring and Logging

Application Logging

Configure structured logging:

```
import logging
from logging.handlers import RotatingFileHandler

# Configure logging
handler = RotatingFileHandler(
    'logs/py3plex-gui.log',
    maxBytes=10 * 1024 * 1024, # 10 MB
    backupCount=5
)
handler.setFormatter(logging.Formatter(
    '[%(asctime)s] %(levelname)s in %(module)s: %(message)s'
))
app.logger.addHandler(handler)
app.logger.setLevel(logging.INFO)
```

Prometheus Metrics

Export metrics for Prometheus:

```
from prometheus_flask_exporter import PrometheusMetrics

metrics = PrometheusMetrics(app)

# Custom metrics
metrics.info('app_info', 'Application info', version='1.0')
```

Health Checks

Implement health check endpoint:

```
@app.route('/health')
def health():
    # Check dependencies
    try:
        # Test database connection, file system, etc.
        return {'status': 'healthy'}, 200
    except Exception as e:
        return {'status': 'unhealthy', 'error': str(e)}, 503
```

7.19.8 Cloud Deployment Examples

AWS EC2

1. Launch EC2 instance (Ubuntu 22.04 LTS)
2. Install dependencies
3. Clone repository
4. Follow production deployment steps above
5. Configure security groups (ports 80, 443)

AWS ECS (Docker)

```
{
  "family": "py3plex-gui",
  "containerDefinitions": [{
    "name": "gui",
    "image": "your-registry/py3plex-gui:latest",
    "memory": 2048,
    "cpu": 1024,
    "essential": true,
    "portMappings": [{
      "containerPort": 5000,
      "protocol": "tcp"
    }],
    "environment": [
      {"name": "FLASK_ENV", "value": "production"}
    ],
    "secrets": [
      {"name": "SECRET_KEY", "valueFrom": "arn:aws:secretsmanager:..."}
    ]
  }]
}
```

Google Cloud Run

```
# Build and push to GCR
gcloud builds submit --tag gcr.io/PROJECT_ID/py3plex-gui

# Deploy
gcloud run deploy py3plex-gui \
  --image gcr.io/PROJECT_ID/py3plex-gui \
  --platform managed \
  --region us-central1 \
  --allow-unauthenticated \
  --set-env-vars FLASK_ENV=production \
  --set-secrets SECRET_KEY=py3plex-secret:latest
```

Azure Container Instances

```
# Create container group
az container create \
  --resource-group py3plex-rg \
  --name py3plex-gui \
  --image your-registry.azurecr.io/py3plex-gui:latest \
  --dns-name-label py3plex-gui \
  --ports 5000 \
  --environment-variables FLASK_ENV=production \
  --secure-environment-variables SECRET_KEY=$SECRET_KEY
```

7.19.9 Performance Tuning

Worker Configuration

```
# Rule of thumb: (2 x CPU cores) + 1
workers=$((2 * $(nproc) + 1))

unicorn \
  --workers $workers \
  --threads 2 \
  --worker-class gthread \
  --timeout 120 \
  --bind 0.0.0.0:5000 \
  gui.app:app
```

Caching

Implement caching for expensive operations:

```
from flask_caching import Cache

cache = Cache(app, config={'CACHE_TYPE': 'simple'})

@app.route('/api/stats/<network_id>')
@cache.cached(timeout=300)
def network_stats(network_id):
    # Expensive computation
    return stats
```

Database for Persistence

For production, consider a database:

```
from flask_sqlalchemy import SQLAlchemy

app.config['SQLALCHEMY_DATABASE_URI'] = 'postgresql://user:pass@localhost/py3plex'
db = SQLAlchemy(app)
```

7.19.10 Backup and Recovery

Automated Backups

```
#!/bin/bash
# backup.sh

BACKUP_DIR="/backups/py3plex"
DATE=$(date +%Y%m%d_%H%M%S)

# Backup uploads
tar -czf "$BACKUP_DIR/uploads_$DATE.tar.gz" /path/to/uploads

# Backup database (if using one)
# pg_dump py3plex > "$BACKUP_DIR/db_$DATE.sql"

# Cleanup old backups (keep last 7 days)
find "$BACKUP_DIR" -type f -mtime +7 -delete
```

Add to crontab:

```
# Daily backup at 2 AM
0 2 * * * /path/to/backup.sh
```

Disaster Recovery

Document recovery procedure:

1. Restore from backup
2. Verify data integrity
3. Restart services
4. Test functionality

7.19.11 Troubleshooting

Common Issues

Port already in use:

```
# Find process using port
sudo lsof -i :5000

# Kill process
sudo kill -9 PID
```

Permission denied:

```
# Fix upload directory permissions
sudo chown -R www-data:www-data /path/to/uploads
```

Out of memory:

```
# Check memory usage
free -h
```

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```
# Adjust worker count or add swap
```

Logs Location

- Application logs: `/var/log/py3plex-gui.log`
- Nginx logs: `/var/log/nginx/`
- Systemd logs: `journalctl -u py3plex-gui`

7.19.12 Related Documentation

- *Py3plex GUI* - Using the GUI
- *Py3plex GUI Architecture* - Technical architecture
- *GUI API Reference* - API endpoints
- *GUI Testing Guide* - Testing the GUI
- *Docker Usage Guide* - Docker details

Security Resources:

- OWASP Flask Security: <https://owasp.org/www-project-web-security-testing-guide/>
- Flask Security Best Practices: <https://flask.palletsprojects.com/en/2.3.x/security/>

7.20 GUI API Reference

This reference documents the REST API endpoints provided by the py3plex web interface.

7.20.1 Overview

The py3plex GUI provides a REST API for:

- Uploading network data
- Running analyses
- Generating visualizations
- Exporting results

Base URL: `http://localhost:5000/api` (development)

Authentication: Currently no authentication (add in production)

Response Format: JSON

7.20.2 General Endpoints

Health Check

Check if the server is running.

Endpoint: GET `/health`

Response:

```
{
  "status": "healthy",
  "version": "0.95",
  "timestamp": "2025-01-23T12:00:00Z"
}
```

Status Codes:

- 200 - Server is healthy
- 503 - Server is unhealthy

Server Info

Get server information.

Endpoint: GET /api/info

Response:

```
{
  "py3plex_version": "0.95",
  "python_version": "3.10.0",
  "available_algorithms": ["louvain", "infomap", "node2vec"],
  "max_upload_size": 104857600
}
```

7.20.3 Network Management

Upload Network

Upload a network file for analysis.

Endpoint: POST /api/upload

Parameters:

- file (form-data, required) - Network file
- input_type (form-data, required) - File format (edgelist, multiedgelist, graphml, etc.)
- directed (form-data, optional) - Boolean, default false

Request Example:

```
curl -X POST http://localhost:5000/api/upload \
-F "file=@network.edgelist" \
-F "input_type=edgelist" \
-F "directed=false"
```

Response:

```
{
  "success": true,
  "network_id": "abc123",
  "num_nodes": 100,
  "num_edges": 450,
  "num_layers": 3,
}
```

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```
{
  "message": "Network uploaded successfully"
}
```

Status Codes:

- 200 - Success
- 400 - Invalid file or parameters
- 413 - File too large
- 500 - Server error

List Networks

List all uploaded networks.

Endpoint: GET /api/networks

Response:

```
{
  "networks": [
    {
      "id": "abc123",
      "filename": "network.edgelist",
      "uploaded_at": "2025-01-23T12:00:00Z",
      "num_nodes": 100,
      "num_edges": 450
    }
  ]
}
```

Get Network Info

Get detailed information about a specific network.

Endpoint: GET /api/networks/<network_id>

Response:

```
{
  "id": "abc123",
  "filename": "network.edgelist",
  "num_nodes": 100,
  "num_edges": 450,
  "num_layers": 3,
  "layers": ["layer1", "layer2", "layer3"],
  "statistics": {
    "density": 0.045,
    "clustering": 0.32
  }
}
```

Delete Network

Delete an uploaded network.

Endpoint: DELETE /api/networks/<network_id>

Response:

```
{
  "success": true,
  "message": "Network deleted successfully"
}
```

7.20.4 Analysis Endpoints

Run Community Detection

Detect communities in a network.

Endpoint: POST /api/analysis/communities

Parameters:

- network_id (required) - Network ID
- algorithm (required) - Algorithm name (louvain, leiden, infomap)
- gamma (optional) - Resolution parameter (default: 1.0)
- omega (optional) - Inter-layer coupling (default: 1.0)

Request:

```
{
  "network_id": "abc123",
  "algorithm": "louvain",
  "gamma": 1.0,
  "omega": 1.0
}
```

Response:

```
{
  "success": true,
  "num_communities": 5,
  "modularity": 0.42,
  "partition": {
    "node1": 0,
    "node2": 0,
    "node3": 1,
    ...
  }
}
```

Compute Statistics

Compute network statistics.

Endpoint: POST /api/analysis/statistics

Parameters:

- `network_id` (required) - Network ID
- `metrics` (required) - List of metrics to compute

Request:

```
{
  "network_id": "abc123",
  "metrics": ["density", "clustering", "degree_distribution"]
}
```

Response:

```
{
  "success": true,
  "statistics": {
    "density": 0.045,
    "clustering": 0.32,
    "degree_distribution": {
      "1": 10,
      "2": 25,
      "3": 30,
      "...": ...
    }
  }
}
```

Compute Centrality

Compute node centrality measures.

Endpoint: POST `/api/analysis/centrality`

Parameters:

- `network_id` (required) - Network ID
- `measure` (required) - Centrality measure (degree, betweenness, pagerank, eigenvector)
- `layer` (optional) - Specific layer to analyze

Request:

```
{
  "network_id": "abc123",
  "measure": "betweenness",
  "layer": "layer1"
}
```

Response:

```
{
  "success": true,
  "centrality": {
    "node1": 0.42,
    "node2": 0.35,
```

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```
"node3": 0.15,  
  ...  
}
```

Generate Embeddings

Generate node embeddings.

Endpoint: POST /api/analysis/embeddings

Parameters:

- network_id (required) - Network ID
- method (required) - Embedding method (node2vec, deepwalk)
- dimensions (optional) - Embedding dimensions (default: 128)
- walk_length (optional) - Walk length (default: 80)
- num_walks (optional) - Walks per node (default: 10)
- p (optional) - Return parameter (default: 1.0)
- q (optional) - In-out parameter (default: 1.0)

Request:

```
{  
  "network_id": "abc123",  
  "method": "node2vec",  
  "dimensions": 128,  
  "walk_length": 80,  
  "num_walks": 10,  
  "p": 1.0,  
  "q": 1.0  
}
```

Response:

```
{  
  "success": true,  
  "embedding_id": "emb456",  
  "dimensions": 128,  
  "num_nodes": 100  
}
```

7.20.5 Visualization Endpoints

Generate Layout

Compute network layout.

Endpoint: POST /api/visualization/layout

Parameters:

- network_id (required) - Network ID

- `algorithm` (required) - Layout algorithm (force, spring, circular, diagonal)
- `iterations` (optional) - Number of iterations (default: 50)

Request:

```
{
  "network_id": "abc123",
  "algorithm": "force",
  "iterations": 100
}
```

Response:

```
{
  "success": true,
  "layout_id": "layout789",
  "positions": {
    "node1": [0.5, 0.3],
    "node2": [0.6, 0.8],
    
  }
}
```

Render Visualization

Generate network visualization image.

Endpoint: POST `/api/visualization/render`

Parameters:

- `network_id` (required) - Network ID
- `layout_id` (optional) - Pre-computed layout ID
- `width` (optional) - Image width (default: 800)
- `height` (optional) - Image height (default: 600)
- `format` (optional) - Output format (png, svg, pdf, default: png)

Request:

```
{
  "network_id": "abc123",
  "layout_id": "layout789",
  "width": 1200,
  "height": 900,
  "format": "png"
}
```

Response:

Returns image file with appropriate Content-Type header.

Status Codes:

- 200 - Success (returns image)
- 400 - Invalid parameters

- 404 - Network or layout not found

7.20.6 Export Endpoints

Export Network

Export network in different formats.

Endpoint: GET /api/export/network/<network_id>

Query Parameters:

- format (required) - Export format (graphml, gml, edgelist, json, arrow)

Example:

```
curl http://localhost:5000/api/export/network/abc123?format=graphml \
-o output.graphml
```

Response:

Returns file in requested format.

Export Analysis Results

Export analysis results as JSON or CSV.

Endpoint: GET /api/export/analysis/<analysis_id>

Query Parameters:

- format (optional) - Export format (json, csv, default: json)

Example:

```
curl http://localhost:5000/api/export/analysis/analysis123?format=csv \
-o results.csv
```

Response:

Returns file in requested format.

7.20.7 Error Handling

Error Response Format

All errors return a consistent JSON format:

```
{
  "success": false,
  "error": "Error message description",
  "error_code": "ERROR_CODE",
  "details": {
    "field": "Additional error details"
  }
}
```


Common Error Codes

HTTP Status	Error Code	Description
400	INVALID_INPUT	Invalid request parameters
400	INVALID_FILE	Invalid or corrupted file
401	UNAUTHORIZED	Authentication required
404	NOT_FOUND	Resource not found
413	FILE_TOO_LARGE	Upload exceeds size limit
500	INTERNAL_ERROR	Server error
503	SERVICE_UNAVAILABLE	Service temporarily unavailable

7.20.8 Rate Limits

Default rate limits (configurable):

- 200 requests per day per IP
- 50 requests per hour per IP
- 10 uploads per hour per IP

Rate limit headers in responses:

```
X-RateLimit-Limit: 50
X-RateLimit-Remaining: 45
X-RateLimit-Reset: 1674480000
```

7.20.9 Usage Examples

Python Client

```
import requests

BASE_URL = "http://localhost:5000/api"

# Upload network
with open("network.edgelist", "rb") as f:
    response = requests.post(
        f"{BASE_URL}/upload",
        files={"file": f},
        data={"input_type": "edgelist", "directed": "false"}
    )

network_id = response.json()["network_id"]
print(f"Uploaded network: {network_id}")

# Run community detection
response = requests.post(
    f"{BASE_URL}/analysis/communities",
    json={
        "network_id": network_id,
        "algorithm": "louvain"
    })
```

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```

    }
)

communities = response.json()
print(f"Found {communities['num_communities']} communities")

```

cURL Examples

```

# Upload network
curl -X POST http://localhost:5000/api/upload \
  -F "file=@network.edgelist" \
  -F "input_type=edgelist" \
  -F "directed=false"

# Get network info
curl http://localhost:5000/api/networks/abc123

# Run analysis
curl -X POST http://localhost:5000/api/analysis/communities \
  -H "Content-Type: application/json" \
  -d '{"network_id":"abc123","algorithm":"louvain"}'

# Export network
curl http://localhost:5000/api/export/network/abc123?format=graphml \
  -o output.graphml

```

JavaScript/Fetch Example

```

const BASE_URL = 'http://localhost:5000/api';

// Upload network
const formData = new FormData();
formData.append('file', fileInput.files[0]);
formData.append('input_type', 'edgelist');
formData.append('directed', 'false');

const response = await fetch(`${BASE_URL}/upload`, {
  method: 'POST',
  body: formData
});

const data = await response.json();
console.log(`Uploaded network: ${data.network_id}`);

// Run analysis
const analysisResponse = await fetch(`${BASE_URL}/analysis/communities`, {
  method: 'POST',
  headers: {
    'Content-Type': 'application/json'
  },
  body: JSON.stringify({

```

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```
    network_id: data.network_id,
    algorithm: 'louvain'
  })
});

const communities = await analysisResponse.json();
console.log(`Found ${communities.num_communities} communities`);
```

7.20.10 Related Documentation

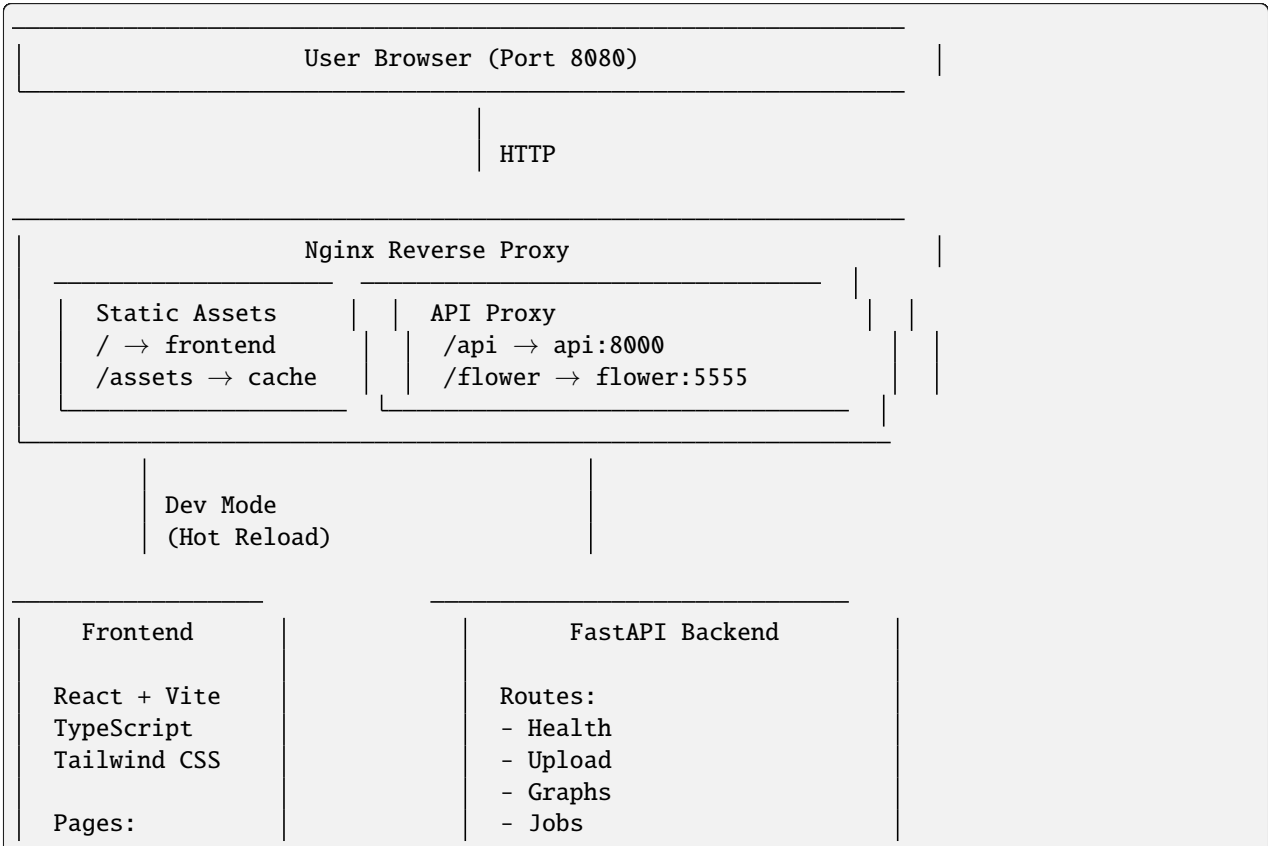
- *Py3plex GUI* - Using the web interface
- *GUI Deployment Guide* - Deploying in production
- *Py3plex GUI Architecture* - Technical architecture
- *GUI Testing Guide* - Testing the API

External Resources:

- REST API Best Practices: <https://restfulapi.net/>
- Flask-RESTful Documentation: <https://flask-restful.readthedocs.io/>

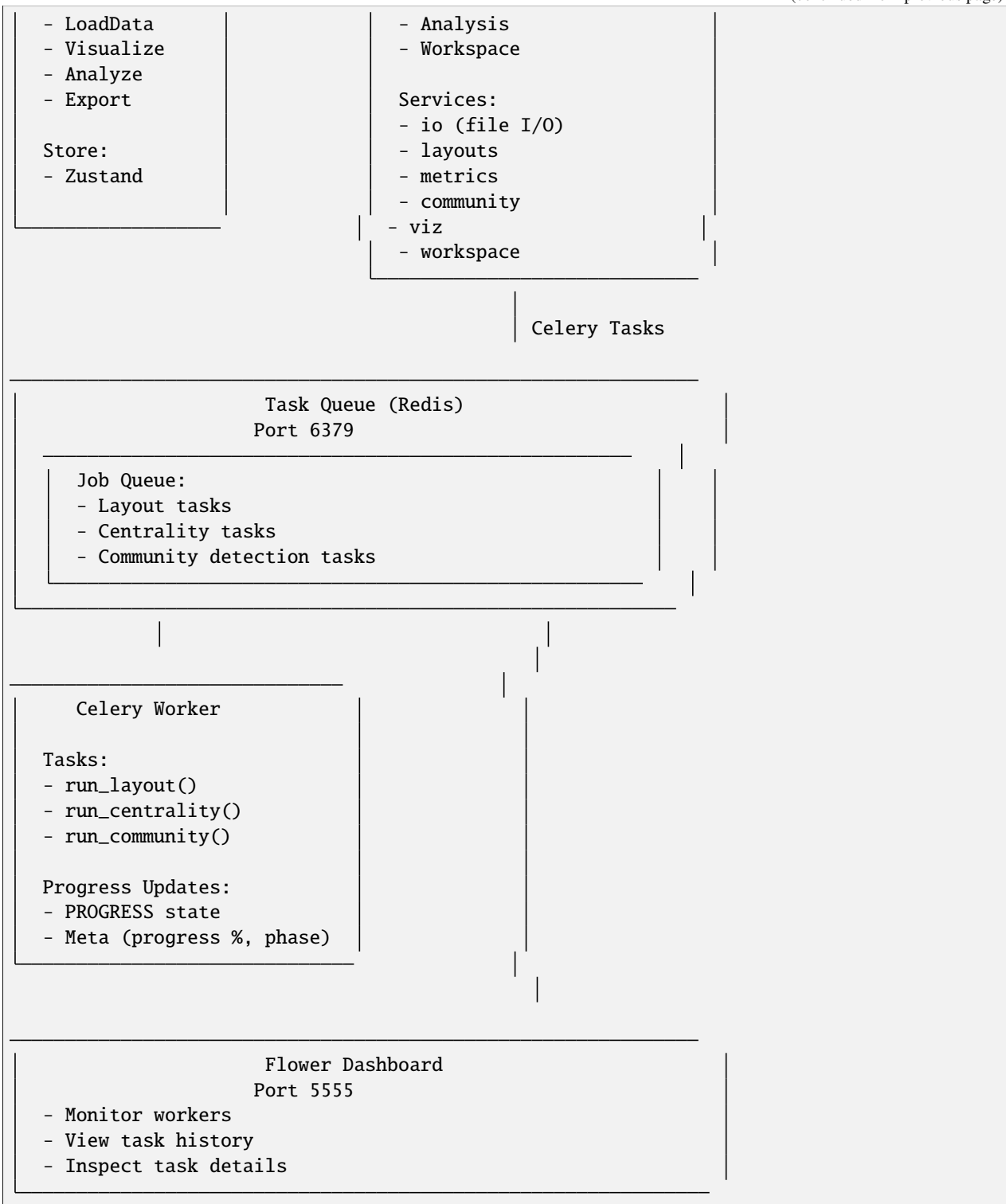
7.21 Py3plex GUI Architecture

7.21.1 System Overview



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7.21.2 Data Flow

Upload & Parse Flow

```

User → Frontend → API /upload → io.save_upload()
                                ↓
                                io.load_graph_from_file()
                                ↓
                                NetworkX graph loaded
                                ↓
                                Stored in GRAPH_REGISTRY
                                ↓
                                Return graph_id to frontend

```

Analysis Job Flow

```

User → Frontend → API /analysis/centrality
                                ↓
                                Create Celery task
                                ↓
                                Return job_id
                                ↓
                                Task queued in Redis
                                ↓
                                Worker picks up task
                                ↓
                                task.update_state(progress=30)
                                ↓
                                services.metrics.compute_centrality()
                                ↓
                                Save results to /data/artifacts/
                                ↓
                                task.update_state(progress=100, result={...})
                                ↓
User ← Frontend ← API /jobs/{job_id} ← Redis result backend

```

7.21.3 Directory Structure

```

gui/
├── docker-compose.yml      # Orchestration
├── compose.gpu.yml         # GPU override
├── Makefile                # Dev commands
├── .env.example            # Config template
├── README                  # User guide
├── TESTING                 # Test guide
├── ARCHITECTURE            # Architecture notes
├──
├── nginx/
│   └── nginx.conf          # Reverse proxy config
├──
├── api/                    # Backend
│   ├── Dockerfile.api
│   └── pyproject.toml

```

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└─ app/	
└─ main.py	# FastAPI app
└─ deps.py	# Dependencies
└─ schemas.py	# Pydantic models
└─ routes/	# Endpoints
└─ health.py	
└─ upload.py	
└─ graphs.py	
└─ jobs.py	
└─ analysis.py	
└─ workspace.py	
└─ services/	# Business logic
└─ io.py	# File I/O
└─ layouts.py	# Layout algorithms
└─ metrics.py	# Centrality metrics
└─ community.py	# Community detection
└─ viz.py	# Visualization data
└─ model.py	# Graph queries
└─ workspace.py	# Save/load
└─ workers/	# Celery
└─ celery_app.py	
└─ tasks.py	
└─ utils/	
└─ logging.py	
└─ worker/	
└─ Dockerfile	# Worker container
└─ frontend/	# UI
└─ Dockerfile.frontend	
└─ package.json	
└─ vite.config.ts	
└─ src/	
└─ main.tsx	# Entry point
└─ App.tsx	# Root component
└─ app.css	# Global styles
└─ lib/	
└─ api.ts	# API client
└─ store.ts	# State management
└─ pages/	
└─ LoadData.tsx	# Upload page
└─ Visualize.tsx	# Viz page
└─ Analyze.tsx	# Analysis page
└─ Export.tsx	# Export page
└─ components/	# Reusable UI
└─ Uploader.tsx	
└─ LayerPanel.tsx	
└─ GraphCanvas.tsx	
└─ JobCenter.tsx	
└─ InspectPanel.tsx	
└─ Toasts.tsx	

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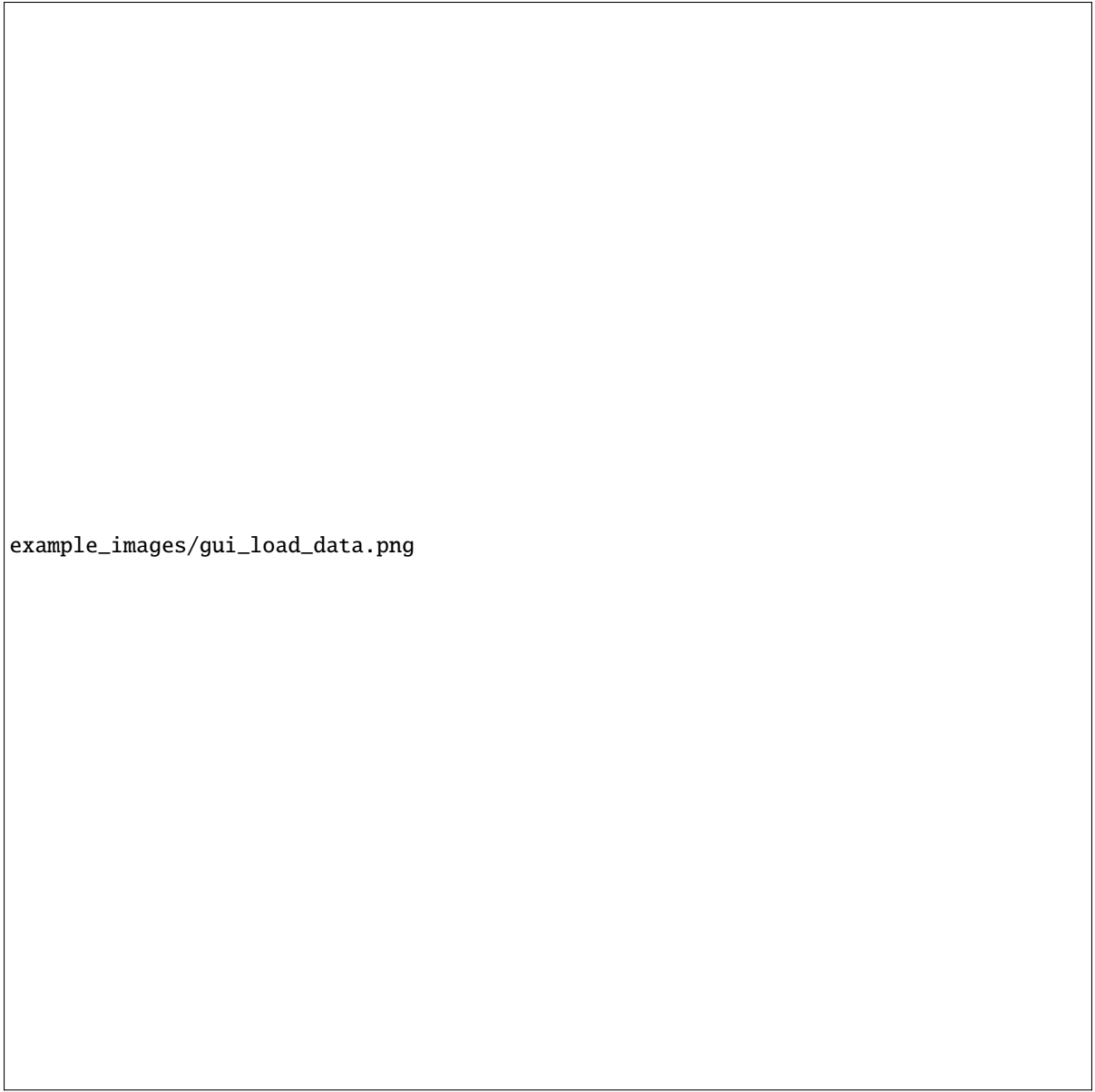
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```
├── ci/                                # Tests
│   ├── api-tests/
│   │   ├── test_health.py
│   │   └── test_upload.py
│   ├── frontend-tests/
│   │   └── smoke.spec.ts
│   └── e2e.playwright.config.ts
├── data/                             # Runtime data (gitignored)
│   ├── uploads/                      # Uploaded files
│   ├── artifacts/                   # Job results
│   └── workspaces/                  # Saved bundles
```

7.21.4 Component Responsibilities

Visual Component Reference

The architecture components are realized in the following user interface pages:



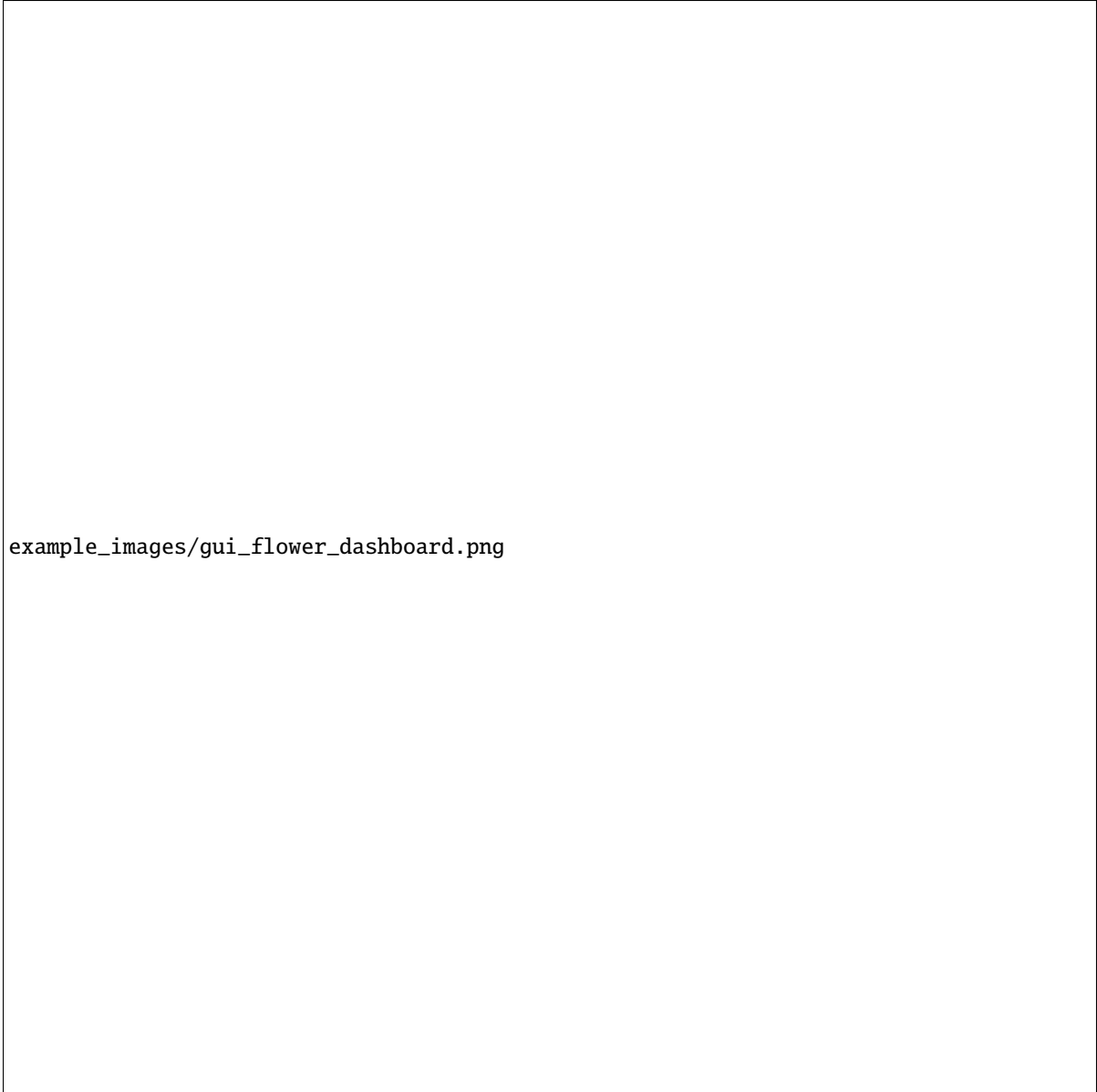
example_images/gui_load_data.png

Figure 1: Frontend - Load Data page utilizing the IO service



example_images/gui_analyze.png

Figure 2: Job orchestration - Analysis page with Celery task execution



example_images/gui_flower_dashboard.png

Figure 3: Worker monitoring - Flower dashboard showing task queue and execution

Frontend

Responsibilities:

- User interaction
- File upload UI
- Real-time job polling
- Graph visualization (placeholder)
- State management (Zustand)

Technologies:

- React 18 (UI framework)
- Vite (build tool, dev server)
- TypeScript (type safety)
- Tailwind CSS (styling)
- Axios (HTTP client)

API (FastAPI)**Responsibilities:**

- REST API endpoints
- Request validation (Pydantic)
- File upload handling
- Job orchestration
- py3plex integration

Key Services:

- `io`: File loading, format detection
- `layouts`: Layout computation (NetworkX)
- `metrics`: Centrality calculations
- `community`: Community detection
- `workspace`: Save/load bundles

Worker (Celery)**Responsibilities:**

- Async job execution
- Progress reporting
- Result persistence
- Resource management

Tasks:

- `run_layout`: Force-directed layouts
- `run_centrality`: Node/edge metrics
- `run_community`: Community detection

Redis**Responsibilities:**

- Job queue (broker)
- Result backend
- Session storage (future)

Nginx

Responsibilities:

- Reverse proxy
- Static file serving
- Gzip compression
- Caching headers
- WebSocket proxy (HMR)

Flower

Responsibilities:

- Worker monitoring
- Task history
- Performance metrics

7.21.5 Data Models

Graph Registry (In-Memory)

```
GRAPH_REGISTRY = {
    "<graph_id>": {
        "graph": nx.Graph(),          # NetworkX graph
        "filepath": "/data/uploads/...", # Original file
        "positions": [NodePosition()], # Layout positions
        "metadata": {}                # Extra info
    }
}
```

Job State (Redis)

```
{
    "job_id": "uuid",
    "status": "running", # queued/running/completed/failed
    "progress": 50,      # 0-100
    "phase": "computing", # human-readable
    "result": {...}      # Output data
}
```

Workspace Bundle (Zip)

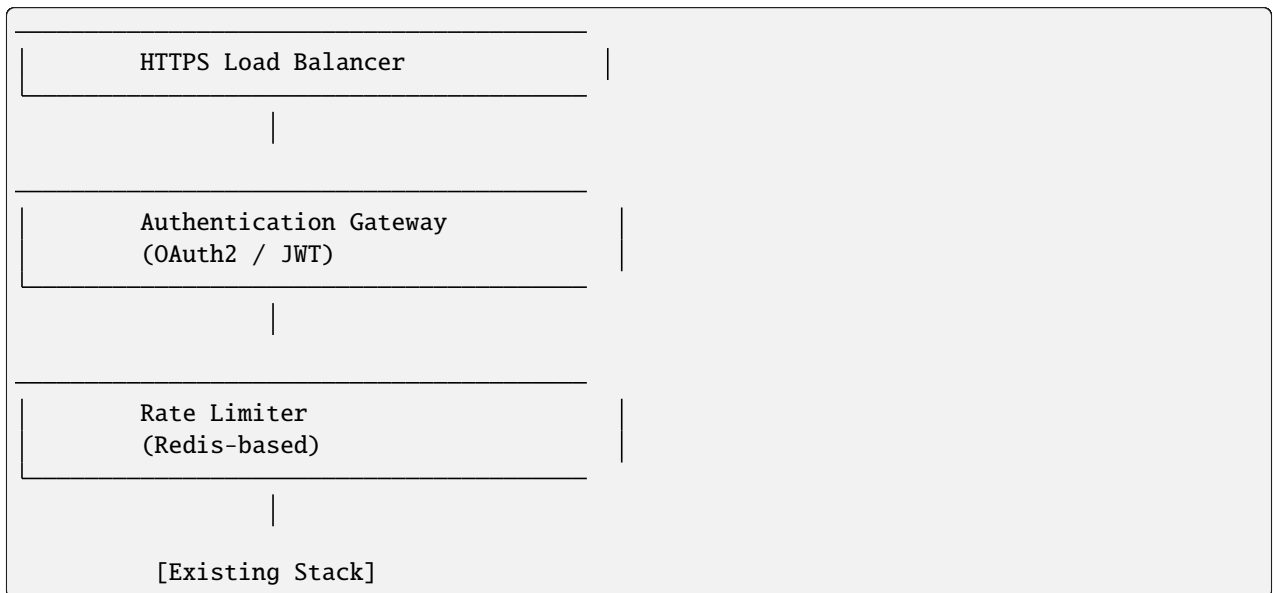
```
workspace_{uuid}.zip
├─ metadata.json      # Graph ID, view state
├─ network.edgelist   # Original file
├─ positions.json     # Layout positions
├─ artifacts/
│   └─ centrality.json
│   └─ community.json
```

7.21.6 Security Architecture

Current (Development Mode)

- ✓ Read-only py3plex mount
- ✓ Isolated data directories
- CORS allows all origins
- No authentication
- No HTTPS
- No rate limiting

Production Hardening (TODO)



7.21.7 Deployment Variants

Local Development (Current)

```
make up # All containers on localhost
```

GPU-Enabled

```
docker compose -f docker-compose.yml -f compose.gpu.yml up
```

Production (Future)

```
# docker-compose.prod.yml
services:
  frontend:
    image: frontend:prod
    # Pre-built static files
```

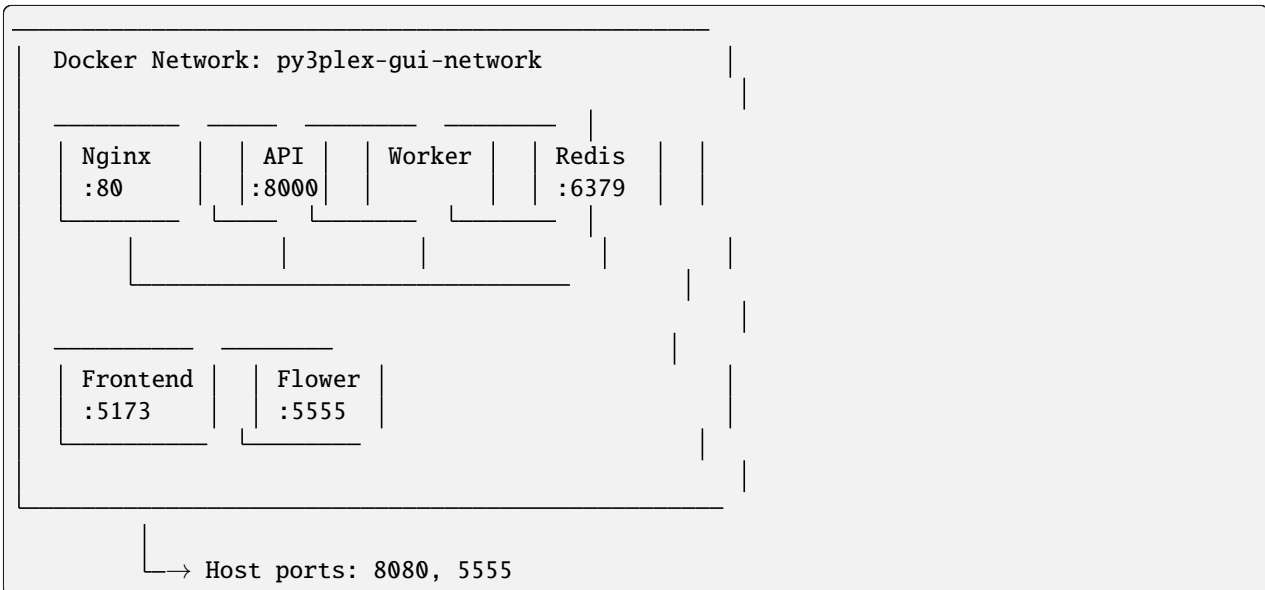
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```
api:
  replicas: 3
  # Load balanced

worker:
  replicas: 5
  # Auto-scaling
```

7.21.8 Network Topology



7.21.9 Volume Mounts

Host	→ Container
../	→ /workspace (ro)
../data/	→ /data
../api/app/	→ /app (dev mode)
../frontend/src/	→ /app/src (dev mode)

7.21.10 Environment Variables

```
# API
API_WORKERS=2           # Uvicorn workers
MAX_UPLOAD_MB=512       # Max file size
DATA_DIR=/data          # Data root

# Celery
CELERY_CONCURRENCY=2     # Worker threads
REDIS_URL=redis://redis:6379/0

# Frontend
VITE_API_URL=http://localhost:8080/api
```

7.21.11 Performance Characteristics

Small Graphs (< 100 nodes)

- Upload: < 1s
- Layout: 2-5s
- Centrality: 1-3s
- Community: 1-3s

Medium Graphs (100-1000 nodes)

- Upload: 1-3s
- Layout: 5-15s
- Centrality: 3-10s
- Community: 3-10s

Large Graphs (> 1000 nodes)

- Consider sampling for preview
- Progressive rendering recommended
- May need GPU acceleration
- Memory: ~1GB per 10k nodes

7.21.12 Future Enhancements

Phase 2

- WebGL visualization
- Real-time collaboration
- Database backend (PostgreSQL)
- Authentication service
- CI/CD pipeline

Phase 3

- GraphQL API
- Plugin system
- Custom algorithms
- Cloud deployment
- Multi-tenancy

Version: 0.1.0

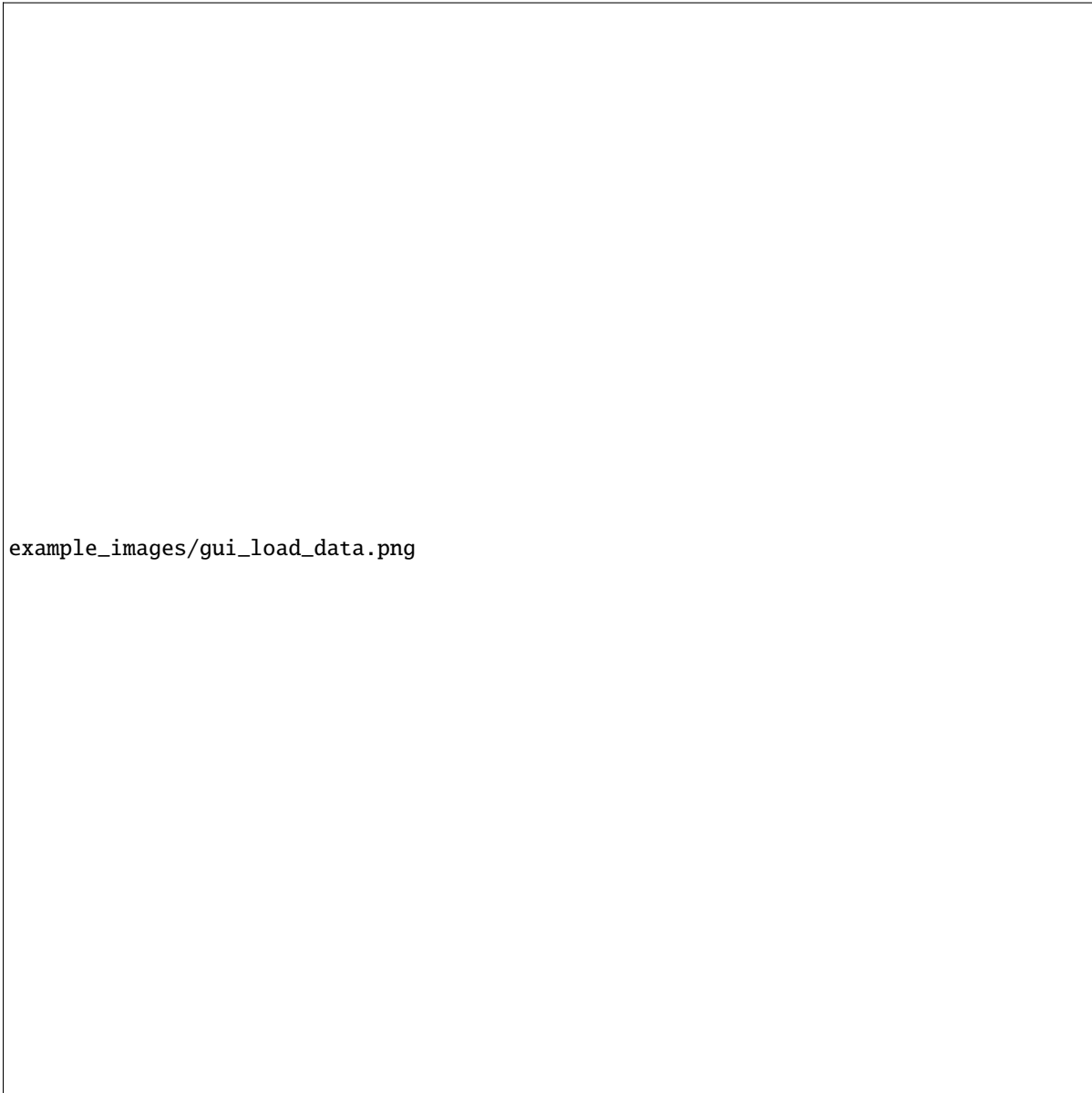
Last Updated: 2025-11-09

7.22 GUI Testing Guide

Complete testing guide for the Py3plex GUI, including automated CI tests and manual validation.

7.22.1 Visual Interface Preview

The Py3plex GUI consists of several key pages that work together for network analysis:



Load Data Page - Upload and parse network files



example_images/gui_visualize.png

Visualize Page - Interactive network view with layer controls



example_images/gui_analyze.png

Analyze Page - Run jobs and monitor progress

7.22.2 Automated Testing (CI)

GitHub Actions Workflow

All GUI tests run automatically on GitHub Actions:

Workflow: `.github/workflows/gui-tests.yml`

Triggers:

- Push to main/master/develop branches (if `gui/` files changed)

- Pull requests targeting main/master/develop (if gui/ files changed)
- Manual workflow dispatch

Test Jobs:

1. API Tests (~5 min)

- Build API, worker, redis containers
- Run pytest suite: ci/api-tests/
- Validate health endpoint
- Test file upload functionality

2. Integration Tests (~10 min)

- Build all services (including nginx, frontend)
- Test full upload → analyze → export flow
- Validate layout job execution
- Test centrality computation
- Verify service health

3. Frontend Build (~5 min)

- Type check TypeScript
- Build production bundle
- Verify dist output

View Results: Check the Actions tab on GitHub or the CI badges in the repository README

Running CI Tests Locally

You can run the same tests locally:

```
cd gui

# API tests
docker compose build api worker redis
docker compose up -d api worker redis
docker compose exec api pytest ci/api-tests/ -v
docker compose down -v

# Integration tests
docker compose up -d --build
# Wait for services
curl -f http://localhost:8080/api/health
# Run manual integration tests (see below)
docker compose down -v

# Frontend build
cd frontend
npm ci
npm run build
```

7.22.3 Manual Testing Guide

Manual testing guide for the Py3plex GUI. Follow these steps to validate the implementation.

Prerequisites

- Docker and Docker Compose installed
- At least 4GB RAM available
- Ports 8080, 5555, 8000, 6379 available

Setup

```
cd gui
cp .env.example .env
make up
```

Expected: All containers start successfully. Check with `docker compose ps`.

Test 1: Health Check

API Health

```
curl http://localhost:8080/api/health
```

Expected Response:

```
{"status": "ok", "version": "0.1.0"}
```

Frontend Access

Open browser to `http://localhost:8080`

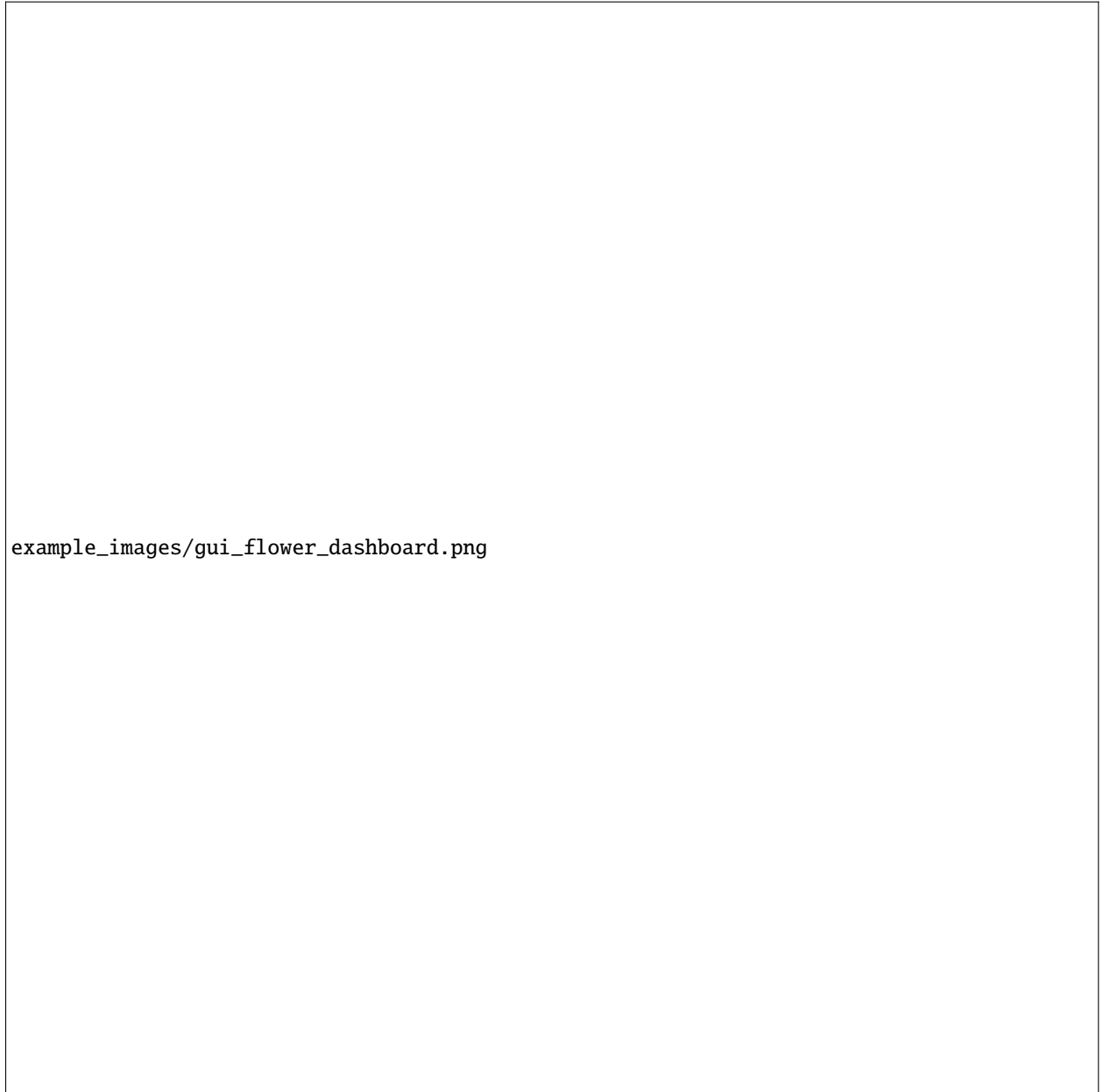
Expected: React app loads with navigation bar showing:

- Load Data
- Visualize
- Analyze
- Export

Flower Dashboard

Open browser to `http://localhost:5555`

Expected: Celery Flower dashboard loads showing workers.



example_images/gui_flower_dashboard.png

Flower monitoring dashboard showing task history and worker status

Test 2: Upload Network

1. Navigate to **Load Data** page
2. Click “Click to upload” or drag and drop `toy_network.edgelist`
3. Click “Upload & Parse”

Expected:

- Upload progress indicator appears

- Network Summary displays:
 - Graph ID (UUID)
 - Filename: `toy_network.edgelist`
 - Nodes: 6
 - Edges: 14
 - Layers: hobby, social, work

Test 3: Visualize Network

1. Click “Visualize Network →” or navigate to **Visualize** page

Expected:

- Page shows three panels: Layers, Network View, Inspect
- Network View shows placeholder with “Graph with X nodes”
- No errors in console

Test 4: Run Layout Job

1. Navigate to **Analyze** page
2. Click “Run Spring Layout” button

Expected:

- Job appears in Job Center below
- Status shows “queued” → “running” → “completed”
- Progress updates (10% → 30% → 80% → 100%)
- Job shows green checkmark when complete

Verify in Flower

1. Open `http://localhost:5555`
2. Check “Tasks” tab

Expected: Layout task appears with status SUCCESS

Test 5: Run Centrality Analysis

1. On **Analyze** page, click “Run Centrality” button

Expected:

- New job appears in Job Center
- Type: “Centrality”
- Status progresses to “completed”
- No errors

Test 6: Run Community Detection

1. On **Analyze** page, click “Detect Communities” button

Expected:

- New job appears in Job Center
- Type: “Community Detection”
- Status progresses to “completed”
- Result includes number of communities found

Test 7: Export Workspace

1. Navigate to **Export** page
2. Enter workspace name: “test-workspace”
3. Click “Save Workspace Bundle”

Expected:

- Success message: “Workspace saved as test-workspace_{uuid}.zip”
- Workspace ID displayed

Verify File Created

```
ls -lh gui/data/workspaces/
```

Expected: Zip file exists with recent timestamp

Test 8: API Direct Testing

Upload via API

```
curl -F "file=@toy_network.edgelist" \
http://localhost:8080/api/upload
```

Expected: JSON response with graph_id

Get Graph Summary

```
GRAPH_ID=<from_previous_step>
curl http://localhost:8080/api/graphs/$GRAPH_ID/summary
```

Expected: JSON with nodes, edges, layers

Start Layout Job

```
curl -X POST \
-H "Content-Type: application/json" \
-d '{"algorithm":"spring","seed":42,"dimensions":2}' \
http://localhost:8080/api/graphs/$GRAPH_ID/layout
```

Expected: JSON response with job_id

Check Job Status

```
JOB_ID=<from_previous_step>
curl http://localhost:8080/api/jobs/$JOB_ID
```

Expected: JSON with status, progress, result

Test 9: Logs Inspection

```
# View all logs
make logs

# Or individual services
docker compose logs api
docker compose logs worker
docker compose logs frontend
```

Expected: No error messages, only INFO level logs

Test 10: Container Health

```
docker compose ps
```

Expected: All services show “healthy” or “running”

Test 11: Data Persistence

1. Upload a network
2. Run a layout job
3. Stop containers: `make down`
4. Restart: `make up`
5. Upload same network again

Expected:

- Uploads work after restart
- Previous artifacts still in `gui/data/`

Test 12: Multiple Jobs

1. Navigate to **Analyze** page
2. Quickly click:
 - Run Spring Layout
 - Run Centrality
 - Detect Communities

Expected:

- All three jobs appear in Job Center
- Jobs process in parallel (check Flower)
- All complete successfully

7.22.4 Common Issues

Port Already in Use

```
# Find and kill process
lsof -ti:8080 | xargs kill -9

# Or change port in docker-compose.yml
ports:
  - "8081:80"
```

Container Won't Start

```
# Check logs
docker compose logs <service>

# Rebuild
make down && make build && make up
```

Permission Errors

```
# Fix data directory permissions
sudo chown -R $(whoami):$(whoami) gui/data/
```

Frontend Can't Reach API

Check nginx config:

```
docker compose exec nginx cat /etc/nginx/nginx.conf
```

Test API directly:

```
curl http://localhost:8000/api/health
```

7.22.5 Cleanup

```
# Stop and remove everything
make down

# Full cleanup including data
make clean
```

7.22.6 Test Checklist

- [] Health endpoints respond
- [] Frontend loads
- [] Flower dashboard accessible
- [] File upload works
- [] Graph summary displays
- [] Layout job completes

- ☐ Centrality job completes
- ☐ Community detection works
- ☐ Workspace save succeeds
- ☐ Job progress updates in real-time
- ☐ Multiple concurrent jobs work
- ☐ Logs show no errors
- ☐ Data persists across restarts

7.22.7 Performance Benchmarks

For reference, on a typical development machine:

- **Startup time:** 30-60 seconds (first build: 3-5 minutes)
- **Upload (toy network):** < 1 second
- **Layout job:** 2-5 seconds
- **Centrality job:** 1-3 seconds
- **Community detection:** 1-3 seconds

Larger networks (1000+ nodes) may take proportionally longer.

7.22.8 Security Testing (Development Mode)

WARNING: Do NOT expose to public internet without:

- ☐ Changing CORS settings
- ☐ Adding authentication
- ☐ Enabling HTTPS
- ☐ Setting rate limits
- ☐ Validating file uploads strictly

7.22.9 Next Steps

If all tests pass:

1. Try with real network datasets
2. Test with large graphs (1000+ nodes)
3. Load test with concurrent users
4. Profile memory usage
5. Test edge cases (malformed files, very large files)

—

Last Updated: 2025-11-09

Version: 0.1.0

7.23 Development Guide

This guide covers **development workflows**, **Makefile commands**, **testing**, and **contributing** to py3plex.

7.23.1 Getting Started

Clone the repository and install in **development mode**:

```
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Setup development environment
make setup

# Install package in editable mode with dev dependencies
make dev-install
```

7.23.2 Makefile Commands

The project includes a **production-grade Makefile** for streamlined development:

```
# View all available commands
make help

# Environment setup
make setup          # Create virtual environment
make dev-install    # Install in editable mode

# Code quality
make format         # Auto-format code
make lint           # Run linters

# Testing
make test           # Run tests with coverage
make coverage       # Open coverage report

# Documentation
make docs           # Build Sphinx documentation

# Build & publish
make clean          # Remove build artifacts
make build           # Build distributions
make publish        # Publish to PyPI

# CI
make ci             # Run full CI checks
```

Key Features:

- **Smart tool detection:** Uses `.venv/bin/` tools if available, otherwise falls back to **global tools**
- **Colorized output:** Clear **success/warning/error** messages
- **Cross-platform:** Works on **Linux** and **macOS**

7.23.3 Testing

Quick testing:

```
python run_tests.py
```

Development testing with pytest:

```
# Run all tests
pytest

# Run with coverage
pytest --cov=py3plex --cov-report=html

# Run specific test file
pytest tests/test_core.py

# Using Makefile (recommended)
make test
```

7.23.4 Code Quality

Tools configured in `pyproject.toml`:

- **Black**: Code formatting
- **Ruff**: Fast linting
- **isort**: Import sorting
- **Mypy**: Type checking
- **Pytest**: Testing with coverage

Before committing:

```
make format # Auto-format code
make lint   # Check code quality
make test   # Run tests
make ci     # Run all checks
```

7.23.5 Contributing

We welcome contributions! Here's how:

1. Fork the repository
2. Create a feature branch
3. Make your changes
4. Run tests and linting: `make ci`
5. Commit with clear messages
6. Push to your fork
7. Open a Pull Request

Code Guidelines:

- Follow existing code style (enforced by Black and Ruff)

- Add tests for new features
- Update documentation as needed
- Keep changes focused and atomic
- Write clear commit messages

7.23.6 Documentation

Build Sphinx documentation:

```
# Using Makefile (recommended)
make docs

# Manual build
cd docfiles
sphinx-build -b html . _build/html
```

The built documentation will be in `docfiles/_build/html/`.

7.23.7 Resources

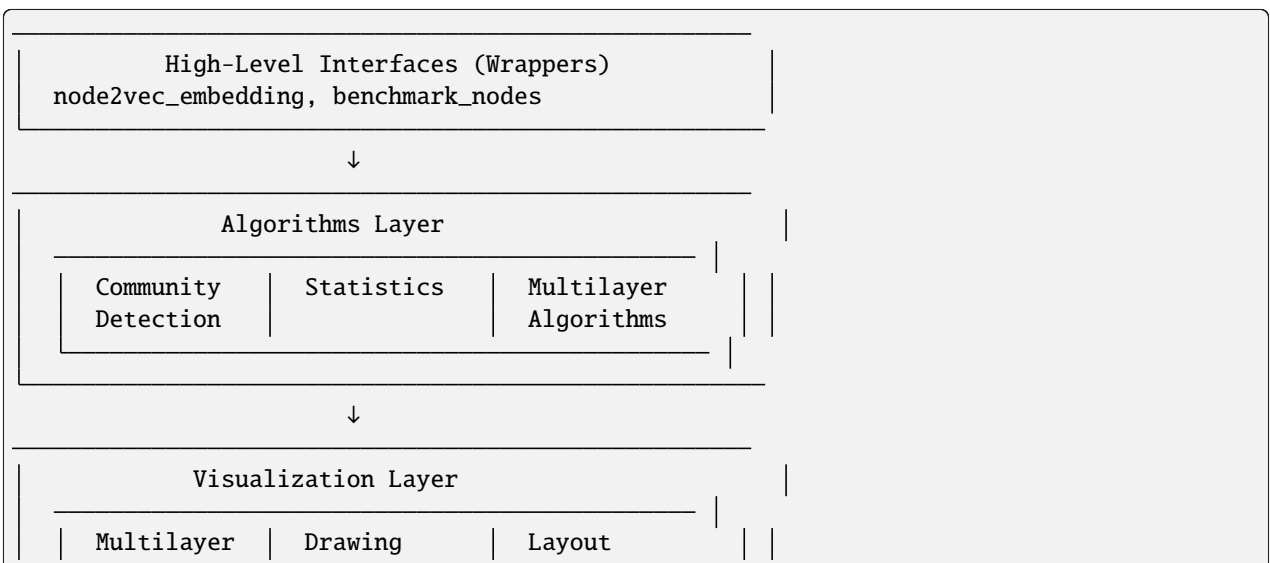
- **Repository:** <https://github.com/SkBlaz/py3plex>
- **Documentation:** <https://skblaz.github.io/py3plex/>
- **Issues:** <https://github.com/SkBlaz/py3plex/issues>
- **Examples:** <https://github.com/SkBlaz/Py3Plex/tree/master/examples>

7.24 Architecture and Design

This document describes the **system architecture**, **design patterns**, and **extension points** of py3plex.

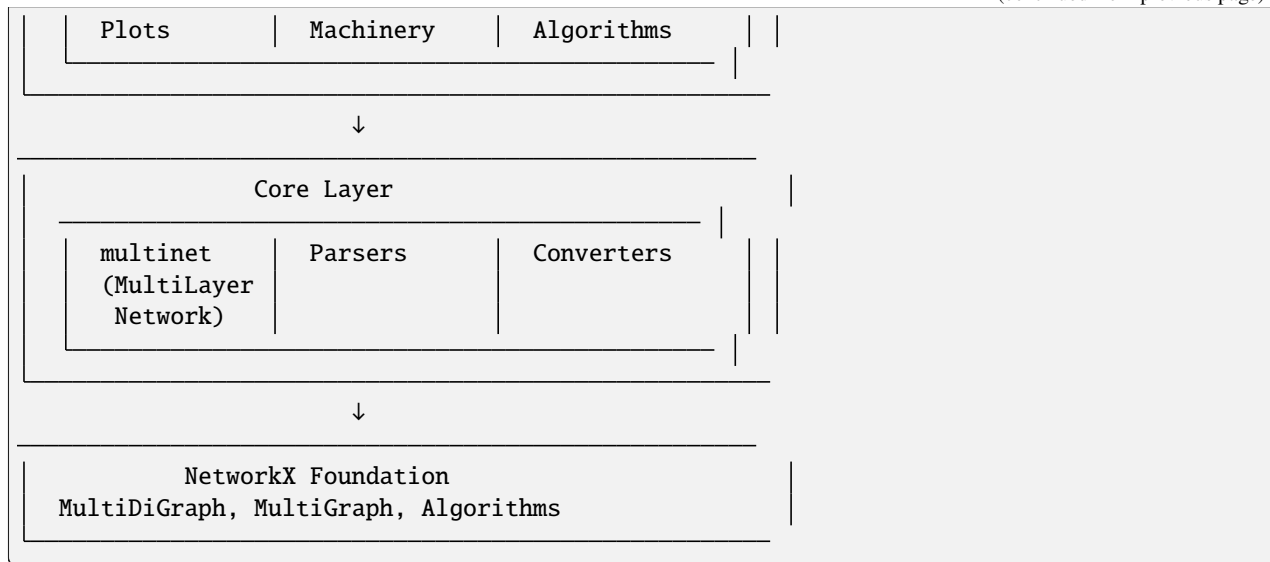
7.24.1 System Overview

py3plex is built as a **modular, layered architecture** with clear separation of concerns:



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7.24.2 Architectural Layers

Core Layer

Purpose: Fundamental data structures and I/O operations

Key Components:

- `multinet.py` - The `multi_layer_network` class
- `parsers.py` - **Input/output** for various formats
- `converters.py` - **Format conversion** utilities
- `random_generators.py` - **Random network** generators
- `HINMINE/` - **Heterogeneous network decomposition**

Responsibilities:

- **Network construction** and manipulation
- **File I/O** (GraphML, GML, GEXF, edge lists, etc.)
- **Layer management**
- **Matrix representations** (adjacency, supra-adjacency)
- **NetworkX integration**

Design Pattern: Facade Pattern - `multi_layer_network` provides a unified interface to complex NetworkX operations

Algorithms Layer

Purpose: Network analysis algorithms optimized for multilayer networks

Key Components:

- `community_detection/` - **Community detection** algorithms
- `statistics/` - **Network statistics** and metrics

- `multilayer_algorithms/` - **Multilayer-specific** algorithms
- `node_ranking/` - **Centrality** and ranking measures
- `general/` - **General-purpose** algorithms (random walks, etc.)

Responsibilities:

- **Community detection** (Louvain, Infomap, Label Propagation)
- **Statistical analysis** (17+ multilayer metrics)
- **Centrality computation** (degree, betweenness, PageRank, etc.)
- **Random walks** and embeddings
- **Network decomposition**

Design Pattern: Strategy Pattern - Different algorithms implement common interfaces

Visualization Layer

Purpose: Network plotting and rendering

Key Components:

- `multilayer.py` - High-level multilayer plotting
- `drawing_machinery.py` - Core drawing primitives
- `layout_algorithms.py` - Layout computation
- `colors.py` - Color scheme generators
- `fa2/` - ForceAtlas2 layout

Responsibilities:

- Diagonal projection plots
- Force-directed layouts
- Matrix visualizations
- Color mapping and legends
- Interactive plots (via Plotly)

Design Pattern: Template Method Pattern - Layout algorithms follow a common template

Wrappers Layer

Purpose: High-level interfaces for common workflows

Key Components:

- `node2vec_embedding.py` - Node2Vec embedding generation
- `benchmark_nodes.py` - Node classification benchmarking

Responsibilities:

- Simplified interfaces for complex workflows
- Integration with external tools
- Benchmarking and evaluation

Design Pattern: Facade Pattern - Simplify complex multi-step operations

7.24.3 Core Data Structure

The multi_layer_network Class

Central to py3plex, this class manages multilayer network state:

```
class multi_layer_network:
    def __init__(self, directed=True, label_delimiter="---", coupling_weight=1.0):
        self.core_network = nx.MultiDiGraph() if directed else nx.MultiGraph()
        self.layer_name_map = {} # Bidirectional mapping
        self.label_delimiter = label_delimiter
        self.coupling_weight = coupling_weight
        self.embedding = None
        self.labels = None
```

Key Attributes:

- core_network - Underlying NetworkX graph
- layer_name_map - Maps layer names to integer IDs
- label_delimiter - Separator for node-layer encoding (default: “---”)
- coupling_weight - Default weight for inter-layer edges
- embedding - Cached node embedding matrix
- labels - Node classification labels

Encoding Scheme:

Nodes are encoded as "{node_id}{delimiter}{layer_id}"

Example: Node ‘A’ in layer ‘social’ becomes "A---social"

This allows NetworkX to handle multilayer structure transparently.

7.24.4 Design Patterns

Facade Pattern

Used in: multi_layer_network, wrappers

Purpose: Provide simplified interface to complex subsystems

```
# Complex underlying operations hidden behind simple interface
network = multinet.multi_layer_network()
network.add_edges(edges, input_type='list') # Handles parsing, encoding, validation
network.basic_stats() # Aggregates multiple NetworkX calls
```

Strategy Pattern

Used in: Algorithms, layout computation

Purpose: Interchangeable algorithms following common interface

```
# Different community detection strategies
def detect_communities(network, method='louvain'):
    strategies = {
        'louvain': community_louvain.best_partition,
        'infomap': community_wrapper.infomap_communities,
```

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```
'label_prop': label_propagation.propagate
}
return strategies[method](network.core_network)
```

Template Method Pattern

Used in: Visualization, layout algorithms

Purpose: Define algorithm skeleton, allow customization in subclasses

```
class LayoutAlgorithm:
    def compute(self, graph):
        self.initialize(graph)
        self.iterate()
        return self.finalize()

    def initialize(self, graph):
        raise NotImplementedError

    def iterate(self):
        raise NotImplementedError

    def finalize(self):
        raise NotImplementedError
```

Dependency Injection

Used in: Configuration, algorithm parameters

Purpose: Inject dependencies rather than hard-coding

```
# Configuration injected rather than hard-coded
from py3plex.config import DEFAULT_COLORS, LAYOUT_PARAMS

def draw_network(network, colors=None, layout_params=None):
    colors = colors or DEFAULT_COLORS
    layout_params = layout_params or LAYOUT_PARAMS
    # Use injected configuration
```

7.24.5 Data Flow

Typical Workflow

1. **Input:** Load or create network

```
network = multinet.multi_layer_network()
network.load_network("data.graphml", input_type="graphml")
```

2. **Processing:** Apply algorithms

```
communities = community_louvain.best_partition(network.core_network)
centrality = calc.multilayer_degree_centrality(network)
```

3. **Analysis:** Compute statistics

```
density = mls.layer_density(network, 'layer1')
correlation = mls.inter_layer_degree_correlation(network, 'L1', 'L2')
```

4. **Visualization:** Render results

```
draw_multilayer_default([network], display=True)
```

5. **Output:** Export results

```
network.save_network("output.graphml", output_type="graphml")
```

State Management

Immutable Operations: Most algorithms don't modify the network

```
# These don't modify the network
centrality = calc.multilayer_degree_centrality(network)
communities = community_louvain.best_partition(network.core_network)
```

Mutable Operations: Some operations modify network state

```
# These modify the network
network.add_edges(new_edges, input_type='list')
network.aggregate_layers(['L1', 'L2'], 'combined')
```

7.24.6 Extension Points

Custom Algorithms

Add new algorithms by following existing patterns:

```
# py3plex/algorithms/my_module/my_algorithm.py
from py3plex.core.multinet import multi_layer_network

def my_centrality(network: multi_layer_network) -> dict:
    """
    Custom centrality measure.

    Parameters
    -----
    network : multi_layer_network
        Input network

    Returns
    -----
    dict
        Node centrality scores
    """
    G = network.core_network
    centrality = {}

    for node in G.nodes():
        # Implement custom logic
```

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```

    centrality[node] = compute_score(node, G)

    return centrality

```

Custom Visualizations

Create custom plots using drawing machinery:

```

from py3plex.visualization import drawing_machinery as dm
import matplotlib.pyplot as plt

def my_custom_plot(network):
    """Custom visualization."""
    fig, ax = plt.subplots(figsize=(10, 8))

    # Compute layout
    pos = dm.compute_layout(network.core_network, 'force')

    # Draw elements
    dm.draw_nodes(ax, network.core_network, pos, node_size=50)
    dm.draw_edges(ax, network.core_network, pos, edge_width=1)
    dm.draw_labels(ax, pos, labels=network.get_node_labels())

    plt.show()

```

Custom Parsers

Add support for new file formats:

```

# py3plex/core/parsers.py
def parse_my_format(input_file, **kwargs):
    """
    Parse custom file format.

    Parameters
    -----
    input_file : str
        Path to input file

    Returns
    -----
    multi_layer_network
        Parsed network
    """
    network = multi_layer_network()

    with open(input_file, 'r') as f:
        for line in f:
            # Parse line and add to network
            pass

    return network

```

7.24.7 Configuration System

Centralized Configuration

py3plex/config.py provides centralized configuration:

```
# Default color palettes (8 options including colorblind-safe)
DEFAULT_COLORS = 'Set1'
COLORBLIND_SAFE = 'colorblind'

# Visualization defaults
DEFAULT_NODE_SIZE = 20
DEFAULT_EDGE_WIDTH = 1.0
DEFAULT_ALPHA = 0.7

# Layout parameters
LAYOUT_PARAMS = {
    'force': {'iterations': 500, 'optimal_distance': 1.0},
    'fa2': {'iterations': 1000, 'gravity': 1.0}
}

# Performance settings
SPARSE_THRESHOLD = 1000 # Use sparse matrices above this node count
MEMORY_WARNING_THRESHOLD = 10000 # Warn for large dense matrices
```

Usage:

```
from py3plex.config import DEFAULT_COLORS, LAYOUT_PARAMS

colors = DEFAULT_COLORS
iterations = LAYOUT_PARAMS['force']['iterations']
```

7.24.8 Testing Architecture

Test Organization

```
tests/
├── test_core_functionality.py      # Core data structure tests
├── test_multilayer_*.py           # Multilayer algorithm tests
├── test_random_walks.py           # Random walk tests
├── test_io_*.py                   # I/O and parsing tests
├── test_config_api.py             # Configuration tests
└── test_utils.py                  # Utility function tests
```

Test Patterns

Unit Tests: Test individual functions in isolation

```
def test_layer_density():
    network = create_test_network()
    density = mls.layer_density(network, 'layer1')
    assert 0 <= density <= 1
```

Integration Tests: Test workflows across modules

```
def test_community_detection_workflow():
    network = load_network("test_data.graphml")
    communities = community_louvain.best_partition(network.core_network)
    assert len(communities) > 0
```

Property-Based Tests: Test invariants

```
def test_centrality_normalization():
    network = create_random_network()
    centrality = calc.multilayer_degree_centrality(network)
    # Centrality values should be normalized
    assert all(0 <= v <= 1 for v in centrality.values())
```

7.24.9 Performance Considerations

Lazy Evaluation

Expensive operations are computed on-demand:

```
class multi_layer_network:
    @property
    def supra_adjacency(self):
        if self._supra_adj_cache is None:
            self._supra_adj_cache = self._compute_supra_adjacency()
        return self._supra_adj_cache
```

Sparse Matrices

Use sparse representations for large networks:

```
def get_supra_adjacency_matrix(self, sparse=True):
    if sparse or len(self.get_nodes()) > SPARSE_THRESHOLD:
        return scipy.sparse.csr_matrix(adj)
    return np.array(adj)
```

Vectorization

Prefer NumPy vectorized operations:

```
# Bad: Python loop
degrees = [sum(1 for _ in G.neighbors(node)) for node in nodes]

# Good: Vectorized
degrees = np.array(list(dict(G.degree()).values()))
```

7.24.10 Logging Infrastructure

Centralized Logging

py3plex/logging_config.py provides structured logging:

```
import logging
from py3plex.logging_config import get_logger
```

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```

logger = get_logger(__name__)

logger.info("Processing network with %d nodes", num_nodes)
logger.warning("Large network detected, using sparse matrices")
logger.error("Invalid layer: %s", layer_name)

```

Log Levels

- **DEBUG:** Detailed diagnostic information
- **INFO:** General informational messages
- **WARNING:** Warning messages (e.g., performance concerns)
- **ERROR:** Error messages
- **CRITICAL:** Critical errors

7.24.11 Error Handling

Custom Exceptions

py3plex/exceptions.py defines domain-specific exceptions:

```

class NetworkError(Exception):
    """Base exception for network errors."""
    pass

class LayerNotFoundError(NetworkError):
    """Raised when layer doesn't exist."""
    pass

class InvalidFormatError(NetworkError):
    """Raised when file format is invalid."""
    pass

```

Usage:

```

from py3plex.exceptions import LayerNotFoundError

def get_layer(self, layer_name):
    if layer_name not in self.layer_name_map:
        raise LayerNotFoundError(f"Layer '{layer_name}' not found")
    return self.layer_name_map[layer_name]

```

7.24.12 Future Architecture

Planned Improvements

1. **Backend Registry:** Support for igraph, cugraph backends
2. **Streaming API:** Process networks larger than memory
3. **Distributed Computing:** Dask/Ray integration for large-scale analysis
4. **Plugin System:** Easy addition of third-party algorithms

5. **Type System:** Full type hints coverage (currently 65%)

7.24.13 See Also

- [Contributing to py3plex](#) - Contributing guidelines
- development - Development workflow
- core - Core API documentation

7.25 Repository Layout

This document explains the structure of the py3plex repository and what each directory contains.

7.25.1 Overview

```
py3plex/
├── py3plex/           # Main package source code
├── docs/              # Built documentation (HTML)
├── docfiles/          # Documentation source (RST files)
├── examples/          # Example scripts
├── tests/             # Test suite
├── gui/               # Web interface
├── datasets/          # Sample datasets
├── benchmarks/        # Performance benchmarks
├── fuzzing/           # Fuzz testing
├── .github/           # GitHub Actions CI/CD
├── pyproject.toml      # Package configuration
├── Makefile           # Development commands
└── README.md          # Project README
```

7.25.2 Main Package (py3plex/)

Source code for the py3plex library:

```
py3plex/
├── __init__.py        # Package initialization
├── core/              # Core data structures
│   ├── multinet.py    # multi_layer_network class
│   ├── parsers.py     # File I/O
│   └── converters.py  # Format conversion
├── algorithms/        # Network algorithms
│   ├── community_detection/
│   ├── statistics/
│   ├── multilayer_algorithms/
│   └── general/
├── visualization/     # Plotting and rendering
│   ├── multilayer.py
│   ├── drawing_machinery.py
│   └── layout_algorithms.py
├── wrappers/          # High-level interfaces
└── io/                # Modern I/O system
```

Key modules:

- `core/multinet.py` - Main `multi_layer_network` class
- `algorithms/` - All analysis algorithms
- `visualization/` - All visualization code
- `io/` - Modern schema-based I/O

7.25.3 Documentation (docfiles/)

Sphinx documentation source files:

```
docfiles/
├── index.rst           # Main documentation index
├── conf.py             # Sphinx configuration
├── Makefile            # Build commands
├── getting_started/    # Beginner guides
├── concepts/           # Conceptual explanations
├── user_guide/         # How-to guides
├── deployment/        # Deployment guides
├── gui/                # GUI documentation
├── dev/               # Developer docs
├── examples/           # Example index
└── reference/         # API reference
```

Building documentation:

```
cd docfiles
make html # Output in _build/html/
```

7.25.4 Examples (examples/)

Executable example scripts demonstrating various features:

```
examples/
├── basic/              # Basic usage
│   ├── example_load_network.py
│   ├── example_basic_stats.py
│   └── example_simple_viz.py
├── visualization/     # Visualization examples
│   ├── example_multilayer_viz.py
│   ├── example_custom_layouts.py
│   └── example_interactive_plots.py
├── analysis/          # Analysis examples
│   ├── example_community_detection.py
│   ├── example_centrality.py
│   └── example_statistics.py
├── advanced/          # Advanced topics
│   ├── example_node2vec.py
│   ├── example_random_walks.py
│   └── example_embeddings.py
└── io/                # I/O examples
    ├── example_arrow_io.py
    └── example_format_conversion.py
```

Running examples:


```
python examples/basic/example_load_network.py
```

7.25.5 Tests (tests/)

Test suite using pytest:

```
tests/
├── test_core.py           # Core functionality
├── test_algorithms.py     # Algorithm tests
├── test_visualization.py  # Visualization tests
├── test_io.py             # I/O tests
├── test_performance_core.py # Performance benchmarks
├── test_fuzzing_properties.py # Property-based tests
└── conftest.py           # Pytest configuration
```

Running tests:

```
make test      # Run all tests
make test-fast # Skip slow tests
make coverage  # With coverage report
```

7.25.6 GUI (gui/)

Web interface for py3plex:

```
gui/
├── app.py           # Flask application
├── routes/          # API endpoints
├── templates/       # HTML templates
├── static/          # CSS, JS, images
├── config.py        # Configuration
└── Dockerfile       # Docker container
```

Running GUI:

```
python gui/app.py
```

7.25.7 Datasets (datasets/)

Sample networks for testing and examples:

```
datasets/
├── test.edgelist      # Simple test network
├── multiedgelist.txt  # Multilayer test
├── karate_club/       # Karate club network
└── bio_networks/     # Biological networks
```

7.25.8 Benchmarks (benchmarks/)

Performance benchmarking suite:

```
benchmarks/
├── bench_core.py      # Core operations
├── bench_algorithms.py # Algorithm performance
├── bench_io.py        # I/O performance
└── bench_aggregation.py # Aggregation benchmarks
```

Running benchmarks:

```
make benchmark
```

7.25.9 CI/CD (.github/)

GitHub Actions workflows:

```
.github/
├── workflows/
│   ├── tests.yml      # Test CI
│   ├── code-quality.yml # Linting/formatting
│   ├── docs.yml       # Documentation build
│   ├── release.yml    # Release automation
└── dependabot.yml     # Dependency updates
```

7.25.10 Configuration Files

Root-level configuration:

- `pyproject.toml` - Package metadata, dependencies, build config
- `Makefile` - Development commands (make test, make docs, etc.)
- `README.md` - Project overview and quick start
- `.gitignore` - Git ignore rules
- `LICENSE` - MIT license
- `CITATION.cff` - Citation information
- `MANIFEST.in` - Package manifest for distribution

7.25.11 Working with the Repository

Initial Setup

```
# Clone repository
git clone https://github.com/SkBlaz/py3plex.git
cd py3plex

# Create virtual environment
python3 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# Install in development mode
pip install -e ".[dev]"
```

Development Workflow

```
# 1. Make changes to code

# 2. Format code
make format

# 3. Run linters
make lint

# 4. Run tests
make test

# 5. Build documentation
make docs

# 6. Commit changes
git add .
git commit -m "Description"
git push
```

7.25.12 Finding Things

“Where is...?”

Core network class?

py3plex/core/multinet.py → multi_layer_network class

Community detection algorithms?

py3plex/algorithms/community_detection/

Visualization code?

py3plex/visualization/

I/O functions?

py3plex/io/ (modern) or py3plex/core/parsers.py (legacy)

Tests for feature X?

tests/test_*.py - grep for the feature name

Example using feature X?

examples/ - check the relevant subdirectory

Documentation source?

docfiles/ - RST files organized by topic

Built documentation?

docs/ - HTML files (generated from docfiles/)

7.25.13 File Naming Conventions

- **Source files:** snake_case.py
- **Test files:** test_<module>.py
- **Example files:** example_<feature>.py
- **Documentation:** <topic>.rst
- **Configuration:** Various (pyproject.toml, conf.py, etc.)

7.25.14 Important Files

For Contributors

- `CONTRIBUTING.md` - Contribution guidelines (if exists)
- `docfiles/dev/contributing.rst` - Contribution documentation
- `Makefile` - Available development commands
- `pyproject.toml` - Package dependencies

For Users

- `README.md` - Quick start guide
- `docfiles/index.rst` - Documentation entry point
- `examples/` - Working code examples
- `LICENSE` - MIT license

For CI/CD

- `.github/workflows/*.yml` - CI pipelines
- `Makefile` - Automated commands
- `pyproject.toml` - Build configuration

7.25.15 Directory Conventions

Source code directories:

- No spaces in names
- Lowercase with underscores
- Descriptive names (`community_detection`, not `cd`)

Test directories:

- Mirror source structure
- One test file per source file when possible

Documentation directories:

- Organize by user journey/topic
- Clear, descriptive names

7.25.16 Package Distribution

When building packages, these files are included:

Included:

- All `.py` files in `py3plex/`
- `LICENSE`
- `README.md`
- Required configuration files

Excluded (via `.gitignore`):

- `__pycache__/`
- `.pytest_cache/`
- `*.pyc`
- `build/, dist/, *.egg-info/`
- Virtual environments
- IDE-specific files

See `MANIFEST.in` for complete inclusion/exclusion rules.

7.25.17 Related Documentation

- *Development Guide* - Development setup and workflow
- *Contributing to py3plex* - How to contribute
- *Architecture and Design* - Internal architecture
- *Installation and Setup* - Installation guide

External Resources:

- Repository: <https://github.com/SkBlaz/py3plex>
- Issues: <https://github.com/SkBlaz/py3plex/issues>
- Pull Requests: <https://github.com/SkBlaz/py3plex/pulls>

7.26 Contributing to py3plex

We welcome contributions to py3plex. This guide explains how to contribute effectively.

7.26.1 Ways to Contribute

You can contribute in many ways:

- Report bugs - Open issues for bugs you encounter
- Suggest features - Propose new features or improvements
- Write documentation - Improve or expand documentation
- Fix bugs - Submit pull requests fixing issues
- Add features - Implement new algorithms or capabilities
- Write tests - Improve test coverage
- Review PRs - Help review pull requests from others

7.26.2 Getting Started

Fork and Clone

1. Fork the repository on GitHub
2. Clone your fork:

```
git clone https://github.com/YOUR_USERNAME/py3plex.git
cd py3plex
```

3. Add upstream remote:

```
git remote add upstream https://github.com/SkBlaz/py3plex.git
```

Development Setup

Install in development mode with all dependencies:

```
# Create virtual environment
python3 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# Install in editable mode with dev dependencies
pip install -e "[dev]"
```

This installs:

- Core dependencies
- Testing tools (pytest, coverage)
- Linting tools (black, ruff, isort, mypy)
- Documentation tools (sphinx)

7.26.3 Development Workflow

Create a Branch

Always create a new branch for your work:

```
git checkout -b feature/my-new-feature
# or
git checkout -b fix/issue-123
```

Make Changes

1. Write your code following our coding standards (see below)
2. Add or update tests
3. Update documentation
4. Run linters and tests

Run Tests

```
# Run all tests
make test

# Or directly with pytest
python run_tests.py
```

Run Linters

```
# Run all linters
make lint

# Or individual tools
make format # Auto-format code (black, isort)
ruff check .
mypy py3plex
```

Commit Changes

Write clear, descriptive commit messages:

```
git add .
git commit -m "Add feature X to support Y

- Implement algorithm for Z
- Add tests for edge cases
- Update documentation"
```

Push and Create PR

```
git push origin feature/my-new-feature
```

Then create a Pull Request on GitHub with:

- Clear title describing the change
- Description of what changed and why
- Reference to related issues (e.g., “Fixes #123”)
- Screenshots for UI changes

7.26.4 Coding Standards

Code Style

We follow PEP 8 with these specifics:

- Line length: 100 characters (not 80)
- Indentation: 4 spaces (no tabs)
- String quotes: Single quotes preferred (‘text’ not “text”)
- Imports: Grouped and sorted (using isort)

Auto-format your code:

```
make format
```

This runs:

- `isort` - Sort and organize imports
- `black` - Format code consistently
- `ruff --fix` - Auto-fix linting issues

Naming Conventions

- Functions/variables: snake_case
- Classes: PascalCase
- Constants: UPPER_SNAKE_CASE
- Private members: `_leading_underscore`

```
# Good
def compute_centrality(network, node_id):
    MAX_ITERATIONS = 100
    centrality_scores = {}
    return centrality_scores

class MultilayerNetwork:
    def __init__(self):
        self._internal_state = {}
```

Docstrings

Use NumPy-style docstrings for all public functions and classes:

```
def multilayer_degree_centrality(network, layer=None, weighted=False):
    """
    Compute degree centrality for multilayer network.

    Parameters
    -----
    network : multi_layer_network
        The multilayer network to analyze.
    layer : str, optional
        Specific layer to analyze. If None, aggregates across all layers.
    weighted : bool, default=False
        If True, compute strength (weighted degree) instead of degree.

    Returns
    -----
    dict
        Dictionary mapping node names to centrality scores.

    Examples
    -----
    >>> network = multinet.multi_layer_network()
    >>> network.add_edges([[ 'A', 'L1', 'B', 'L1', 1]])
    >>> centrality = multilayer_degree_centrality(network)
    >>> centrality['A---L1']
    1.0

    See Also
    -----
    multilayer_betweenness_centrality : Betweenness centrality

    Notes
```

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```

-----
For directed networks, this computes out-degree by default.

References
-----
.. [1] De Domenico, M., et al. (2013). Mathematical formulation of
      multilayer networks. Physical Review X, 3(4), 041022.
"""
pass

```

Type Hints

Add type hints to improve code clarity:

```

from typing import Dict, List, Optional
from py3plex.core.multinet import multi_layer_network

def compute_statistics(
    network: multi_layer_network,
    layers: Optional[List[str]] = None
) -> Dict[str, float]:
    """Compute network statistics."""
    pass

```

7.26.5 Testing Guidelines

Test Requirements

All new code should include tests:

- Unit tests for individual functions
- Integration tests for workflows
- Edge cases for boundary conditions
- Documentation tests for examples in docstrings

Writing Tests

Use pytest for testing:

```

# tests/test_my_feature.py
import pytest
from py3plex.core import multinet
from py3plex.algorithms.statistics import multilayer_statistics as mls

def test_layer_density():
    """Test layer density calculation."""
    # Setup
    network = multinet.multi_layer_network()
    network.add_edges([
        ['A', 'L1', 'B', 'L1', 1],
        ['B', 'L1', 'C', 'L1', 1],

```

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```
], input_type='list')

# Execute
density = mls.layer_density(network, 'L1')

# Assert
assert 0 <= density <= 1
assert density == pytest.approx(0.333, abs=0.01)

def test_invalid_layer():
    """Test error handling for invalid layer."""
    network = multinet.multi_layer_network()

    with pytest.raises(ValueError):
        mls.layer_density(network, 'nonexistent')
```

Test Coverage

Aim for high test coverage:

```
# Run tests with coverage
pytest --cov=py3plex --cov-report=html

# View coverage report
open htmlcov/index.html # macOS
# or
xdg-open htmlcov/index.html # Linux
```

Target: >80% coverage for new code

7.26.6 Documentation

Update Documentation

When adding features, update:

1. **Docstrings** in the code
2. **RST files** in docfiles/ if adding major features
3. **Examples** in examples/ directory
4. **README** if changing installation or basic usage

Build Documentation

```
cd docfiles
make html

# View documentation
open _build/html/index.html
```

Documentation Style

- Clear and concise - Use simple language
- Code examples - Include working examples
- Cross-references - Link to related documentation
- Visual aids - Add diagrams where helpful

7.26.7 Pull Request Guidelines

Before Submitting

[OK] Code follows style guide (`make lint` passes)

[OK] All tests pass (`make test` passes)

[OK] New code has tests

[OK] Documentation is updated

[OK] Commit messages are clear

[OK] Branch is up to date with main

PR Description

Include in your PR description:

- What changed
- Why the change is needed
- How you implemented it
- Testing done
- Screenshots for visual changes
- Breaking changes if any

Example PR template:

```
## Description

Implements multilayer PageRank centrality following [citation].

## Motivation

Closes #123 - needed for analyzing directed multilayer networks.

## Changes

- Add `multilayer_pagerank()` function
- Add tests with 90% coverage
- Add example in `example_centrality.py`
- Update documentation in `centrality.rst`

## Testing

- [x] Unit tests pass
```

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- [x] Integration test on real network
- [x] Tested on Python 3.8, 3.10, 3.12
- [x] Linting passes

Breaking Changes

None.

Review Process

1. Maintainers will review your PR
2. Address any feedback or requested changes
3. Once approved, your PR will be merged
4. Your contribution will be in the next release!

7.26.8 Reporting Issues

Bug Reports

When reporting bugs, include:

- Description of the bug
- Steps to reproduce
- Expected behavior
- Actual behavior
- Python version (`python --version`)
- py3plex version
- Operating system
- Minimal code example

Example:

Bug Description

``layer_density()`` returns negative values for certain graphs.

Steps to Reproduce

```
```python
network = multinet.multi_layer_network()
network.add_edges([['A', 'L1', 'B', 'L1', -1]])
density = mls.layer_density(network, 'L1')
print(density) # -0.5
```
```

Expected

Density should be between 0 and 1, or raise `ValueError` for negative weights.

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Actual

Returns -0.5

Environment

- Python 3.10.5
- py3plex 0.95a
- Ubuntu 22.04

Feature Requests

For feature requests, describe:

- Use case - What problem does it solve?
- Proposed solution - How should it work?
- Alternatives - Other approaches considered
- Additional context - Examples, papers, etc.

7.26.9 Code of Conduct

- Be respectful and inclusive
- Welcome newcomers
- Focus on what is best for the community
- Show empathy towards other community members

Contributors are expected to follow these principles to foster a welcoming and productive community.

7.26.10 License

By contributing, you agree that your contributions will be licensed under the MIT License.

7.26.11 Recognition

Contributors are recognized in:

- GitHub contributors page
- Release notes
- acknowledgements section in the documentation

7.26.12 Getting Help

- Questions: Open a GitHub Discussion
- Chat: Join our community chat (link in README)
- Email: Contact maintainers at blaz.skrlj@ijs.si

7.26.13 Next Steps

- Read development for development workflow details
- See architecture for system architecture
- Browse [existing issues](#)
- Check [good first issues](#)

Thank you for contributing to py3plex!

7.27 Examples

This page provides a curated list of example scripts demonstrating py3plex features. All examples are available in the [examples/ directory](#) of the repository.

7.27.1 Getting Started Examples

Basic Network Creation

- `example_load_network.py` - Loading networks from files
- `example_create_network.py` - Creating networks from scratch
- `example_basic_stats.py` - Computing basic statistics

Quick Tutorials

- `tutorial_10min.py` - Executable version of the 10-minute tutorial
- `example_quick_start.py` - Quick start examples

7.27.2 Visualization Examples

Basic Visualization

- `example_multilayer_visualization.py` - Basic multilayer plots
- `example_simple_viz.py` - Simple visualization examples
- `example_hairball.py` - Hairball plots

Advanced Visualization

- `example_custom_layouts.py` - Custom layout algorithms
- `example_diagonal_plot.py` - Diagonal projection for multilayer networks
- `example_interactive_plots.py` - Interactive visualizations with Plotly
- `example_layer_visualization.py` - Layer-by-layer visualization

Community Visualization

- `example_community_viz.py` - Visualizing detected communities
- `example_colored_networks.py` - Custom node/edge coloring

7.27.3 Community Detection Examples

Single-Layer Community Detection

- `example_community_detection.py` - Louvain and Infomap
- `example_louvain.py` - Louvain algorithm examples

- `example_label_propagation.py` - Semi-supervised community detection

Multilayer Community Detection

- `example_multilayer_communities.py` - Multilayer Louvain
- `example_modularity.py` - Modularity optimization
- `example_overlapping_communities.py` - Overlapping community detection

7.27.4 Network Statistics Examples

Basic Statistics

- `example_multilayer_statistics.py` - Multilayer-specific statistics
- `example_layer_comparison.py` - Comparing layers
- `example_node_metrics.py` - Node-level metrics

Centrality Measures

- `example_centrality.py` - Degree, betweenness, PageRank
- `example_multilayer_centrality.py` - Multilayer centrality measures
- `example_versatility.py` - Versatility centrality

Network Comparison

- `example_network_comparison.py` - Comparing multiple networks
- `example_statistical_tests.py` - Statistical comparison

7.27.5 I/O and Data Format Examples

Loading Networks

- `example_IO.py` - Various input formats
- `example_edgelist_loading.py` - Edge list formats
- `example_graphml.py` - GraphML format
- `example_csv_loading.py` - CSV with sidecars

Modern I/O (Arrow/Parquet)

- `example_arrow_io.py` - Apache Arrow format
- `example_parquet.py` - Parquet format
- `example_schema_graph.py` - Schema-based I/O

Format Conversion

- `example_format_conversion.py` - Converting between formats
- `example_export.py` - Exporting networks

7.27.6 Random Walks and Embeddings

Random Walks

- `example_random_walks.py` - Basic and Node2Vec walks
- `example_walkers.py` - Different walk strategies

- `example_metapath_walks.py` - Meta-path based walks

Node Embeddings

- `example_n2v_embedding.py` - Node2Vec embeddings
- `example_deepwalk.py` - DeepWalk embeddings
- `example_embeddings.py` - Various embedding methods

Embedding Applications

- `example_link_prediction.py` - Link prediction with embeddings
- `example_node_classification.py` - Node classification
- `example_clustering_embeddings.py` - Clustering with embeddings

7.27.7 Network Decomposition

- `example_network_decomposition.py` - Meta-path feature extraction
- `example_feature_extraction.py` - Network feature engineering
- `example_tensor_decomposition.py` - Tensor decomposition methods

7.27.8 Network Manipulation

- `example_manipulation.py` - Network operations (add, remove, filter)
- `example_subnetworks.py` - Extracting subnetworks
- `example_layer_operations.py` - Layer-specific operations
- `example_aggregation.py` - Aggregating layers

7.27.9 Algorithms and Analysis

Network Dynamics

- `example_spreading.py` - Network traversal and spreading processes
- `example_sir_epidemic.py` - SIR epidemic simulation
- `example_diffusion.py` - Diffusion processes

Specialized Algorithms

- `example_ricci_curvature.py` - Ricci curvature computation
- `example_motifs.py` - Network motif discovery
- `example_path_analysis.py` - Path-based analysis

Machine Learning

- `example_ml_features.py` - Feature extraction for ML
- `example_graph_kernels.py` - Graph kernel methods
- `example_supervised_learning.py` - Supervised learning on networks

7.27.10 NetworkX Integration

- `example_networkx_wrapper.py` - Using NetworkX functions
- `example_networkx_interop.py` - NetworkX interoperability
- `example_convert_networkx.py` - Converting to/from NetworkX

7.27.11 Benchmarking

- `example_benchmarking.py` - Performance benchmarking
- `example_scalability.py` - Scalability testing
- `example_memory_profiling.py` - Memory usage analysis

7.27.12 GUI and API

- `example_api_usage.py` - Using the REST API
- `example_gui_integration.py` - GUI integration examples
- `example_batch_processing.py` - Batch processing with CLI

7.27.13 Real-World Datasets

Biological Networks

- `example_protein_interaction.py` - Protein-protein interaction networks
- `example_gene_regulation.py` - Gene regulatory networks
- `example_metabolic_networks.py` - Metabolic pathways

Social Networks

- `example_social_multiplex.py` - Multiplex social networks
- `example_citation_network.py` - Citation networks
- `example_collaboration.py` - Collaboration networks

Transportation

- `example_multimodal_transport.py` - Multi-modal transportation
- `example_flight_network.py` - Airline networks

7.27.14 Running Examples

All examples can be run directly with Python:

```
# Basic usage
python examples/example_multilayer_visualization.py

# With custom data
python examples/example_load_network.py path/to/your/network.edgelist

# From repository root
cd py3plex
python examples/basic/example_load_network.py
```

Many examples accept command-line arguments:

```
python examples/example_community_detection.py --algorithm louvain --input data.graphml
```

7.27.15 Example Template

Use this template for your own scripts:

```
"""
Example: Your Feature
=====

Description of what this example demonstrates.

Usage:
    python example_your_feature.py
"""

from py3plex.core import multinet
from py3plex.visualization.multilayer import draw_multilayer_default

def main():
    # Load or create network
    network = multinet.multi_layer_network()
    network.add_edges([
        ['A', 'layer1', 'B', 'layer1', 1]
    ], input_type="list")

    # Your analysis
    network.basic_stats()

    # Visualization
    draw_multilayer_default([network], display=True)

if __name__ == "__main__":
    main()
```

7.27.16 Contributing Examples

To contribute an example:

1. Create a well-documented script in `examples/`
2. Use the template above
3. Test that it runs without errors
4. Add it to this index with a brief description
5. Submit a pull request

See [Contributing to py3plex](#) for detailed guidelines.

7.27.17 Related Documentation

- *5-Minute Quickstart* - Quick start guide
- *10-Minute Tutorial: Getting Started with py3plex* - 10-minute tutorial
- *Working with Networks* - Working with networks
- *Visualization Guide* - Visualization guide

Repository:

- Examples directory: <https://github.com/SkBlaz/py3plex/tree/master/examples>
- Submit examples: <https://github.com/SkBlaz/py3plex/pulls>

7.28 Algorithm Roadmap

This page provides a **comprehensive overview** of all algorithms implemented in py3plex, organized by category and showing relationships between different methods.

Purpose: Help you quickly find the right algorithm for your multilayer network analysis task.

7.28.1 Algorithm Categories

The algorithms in py3plex are organized into these main categories:

1. *Community Detection Algorithms* - Finding community structure
2. *Centrality Measures* - Computing node importance
3. *Network Statistics* - Network-level metrics
4. *Node Ranking & Classification* - Ranking and classification methods
5. *Random Walks & Embeddings* - Random walk-based methods
6. *Visualization Algorithms* - Layout and rendering
7. *Benchmark & Utilities* - Testing and evaluation

7.28.2 Community Detection Algorithms

Module: `py3plex.algorithms.community_detection`

These algorithms identify groups of densely connected nodes (communities) in multilayer networks.

Louvain Algorithm

Path: `community_detection.community_louvain`

Functions:

- `best_partition(graph, partition, weight, resolution, randomize, random_state)` - Main entry point
- `generate_dendrogram(graph, partition, weight, resolution, randomize, random_state)` - Hierarchical communities
- `modularity(partition, graph, weight)` - Calculate modularity score
- `partition_at_level(dendrogram, level)` - Extract partition at specific level
- `induced_graph(partition, graph, weight)` - Create quotient graph

Best for: Fast community detection on large single-layer networks

Complexity: $O(n \log n)$

Related algorithms:

- *Multilayer Louvain* (multilayer extension)
- *Leiden Algorithm* (improved Louvain)

Infomap Algorithm

Path: `community_detection.community_wrapper`

Functions:

- `infomap_communities(network_input, binary_path, arguments)` - Main entry point
- `run_infomap(network, binary_path, arguments)` - Execute Infomap binary
- `parse_infomap(outfile)` - Parse Infomap output

Best for: Flow-based community detection, hierarchical structure

Complexity: $O(m \log n)$ where m is edges, n is nodes

Related algorithms:

- *Louvain Algorithm* (faster alternative)

Multilayer Louvain

Path: `community_detection.multilayer_modularity`

Functions:

- `louvain_multilayer(supra_adj, omega, resolution, seed)` - Multilayer Louvain
- `multilayer_modularity(supra_adj, partition, omega)` - Calculate multilayer modularity
- `build_supra_modularity_matrix(network, omega)` - Build modularity matrix

Best for: Community detection across multiple layers with inter-layer coupling

Complexity: $O(n^2 L)$ where L is number of layers

Algorithm details: Implements Mucha et al. (2010) multilayer modularity optimization

Related algorithms:

- *Louvain Algorithm* (single-layer version)
- *Leiden Algorithm* (alternative optimizer)

Leiden Algorithm

Path: `community_detection.leiden_multilayer`

Functions:

- `leiden_multilayer(supra_adj, omega, resolution, n_iterations, seed)` - Main entry point

Data structures:

- `LeidenResult` - Holds partition results with modularity score

Best for: More stable community detection than Louvain

Complexity: $O(n \log n)$ with better quality guarantees

Related algorithms:

- *Louvain Algorithm* (predecessor)
- *Multilayer Louvain* (multilayer extension)

NoRC Algorithm

Path: `community_detection.NoRC` and `community_detection.community_wrapper`

Functions:

- `NoRC_communities(network, alpha)` - Wrapper function
- `NoRC_communities_main(matrix, alpha)` - Core implementation
- `page_rank_kernel(index_row)` - PageRank computation
- `create_tree(centers)` - Build community hierarchy

Best for: Networks with overlapping communities

Related algorithms:

- *Community Ranking* (related approach)

Community Measures

Path: `community_detection.community_measures`

Functions:

- `modularity(network, partition, weight)` - Calculate modularity
- `size_distribution(network_partition)` - Community size distribution
- `number_of_communities(network_partition)` - Count communities

Purpose: Evaluate quality of community partitions

Related algorithms: All community detection algorithms above

Community Ranking

Path: `community_detection.community_ranking`

Functions:

- `page_rank_kernel(index_row)` - PageRank-based ranking
- `create_tree(centers)` - Build ranking tree
- `return_infomap_communities(network)` - Get Infomap communities

Best for: Ranking nodes within detected communities

Related algorithms:

- *Node Ranking* (general ranking)

Multilayer Benchmarks

Path: `community_detection.multilayer_benchmark`

Functions:

- `generate_multilayer_lfr(n_nodes, n_layers, avg_degree, ...)` - LFR benchmark

- `generate_coupled_er_multilayer(n_nodes, n_layers, p_intra, p_inter)` - ER benchmark
- `generate_sbm_multilayer(sizes, p_matrix, n_layers, ...)` - SBM benchmark

Purpose: Generate synthetic multilayer networks with known community structure for testing

Related algorithms: All community detection algorithms

7.28.3 Centrality Measures

Module: `py3plex.algorithms.multilayer_algorithms.centrality`

These algorithms measure the importance or influence of nodes in multilayer networks.

MultilayerCentrality Class

Path: `multilayer_algorithms.centrality.MultilayerCentrality`

Main class for computing various centrality measures.

Basic Centrality Measures

Methods:

- `overlapping_degree_centrality()` - Sum of degrees across all layers
- `layer_specific_degree_centrality(layer)` - Degree within specific layer
- `multilayer_degree_centrality()` - Supra-adjacency degree
- `average_degree_centrality()` - Average degree across layers

Best for: Quick importance ranking, well-connected node identification

Complexity: $O(n + m)$ where n is nodes, m is edges

Related measures:

- *Versatility Centrality* (cross-layer activity)

Eigenvector-Based Measures

Methods:

- `multiplex_eigenvector_centrality(max_iter, tol)` - Eigenvector centrality
- `pagerank_centrality(alpha, max_iter, tol)` - PageRank
- `katz_centrality(alpha, beta, max_iter, tol)` - Katz centrality
- `communicability_centrality(beta)` - Communicability centrality

Best for: Influence measure, recursive importance, prestige ranking

Complexity: $O(m)$ per iteration, typically converges quickly

Related measures:

- *Hubs and Authorities (HITS)* (directed networks)

Path-Based Measures

Methods:

- `multilayer_betweenness centrality()` - Betweenness across layers
- `multilayer_closeness centrality()` - Closeness across layers

Best for: Bridge identification, broadcasting efficiency

Complexity: $O(nm)$ for unweighted, $O(nm + n^2 \log n)$ for weighted

Warning: Expensive for large networks (>1000 nodes)

Related measures:

- *Utility Functions* (batch computation)

Hubs and Authorities (HITS)

Path: `node_ranking.node_ranking.hubs_and_authorities`

Best for: Ranking nodes in directed networks (authority = important destination, hub = important source)

Related measures:

- PageRank in *Centrality Measures*

Utility Functions

Functions:

- `compute_all_centralities(network, include_path_based, include_advanced)` - Compute multiple measures

Purpose: Batch computation of multiple centrality measures

Versatility Centrality

Path: `algorithms.statistics.multilayer_statistics.versatility_centrality`

Function:

- `versatility_centrality(network, centrality_type)` - Cross-layer versatility measure

Best for: Measuring how uniformly a node's importance is distributed across layers

Complexity: $O(m \times L)$ where L is number of layers

Related measures:

- All centrality measures in *Centrality Measures*

Supra Matrix Function Centralities

Path: `multilayer_algorithms.supra_matrix_function_centrality`

Functions:

- `communicability_centrality(supra_matrix, beta)` - Matrix exponential based
- `katz_centrality(supra_matrix, alpha, beta, max_iter, tol)` - Resolvent based

Best for: Advanced centrality using matrix functions on supra-adjacency

Complexity: Depends on matrix operations, typically $O(n^2)$ or $O(n^3)$

Related measures:

- *Centrality Measures* (standard measures)

MultiXRank

Path: `multilayer_algorithms.multixrank`

Functions:

- `multixrank_from_py3plex_networks(networks, config)` - MultiXRank algorithm

Best for: Ranking in multiplex heterogeneous networks with bipartite layers

Algorithm details: Extends PageRank to multiplex/heterogeneous networks

Related algorithms:

- PageRank in *Centrality Measures*

7.28.4 Network Statistics

Module: `py3plex.algorithms.statistics`

These algorithms compute various statistical properties of multilayer networks.

Multilayer Statistics

Path: `statistics.multilayer_statistics`

Layer-level measures:

- `layer_density(network, layer)` - Density of specific layer
- `edge_overlap(network, layer_i, layer_j)` - Edge overlap between layers
- `inter_layer_coupling_strength(network, layer_i, layer_j)` - Coupling strength
- `inter_layer_degree_correlation(network, layer_i, layer_j)` - Degree correlation
- `inter_layer_assortativity(network, layer_i, layer_j)` - Assortativity between layers
- `layer_similarity(network, layer_i, layer_j, method)` - Layer similarity

Node-level measures:

- `node_activity(network, node)` - Activity across layers
- `degree_vector(network, node, weighted)` - Degree vector across layers
- `multilayer_clustering_coefficient(network, node)` - Multilayer clustering

Network-level measures:

- `interdependence(network, sample_size)` - Network interdependence
- `supra_laplacian_spectrum(network, k)` - Spectral properties
- `algebraic_connectivity(network)` - Second smallest Laplacian eigenvalue

Best for: Understanding multilayer network structure and properties

Complexity: Varies by measure, typically $O(n)$ to $O(n^2)$

Related algorithms:

- *Multilayer Statistics (Detailed)* (simpler measures)
- *Topology Analysis* (topological properties)

Multilayer Statistics (Detailed)

See the full section above for complete details on multilayer statistics functions.

Basic Statistics

Path: `statistics.basic_statistics`

Functions:

- `identify_n_hubs(network, n, metric)` - Find top-n hub nodes
- `core_network_statistics(network)` - Comprehensive network statistics

Best for: Quick network overview, hub identification

Complexity: $O(n + m)$ for most measures

Related algorithms:

- *Multilayer Statistics (Detailed)* (advanced measures)

Topology Analysis

Path: `statistics.topology`

Functions for analyzing topological properties of networks.

Purpose: Study structural properties like degree distributions, connected components, etc.

Related algorithms:

- *Multilayer Statistics (Detailed)*

Correlation Networks

Path: `statistics.correlation_networks`

Functions:

- `default_correlation_to_network(correlation_matrix, threshold)` - Convert correlation matrix to network
- `pick_threshold(matrix)` - Automatically select threshold

Best for: Creating networks from correlation/similarity matrices

Related algorithms:

- *Network Statistics* (analysis after construction)

Enrichment Analysis

Path: `statistics.enrichment_modules`

Functions:

- `compute_enrichment(term_dataset, annotation_database, ...)` - General enrichment
- `fet_enrichment_generic(...)` - Fisher's exact test enrichment
- `fet_enrichment_terms(...)` - Term enrichment
- `fet_enrichment_uniprot(...)` - UniProt annotation enrichment
- `calculate_pval(term, alternative)` - p-value calculation
- `multiple_test_correction(input_dataset)` - FDR correction

Best for: Finding over-represented terms/annotations in network communities

Related algorithms:

- *Community Detection Algorithms* (provides communities for enrichment)

Statistical Comparison

Path: `statistics.stats_comparison`

Functions: Various statistical tests for comparing networks or algorithms

Related modules:

- `statistics.bayesian_tests` - Bayesian statistical tests
- `statistics.bayesian_distances` - Bayesian distance measures
- `statistics.critical_distances` - Critical difference diagrams

Best for: Comparing performance of different algorithms or network properties

Power Law Analysis

Path: `statistics.powerlaw`

Functions for testing and fitting power law distributions in networks.

Best for: Analyzing degree distributions, checking for scale-free properties

Related algorithms:

- *Multilayer Statistics (Detailed)* (degree distributions)

7.28.5 Node Ranking & Classification

Module: `py3plex.algorithms.node_ranking` and `py3plex.algorithms.network_classification`

Node Ranking

Path: `node_ranking.node_ranking`

Functions:

- `sparse_page_rank(graph, alpha, personalization, max_iter, tol)` - Sparse PageRank
- `hubs_and_authorities(graph)` - HITS algorithm
- `hub_matrix(graph)` - Hub matrix computation
- `authority_matrix(graph)` - Authority matrix computation
- `stochastic_normalization(matrix)` - Normalize for random walks
- `stochastic_normalization_hin(matrix)` - Normalize for heterogeneous networks

Best for: Ranking nodes by importance in directed networks

Complexity: $O(m)$ per iteration for PageRank, $O(nm)$ for HITS

Related algorithms:

- PageRank in *Centrality Measures*
- *Label Propagation* (classification)

Label Propagation

Path: `network_classification.label_propagation`

Functions:

- `label_propagation(adjacency, labels, alpha, max_iter, tol, normalization)` - Main algorithm
- `validate_label_propagation(...)` - Validation function
- `label_propagation_normalization(matrix)` - Normalization
- `normalize_initial_matrix_freq(mat)` - Frequency normalization
- `normalize_amplify_freq(mat)` - Amplified normalization
- `normalize_exp(mat)` - Exponential normalization

Best for: Semi-supervised node classification

Complexity: $O(m \times i)$ where i is number of iterations

Related algorithms:

- *Personalized PageRank (PPR)* (personalized version)
- Label propagation in *Community Detection Algorithms*

Personalized PageRank (PPR)

Path: `network_classification.PPR`

Functions:

- `construct_PPR_matrix(graph_matrix, parallel)` - Build PPR matrix
- `construct_PPR_matrix_targets(graph_matrix, targets, parallel)` - PPR for specific targets
- `validate_ppr(...)` - Validation function

Best for: Local network exploration, personalized node ranking

Complexity: $O(n^2)$ for dense computation, can be approximated

Related algorithms:

- *Label Propagation* (similar propagation mechanism)
- PageRank in *Node Ranking*

7.28.6 Random Walks & Embeddings

Module: `py3plex.algorithms.general` and `py3plex.wrappers`

Random Walk Primitives

Path: `general.walkers`

Functions:

- `basic_random_walk(graph, start_node, walk_length)` - Simple random walk
- `node2vec_walk(graph, start_node, walk_length, p, q)` - Biased random walk
- `generate_walks(graph, num_walks, walk_length, strategy, ...)` - Generate multiple walks

- `general_random_walk(G, start_node, iterations, teleportation_prob)` - Random walk with restart
- `layer_specific_random_walk(network, start_node, start_layer, walk_length, ...)` - Multi-layer walk

Best for: Foundation for embedding algorithms, network exploration

Complexity: $O(L)$ per walk where L is walk length

Related algorithms:

- *Node2Vec Embedding* (uses these walks)
- *DeepWalk Embedding* (uses these walks)

Node2Vec Embedding

Path: `wrappers.node2vec_embedding`

Functions:

- `generate_embeddings(graph, dimensions, walk_length, num_walks, p, q, ...)` - Generate embeddings
- `train_node2vec(graph, ...)` - Training wrapper

Best for: Feature learning, link prediction, node classification

Complexity: $O(r \times l \times V)$ where r is walks per node, l is walk length, V is number of vertices

Parameters:

- p : Return parameter (controls backtracking)
- q : In-out parameter (controls exploration vs exploitation)

Related algorithms:

- *DeepWalk Embedding* (special case with $p=1, q=1$)
- *Random Walk Primitives* (underlying walks)

DeepWalk Embedding

Implementation: Node2Vec with $p=1, q=1$

Best for: Simpler alternative to Node2Vec with uniform random walks

Related algorithms:

- *Node2Vec Embedding* (generalization)

Entanglement Analysis

Path: `multilayer_algorithms.entanglement`

Functions:

- `compute_entanglement_analysis(network)` - Full entanglement analysis
- `build_occurrence_matrix(network)` - Build node-layer occurrence matrix
- `compute_blocks(c_matrix)` - Compute block structure
- `compute_entanglement(block_matrix)` - Calculate entanglement metrics

Best for: Analyzing node-layer participation patterns

Purpose: Measure how nodes are entangled across layers

Related algorithms:

- *Versatility Centrality* (similar cross-layer analysis)

7.28.7 Visualization Algorithms

Module: `py3plex.visualization`

Layout Algorithms

Path: `visualization.layout_algorithms`

Available layouts:

- Force-directed layouts (Spring, Fruchterman-Reingold)
- Circular layout
- Spectral layout
- Hierarchical layout
- Diagonal projection (multilayer-specific)

Best for: Positioning nodes for visualization

Related algorithms:

- *Multilayer Visualization* (uses these layouts)

Multilayer Visualization

Path: `visualization.multilayer`

Main functions:

- `draw_multilayer_default(networks, ...)` - Default multilayer drawing
- `hairball_plot(network, layout_algorithm)` - Single-layer projection
- Diagonal projection plotting
- Layer-separated visualization

Best for: Visualizing multilayer network structure

Scalability:

- Diagonal projection: 10,000+ nodes
- Force-directed: <5,000 nodes
- Matrix view: 100,000+ nodes

Related algorithms:

- *Layout Algorithms* (positioning)

Drawing Machinery

Path: `visualization.drawing_machinery`

Low-level drawing functions and utilities.

Purpose: Support functions for rendering networks

Color Schemes

Path: `visualization.colors`

Functions for generating color schemes for communities, layers, node types.

Related algorithms:

- *Community Detection Algorithms* (uses colors)
- *Multilayer Visualization* (uses colors)

Embedding Visualization

Path: `visualization.embedding_visualization`

Functions for visualizing node embeddings in 2D/3D space.

Best for: Visualizing results of embedding algorithms

Related algorithms:

- *Node2Vec Embedding*
- *DeepWalk Embedding*

7.28.8 Benchmark & Utilities

Network Classification Benchmark

Path: `general.benchmark_classification`

Functions:

- `evaluate_oracle_F1(probs, Y_real)` - Evaluate classification performance

Best for: Evaluating node classification algorithms

Related algorithms:

- *Label Propagation*
- *Node2Vec Embedding*

Benchmark Visualizations

Path: `visualization.benchmark_visualizations`

Functions for visualizing algorithm comparison results.

Related algorithms:

- All algorithms (for benchmarking)

7.28.9 Algorithm Selection Guide

This section helps you choose the right algorithm category based on your task.

By Task Type

Community Detection:

→ See *Community Detection Algorithms*

Node Importance:

→ See *Centrality Measures*

Node Classification:

→ See *Label Propagation* or *Node2Vec Embedding*

Network Comparison:

→ See *Network Statistics*

Feature Learning:

→ See *Node2Vec Embedding*

Visualization:

→ See *Visualization Algorithms*

By Network Size

Small (<1,000 nodes):

- All algorithms work well
- Can use expensive path-based measures
- All visualization methods

Medium (1,000-10,000 nodes):

- Use Louvain over Infomap
- Avoid full betweenness centrality
- Use diagonal projection for visualization

Large (10,000-100,000 nodes):

- Degree-based measures only
- Louvain or label propagation
- Matrix visualizations

Very Large (>100,000 nodes):

- Degree, PageRank only
- Label propagation for communities
- Sparse matrix operations essential

By Network Type

Single-layer networks:

- Standard algorithms: Louvain, PageRank, betweenness
- See *Community Detection Algorithms* and *Centrality Measures*

Multiplex/multilayer networks:

- Multilayer-specific: *Multilayer Louvain*, *Versatility Centrality*
- Layer statistics: *Multilayer Statistics (Detailed)*
- Cross-layer measures

Directed networks:

- PageRank, HITS algorithm
- See *Node Ranking*

Weighted networks:

- All algorithms support weights
- Specify weight parameter

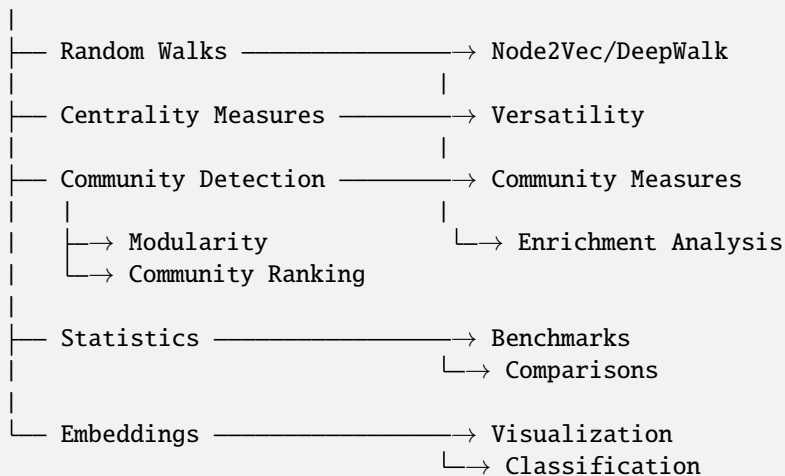
Temporal networks:

- Layer-based representation
- Time-respecting random walks: *Random Walk Primitives*

7.28.10 Algorithm Dependencies

This diagram shows how algorithms depend on each other:

Basic Network Operations



7.28.11 Common Algorithm Workflows

Community Detection Workflow

```
# 1. Detect communities
from py3plex.algorithms.community_detection import louvain_multilayer
communities = louvain_multilayer(network.get_supra_adjacency_matrix())

# 2. Evaluate quality
from py3plex.algorithms.community_detection import multilayer_modularity
quality = multilayer_modularity(network.get_supra_adjacency_matrix(), communities)

# 3. Analyze communities
from py3plex.algorithms.statistics import multilayer_statistics as mls
```

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```

from py3plex.algorithms.statistics.enrichment_modules import compute_enrichment

# 4. Visualize
from py3plex.visualization.multilayer import draw_multilayer_default
draw_multilayer_default([network], communities=communities)

```

Centrality Analysis Workflow

```

# 1. Compute centralities
from py3plex.algorithms.multilayer_algorithms.centrality import MultilayerCentrality
calc = MultilayerCentrality(network)

# 2. Multiple measures
degree = calc.overlapping_degree_centrality()
pagerank = calc.pagerank_centrality()
betweenness = calc.multilayer_betweenness_centrality()

# 3. Cross-layer analysis
from py3plex.algorithms.statistics import multilayer_statistics as mls
versatility = mls.versatility_centrality(network, centrality_type='degree')

# 4. Identify hubs
from py3plex.algorithms.statistics.basic_statistics import identify_n_hubs
hubs = identify_n_hubs(network, n=10, metric='degree')

```

Embedding Workflow

```

# 1. Generate walks
from py3plex.algorithms.general.walkers import generate_walks
walks = generate_walks(network.core_network, num_walks=10,
                       walk_length=80, strategy='node2vec')

# 2. Learn embeddings
from py3plex.wrappers.node2vec_embedding import generate_embeddings
embeddings = generate_embeddings(network.core_network,
                                dimensions=128, p=1.0, q=1.0)

# 3. Use for classification
from py3plex.algorithms.network_classification.label_propagation import label_propagation
# Or use embeddings directly with sklearn

```

Statistical Analysis Workflow

```

# 1. Layer-level statistics
from py3plex.algorithms.statistics import multilayer_statistics as mls
density = mls.layer_density(network, 'layer1')
overlap = mls.edge_overlap(network, 'layer1', 'layer2')
correlation = mls.inter_layer_degree_correlation(network, 'layer1', 'layer2')

# 2. Node-level statistics
for node in important_nodes:

```

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```
activity = mls.node_activity(network, node)
degree_vec = mls.degree_vector(network, node)

# 3. Network-level statistics
interdep = mls.interdependence(network)
connectivity = mls.algebraic_connectivity(network)
```

7.28.12 Further Reading

For detailed usage examples and parameter descriptions:

- *Algorithm Landscape* - Algorithm selection guide with complexity analysis
- *Community Detection Tutorial* - Community detection examples
- *Network Statistics* - Centrality computation examples
- *Performance and Scalability Best Practices* - Performance optimization tips
- *API Documentation* - Complete API reference

7.28.13 Citation Information

When using specific algorithm families, please cite the original papers:

Louvain & Multilayer Modularity:

Blondel et al. (2008), Mucha et al. (2010)

Node2Vec:

Grover & Leskovec (2016)

Infomap:

Rosvall & Bergstrom (2008)

See citation for complete citation information and BibTeX entries.

7.29 API Documentation

This section contains the complete API documentation for py3plex, automatically generated from docstrings.

For auto-generated module documentation, run:

```
cd docfiles
sphinx-apidoc -o AUTOGEN_results -f ../py3plex
make html
```

7.29.1 Core Modules

```
py3plex.core.multinet.ensure(*args, **kwargs)
```

```
py3plex.core.multinet.invariant(*args, **kwargs)
```

```
class py3plex.core.multinet.multi_layer_network(verbose: bool = True, network_type: str =
    'multilayer', directed: bool = True, dummy_layer: str
    = 'null', label_delimiter: str = '---', coupling_weight:
    int | float = 1)
```

Bases: `object`

Main class for multilayer network analysis and manipulation.

This class provides a comprehensive toolkit for creating, analyzing, and visualizing multilayer networks where nodes can exist in multiple layers and edges can connect nodes within or across layers.

Supported Network Types:

- **multilayer**: General multilayer networks with arbitrary layer structure
- **multiplex**: Special case where all layers share the same nodes, with automatic coupling edges between corresponding nodes across layers

Key Features:

- Dict-based API for adding nodes and edges (see `add_nodes()` and `add_edges()`)
- NetworkX interoperability via `to_networkx()` and `from_networkx()`
- Multiple I/O formats (edgelist, GML, GraphML, gpickle, etc.)
- Visualization methods for multilayer layouts
- Community detection and centrality analysis
- Random walk and embedding generation

Hypergraph Support:

This class does NOT natively support true hypergraphs (edges connecting more than two nodes). For hypergraph-like structures, consider: - Using bipartite projections (nodes and hyperedges as separate node types) - The incidence gadget encoding via `to_homogeneous_hypergraph()` - External hypergraph libraries with conversion utilities

Notes

- Nodes in multilayer networks are represented as `(node_id, layer)` tuples
- Use `add_nodes()` and `add_edges()` with dict format for easiest interaction
- See examples/ directory for usage patterns and best practices

Examples

```
>>> # Create a basic multilayer network
>>> net = multi_layer_network(network_type='multilayer', directed=False)
>>>
>>> # Add nodes to different layers
>>> net.add_nodes([
...     {'source': 'A', 'type': 'social'},
...     {'source': 'B', 'type': 'social'},
...     {'source': 'A', 'type': 'email'} # Same node, different layer
... ])
>>>
>>> # Add edges (intra-layer and inter-layer)
>>> net.add_edges([
...     {'source': 'A', 'target': 'B',
...      'source_type': 'social', 'target_type': 'social'},
...     {'source': 'A', 'target': 'A',
...      'source_type': 'social', 'target_type': 'email'}
```

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```
... ])
>>>
>>> print(net) # Shows network statistics
<multi_layer_network: type=multilayer, directed=False, nodes=3, edges=2, layers=2>
```

add_dummy_layers()

Internal function, for conversion between objects

add_edges(*edge_dict_list: List[Dict] | List[List] | Tuple, input_type: str = 'dict'*) → None

Add edges to the multilayer network.

This method supports multiple input formats for specifying edges between nodes in different layers. The most common format is dict-based.

Parameters

- **edge_dict_list** – Edge data in one of the supported formats (see below)
- **input_type** – Format of edge data ('dict', 'list', or 'px_edge')

Supported Formats:

Dict format (recommended): ``python {

```
    'source': 'node1', # Source node ID 'target': 'node2', # Target node ID 'source_type':
    'layer1', # Source layer name 'target_type': 'layer2', # Target layer name (can be same
    as source) 'weight': 1.0, # Optional: edge weight 'type': 'interaction' # Optional: edge
    type/label
```

List format: [node1, layer1, node2, layer2]

px_edge format: ((node1, layer1), (node2, layer2), {'weight': 1.0})

Examples

```
>>> # Add single intra-layer edge
>>> net = multi_layer_network()
>>> net.add_edges([
...     'source': 'A',
...     'target': 'B',
...     'source_type': 'protein',
...     'target_type': 'protein'
... ])
```

```
>>> # Add inter-layer edge with weight
>>> net.add_edges([
...     'source': 'gene1',
...     'target': 'protein1',
...     'source_type': 'genes',
...     'target_type': 'proteins',
...     'weight': 0.95,
...     'type': 'expression'
... ])
```

```
>>> # Add multiple edges at once
>>> edges = [
```

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```

...     {'source': 'A', 'target': 'B', 'source_type': 'layer1', 'target_type':
↪ 'layer1'},
...     {'source': 'B', 'target': 'C', 'source_type': 'layer1', 'target_type':
↪ 'layer1'}
... ]
>>> net.add_edges(edges)

```

Raises

Exception – If `input_type` is not one of ‘dict’, ‘list’, or ‘px_edge’

Notes

- For intra-layer edges, use the same layer for `source_type` and `target_type`
- For inter-layer edges, use different layers
- Edge weights default to 1.0 if not specified

add_nodes(*node_dict_list: List[Dict] | Dict, input_type: str = 'dict')* → None

Add nodes to the multilayer network.

Nodes in a multilayer network are identified by both their ID and the layer they belong to. This method adds nodes using a dict-based format.

Parameters

- **node_dict_list** – Node data as a dict or list of dicts (see format below)
- **input_type** – Format of node data (currently only ‘dict’ is supported)

Dict Format:

```

python {
    'source': 'node_id', # Node identifier (can be string or number) 'type': 'layer_name', # Layer
this node belongs to 'weight': 1.0, # Optional: node weight/importance 'label': 'display' #
Optional: display label # ... any other node attributes

```

Examples

```

>>> # Add single node
>>> net = multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])

```

```

>>> # Add multiple nodes to the same layer
>>> nodes = [
...     {'source': 'A', 'type': 'protein'},
...     {'source': 'B', 'type': 'protein'},
...     {'source': 'C', 'type': 'protein'}
... ]
>>> net.add_nodes(nodes)

```

```

>>> # Add nodes with attributes
>>> net.add_nodes([
...     'source': 'gene1',

```

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```
...     'type': 'genes',
...     'weight': 0.8,
...     'label': 'BRCA1',
...     'chromosome': '17'
... })
```

```
>>> # Add nodes to multiple layers
>>> multi_layer_nodes = [
...     {'source': 'entity1', 'type': 'layer1'},
...     {'source': 'entity1', 'type': 'layer2'}, # Same entity, different layer
...     {'source': 'entity2', 'type': 'layer1'}
... ]
>>> net.add_nodes(multi_layer_nodes)
```

Notes

- The same node ID can exist in multiple layers
- Each (node_id, layer) combination is treated as a unique node
- Additional attributes beyond ‘source’ and ‘type’ are preserved
- Nodes must be added before edges referencing them

aggregate_edges(*metric='count', normalize_by='degree'*)

Edge aggregation method

Count weights across layers and return a weighted network

Parameters

- **param1** – aggregation operator (count is default)
- **param2** – normalization of the values

Returns

A simplified network.

basic_stats(*target_network=None*)

A method for obtaining a network’s statistics.

Displays: - Basic network info (nodes, edges) - Total unique nodes (counting each (node, layer) as unique)
- Unique node IDs (across all layers) - Per-layer node counts

compute_ollivier_ricci(*mode: str = 'core', layers: List[Any] | None = None, alpha: float = 0.5, weight_attr: str = 'weight', curvature_attr: str = 'ricciCurvature', verbose: str = 'ERROR', backend_kwargs: Dict[str, Any] | None = None, inplace: bool = True, interlayer_weight: float = 1.0*) → Dict[str, Any]

Compute Ollivier-Ricci curvature on the multilayer network.

This method provides flexible computation of Ollivier-Ricci curvature at different levels of the multilayer network:

- **core mode**: Compute curvature on the aggregated (flattened) network
- **layers mode**: Compute curvature separately for each layer
- **supra mode**: Compute curvature on the full supra-graph including both intra-layer and inter-layer edges

Parameters

- **mode** – Scope of computation. Options: “core”, “layers”, “supra”.
- **layers** – List of layer identifiers to process (only for mode=“layers”). If None, all layers are processed.
- **alpha** – Ollivier-Ricci parameter in [0, 1] controlling the mass distribution. Default: 0.5.
- **weight_attr** – Name of edge attribute containing weights. Default: “weight”.
- **curvature_attr** – Name of edge attribute to store curvature values. Default: “ricciCurvature”.
- **verbose** – Verbosity level. Options: “INFO”, “DEBUG”, “ERROR”. Default: “ERROR”.
- **backend_kwargs** – Additional keyword arguments for OllivierRicci constructor.
- **inplace** – If True, update internal graphs. If False, return new graphs without modifying the network. Default: True.
- **interlayer_weight** – Weight for inter-layer coupling edges (only for mode=“supra”). Default: 1.0.

Returns

Dictionary mapping scope identifiers to NetworkX graphs with computed curvatures: - mode=“core”: {“core”: graph_with_curvature} - mode=“layers”: {layer_id: graph_with_curvature, ...} - mode=“supra”: {“supra”: supra_graph_with_curvature}

Raises

- **RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.
- **ValueError** – If mode is invalid or layers contains invalid identifiers.

Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>>
>>> # Compute on aggregated network
>>> result = net.compute_ollivier_ricci(mode="core")
>>>
>>> # Compute per layer
>>> result = net.compute_ollivier_ricci(mode="layers")
>>>
>>> # Compute on supra-graph
>>> result = net.compute_ollivier_ricci(mode="supra", inplace=False)
```

compute_ollivier_ricci_flow(mode: str = 'core', layers: List[Any] | None = None, alpha: float = 0.5, iterations: int = 10, method: str = 'OTD', weight_attr: str = 'weight', curvature_attr: str = 'ricciCurvature', verbose: str = 'ERROR', backend_kwargs: Dict[str, Any] | None = None, inplace: bool = True, interlayer_weight: float = 1.0) → Dict[str, Any]

Compute Ollivier-Ricci flow on the multilayer network.

Ricci flow iteratively adjusts edge weights based on their Ricci curvature, effectively revealing and enhancing community structure. After Ricci flow, edges with negative curvature (community boundaries) have reduced weights, while edges with positive curvature have increased weights.

Parameters

- **mode** – Scope of computation. Options: “core”, “layers”, “supra”.
- **layers** – List of layer identifiers to process (only for mode=“layers”). If None, all layers are processed.
- **alpha** – Ollivier-Ricci parameter in [0, 1]. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **method** – Ricci flow method. Options: “OTD” (Optimal Transport Distance, recommended), “ATD” (Average Transport Distance). Default: “OTD”.
- **weight_attr** – Name of edge attribute containing weights. After Ricci flow, these weights are updated to reflect the flow metric.
- **curvature_attr** – Name of edge attribute for curvature values. Default: “ricciCurvature”.
- **verbose** – Verbosity level. Options: “INFO”, “DEBUG”, “ERROR”. Default: “ERROR”.
- **backend_kwargs** – Additional keyword arguments for OllivierRicci constructor.
- **inplace** – If True, update internal graphs. If False, return new graphs. Default: True.
- **interlayer_weight** – Weight for inter-layer coupling edges (only for mode=“supra”). Default: 1.0.

Returns

Dictionary mapping scope identifiers to NetworkX graphs with Ricci flow applied: - mode=“core”: {“core”: graph_with_flow} - mode=“layers”: {layer_id: graph_with_flow, ...} - mode=“supra”: {“supra”: supra_graph_with_flow}

Raises

- **RicciBackendNotAvailable** – If GraphRicciCurvature is not installed.
- **ValueError** – If mode is invalid or layers contains invalid identifiers.

Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>>
>>> # Apply Ricci flow to aggregated network
>>> result = net.compute_ollivier_ricci_flow(mode="core", iterations=20)
>>>
>>> # Apply to each layer
>>> result = net.compute_ollivier_ricci_flow(mode="layers", iterations=10)
```

`edges_from_temporal_table(edge_df)`

Convert a temporal edge DataFrame to edge tuple list.

Extracts edges from a pandas DataFrame with temporal/activity information and converts them to a list of edge tuples suitable for network construction.

Parameters

edge_df – pandas DataFrame with columns: - node_first: Source node identifier - node_second: Target node identifier - layer_name: Layer identifier

Returns

List of edge tuples in format:

(node_first, node_second, layer_first, layer_second, weight) where weight is always 1

Return type

list

Notes

- All values are converted to strings
- All edges are assigned weight=1
- Both source and target are assumed to be in the same layer

Examples

```
>>> import pandas as pd
>>> net = multi_layer_network()
>>> df = pd.DataFrame({
...     'node_first': ['A', 'B'],
...     'node_second': ['B', 'C'],
...     'layer_name': ['L1', 'L1']
... })
>>> result = net.edges_from_temporal_table(df)
>>> len(result) >= 2
True
```

See also

fill_tmp_with_edges: Add these edges to layer graphs

fill_tmp_with_edges(edge_df)

Fill temporary layer graphs with edges from a DataFrame.

Populates the emptied layer graphs (created by remove_layer_edges) with edges from a temporal/activity DataFrame. Useful for temporal network analysis where edge sets change over time.

Parameters

edge_df – pandas DataFrame with columns: - node_first: Source node identifier - node_second: Target node identifier - layer_name: Layer identifier

Notes

- Requires remove_layer_edges() to be called first
- Edges are grouped by layer
- Modifies self.tmp_layers in place

- Each edge is stored as ((node_first, layer), (node_second, layer))

Raises

AttributeError – If self.tmp_layers doesn't exist (call remove_layer_edges first)

Examples

These examples require proper network setup and are for illustration only.

```
>>> import pandas as pd
>>> net = multi_layer_network()
>>> net.split_to_layers()
>>> net.remove_layer_edges()
>>> df = pd.DataFrame({
...     'node_first': ['A', 'B'],
...     'node_second': ['B', 'C'],
...     'layer_name': ['L1', 'L1']
... })
>>> net.fill_tmp_with_edges(df)
```

See also

remove_layer_edges: Creates empty layer graphs edges_from_temporal_table: Convert DataFrame to edge list

from_homogeneous_hypergraph(*H*)

Decode a homogeneous graph created by to_homogeneous_hypergraph.

This method reconstructs a multiplex network from its incidence gadget encoding. It identifies edge-nodes by their degree and cycle structure, then reconstructs the original layers based on cycle lengths (prime numbers).

Parameters

H (*networkx.Graph*) – Homogeneous graph created by to_homogeneous_hypergraph().

Returns

- *dict* – Dictionary mapping layer names to lists of edges: {layer: [(u, v), ...]}
- *Examples* – Example requires proper network setup - for illustration only.

```
>>> network = multi_layer_network()
>>> network.add_layer("A")
>>> network.add_nodes([("1", "A"), ("2", "A")])
>>> network.add_edges([(("1", "A"), ("2", "A"))])
>>> H, node_map, edge_info = network.to_homogeneous_hypergraph()
>>> recovered = network.from_homogeneous_hypergraph(H)
>>> print(recovered)
{'layer_with_prime_2': [('1', '2')]}
```

Notes

The decoded layer names indicate the prime number used for encoding: - “layer_with_prime_2” corresponds to the first layer - “layer_with_prime_3” corresponds to the second layer, etc.

classmethod `from_networkx`(*G*: *Graph*, *network_type*: *str* = 'multilayer', *directed*: *bool* | *None* = *None*)
 → *multi_layer_network*

Create a `multi_layer_network` from a NetworkX graph.

This class method converts a NetworkX graph into a py3plex `multi_layer_network`. For multilayer networks, nodes should be tuples of (node_id, layer).

Parameters

- **G** – NetworkX graph to convert
- **network_type** – Type of network ('multilayer' or 'multiplex')
- **directed** – Whether to treat the network as directed. If None, inferred from G.

Returns

A new `multi_layer_network` instance

Return type

multi_layer_network

Examples

```
>>> import networkx as nx
>>> G = nx.Graph()
>>> G.add_nodes_from([('A', 'layer1'), ('B', 'layer1')])
>>> G.add_edge(('A', 'layer1'), ('B', 'layer1'))
>>> net = multi_layer_network.from_networkx(G)
>>> print(net)
<multi_layer_network: type=multilayer, directed=False, nodes=2, edges=1,
↳ layers=1>
```

Notes

- For proper multilayer behavior, ensure nodes are (node_id, layer) tuples
- Edge attributes are preserved during conversion
- The input graph is copied, not referenced

get_decomposition(*heuristic*='all', *cycle*=None, *parallel*=False, *alpha*=1, *beta*=0)

Core method for obtaining a network's decomposition in terms of relations

get_decomposition_cycles(*cycle*=None)

A supporting method for obtaining decomposition triplets

get_degrees()

A simple wrapper which computes node degrees.

get_edges(*data*: *bool* = False, *multiplex_edges*: *bool* = False) → Any

A method for obtaining a network's edges

Parameters

- **data** – If True, return edge data along with edge tuples
- **multiplex_edges** – If True, include coupling edges in multiplex networks

Yields

Edge tuples, optionally with data

Raises

Exception – If network type is not specified

get_label_matrix()

Return network labels

get_layers(*style='diagonal', compute_layouts='force', layout_parameters=None, verbose=True*)

A method for obtaining layerwise distributions

get_neighbors(*node_id: str, layer_id: str | None = None*) → Any

Get neighbors of a node in a specific layer.

Parameters

- **node_id** – Node identifier
- **layer_id** – Layer identifier (optional)

Returns

Iterator of neighbor nodes

get_nodes(*data: bool = False*) → Any

A method for obtaining a network's nodes

Parameters

data – If True, return node data along with node identifiers

Yields

Node identifiers, optionally with data

get_nx_object()

Return only core network with proper annotations

get_supra_adjacency_matrix(*mtype='sparse'*)

Get sparse representation of the supra matrix.

Parameters

mtype – 'sparse' or 'dense' - matrix representation type

Returns

Supra-adjacency matrix in requested format

Warning

For large multilayer networks, dense matrices can consume significant memory $(N \cdot L)^2 \cdot 8$ bytes for float64.

get_tensor(*sparsity_type='bsr'*)

TODO

get_unique_entity_counts()

Count unique entities in the network.

Returns

(**total_unique_nodes**, **unique_node_ids**, **nodes_per_layer**)

- **total_unique_nodes**: count of unique (node, layer) tuples
- **unique_node_ids**: count of unique node IDs (across all layers)

- `nodes_per_layer`: dict mapping layer to count of nodes in that layer

Return type

tuple

invert(*override_core=False*)

invert the nodes to edges. Get the “edge graph”. Each node is here an edge.

load_embedding(*embedding_file*)

Embedding loading method

load_layer_name_mapping(*mapping_name, header=False*)

Layer-node mapping loader method

Parameters**param1** – The name of the mapping file.**Returns**`self.layer_name_map` is filled, returns nothing.**load_network**(*input_file: str | None = None, directed: bool = False, input_type: str = 'gml', label_delimiter: str = '---'*) → *multi_layer_network*

Main network loader

This method loads and prepares a given network.

Parameters

- **input_file** – Path to the network file to load
- **directed** – Whether the network is directed
- **input_type** – Format of the input file ('gml', 'graphml', 'edgelist', 'gpickle', etc.)
- **label_delimiter** – Delimiter used to separate layer names in node labels

ReturnsSelf for method chaining. Populates `self.core_network`, `self.labels`, and `self.activity`**load_network_activity**(*activity_file*)

Network activity loader

Args:`param1`: The name of the generic activity file -> 65432 61888 1377688175 RE, `n1` `n2` timestamp and layer name. Note that layer node mappings MUST be loaded in order to map nodes to activity properly.**Returns:**`self.activity` is filled.**load_temporal_edge_information**(*input_file=None, input_type='edge_activity', directed=False, layer_mapping=None*)

A method for loading temporal edge information

merge_with(*target_px_object*)

Merge two px objects.

monitor(*message*)

A simple monitor method for logging

monoplex_nx_wrapper(*method*, *kwargs*=None)

A generic networkx function wrapper.

Parameters

- **method** (*str*) – Name of the NetworkX function to call (e.g., ‘degree_centrality’, ‘betweenness_centrality’)
- **kwargs** (*dict*, *optional*) – Keyword arguments to pass to the NetworkX function. For example, for betweenness_centrality you can pass: - weight: Edge attribute to use as weight - normalized: Whether to normalize betweenness values - distance: Edge attribute to use as distance (for closeness_centrality)

Returns

The result of the NetworkX function call.

Raises

AttributeError – If the specified method does not exist in NetworkX.

Example

```
# Unweighted betweenness centrality centralities = network.monoplex_nx_wrapper("betweenness_centrality")
# Weighted betweenness centrality centralities = network.monoplex_nx_wrapper("betweenness_centrality",
kwargs={"weight": "weight"})
# With multiple parameters centralities = network.monoplex_nx_wrapper("betweenness_centrality",
kwargs={"weight": "weight", "normalized": True})
```

read_ground_truth_communities(*cfile*)

Parse ground truth community file and make mappings to the original nodes. This works based on node ID mappings, exact node,layer tuples are to be added. :param param1: ground truth communities.

Returns

self.ground_truth_communities

remove_edges(*edge_dict_list*: List[Dict] | List[List], *input_type*: str = 'list') → None

A method for removing edges..

Parameters

- **edge_dict_list** – Edge data in dict or list format
- **input_type** – Format of edge data ('dict' or 'list')

Raises

Exception – If input_type is not valid

remove_layer_edges()

Remove all edges from separate layer graphs while keeping nodes.

This method creates empty copies of each layer graph with all nodes intact but no edges. Useful for reconstructing networks with different edge sets or for temporal network analysis.

Notes

- Requires split_to_layers() to be called first
- Stores empty layer graphs in self.tmp_layers
- Original graphs in self.separate_layers remain unchanged
- All nodes and their attributes are preserved

Raises

RuntimeError – If `split_to_layers()` hasn't been called yet

See also

`split_to_layers`: Must be called before this method
`fill_tmp_with_edges`: Add edges back to emptied layers

remove_nodes(*node_dict_list*, *input_type*='dict')

Remove nodes from the network

save_network(*output_file*=None, *output_type*='edgelist')

Save the network to a file in various formats.

This method exports the multilayer network to different file formats for persistence, sharing, or use with other tools.

Parameters

- **output_file** – Path where the network should be saved
- **output_type** – Format for saving ('edgelist', 'multiedgelist', 'multiedgelist_encoded', or 'gpickle')

Supported Formats:

- 'edgelist': Simple edge list format (standard NetworkX)
- 'multiedgelist': Multilayer edge list with layer information
- 'multiedgelist_encoded': Multilayer edge list with integer encoding
- 'gpickle': Python pickle format (preserves all attributes)

Examples

```
>>> net = multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> net.add_edges([{'source': 'A', 'target': 'B',
...                 'source_type': 'layer1', 'target_type': 'layer1'}])
>>> net.save_network('network.txt', output_type='multiedgelist')
```

```
>>> # For faster I/O with all metadata preserved
>>> net.save_network('network.gpickle', output_type='gpickle')
```

Notes

- 'gpickle' format preserves all node/edge attributes
- 'multiedgelist_encoded' creates `node_map` and `layer_map` attributes
- Edge weights and types are preserved in supported formats

serialize_to_edgelist(*edgelist_file*='./tmp/tmpedgelist.txt', *tmp_folder*='tmp', *out_folder*='out', *multiplex*=False)

Serialize the multilayer network to an edgelist file.

Converts the network to a numeric edgelist format suitable for external tools and algorithms that require integer node/layer identifiers.

Parameters

- **edgelist_file** – Path to output edgelist file (default: “./tmp/tmpedgelist.txt”)
- **tmp_folder** – Temporary folder for intermediate files (default: “tmp”)
- **out_folder** – Output folder for results (default: “out”)
- **multiplex** – If True, use multiplex format (node layer node layer weight) If False, use simple edgelist format (node1 node2 weight)

Returns

Inverse node mapping (numeric_id -> original_node_tuple)

Use this to decode results from external algorithms

Return type

dict

File Formats:

- Multiplex format: node1_id layer1_id node2_id layer2_id weight
- Simple format: node1_id node2_id weight

Notes

- Creates tmp_folder and out_folder if they don’t exist
- Nodes are mapped to sequential integers starting from 0
- Layers are mapped to sequential integers starting from 0 (multiplex mode)
- All edges have weight 1 unless explicitly specified

Examples

Example requires file output - for illustration only.

```
>>> net = multi_layer_network()
>>> # ... build network ...
>>> node_mapping = net.serialize_to_edgelist(
...     edgelist_file='network.txt',
...     multiplex=True
... )
>>> # Use node_mapping to decode results
>>> original_node = node_mapping[0] # Get original node for id 0
```

See also

load_network: Load networks from file save_network: Alternative serialization method

sparse_to_px(*directed=None*)

Convert sparse matrix to py3plex format

Parameters

directed – Whether the network is directed (uses self.directed if None)

split_to_layers(*style='diagonal', compute_layouts='force', layout_parameters=None, verbose=True, multiplex=False, convert_to_simple=False*)

A method for obtaining layerwise distributions

subnetwork(*input_list=None, subset_by='node_layer_names'*)

Construct a subgraph based on a set of nodes.

summary()

Generate a summary of network statistics.

Computes and returns key metrics about the multilayer network structure.

Returns

Network statistics including:

- 'Number of layers': Count of unique layers
- 'Nodes': Total number of nodes
- 'Edges': Total number of edges
- 'Mean degree': Average node degree
- 'CC': Number of connected components

Return type

dict

Examples

```
>>> net = multi_layer_network()
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> net.add_edges([{'source': 'A', 'target': 'B',
...                 'source_type': 'layer1', 'target_type': 'layer1'}])
>>> stats = net.summary()
>>> print(f"Network has {stats['Nodes']} nodes and {stats['Edges']} edges")
Network has 2 nodes and 1 edges
```

Notes

- Connected components are computed on the undirected version
- Mean degree is averaged across all nodes in all layers

test_scale_free()

Test the scale-free-ness of the network

to_homogeneous_hypergraph()

Transform a multiplex network into a homogeneous graph using incidence gadget encoding.

This method encodes the multiplex structure where each layer is represented by a unique prime number signature. Each edge becomes an edge-node connected to its endpoints and a cycle of length prime-1 that encodes the layer.

Returns

- *tuple* (*H*, *node_mapping*, *edge_info*) –

H

[networkx.Graph] Homogeneous unlabeled graph encoding the multiplex structure.

node_mapping

[dict] Maps each original node to its vertex-node in H.

edge_info

[dict] Mapping from each edge-node in H to its (layer, endpoints) tuple.

- *Examples* – Example requires sympy dependency - for illustration only.
- `>>> network = multi_layer_network(directed=False) # doctest (+SKIP)`
- `>>> network.add_nodes([{'source': '1', 'type': 'A'}, {'source': '2', 'type': 'A'}], input_type='dict') # doctest: +SKIP)`
- `>>> network.add_edges([{'source': '1', 'target': '2', 'source_type': 'A', 'target_type': 'A'}], input_type='dict') # doctest: +SKIP)`
- `>>> H, node_map, edge_info = network.to_homogeneous_hypergraph() # doctest (+SKIP)`
- `>>> print(f"Homogeneous graph has {len(H.nodes())} nodes") # doctest (+SKIP)`

Notes

This transformation uses prime-based signatures to encode layers: - Each layer is assigned a unique prime number (2, 3, 5, 7, ...) - Each edge in layer with prime p is connected to a cycle of length p - The cycle structure uniquely identifies the layer

to_json()

A method for exporting the graph to a json file

Args: self

to_networkx() → Graph

Convert the multilayer network to a NetworkX graph.

Returns a copy of the core network as a NetworkX graph. The returned graph preserves all node and edge attributes, including layer information for multilayer networks (where nodes are typically (node_id, layer) tuples).

Returns

A NetworkX graph (MultiGraph or MultiDiGraph depending on network type)

Return type

nx.Graph

Examples

```
>>> net = multi_layer_network(directed=False)
>>> net.add_nodes([{'source': 'A', 'type': 'layer1'}])
>>> nx_graph = net.to_networkx()
>>> print(type(nx_graph))
<class 'networkx.classes.multigraph.MultiGraph'>
```

Notes

- For multilayer networks, nodes are tuples: (node_id, layer)
- All edge attributes (weight, type, etc.) are preserved
- The returned graph is a copy, not a reference

to_sparse_matrix(*replace_core=False, return_only=False*)

Conver the matrix to scipy-sparse version. This is useful for classification.

visualize_matrix(*kwargs=None*)

Plot the matrix – this plots the supra-adjacency matrix

visualize_network(*style='diagonal', parameters_layers=None, parameters_multiedges=None, show=False, compute_layouts='force', layouts_parameters=None, verbose=True, orientation='upper', resolution=0.01, axis=None, fig=None, no_labels=False, linewidth=1.7, alphachannel=0.3, linepoints='-', legend=False*)

Visualize the multilayer network.

Supports multiple visualization styles: - ‘diagonal’: Layer-centric diagonal layout with inter-layer edges - ‘hairball’: Aggregate hairball plot of all layers - ‘flow’ or ‘alluvial’: Layered flow visualization with horizontal bands - ‘sankey’: Sankey diagram showing inter-layer flow strength

Parameters

- **style** – Visualization style (‘diagonal’, ‘hairball’, ‘flow’, ‘alluvial’, or ‘sankey’)
- **parameters_layers** – Custom parameters for layer drawing
- **parameters_multiedges** – Custom parameters for edge drawing
- **show** – Show plot immediately
- **compute_layouts** – Layout algorithm (currently unused)
- **layouts_parameters** – Layout parameters (currently unused)
- **verbose** – Enable verbose output
- **orientation** – Edge orientation for diagonal style
- **resolution** – Resolution for edge curves
- **axis** – Optional matplotlib axis to draw on
- **fig** – Optional matplotlib figure (currently unused)
- **no_labels** – Hide network labels
- **linewidth** – Width of edge lines
- **alphachannel** – Alpha channel for edge transparency
- **linepoints** – Line style for edges
- **legend** – Show legend (for hairball style)

Returns

Matplotlib axis object

Raises

Exception – If style is not recognized

Performance Notes:

For large networks (>500 nodes), visualization performance may degrade: - Layout computation can be slow ($O(n^2)$ for force-directed layouts) - Rendering many edges is memory and CPU intensive - Consider filtering or sampling for exploratory visualization - Use simpler layouts or increase layout iteration limits

Approximate rendering times on typical hardware: - 100 nodes: <1 second - 500 nodes: 5-10 seconds - 1000 nodes: 30-60 seconds - 5000+ nodes: Several minutes, may run out of memory

visualize_ricci_core(*alpha: float = 0.5, iterations: int = 10, layout_type: str = 'mds', dim: int = 2, **kwargs*)

Visualize the aggregated core network using Ricci-flow-based layout.

This method is a high-level wrapper for Ricci-flow-based visualization of the core (aggregated) network. It automatically computes Ricci flow if not already done and creates an informative layout that emphasizes geometric structure and communities.

Parameters

- **alpha** – Ollivier-Ricci parameter for flow computation. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm (“mds”, “spring”, “spectral”). Default: “mds”.
- **dim** – Dimensionality of layout (2 or 3). Default: 2.
- ****kwargs** – Additional arguments passed to `visualize_multilayer_ricci_core`.

Returns

Tuple of (figure, axes, positions_dict).

Raises

RicciBackendNotAvailable – If `GraphRicciCurvature` is not installed.

Examples

Example requires `GraphRicciCurvature` library - for illustration only.

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.add_edges([
...     ['A', 'layer1', 'B', 'layer1', 1],
...     ['B', 'layer1', 'C', 'layer1', 1],
... ], input_type="list")
>>> fig, ax, pos = net.visualize_ricci_core()
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

See also

`visualize_ricci_layers`: Per-layer visualization with Ricci flow `visualize_ricci_supra`: Supra-graph visualization with Ricci flow

visualize_ricci_layers(*layers: List[Any] | None = None, alpha: float = 0.5, iterations: int = 10, layout_type: str = 'mds', share_layout: bool = True, **kwargs*)

Visualize individual layers using Ricci-flow-based layouts.

This method creates visualizations of individual layers with layouts derived from Ricci flow. Layers can share a common coordinate system (for easier comparison) or have independent layouts.

Parameters

- **layers** – List of layer identifiers to visualize. If None, uses all layers.
- **alpha** – Ollivier-Ricci parameter. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm. Default: “mds”.
- **share_layout** – If True, use shared coordinates across layers. Default: True.
- ****kwargs** – Additional arguments passed to `visualize_multilayer_ricci_layers`.

Returns

Tuple of (figure, layer_positions_dict).

Raises

RicciBackendNotAvailable – If `GraphRicciCurvature` is not installed.

Examples

```
>>> fig, pos_dict = net.visualize_ricci_layers(
...     arrangement="grid", share_layout=True
... )
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

See also

`visualize_ricci_core`: Core network visualization with Ricci flow `visualize_ricci_supra`: Supra-graph visualization with Ricci flow

visualize_ricci_supra(*alpha*: float = 0.5, *iterations*: int = 10, *layout_type*: str = 'mds', *dim*: int = 2, ****kwargs**)

Visualize the full supra-graph using Ricci-flow-based layout.

This method visualizes the complete multilayer structure including both intra-layer edges (within layers) and inter-layer edges (coupling between layers) using a layout derived from Ricci flow.

Parameters

- **alpha** – Ollivier-Ricci parameter. Default: 0.5.
- **iterations** – Number of Ricci flow iterations. Default: 10.
- **layout_type** – Layout algorithm. Default: “mds”.
- **dim** – Dimensionality (2 or 3). Default: 2.
- ****kwargs** – Additional arguments passed to `visualize_multilayer_ricci_supra`.

Returns

Tuple of (figure, axes, positions_dict).

Raises

RicciBackendNotAvailable – If `GraphRicciCurvature` is not installed.

Examples

```
>>> fig, ax, pos = net.visualize_ricci_supra(dim=3)
>>> import matplotlib.pyplot as plt
>>> plt.show()
```

See also

`visualize_ricci_core`: Core network visualization with Ricci flow `visualize_ricci_layers`: Per-layer visualization with Ricci flow

`py3plex.core.multinet.require(*args, **kwargs)`

`py3plex.core.parsers.ensure(*args, **kwargs)`

`py3plex.core.parsers.load_edge_activity_file(fname: str, layer_mapping: str | None = None) → DataFrame`

`py3plex.core.parsers.load_edge_activity_raw(activity_file: str, layer_mappings: dict) → DataFrame`

Basic parser for loading generic activity files. Here, temporal edges are given as tuples -> this can be easily transformed for example into a pandas dataframe!

Parameters

- **activity_file** – Path to activity file
- **layer_mappings** – Dictionary mapping layer names to IDs

Returns

DataFrame with edge activity data

`py3plex.core.parsers.load_temporal_edge_information(input_network: str, input_type: str, layer_mapping: str | None = None) → DataFrame | None`

`py3plex.core.parsers.parse_detangler_json(file_path: str, directed: bool = False) → Tuple[MultiGraph | MultiDiGraph, None]`

Parser for generic Detangler files :param file_path: Path to Detangler JSON file :param directed: Whether to create directed graph

`py3plex.core.parsers.parse_edgelist_multi_types(input_name: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]`

Parse an edgelist file with multiple edge types.

Reads a text file where each line represents an edge, optionally with weights and edge types. Lines starting with '#' are treated as comments.

File Format:

node1 node2 [weight] [edge_type]

Lines starting with '#' are ignored (comments)

Parameters

- **input_name** – Path to edgelist file
- **directed** – Whether to create a directed graph

Returns

(parsed_graph, None for labels)

Return type

Tuple[Union[nx.MultiGraph, nx.MultiDiGraph], None]

Notes

- All nodes are assigned to a “null” layer
- Default weight is 1 if not specified
- Edge type is optional (4th column)
- Handles both 2-column (node pairs) and 3+ column formats

Examples

```
>>> # File content:
>>> # A B 1.0 friendship
>>> # B C 2.0 collaboration
>>> graph, _ = parse_edgelist_multi_types('edges.txt', directed=False)
```

py3plex.core.parsers.**parse_embedding**(input_name: str) → Tuple[ndarray, ndarray]

Loader for generic embedding as outputed by GenSim

Parameters

input_name – Path to embedding file

Returns

Tuple of (embedding matrix, embedding indices)

py3plex.core.parsers.**parse_gml**(file_name: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]

Parse a gml network.

Parameters

- **file_name** – Path to GML file
- **directed** – Whether to create directed graph

Returns

Tuple of (multigraph, possible labels)

Contracts:

- Precondition: file_name must be a non-empty string
- Postcondition: result graph is not None
- Postcondition: result is a MultiGraph or MultiDiGraph

py3plex.core.parsers.**parse_gpickle**(file_name: str, directed: bool = False, layer_separator: str | None = None) → Tuple[MultiGraph | MultiDiGraph, None]

A parser for generic Gpickle as stored by Py3plex.

Parameters

- **file_name** – Path to gpickle file
- **directed** – Whether to create directed graph

- **layer_separator** – Optional separator for layer parsing

Contracts:

- Precondition: `file_name` must be a non-empty string
- Postcondition: result graph is not `None`
- Postcondition: result is a `MultiGraph` or `MultiDiGraph`

`py3plex.core.parsers.parse_gpickle_biomine(file_name: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]`

Gpickle parser for biomine graphs :param `file_name`: Path to gpickle containing BioMine data :param `directed`: Whether to create directed graph

`py3plex.core.parsers.parse_matrix(file_name: str, directed: bool) → Tuple[Any, Any]`

Parser for matrices.

Parameters

- **file_name** – Path to .mat file
- **directed** – Whether the graph is directed

Returns

Tuple of (network, group) from the .mat file

Contracts:

- Precondition: `file_name` must be a non-empty string
- Postcondition: network must not be `None`

`py3plex.core.parsers.parse_matrix_to_nx(file_name: str, directed: bool) → Graph | DiGraph`

Parser for matrices to NetworkX graph.

Parameters

- **file_name** – Path to .mat file
- **directed** – Whether to create directed graph

Returns

NetworkX Graph or DiGraph

`py3plex.core.parsers.parse_multi_edgelist(input_name: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]`

A generic multiedgelist parser n l n l w :param `input_name`: Path to text file containing multiedges :param `directed`: Whether to create directed graph

`py3plex.core.parsers.parse_multiedge_tuple_list(network: list, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]`

Parse a list of edge tuples into a multilayer network.

Parameters

- **network** – List of edge tuples (node_first, node_second, layer_first, layer_second, weight)
- **directed** – Whether to create directed graph

`py3plex.core.parsers.parse_multiplex_edges(input_name: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None]`

Parse a multiplex edgelist file where each line specifies layer and edge.

File Format:

layer node1 node2 [weight]

Each line: layer_id source_node target_node [optional_weight]

Parameters

- **input_name** – Path to multiplex edgelist file
- **directed** – Whether to create a directed graph

Returns

(parsed_graph, None for labels)

Return type

Tuple[Union[nx.MultiGraph, nx.MultiDiGraph], None]

Notes

- Each edge belongs to a specific layer (first column)
- Nodes are represented as (node_id, layer) tuples
- Default weight is 1 if not specified
- All edges have type='default' attribute
- Automatically couples nodes across layers for multiplex structure

Examples

```
>>> # File content:
>>> # layer1 A B 1.5
>>> # layer2 A B 2.0
>>> # layer1 B C 1.0
>>> graph, _ = parse_multiplex_edges('multiplex.txt', directed=False)
>>> # Creates nodes: (A, layer1), (A, layer2), (B, layer1), etc.
```

`py3plex.core.parsers.parse_multiplex_folder(input_folder: str, directed: bool) → Tuple[MultiGraph | MultiDiGraph, None, DataFrame]`

Parse a folder containing multiplex network files.

Expects a folder with specific file formats for edges, layers, and optional activity.

Expected Files:

- *.edges: Edge information (format: layer_id node1 node2 weight)
- layers.txt: Layer definitions (format: layer_id layer_name)
- activity.txt: Optional temporal activity (format: node1 node2 timestamp layer_name)

Parameters

- **input_folder** – Path to folder containing multiplex network files
- **directed** – Whether to create a directed graph

Returns

- Union[nx.MultiGraph, nx.MultiDiGraph]: Parsed multilayer graph
- None: Placeholder for labels (not used)
- pd.DataFrame: Time series activity data (empty if no activity.txt)

Return type

Tuple containing

Notes

- Uses glob to find files with specific extensions
- Layer mapping is built from layers.txt
- Activity data is optional and returned as pandas DataFrame
- Nodes are represented as (node_id, layer_id) tuples

Examples

```
>>> # Folder structure:
>>> # my_network/
>>> #   network.edges
>>> #   layers.txt
>>> #   activity.txt (optional)
>>> graph, _, activity_df = parse_multiplex_folder('my_network/', directed=False)
```

```
py3plex.core.parsers.parse_network(input_name: str | Any, f_type: str = 'gml', directed: bool = False,
                                   label_delimiter: str | None = None, network_type: str = 'multilayer')
    → Tuple[Any, Any, Any]
```

A wrapper method for available parsers!

Parameters

- **input_name** – Path to network file or network object
- **f_type** – Type of file format to parse
- **directed** – Whether to create directed graph
- **label_delimiter** – Optional delimiter for labels
- **network_type** – Type of network (multilayer or multiplex)

Returns

Tuple of (parsed_network, labels, time_series)

```
py3plex.core.parsers.parse_nx(nx_object: Graph, directed: bool) → Tuple[Graph, None]
```

Core parser for networkx objects.

Parameters

- **nx_object** – A networkx graph
- **directed** – Whether the graph is directed

Returns

Tuple of (graph, None)

Contracts:

- Precondition: nx_object must not be None

- Precondition: nx_object must be a NetworkX graph
- Postcondition: result graph is not None

`py3plex.core.parsers.parse_simple_edgelist(input_name: str, directed: bool) → Tuple[Graph | DiGraph, None]`

Simple edgelist n n w :param input_name: Path to text file :param directed: Whether to create directed graph

`py3plex.core.parsers.parse_spin_edgelist(input_name: str, directed: bool) → Tuple[Graph, None]`

Parse SPIN format edgelist file.

SPIN format includes node pairs with edge tags and optional weights.

File Format:

node1 node2 tag [weight]

Each line: source_node target_node edge_tag [optional_weight]

Parameters

- **input_name** – Path to SPIN edgelist file
- **directed** – Whether to create directed graph (currently creates undirected)

Returns

(parsed_graph, None for labels)

Return type

Tuple[nx.Graph, None]

Notes

- Currently always returns nx.Graph (undirected) regardless of directed parameter
- Edge tag is stored in edge ‘type’ attribute
- Default weight is 1 if not specified (4th column)

Examples

```
>>> # File content:
>>> # A B protein_interaction 0.95
>>> # B C gene_regulation 0.80
>>> graph, _ = parse_spin_edgelist('spin_edges.txt', directed=False)
```

`py3plex.core.parsers.require(*args, **kwargs)`

`py3plex.core.parsers.save_edgelist(input_network: Graph, output_file: str, attributes: bool = False) → None`

Save network to edgelist format.

For multilayer networks (where nodes are tuples of (node_id, layer)), saves in format: node1 layer1 node2 layer2

For regular networks, saves in format: node1 node2

`py3plex.core.parsers.save_gpickle(input_network: Any, output_file: str) → None`

`py3plex.core.parsers.save_multiedgelist(input_network: Any, output_file: str, attributes: bool = False, encode_with_ints: bool = False) → Tuple[Dict[Any, str], Dict[Any, str]] | None`

Save multiedgelist – as n1, l1, n2, l2, w

Returns

When `encode_with_ints` is `True`, returns tuple of (`node_encodings`, `type_encodings`) Otherwise returns `None`

`py3plex.core.converters.compute_layout`(*network: Graph, compute_layouts: str, layout_parameters: Dict[str, Any] | None, verbose: bool*) → `Graph`

Compute and normalize layout for a network.

Parameters

- **network** – NetworkX graph to compute layout for
- **compute_layouts** – Layout algorithm to use ('force', 'random', 'custom_coordinates')
- **layout_parameters** – Optional parameters for layout algorithms
- **verbose** – Whether to print verbose output

Returns

Network with 'pos' attribute added to nodes

Contracts:

- Precondition: network must not be `None` and must have at least one node
- Precondition: `compute_layouts` must be a valid algorithm name
- Postcondition: all nodes have 'pos' attribute (layout preserves nodes)

`py3plex.core.converters.ensure`(*args, **kwargs)

`py3plex.core.converters.prepare_for_parsing`(*multinet*)

Compute layout for a hairball visualization

Parameters

param1 (*obj*) – multilayer object

Returns

(names, prepared network)

Return type

tuple

`py3plex.core.converters.prepare_for_visualization`(*multinet: Graph, network_type: str = 'multilayer', compute_layouts: str = 'force', layout_parameters: Dict[str, Any] | None = None, verbose: bool = True, multiplex: bool = False*) → `Tuple[List[Any], List[Graph], Any]`

This functions takes a multilayer object and returns individual layers, their names, as well as multilayer edges spanning over multiple layers.

Parameters

- **multinet** – multilayer network object
- **network_type** – "multilayer" or "multiplex"
- **compute_layouts** – Layout algorithm ('force', 'random', etc.)
- **layout_parameters** – Optional layout parameters
- **verbose** – Whether to print progress information
- **multiplex** – Whether to treat as multiplex network

Returns

(layer_names, layer_networks_list, multiedges)

- layer_names: List of layer names
- layer_networks_list: List of NetworkX graph objects for each layer
- multiedges: Dictionary of edges spanning multiple layers

Return type

tuple

py3plex.core.converters.**prepare_for_visualization_hairball**(*multinet*, *compute_layouts=False*)

Compute layout for a hairball visualization

Parameters

param1 (*obj*) – multilayer object

Returns

(names, prepared network)

Return type

tuple

py3plex.core.converters.**require**(*args, **kwargs)

py3plex.core.random_generators.**ensure**(*args, **kwargs)

py3plex.core.random_generators.**random_multilayer_ER**(*n: int*, *l: int*, *p: float*, *directed: bool = False*) → Any

Generate random multilayer Erdős-Rényi network.

Parameters

- **n** – Number of nodes (must be positive)
- **l** – Number of layers (must be positive)
- **p** – Edge probability in [0, 1]
- **directed** – If True, generate directed network

Returns

multi_layer_network object

Contracts:

- Precondition: $n > 0$ - must have at least one node
- Precondition: $l > 0$ - must have at least one layer
- Precondition: $0 \leq p \leq 1$ - probability must be valid
- Postcondition: result is not None - must return valid network

py3plex.core.random_generators.**random_multiplex_ER**(*n: int*, *l: int*, *p: float*, *directed: bool = False*) → Any

Generate random multiplex Erdős-Rényi network.

Parameters

- **n** – Number of nodes (must be positive)
- **l** – Number of layers (must be positive)

- **p** – Edge probability in [0, 1]
- **directed** – If True, generate directed network

Returns

multi_layer_network object

Contracts:

- Precondition: $n > 0$ - must have at least one node
- Precondition: $l > 0$ - must have at least one layer
- Precondition: $0 \leq p \leq 1$ - probability must be valid
- Postcondition: result is not None - must return valid network

`py3plex.core.random_generators.random_multiplex_generator(n: int, m: int, d: float = 0.9) → MultiGraph`

Generate a multiplex network from a random bipartite graph.

Parameters

- **n** – Number of nodes (must be positive)
- **m** – Number of layers (must be positive)
- **d** – Layer dropout to avoid cliques, range [0..1] (default: 0.9)

Returns

Generated multiplex network as a MultiGraph

Contracts:

- Precondition: $n > 0$ - must have at least one node
- Precondition: $m > 0$ - must have at least one layer
- Precondition: $0 \leq d \leq 1$ - dropout must be valid probability
- Postcondition: result is not None
- Postcondition: result is a NetworkX MultiGraph

`py3plex.core.random_generators.require(*args, **kwargs)`

Supporting methods for parsers and converters.

This module provides utility functions for network parsing and conversion, including layer splitting, multiplex edge addition, and GAF parsing.

`py3plex.core.supporting.add_mpx_edges(input_network: Graph) → Graph`

Add multiplex edges between corresponding nodes across layers.

Multiplex edges connect nodes representing the same entity across different layers of a multilayer network.

Parameters

input_network – NetworkX graph with multilayer structure.

Returns

Network with added multiplex edges between corresponding nodes.

Contracts:

- Precondition: input_network must not be None

- Precondition: `input_network` must be a NetworkX graph
- Postcondition: result is a NetworkX graph

Example

```
>>> network = nx.Graph()
>>> network.add_node(('A', 'layer1'))
>>> network.add_node(('A', 'layer2'))
>>> network = add_mpx_edges(network)
```

`py3plex.core.supporting.ensure(*args, **kwargs)`

`py3plex.core.supporting.parse_gaf_to_uniprot_GO(gaf_mappings: str, filter_terms: int | None = None)`
 $\rightarrow \text{Dict}[\text{str}, \text{List}[\text{str}]]$

Parse Gene Association File (GAF) to map UniProt IDs to GO terms.

Parameters

- **gaf_mappings** – Path to GAF file.
- **filter_terms** – Optional minimum occurrence threshold for GO terms.

Returns

Dictionary mapping UniProt IDs to lists of associated GO terms.

Example

```
>>> mappings = parse_gaf_to_uniprot_GO("gaf_file.gaf", filter_terms=5)
```

`py3plex.core.supporting.require(*args, **kwargs)`

`py3plex.core.supporting.split_to_layers(input_network: Graph)  $\rightarrow$  Dict[Any, Graph]`

Split a multilayer network into separate layer subgraphs.

Parameters

input_network – NetworkX graph containing nodes from multiple layers.

Returns

Dictionary mapping layer names to their corresponding subgraphs.

Contracts:

- Precondition: `input_network` must not be None
- Precondition: `input_network` must be a NetworkX graph
- Postcondition: result is a dictionary
- Postcondition: all values are NetworkX graphs

Example

```
>>> network = nx.Graph()
>>> network.add_node(('A', 'layer1'))
>>> network.add_node(('B', 'layer2'))
>>> layers = split_to_layers(network)
```

NetworkX compatibility layer for py3plex. This module provides compatibility functions for different NetworkX versions.

`py3plex.core.nx_compat.ensure(*args, **kwargs)`

`py3plex.core.nx_compat.is_string_like(obj: Any) → bool`

Check if obj is string-like (compatible with NetworkX < 3.0).

Parameters

obj – Object to check

Returns

True if string-like

Return type

bool

`py3plex.core.nx_compat.nx_from_scipy_sparse_matrix(A: Any, parallel_edges: bool = False, create_using: Graph | None = None, edge_attribute: str = 'weight') → Graph`

Create a graph from scipy sparse matrix (compatible with NetworkX < 3.0 and >= 3.0).

Parameters

- **A** – scipy sparse matrix
- **parallel_edges** – Whether to create parallel edges (ignored in NetworkX 3.0+)
- **create_using** – Graph type to create
- **edge_attribute** – Edge attribute name for weights

Returns

NetworkX graph

Contracts:

- Precondition: A must not be None
- Precondition: edge_attribute must be a non-empty string
- Postcondition: returns a NetworkX graph

`py3plex.core.nx_compat.nx_info(G: Graph) → str`

Get network information (compatible with NetworkX < 3.0 and >= 3.0).

Parameters

G – NetworkX graph

Returns

Network information

Return type

str

Contracts:

- Precondition: G must not be None and must be a NetworkX graph
- Postcondition: returns a non-empty string

`py3plex.core.nx_compat.nx_read_gpickle(path: str) → Graph`

Read a graph from a pickle file (compatible with NetworkX < 3.0 and >= 3.0).

Parameters

path – File path

Returns

NetworkX graph

Contracts:

- Precondition: path must be a non-empty string
- Postcondition: returns a NetworkX graph

`py3plex.core.nx_compat.nx_to_scipy_sparse_matrix(G: Graph, nodelist: list | None = None, dtype: Any | None = None, weight: str = 'weight', format: str = 'csr') → Any`

Convert graph to scipy sparse matrix (compatible with NetworkX < 3.0 and >= 3.0).

Parameters

- **G** – NetworkX graph
- **nodelist** – List of nodes
- **dtype** – Data type
- **weight** – Edge weight attribute
- **format** – Sparse matrix format

Returns

scipy sparse matrix

Contracts:

- Precondition: G must not be None and must be a NetworkX graph
- Precondition: weight must be a non-empty string
- Precondition: format must be a non-empty string
- Postcondition: returns a non-None sparse matrix

`py3plex.core.nx_compat.nx_write_gpickle(G: Graph, path: str) → None`

Write a graph to a pickle file (compatible with NetworkX < 3.0 and >= 3.0).

Parameters

- **G** – NetworkX graph
- **path** – File path

Contracts:

- Precondition: G must not be None and must be a NetworkX graph
- Precondition: path must be a non-empty string

`py3plex.core.nx_compat.require(*args, **kwargs)`

HINMINE Network Decomposition

`py3plex.core.HINMINE.decomposition.aggregate_sum(input_thing, classes, universal_set)`

`py3plex.core.HINMINE.decomposition.aggregate_weighted_sum(input_thing, classes, universal_set)`

```
py3plex.core.HINMINE.decomposition.calculate_importance_chi(classes, universal_set, linked_nodes,
                                                            n, **kwargs)
```

Calculates importance of a single midpoint using chi-squared weighing. :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_delta(classes, universal_set,
                                                              linked_nodes, n, **kwargs)
```

Calculates importance of a single midpoint using delta-idf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_gr(classes, universal_set, linked_nodes, n,
                                                            **kwargs)
```

Calculates importance of a single midpoint using the GR (gain ratio) :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_idf(classes, universal_set, linked_nodes,
                                                             n, **kwargs)
```

Calculates importance of a single midpoint using idf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_ig(classes, universal_set, linked_nodes, n,
                                                            **kwargs)
```

Calculates importance of a single midpoint using IG (information gain) weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_okapi(classes, universal_set,
                                                              linked_nodes, n, degrees=None,
                                                              avgdegree=None)
```

```
py3plex.core.HINMINE.decomposition.calculate_importance_rf(classes, universal_set, linked_nodes, n,
                                                            **kwargs)
```

Calculates importance of a single midpoint using rf weighing :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_tf(classes, universal_set, linked_nodes, n,
                                                            **kwargs)
```

Calculates importance of a single midpoint using term frequency weighing. :param classes: List of all classes :param universal_set: Set of all indices to consider :param linked_nodes: Set of all nodes linked by the midpoint :param n: Number of elements of universal set :return: List of weights of the midpoint for each label in class

```
py3plex.core.HINMINE.decomposition.calculate_importance_w2w(classes, universal_set, linked_nodes,
                                                             n, **kwargs)
```

```
py3plex.core.HINMINE.decomposition.calculate_importances(midpoints, classes, universal_set, method,
                                                         degrees=None, avgdegree=None)
```

```
py3plex.core.HINMINE.decomposition.chi_value(actual_pos_num, predicted_pos_num, tp, n)
```

```

py3plex.core.HINMINE.decomposition.get_aggregation_method(method_name)
py3plex.core.HINMINE.decomposition.get_calculation_method(method_name)
py3plex.core.HINMINE.decomposition.gr_value(actual_pos_num, predicted_pos_num, tp, n)
py3plex.core.HINMINE.decomposition.hinmine_decompose(network, heuristic, cycle=None,
                                                    parallel=False)
py3plex.core.HINMINE.decomposition.hinmine_get_cycles(network, cycle=None)
py3plex.core.HINMINE.decomposition.ig_value(actual_pos_num, predicted_pos_num, tp, n)
py3plex.core.HINMINE.decomposition.np_calculate_importance_chi(predicted, label_matrix,
                                                                actual_pos_nums)
py3plex.core.HINMINE.decomposition.np_calculate_importance_tf(predicted, label_matrix)
py3plex.core.HINMINE.decomposition.rf_value(predicted_pos_num, tp)
py3plex.core.HINMINE.IO.load_hinmine_object(infile, label_delimiter='---', weight_tag=False,
                                           targets=True)

```

7.29.2 Configuration and Utilities

Centralized configuration for py3plex.

This module provides default settings for visualization, layout algorithms, and other configurable aspects of the library. Users can override these settings by modifying the values after import.

Example

```

>>> from py3plex import config
>>> config.DEFAULT_NODE_SIZE = 15
>>> config.DEFAULT_EDGE_ALPHA = 0.5

```

`py3plex.config.get_color_palette(name: str | None = None) → List[str]`

Get a color palette by name.

Parameters

name – Palette name. If None, returns the default palette.

Returns

List of color hex codes.

Raises

ValueError – If palette name is not recognized.

Example

```

>>> from py3plex.config import get_color_palette
>>> colors = get_color_palette("rainbow")
>>> print(colors[0])
' #FF6B6B '

```

`py3plex.config.reset_to_defaults() → None`

Reset all configuration values to their defaults.

This is useful for testing or when you want to start fresh.

Example

```
>>> from py3plex import config
>>> config.DEFAULT_NODE_SIZE = 20
>>> config.reset_to_defaults()
>>> print(config.DEFAULT_NODE_SIZE)
10
```

Utility functions for py3plex.

This module provides common utilities used across the library, including random state management for reproducibility and deprecation warnings.

`py3plex.utils.deprecated(reason: str, version: str | None = None, alternative: str | None = None) → Callable[[Callable], Callable]`

Decorator to mark functions/methods as deprecated.

This decorator will issue a DeprecationWarning when the decorated function is called, providing information about why it's deprecated and what to use instead.

Parameters

- **reason** – Explanation of why the function is deprecated
- **version** – Version in which the function was deprecated (optional)
- **alternative** – Suggested alternative function/method (optional)

Returns

Decorator function

Example

```
>>> @deprecated(
...     reason="This function is obsolete",
...     version="0.95a",
...     alternative="new_function()"
... )
... def old_function():
...     pass
```

`py3plex.utils.ensure(*args, **kwargs)`

`py3plex.utils.get_background_knowledge_dir() → str`

Get the absolute path to the background knowledge directory.

Returns

Absolute path to the background_knowledge directory

Return type

str

Examples

```
>>> from py3plex.utils import get_background_knowledge_dir
>>> dir_path = get_background_knowledge_dir()
```

`py3plex.utils.get_background_knowledge_path(filename: str) → str`

Get the absolute path to a background knowledge file or directory.

Convenience wrapper around `get_data_path()` specifically for background knowledge files.

Parameters

filename – Name or relative path of the background knowledge file. Use empty string or ‘.’ to get the background_knowledge directory itself.

Returns

Absolute path to the background knowledge file or directory

Return type

str

Examples

```
>>> from py3plex.utils import get_background_knowledge_path
>>> path = get_background_knowledge_path("bk.n3")
>>> dir_path = get_background_knowledge_path(".")
```

`py3plex.utils.get_data_path(relative_path: str) → str`

Get the absolute path to a data file in the repository.

This function searches for data files in multiple locations to support both: - Running examples from a cloned repository - Running scripts/notebooks from any directory with datasets locally available

Search order: 1. Relative to the calling script’s directory (for examples in cloned repo) 2. Relative to current working directory (for notebooks/user scripts) 3. Relative to py3plex package location (for editable installs)

Parameters

relative_path – Path relative to repository root (e.g., “datasets/intact02.gpickle”)

Returns

Absolute path to the file

Return type

str

Raises

Py3plexIOError – If the file cannot be found in any search location

Examples

```
>>> from py3plex.utils import get_data_path
>>> path = get_data_path("datasets/intact02.gpickle")
>>> os.path.exists(path)
True
```

Note

When py3plex is installed via pip, datasets are not included in the package. Users should either: - Clone the repository and run examples from there - Download datasets separately and place them relative to their scripts - Use current working directory with datasets folder

`py3plex.utils.get_dataset_path(filename: str) → str`

Get the absolute path to a dataset file.

Convenience wrapper around `get_data_path()` specifically for dataset files.

Parameters

filename – Name or relative path of the dataset file

Returns

Absolute path to the dataset file

Return type

str

Examples

```
>>> from py3plex.utils import get_dataset_path
>>> path = get_dataset_path("intact02.gpickle")
>>> os.path.exists(path)
True
```

`py3plex.utils.get_example_image_path(filename: str) → str`

Get the absolute path to an example image file.

Convenience wrapper around `get_data_path()` specifically for example image files.

Parameters

filename – Name or relative path of the image file

Returns

Absolute path to the example image file

Return type

str

Examples

```
>>> from py3plex.utils import get_example_image_path
>>> path = get_example_image_path("intact_10_BH.png")
```

`py3plex.utils.get_multilayer_dataset_path(relative_path: str) → str`

Get the absolute path to a multilayer dataset file.

Convenience wrapper around `get_data_path()` specifically for multilayer dataset files.

Parameters

relative_path – Relative path within multilayer_datasets directory

Returns

Absolute path to the multilayer dataset file

Return type

str

Examples

```
>>> from py3plex.utils import get_multilayer_dataset_path
>>> path = get_multilayer_dataset_path("MLKing/MLKing2013_multiplex.edges")
```

`py3plex.utils.get_rng(seed: int | Generator | None = None) → Generator`

Get a NumPy random number generator with optional seed.

This provides a unified interface for random state management across the library, ensuring reproducibility when a seed is provided.

Parameters

seed – Random seed for reproducibility. Can be: - None: Use default unseeded generator - int: Seed value for the generator - `np.random.Generator`: Pass through existing generator

Returns

Initialized random number generator

Return type

`np.random.Generator`

Examples

```
>>> rng = get_rng(42)
>>> rng.random() # Reproducible random number
0.7739560485559633
```

```
>>> rng1 = get_rng(42)
>>> rng2 = get_rng(42)
>>> rng1.random() == rng2.random()
True
```

```
>>> existing_rng = np.random.default_rng(123)
>>> rng = get_rng(existing_rng)
>>> rng is existing_rng
True
```

Contracts:

- Postcondition: result is a NumPy random Generator

Note

Uses `numpy.random.Generator` (modern API introduced in NumPy 1.17) rather than the legacy `numpy.random.RandomState` API.

Negative seeds are converted to positive values by taking absolute value to ensure compatibility with NumPy's `SeedSequence`.

`py3plex.utils.require(*args, **kwargs)`

`py3plex.utils.validate_multilayer_input(network_data: Any) → None`

Validate multilayer network input data.

Performs sanity checks on multilayer network structures to catch common errors early.

Parameters

network_data – Network data to validate (can be various formats)

Raises

ValueError – If the network data is invalid

Contracts:

- Precondition: `network_data` must not be `None`

Example

```
>>> from py3plex.utils import validate_multilayer_input
>>> validate_multilayer_input(my_network)
```

`py3plex.utils.warn_if_deprecated(feature_name: str, reason: str, alternative: str | None = None) → None`

Issue a deprecation warning for a feature.

This is useful for deprecating specific usage patterns or parameter combinations rather than entire functions.

Parameters

- **feature_name** – Name of the deprecated feature
- **reason** – Explanation of why it's deprecated
- **alternative** – Suggested alternative (optional)

Example

```
>>> def my_function(old_param=None, new_param=None):
...     if old_param is not None:
...         warn_if_deprecated(
...             "old_param",
...             "This parameter is no longer used",
...             "new_param"
...         )
```

Custom exception types for the py3plex library.

This module defines domain-specific exceptions to provide clear error messages and enable better error handling throughout the library.

exception `py3plex.exceptions.AlgorithmError`

Bases: `Py3plexException`

Exception raised when an algorithm execution fails.

exception `py3plex.exceptions.CentralityComputationError`

Bases: `AlgorithmError`

Exception raised when centrality computation fails.

exception `py3plex.exceptions.CommunityDetectionError`

Bases: `AlgorithmError`

Exception raised when community detection fails.

exception `py3plex.exceptions.ConversionError`

Bases: `Py3plexException`

Exception raised when format conversion fails.

exception `py3plex.exceptions.DecompositionError`

Bases: `AlgorithmError`

Exception raised when network decomposition fails.

exception `py3plex.exceptions.EmbeddingError`

Bases: [*AlgorithmError*](#)

Exception raised when embedding generation fails.

exception `py3plex.exceptions.ExternalToolError`

Bases: [*Py3plexException*](#)

Exception raised when external tool execution fails.

exception `py3plex.exceptions.IncompatibleNetworkError`

Bases: [*Py3plexException*](#)

Exception raised when network format is incompatible with an operation.

exception `py3plex.exceptions.InvalidEdgeError`

Bases: [*Py3plexException*](#)

Exception raised when an invalid edge is specified.

exception `py3plex.exceptions.InvalidLayerError`

Bases: [*Py3plexException*](#)

Exception raised when an invalid layer is specified.

exception `py3plex.exceptions.InvalidNodeError`

Bases: [*Py3plexException*](#)

Exception raised when an invalid node is specified.

exception `py3plex.exceptions.NetworkConstructionError`

Bases: [*Py3plexException*](#)

Exception raised when network construction fails.

exception `py3plex.exceptions.ParsingError`

Bases: [*Py3plexException*](#)

Exception raised when parsing input data fails.

exception `py3plex.exceptions.Py3plexException`

Bases: `Exception`

Base exception class for all py3plex-specific exceptions.

exception `py3plex.exceptions.Py3plexFormatError`

Bases: [*Py3plexException*](#)

Exception raised when input format is invalid or cannot be parsed.

exception `py3plex.exceptions.Py3plexIOError`

Bases: [*Py3plexException*](#)

Exception raised when I/O operations fail (file reading, writing, etc.).

exception `py3plex.exceptions.Py3plexLayoutError`

Bases: [*Py3plexException*](#)

Exception raised when layout computation or visualization positioning fails.

exception `py3plex.exceptions.Py3plexMatrixError`

Bases: `Py3plexException`

Exception raised when matrix operations fail or matrix is invalid.

exception `py3plex.exceptions.VisualizationError`

Bases: `Py3plexException`

Exception raised when visualization operations fail.

7.29.3 Algorithms

Community Detection

```
py3plex.algorithms.community_detection.community_wrapper.NoRC_communities(network,
                                                                           verbose=True,
                                                                           cluster-
                                                                           ing_scheme='kmeans',
                                                                           output='mapping',
                                                                           prob_threshold=0.001,
                                                                           parallel_step=8,
                                                                           commu-
                                                                           nity_range=None,
                                                                           fine_range=3)

py3plex.algorithms.community_detection.community_wrapper.infomap_communities(graph: Graph,
                                                                              binary: str =
                                                                              './infomap',
                                                                              edgelist_file: str
                                                                              =
                                                                              './tmp/tmpedgelist.txt',
                                                                              multiplex: bool
                                                                              = False,
                                                                              verbose: bool =
                                                                              False,
                                                                              overlapping:
                                                                              bool = False,
                                                                              iterations: int =
                                                                              200, output: str
                                                                              = 'mapping',
                                                                              seed: int | None
                                                                              = None) →
                                                                              Dict[Any, int] |
                                                                              Dict[Any,
                                                                              List[int]]
```

Detect communities using the Infomap algorithm.

Parameters

- **graph** – Input graph (NetworkX graph or `multi_layer_network`)
- **binary** – Path to Infomap binary (default: “./infomap”)
- **edgelist_file** – Temporary file for edgelist (default: “./tmp/tmpedgelist.txt”)
- **multiplex** – Whether to use multiplex mode (default: False)
- **verbose** – Whether to show verbose output (default: False)

- **overlapping** – Whether to detect overlapping communities (default: False)
- **iterations** – Number of iterations (default: 200)
- **output** – Output format - “mapping” or “partition” (default: “mapping”)
- **seed** – Random seed for reproducibility (default: None) Note: Requires Infomap binary that supports `--seed` parameter

Returns

Dict mapping nodes to community IDs (if `output="mapping"`) or Dict mapping community IDs to lists of nodes (if `output="partition"`)

Raises

- **FileNotFoundError** – If Infomap binary is not found
- **PermissionError** – If Infomap binary is not executable

Examples

```
>>> # Using with seed for reproducibility
>>> partition = infomap_communities(graph, seed=42)
>>>
>>> # Get partition format instead of mapping
>>> communities = infomap_communities(graph, output="partition")
```

```
py3plex.algorithms.community_detection.community_wrapper.louvain_communities(network, out-
                                                                    put='mapping')
```

```
py3plex.algorithms.community_detection.community_wrapper.parse_infomap(outfile)
```

```
py3plex.algorithms.community_detection.community_wrapper.run_infomap(infile: str, multiplex: bool
                                                                    = True, overlapping: bool
                                                                    = False, binary: str =
                                                                    './infomap', verbose: bool
                                                                    = True, iterations: int =
                                                                    1000, seed: int | None =
                                                                    None) → None
```

Multilayer Modularity Maximization (Mucha et al., 2010)

This module implements multilayer modularity quality function and optimization algorithms for community detection in multilayer/multiplex networks.

References

Mucha et al., “Community Structure in Time-Dependent, Multiscale, and Multiplex Networks”, Science 328:876-878 (2010)

```
py3plex.algorithms.community_detection.multilayer_modularity.build_supra_modularity_matrix(network:  
Any,  
gamma:  
float  
|  
Dict[Any,  
float]  
=  
1.0,  
omega:  
float  
|  
ndar-  
ray  
=  
1.0,  
weight:  
str  
=  
'weight')  
→  
Tu-  
ple[ndarray,  
List[Tuple[Any,  
Any]]]
```

Build the supra-modularity matrix B for multilayer network.

The supra-modularity matrix is: $B_{\{i,j\}} = (A_{ij} - \frac{k_i k_j}{2m}) + \omega_{ij}$

This matrix can be used for spectral community detection methods.

Parameters

- **network** – py3plex multi_layer_network object
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute

Returns

Tuple of (modularity_matrix, node_layer_list) - modularity_matrix: Supra-modularity matrix B ($NL \times NL$) - node_layer_list: List of (node, layer) tuples corresponding to matrix indices

```

py3plex.algorithms.community_detection.multilayer_modularity.louvain_multilayer(network:
    Any,
    gamma:
    float |
    Dict[Any,
    float] = 1.0,
    omega:
    float |
    ndarray =
    1.0, weight:
    str =
    'weight',
    max_iter:
    int = 100,
    random_state:
    int | None =
    None) →
    Dict[Tuple[Any,
    Any], int]

```

Generalized Louvain algorithm for multilayer networks.

This implements the multilayer Louvain method as described in Mucha et al. (2010), which greedily maximizes the multilayer modularity quality function.

Complexity:

- **Time:** $O(n \times L \times d \times k)$ per iteration, where:
 - n = number of nodes per layer
 - L = number of layers
 - d = average degree
 - k = number of communities
- Typical: $O(n \times L)$ iterations until convergence
- Worst case: $O((n \times L)^2)$ for dense networks
- Space: $O((n \times L)^2)$ for supra-adjacency matrix (use sparse for large networks)

Parameters

- **network** – py3plex multi_layer_network object
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute
- **max_iter** – Maximum number of iterations
- **random_state** – Random seed for reproducibility

Returns

Dictionary mapping (node, layer) tuples to community IDs

Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.algorithms.community_detection.multilayer_modularity import _
↳ louvain_multilayer
>>>
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
...     ['A', 'L2', 'C', 'L2', 1]
... ], input_type='list')
>>>
>>> communities = louvain_multilayer(network, gamma=1.0, omega=1.0, random_state=42)
>>> print(communities)
```

Note

For reproducible results, always set `random_state` parameter.

`py3plex.algorithms.community_detection.multilayer_modularity.multilayer_modularity(network: Any, communities: Dict[Tuple[Any, Any], int], gamma: float | Dict[Any, float] = 1.0, omega: float | ndarray = 1.0, weight: str = 'weight') → float`

Calculate multilayer modularity quality function (Mucha et al., 2010).

The multilayer modularity is defined as: $Q = (1/2) \sum_{ij} [(A^l_{ij} - P^l_{ij}) + g_l] (g_i, g_j)$

where: - A^l_{ij} is the adjacency matrix of layer - P^l_{ij} is the null model (e.g., Newman-Girvan: $k_i k_j / 2m_l$) - g_l is the resolution parameter for layer - g_l is the inter-layer coupling strength - $g_l = 1$ if $l = l'$, else 0 (Kronecker delta) - $g_l = 1$ if $i=j$, else 0 - $(g_i, g_j) = 1$ if node i in layer l and node j in layer l' are in same community - w is the total edge weight in the supra-network

Parameters

- **network** – py3plex `multi_layer_network` object
- **communities** – Dictionary mapping (node, layer) tuples to community IDs

- **gamma** – Resolution parameter(s). Can be: - Single float: same resolution for all layers - Dict[layer, float]: layer-specific resolution parameters
- **omega** – Inter-layer coupling strength. Can be: - Single float: uniform coupling between all layer pairs - np.ndarray: layer-pair specific coupling matrix ($L \times L$)
- **weight** – Edge weight attribute (default: “weight”)

Returns

Modularity value Q [-1, 1]

Examples

```
>>> from py3plex.core import multinet
>>> from py3plex.algorithms.community_detection.multilayer_modularity import _
    ↪ multilayer_modularity
>>>
>>> # Create a simple multilayer network
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1],
...     ['A', 'L2', 'C', 'L2', 1]
... ], input_type='list')
>>>
>>> # Assign communities
>>> communities = {
...     ('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1,
...     ('A', 'L2'): 0, ('C', 'L2'): 0
... }
>>>
>>> # Calculate modularity
>>> Q = multilayer_modularity(network, communities, gamma=1.0, omega=1.0)
>>> print(f"Modularity: {Q:.3f}")
```

This module implements community detection.

class py3plex.algorithms.community_detection.community_louvain.**Status**

Bases: object

To handle several data in one struct.

Could be replaced by named tuple, but don't want to depend on python 2.6

copy()

Perform a deep copy of status

degrees: dict = {}

gdegrees: dict = {}

init(graph, weight, part=None)

Initialize the status of a graph with every node in one community

internals: dict = {}

node2com: dict = {}

```
total_weight = 0
```

```
py3plex.algorithms.community_detection.community_louvain.best_partition(graph: Graph,  
                                                                        partition: Dict | None  
                                                                        = None, weight: str =  
                                                                        'weight', resolution:  
                                                                        float = 1.0,  
                                                                        randomize: bool =  
                                                                        False) → Dict
```

Compute the partition of the graph nodes which maximises the modularity (or try..) using the Louvain heuristics

This is the partition of highest modularity, i.e. the highest partition of the dendrogram generated by the Louvain algorithm.

Parameters

- **graph** (*networkx.Graph*) – the networkx graph which is decomposed
- **partition** (*dict, optional*) – the algorithm will start using this partition of the nodes. It's a dictionary where keys are their nodes and values the communities
- **weight** (*str, optional*) – the key in graph to use as weight. Default to 'weight'
- **resolution** (*double, optional*) – Will change the size of the communities, default to 1. represents the time described in “Laplacian Dynamics and Multiscale Modular Structure in Networks”, R. Lambiotte, J.-C. Delvenne, M. Barahona
- **randomize** (*boolean, optional*) – Will randomize the node evaluation order and the community evaluation order to get different partitions at each call

Returns

partition – The partition, with communities numbered from 0 to number of communities

Return type

dictionary

Raises

NetworkXError – If the graph is not Eulerian.

See also

[*generate_dendrogram*](#)

Notes

Uses Louvain algorithm

References

large networks. J. Stat. Mech 10008, 1-12(2008).

Examples

```
>>> #Basic usage  
>>> G=nx.erdos_renyi_graph(100, 0.01)  
>>> part = best_partition(G)
```



```

>>> #other example to display a graph with its community :
>>> #better with karate_graph() as defined in networkx examples
>>> #erdos renyi don't have true community structure
>>> G = nx.erdos_renyi_graph(30, 0.05)
>>> #first compute the best partition
>>> partition = community.best_partition(G)
>>> #drawing
>>> size = float(len(set(partition.values())))
>>> pos = nx.spring_layout(G)
>>> count = 0.
>>> for com in set(partition.values()) :
>>>     count += 1.
>>>     list_nodes = [nodes for nodes in partition.keys()
>>>                    if partition[nodes] == com]
>>>     nx.draw_networkx_nodes(G, pos, list_nodes, node_size = 20,
>>>                            node_color = str(count / size))
>>> nx.draw_networkx_edges(G, pos, alpha=0.5)
>>> plt.show()

```

py3plex.algorithms.community_detection.community_louvain.**generate_dendrogram**(*graph*: Graph, *part_init*: Dict | None = None, *weight*: str = 'weight', *resolution*: float = 1.0, *randomize*: bool = False) → List[Dict]

Find communities in the graph and return the associated dendrogram

A dendrogram is a tree and each level is a partition of the graph nodes. Level 0 is the first partition, which contains the smallest communities, and the best is len(dendrogram) - 1. The higher the level is, the bigger are the communities

Parameters

- **graph** (*networkx.Graph*) – the networkx graph which will be decomposed
- **part_init** (*dict*, *optional*) – the algorithm will start using this partition of the nodes. It's a dictionary where keys are their nodes and values the communities
- **weight** (*str*, *optional*) – the key in graph to use as weight. Default to 'weight'
- **resolution** (*double*, *optional*) – Will change the size of the communities, default to 1. represents the time described in “Laplacian Dynamics and Multiscale Modular Structure in Networks”, R. Lambiotte, J.-C. Delvenne, M. Barahona

Returns

dendrogram – a list of partitions, ie dictionaries where keys of the i+1 are the values of the i. and where keys of the first are the nodes of graph

Return type

list of dictionaries

Raises

TypeError – If the graph is not a networkx.Graph

See also[*best_partition*](#)**Notes**

Uses Louvain algorithm

References

networks. J. Stat. Mech 10008, 1-12(2008).

Examples

```

>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> dendo = generate_dendrogram(G)
>>> for level in range(len(dendo) - 1) :
>>>     print("partition at level", level,
>>>           "is", partition_at_level(dendo, level))
:param weight:
:type weight:

```

`py3plex.algorithms.community_detection.community_louvain.induced_graph`(*partition: Dict, graph: Graph, weight: str = 'weight'*) → Graph

Produce the graph where nodes are the communities

there is a link of weight w between communities if the sum of the weights of the links between their elements is w

Parameters

- **partition** (*dict*) – a dictionary where keys are graph nodes and values the part the node belongs to
- **graph** (*networkx.Graph*) – the initial graph
- **weight** (*str, optional*) – the key in graph to use as weight. Default to ‘weight’

Returns

g – a networkx graph where nodes are the parts

Return type

networkx.Graph

Examples

```

>>> n = 5
>>> g = nx.complete_graph(2*n)
>>> part = dict([])
>>> for node in g.nodes() :
>>>     part[node] = node % 2
>>> ind = induced_graph(part, g)
>>> goal = nx.Graph()
>>> goal.add_weighted_edges_from([(0,1,n*n),(0,0,n*(n-1)/2), (1, 1, n*(n-1)/2)]) #_
↪ NOQA

```

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```
>>> nx.is_isomorphic(int, goal)
True
```

`py3plex.algorithms.community_detection.community_louvain.load_binary(data)`

Load binary graph as used by the cpp implementation of this algorithm

`py3plex.algorithms.community_detection.community_louvain.modularity(partition: Dict, graph: Graph, weight: str = 'weight') → float`

Compute the modularity of a partition of a graph

Parameters

- **partition** (*dict*) – the partition of the nodes, i.e a dictionary where keys are their nodes and values the communities
- **graph** (*networkx.Graph*) – the networkx graph which is decomposed
- **weight** (*str, optional*) – the key in graph to use as weight. Default to ‘weight’

Returns

modularity – The modularity

Return type

float

Raises

- **KeyError** – If the partition is not a partition of all graph nodes
- **ValueError** – If the graph has no link
- **TypeError** – If graph is not a networkx.Graph

References

structure in networks. Physical Review E 69, 26113(2004).

Examples

```
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> part = best_partition(G)
>>> modularity(part, G)
```

`py3plex.algorithms.community_detection.community_louvain.partition_at_level(dendrogram: List[Dict], level: int) → Dict`

Return the partition of the nodes at the given level

A dendrogram is a tree and each level is a partition of the graph nodes. Level 0 is the first partition, which contains the smallest communities, and the best is `len(dendrogram) - 1`. The higher the level is, the bigger are the communities

Parameters

- **dendrogram** (*list of dict*) – a list of partitions, ie dictionaries where keys of the `i+1` are the values of the `i`.
- **level** (*int*) – the level which belongs to `[0..len(dendrogram)-1]`

Returns

partition – A dictionary where keys are the nodes and the values are the set it belongs to

Return type

dictionnary

Raises

KeyError – If the dendrogram is not well formed or the level is too high

See also

[*best_partition*](#), [*generate_dendrogram*](#)

Examples

```
>>> G=nx.erdos_renyi_graph(100, 0.01)
>>> dendrogram = generate_dendrogram(G)
>>> for level in range(len(dendrogram) - 1) :
>>>     print("partition at level", level, "is", partition_at_level(dendrogram,
↪level)) # NOQA
```

Community quality measures and metrics.

This module provides various measures for assessing the quality of community partitions in networks, including modularity, size distribution, and other statistical metrics.

`py3plex.algorithms.community_detection.community_measures.modularity`(*G: Graph, communities: Dict[Any, List[Any]], weight: str = 'weight'*) → float

Calculate modularity of a graph partition.

Parameters

- **G** – NetworkX graph
- **communities** – Dictionary mapping community IDs to node lists
- **weight** – Edge weight attribute (default: “weight”)

Returns

Modularity value

`py3plex.algorithms.community_detection.community_measures.number_of_communities`(*network_partition: Dict[Any, List[Any]]*) → int

Count number of communities in a partition.

Parameters

network_partition – Dictionary mapping community IDs to node lists

Returns

Number of communities

`py3plex.algorithms.community_detection.community_measures.size_distribution`(*network_partition: Dict[Any, List[Any]]*) → ndarray

Calculate size distribution of communities.

Parameters

network_partition – Dictionary mapping community IDs to node lists

Returns

Array of community sizes

Multilayer Synthetic Graph Generation for Community Detection Benchmarks

This module implements synthetic multilayer/multiplex graph generators with ground-truth community structure for benchmarking community detection algorithms.

Includes: - Multilayer LFR (Lancichinetti-Fortunato-Radicchi) benchmark - Coupled/Interdependent Erdős-Rényi models - Support for overlapping communities across layers - Support for partial node presence across layers

References

Lancichinetti et al., “Benchmark graphs for testing community detection algorithms”, Phys. Rev. E 78, 046110 (2008)

Granell et al., “Benchmark model to assess community structure in evolving networks”, Phys. Rev. E 92, 012805 (2015)

```
py3plex.algorithms.community_detection.multilayer_benchmark.generate_coupled_er_multilayer(n:
                                                    int,
                                                    lay-
                                                    ers:
                                                    List[str],
                                                    p:
                                                    float
                                                    |
                                                    List[float]
                                                    =
                                                    0.1,
                                                    omega:
                                                    float
                                                    =
                                                    1.0,
                                                    cou-
                                                    pling_probabili
                                                    float
                                                    =
                                                    1.0,
                                                    di-
                                                    rected:
                                                    bool
                                                    =
                                                    False,
                                                    seed:
                                                    int
                                                    |
                                                    None
                                                    =
                                                    None)
                                                    →
                                                    Any
```

Generate coupled/interdependent Erdős-Rényi multilayer networks.

Creates random Erdős-Rényi graphs in each layer and couples nodes across layers with specified coupling strength and probability.

Parameters

- **n** – Number of nodes
- **layers** – List of layer names
- **p** – Edge probability per layer. Can be float (same for all layers) or list of floats per layer.
- **omega** – Inter-layer coupling strength (weight of identity links)
- **coupling_probability** – Probability that a node has inter-layer coupling. Range: [0, 1] - 1.0 = all nodes coupled (full multiplex) - <1.0 = partial coupling (interdependent networks)
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

Returns

py3plex multi_layer_network object

Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import _  
↳ generate_coupled_er_multilayer  
>>>  
>>> # Full multiplex ER network  
>>> network = generate_coupled_er_multilayer(  
...     n=100,  
...     layers=['L1', 'L2', 'L3'],  
...     p=0.1,  
...     omega=1.0,  
...     coupling_probability=1.0  
... )  
>>>  
>>> # Partially coupled (interdependent)  
>>> network = generate_coupled_er_multilayer(  
...     n=100,  
...     layers=['L1', 'L2'],  
...     p=0.1,  
...     omega=0.5,  
...     coupling_probability=0.5 # Only 50% nodes coupled  
... )
```

```

py3plex.algorithms.community_detection.multilayer_benchmark.generate_multilayer_lfr(n: int,
                                                                 layers:
                                                                 List[str],
                                                                 tau1:
                                                                 float =
                                                                 2.0,
                                                                 tau2:
                                                                 float =
                                                                 1.5,
                                                                 mu:
                                                                 float |
                                                                 List[float]
                                                                 = 0.1,
                                                                 avg_degree:
                                                                 float |
                                                                 List[float]
                                                                 = 10.0,
                                                                 min_community:
                                                                 int =
                                                                 20,
                                                                 max_community:
                                                                 int |
                                                                 None
                                                                 =
                                                                 None,
                                                                 com-
                                                                 mu-
                                                                 nity_persistence:
                                                                 float =
                                                                 1.0,
                                                                 node_overlap:
                                                                 float =
                                                                 1.0,
                                                                 over-
                                                                 lap-
                                                                 ping_nodes:
                                                                 int =
                                                                 0,
                                                                 over-
                                                                 lap-
                                                                 ping_membership:
                                                                 int =
                                                                 2, di-
                                                                 rected:
                                                                 bool =
                                                                 False,
                                                                 seed:
                                                                 int |
                                                                 None
                                                                 =
                                                                 None)
→ Tuple[Any,
Dict[Tuple[Any,
str],
Set[int]]]

```

Generate multilayer LFR benchmark networks with controllable community structure.

This extends the LFR benchmark to multilayer networks, allowing control over:

- Community persistence across layers (how many nodes keep their community)
- Node overlap across layers (which nodes appear in which layers)
- Overlapping communities (nodes belonging to multiple communities)

Parameters

- **n** – Number of nodes
- **layers** – List of layer names
- **tau1** – Power-law exponent for degree distribution (typically 2-3)
- **tau2** – Power-law exponent for community size distribution (typically 1-2)
- **mu** – Mixing parameter (fraction of edges outside community). Can be float (same for all layers) or list of floats per layer. Range: [0, 1], where 0 = perfect communities, 1 = random
- **avg_degree** – Average degree per layer. Can be float (same for all layers) or list of floats per layer.
- **min_community** – Minimum community size
- **max_community** – Maximum community size (default: $n/2$)
- **community_persistence** – Probability that a node keeps its community from one layer to the next. Range: [0, 1] - 1.0 = identical communities across all layers - 0.0 = completely independent communities per layer
- **node_overlap** – Fraction of nodes present in all layers. Range: [0, 1] - 1.0 = all nodes in all layers (full multiplex) - <1.0 = some nodes absent from some layers
- **overlapping_nodes** – Number of nodes that belong to multiple communities within each layer
- **overlapping_membership** – Number of communities each overlapping node belongs to
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

Returns

Tuple of (network, ground_truth_communities) - network: py3plex multi_layer_network object
- ground_truth_communities: Dict mapping (node, layer) to Set of community IDs

Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import _  
↳ generate_multilayer_lfr  
>>>  
>>> # Generate with identical communities across layers  
>>> network, communities = generate_multilayer_lfr(  
...     n=100,  
...     layers=['L1', 'L2', 'L3'],  
...     mu=0.1,  
...     community_persistence=1.0  
... )  
>>>  
>>> # Generate with evolving communities  
>>> network, communities = generate_multilayer_lfr(  

```

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```

...     n=100,
...     layers=['T0', 'T1', 'T2'],
...     mu=0.1,
...     community_persistence=0.7 # 70% nodes keep community
... )

```

```

py3plex.algorithms.community_detection.multilayer_benchmark.generate_sbm_multilayer(n: int,
                                                                                      layers:
                                                                                      List[str],
                                                                                      com-
                                                                                      muni-
                                                                                      ties:
                                                                                      List[Set[int]],
                                                                                      p_in:
                                                                                      float |
                                                                                      List[float]
                                                                                      = 0.3,
                                                                                      p_out:
                                                                                      float |
                                                                                      List[float]
                                                                                      =
                                                                                      0.05,
                                                                                      com-
                                                                                      mu-
                                                                                      nity_persistence:
                                                                                      float =
                                                                                      1.0, di-
                                                                                      rected:
                                                                                      bool =
                                                                                      False,
                                                                                      seed:
                                                                                      int |
                                                                                      None
                                                                                      =
                                                                                      None)
→ Tuple[Any,
Dict[Tuple[Any,
str],
int]]

```

Generate multilayer stochastic block model (SBM) networks.

Creates networks where nodes are divided into communities with different intra- and inter-community connection probabilities.

Parameters

- **n** – Number of nodes
- **layers** – List of layer names
- **communities** – List of node sets defining initial communities
- **p_in** – Intra-community edge probability per layer
- **p_out** – Inter-community edge probability per layer

- **community_persistence** – Probability nodes keep their community across layers
- **directed** – Whether to generate directed networks
- **seed** – Random seed for reproducibility

Returns

Tuple of (network, ground_truth_communities) - network: py3plex multi_layer_network object
- ground_truth_communities: Dict mapping (node, layer) to community ID

Examples

```
>>> from py3plex.algorithms.community_detection.multilayer_benchmark import _  
↳ generate_sbm_multilayer  
>>>  
>>> # Define initial communities  
>>> communities = [  
...     {0, 1, 2, 3, 4}, # Community 0  
...     {5, 6, 7, 8, 9} # Community 1  
... ]  
>>>  
>>> network, ground_truth = generate_sbm_multilayer(  
...     n=10,  
...     layers=['L1', 'L2'],  
...     communities=communities,  
...     p_in=0.7,  
...     p_out=0.1,  
...     community_persistence=0.8  
... )
```

Statistics

Multilayer Network Statistics

This module implements various statistics for multilayer and multiplex networks, following standard definitions from multilayer network analysis literature.

References

- Kivelä et al. (2014), “Multilayer networks”, J. Complex Networks 2(3), 203-271
- De Domenico et al. (2013), “Mathematical formulation of multilayer networks”, PRX 3, 041022
- Mucha et al. (2010), “Community Structure in Time-Dependent, Multiscale, and Multiplex Networks”, Science 328, 876-878

Authors: py3plex contributors Date: 2025

py3plex.algorithms.statistics.multilayer_statistics.**algebraic_connectivity**(network: Any) → float

Calculate algebraic connectivity (λ_2).

Formula: $\lambda_2()$

Second smallest eigenvalue of the supra-Laplacian (Fiedler value).

Indicates global connectivity and diffusion efficiency of the multilayer system.

Properties:

$\lambda_0 = 0$ always (associated with constant eigenvector) $\lambda_1 > 0$ if and only if the multilayer network is connected
 Larger λ_1 indicates better connectivity and faster diffusion/synchronization

Parameters

network – py3plex multi_layer_network object

Returns

Second smallest eigenvalue (Fiedler value)

Examples

```
>>> alg_conn = algebraic_connectivity(network)
```

Reference:

Fiedler (1973), Sole-Ribalta et al. (2013)

```
py3plex.algorithms.statistics.multilayer_statistics.community_participation_coefficient(network:
                                                                                          Any,
                                                                                          com-
                                                                                          mu-
                                                                                          ni-
                                                                                          ties:
                                                                                          Dict[Tuple[Any,
                                                                                          Any],
                                                                                          int],
                                                                                          node:
                                                                                          Any)
                                                                                          →
                                                                                          float
```

Calculate participation coefficient for a node across community structure.

Measures how evenly a node's connections are distributed across different communities, across all layers. A node with connections to many communities has high participation.

Formula: $P = 1 - (k_s / k)^2$

where k_s is the number of connections node i has to community s , and k is the total degree of node i across all layers.

Parameters

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID
- **node** – Node identifier (not node-layer tuple)

Returns

Participation coefficient value between 0 and 1

Examples

```
>>> communities = detect_communities(network)
>>> pc = community_participation_coefficient(network, communities, 'Alice')
>>> print(f"Participation: {pc:.3f}")
```

Reference:

Guimerà & Amaral (2005), “Functional cartography of complex metabolic networks”

```
py3plex.algorithms.statistics.multilayer_statistics.community_participation_entropy(network: Any, communities: Dict[Tuple[Any, Any], int], node: Any) → float
```

Calculate participation entropy for a node across community structure.

Shannon entropy-based measure of how evenly a node distributes its connections across different communities. Higher entropy indicates more diverse community participation.

Formula: $H = - \sum_s (k_s / k) \log(k_s / k)$

where k_s is connections to community s , k is total degree.

Parameters

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID
- **node** – Node identifier (not node-layer tuple)

Returns

Entropy value (higher = more diverse participation)

Examples

```
>>> entropy = community_participation_entropy(network, communities, 'Alice')
>>> print(f"Participation entropy: {entropy:.3f}")
```

Reference:

Based on Shannon entropy applied to community structure

```
py3plex.algorithms.statistics.multilayer_statistics.compute_modularity_score(network: Any, communities: Dict[Tuple[Any, Any], int], gamma: float = 1.0, omega: float = 1.0) → float
```

Compute explicit multislice modularity score.

Direct computation of the modularity quality function for a given community partition, without running detection algorithms.

Formula: $Q = (1/2) \sum_c [(A - \sum_k k_k / (2m)) + \cdot] (c, c)$

Parameters

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) to community ID
- **gamma** – Resolution parameter (default: 1.0)
- **omega** – Inter-layer coupling strength (default: 1.0)

Returns

Modularity score Q (higher is better)

Examples

```
>>> communities = {('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1}
>>> Q = compute_modularity_score(network, communities)
>>> print(f"Modularity: {Q:.3f}")
```

Reference:

Mucha et al. (2010), Science 328, 876-878

```
py3plex.algorithms.statistics.multilayer_statistics.cross_layer_mutual_information(network:
Any,
layer_i:
str,
layer_j:
str,
bins: int
= 10)
→ float
```

Calculate cross-layer mutual information ($I(L; L)$).

Formula: $I(L; L) = H(L) + H(L) - H(L, L)$

Measures statistical dependence between degree distributions in two layers; quantifies how much knowing a node's degree in one layer tells us about its degree in another layer.

Variables:

$H(L)$ = entropy of degree distribution in layer i $H(L)$ = entropy of degree distribution in layer j $H(L, L)$ = joint entropy of degree distributions

Properties:

$I = 0$ when layers are independent $I > 0$ indicates statistical dependence (higher = stronger) $I(L; L) \min(H(L), H(L))$

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier
- **bins** – Number of bins for discretizing degree distributions

Returns

Mutual information value in bits

Examples

```
>>> mi = cross_layer_mutual_information(network, 'L1', 'L2', bins=10)
>>> print(f"Mutual information: {mi:.3f} bits")
```

Reference:

Cover & Thomas (2006), “Elements of Information Theory” De Domenico et al. (2015), “Structural reducibility”

`py3plex.algorithms.statistics.multilayer_statistics.cross_layer_redundancy_entropy`(*network*: Any) → float

Calculate cross-layer redundancy entropy ($H_{\text{redundancy}}$).

Formula: $H_r = -r \log_2(r)$, where r is the normalized edge overlap between layers i and j .

Measures diversity in structural redundancy across layer pairs; high entropy indicates varied overlap patterns, low entropy indicates uniform redundancy.

Variables:

$r = \text{edge_overlap}(i,j) / \text{edge_overlap}(.)$ Normalized overlap proportion for each layer pair

Parameters

network – py3plex multi_layer_network object

Returns

Entropy value in bits

Examples

```
>>> entropy = cross_layer_redundancy_entropy(network)
>>> print(f"Cross-layer redundancy entropy: {entropy:.3f}")
```

Reference:

Bianconi (2018), “Multilayer Networks: Structure and Function”

`py3plex.algorithms.statistics.multilayer_statistics.degree_vector`(*network*: Any, *node*: Any, *weighted*: bool = False) → Dict[str, float]

Calculate degree vector (k).

Formula: $k = (k^1, k^2, \dots, k)$

Node degree in each layer; can be analyzed via mean, variance, or entropy to capture node versatility.

Variables:

k = degree of node i in layer For undirected: $k = A$

Parameters

- **network** – py3plex multi_layer_network object
- **node** – Node identifier
- **weighted** – If True, return strength instead of degree

Returns

Dictionary mapping layer to degree/strength

Examples

```
>>> degrees = degree_vector(network, 'A')
>>> print(f"Degree in layer L1: {degrees['L1']}")
```

Reference:

Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

`py3plex.algorithms.statistics.multilayer_statistics.edge_overlap(network: Any, layer_i: str, layer_j: str) → float`

Calculate edge overlap ($\wedge$).

Formula: $\wedge = |E_i \cap E_j| / |E_i \cup E_j|$

Jaccard similarity of edge sets between two layers; measures structural redundancy.

Variables:

E_i = set of edges in layer i E_j = set of edges in layer j $| \cdot |$ = cardinality (number of elements)

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

Returns

Overlap coefficient between 0 and 1 (Jaccard similarity)

Examples

```
>>> overlap = edge_overlap(network, 'L1', 'L2')
```

Reference:

Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

`py3plex.algorithms.statistics.multilayer_statistics.entropy_of_multiplexity(network: Any) → float`

Calculate entropy of multiplexity (H).

Formula: $H = - \sum p \log_2(p)$, where $p = E_i / E$

Shannon entropy of layer contributions; measures layer diversity.

Variables:

p = proportion of edges in layer i E_i = number of edges in layer i $\log_2$ gives entropy in bits

Properties:

$H = 0$ when all edges are in one layer (minimum entropy/diversity) $H = \log_2(L)$ when edges are uniformly distributed across L layers (maximum entropy)

Parameters

network – py3plex multi_layer_network object

Returns

Entropy value in bits

Examples

```
>>> entropy = entropy_of_multiplexity(network)
```

Reference:

De Domenico et al. (2013), Shannon (1948)

```
py3plex.algorithms.statistics.multilayer_statistics.inter_layer_assortativity(network: Any,  
layer_i: str,  
layer_j: str)  
→ float
```

Calculate inter-layer assortativity (r).

Formula: $r^{\wedge} = \text{cov}(k^{\wedge}, k^{\wedge}) / (\) = \text{corr}(k^{\wedge}, k^{\wedge})$

Measures whether nodes with similar degrees tend to connect across different layers.

Variables:

$k^{\wedge}$ = degree vector in layer $k^{\wedge}$ = degree vector in layer , = standard deviations of degrees in layers and
Equivalent to Pearson correlation of degree vectors

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

Returns

Assortativity coefficient

Examples

```
>>> assort = inter_layer_assortativity(network, 'L1', 'L2')
```

Reference:

Newman (2002), Nicosia & Latora (2015)

```
py3plex.algorithms.statistics.multilayer_statistics.inter_layer_coupling_strength(network:  
Any,  
layer_i:  
str,  
layer_j:  
str) →  
float
```

Calculate inter-layer coupling strength ($C^{\wedge}$).

Formula: $C^{\wedge} = (1/N_{_}) w^{\wedge}$

Average weight of inter-layer connections between corresponding nodes in two layers. Quantifies cross-layer connectivity.

Variables:

$N_{_}$ = number of nodes present in both layers and $w^{\wedge}$ = weight of inter-layer edge connecting node i in layer to node i in layer

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

Returns

Average coupling strength

Examples

```
>>> coupling = inter_layer_coupling_strength(network, 'L1', 'L2')
```

Reference:

De Domenico et al. (2013), Physical Review X 3(4), 041022

Contracts:

- Precondition: network must not be None
- Precondition: layer_i and layer_j must be non-empty strings
- Postcondition: result is non-negative (weights are non-negative)
- Postcondition: result is not NaN

py3plex.algorithms.statistics.multilayer_statistics.**inter_layer_degree_correlation**(*network:*
Any,
layer_i:
str,
layer_j:
str) →
float

Calculate inter-layer degree correlation ($r^{\wedge}$).Formula: $r^{\wedge} = (k - k)(k - k) / [(k - k)^2 ((k - k)^2)]$

Pearson correlation of node degrees between two layers; reveals if highly connected nodes in one layer are also central in others.

Variables: k = degree of node i in layer k = mean degree in layer Sum over nodes present in both layers**Parameters**

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()

Returns

Pearson correlation coefficient between -1 and 1

Examples

```
>>> corr = inter_layer_degree_correlation(network, 'L1', 'L2')
```

Reference:

Battiston et al. (2014), Nicosia & Latora (2015)

```
py3plex.algorithms.statistics.multilayer_statistics.inter_layer_dependence_entropy(network: Any,
                                                                                    layer_i: str,
                                                                                    layer_j: str) → float
```

Calculate inter-layer dependence entropy (H_{dep}).

Formula: $H_{dep} = -p \log_2(p)$, where p is the proportion of inter-layer edges for each node n connecting layers i and j .

Measures heterogeneity in how nodes couple the two layers; high entropy indicates diverse coupling patterns, low entropy indicates uniform coupling.

Variables:

p = proportion of inter-layer edges incident to node n Total over all nodes connecting the two layers

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

Returns

Entropy value in bits

Examples

```
>>> entropy = inter_layer_dependence_entropy(network, 'L1', 'L2')
>>> print(f"Inter-layer dependence entropy: {entropy:.3f}")
```

Reference:

De Domenico et al. (2015), “Ranking in interconnected multilayer networks”

```
py3plex.algorithms.statistics.multilayer_statistics.interdependence(network: Any, sample_size: int = 100) → float
```

Calculate interdependence ().

Formula: $= \langle d \rangle / \langle d \rangle$

Quantifies how much shortest-path communication depends on inter-layer connections.

Variables:

d = shortest path from node i to node j in the full multilayer network $d = (1/L) \sum d_{ij}$ d is the average shortest path across individual layers $\langle \cdot \rangle$ = average over sampled node pairs

Interpretation:

< 1 : multilayer connectivity reduces path lengths (positive interdependence) $= 1$: inter-layer connections provide little benefit > 1 : multilayer structure increases path lengths (rare)

Parameters

- **network** – py3plex multi_layer_network object
- **sample_size** – Number of node pairs to sample for estimation

Returns

Interdependence ratio

Examples

```
>>> interdep = interdependence(network, sample_size=50)
```

Reference:

Gomez et al. (2013), Buldyrev et al. (2010)

`py3plex.algorithms.statistics.multilayer_statistics.interlayer_degree_correlation_matrix`(*network*: Any) → Tuple[ndarray, list]

Calculate inter-layer degree correlation matrix.

Computes Pearson correlation coefficients for node degrees between all pairs of layers, organized as a symmetric correlation matrix.

Formula: $\text{Matrix}[i,j] = r^{ij} = \text{corr}(k^i, k^j)$

where k^i and k^j are degree vectors for layers i and j over common nodes.

Properties:

- Diagonal elements are 1.0 (self-correlation)
- Off-diagonal elements in $[-1, 1]$
- Symmetric matrix
- Positive values indicate positive degree correlation
- Negative values indicate negative degree correlation

Parameters

network – py3plex multi_layer_network object

Returns

- `correlation_matrix`: 2D numpy array of shape (num_layers, num_layers)
- `layer_labels`: List of layer names corresponding to matrix indices

Return type

Tuple of (correlation_matrix, layer_labels)

Examples

```
>>> corr_matrix, layers = interlayer_degree_correlation_matrix(network)
>>> import matplotlib.pyplot as plt
>>> import seaborn as sns
>>> sns.heatmap(corr_matrix, annot=True, xticklabels=layers,
...             yticklabels=layers, cmap='coolwarm', center=0,
...             vmin=-1, vmax=1)
>>> plt.title('Inter-layer Degree Correlation Matrix')
>>> plt.show()
```

Reference:

Nicosia & Latora (2015), “Measuring and modeling correlations in multiplex networks” Battiston et al. (2014), “Structural measures for multiplex networks”

`py3plex.algorithms.statistics.multilayer_statistics.layer_connectivity_entropy`(*network*: Any, *layer*: str) → float

Calculate entropy of layer connectivity ($H_{\text{connectivity}}$).

Formula: $H_c = - (k/k) \log_2(k/k)$

Shannon entropy of degree distribution within a layer; measures heterogeneity of node connectivity patterns.

Variables:

k = degree of node i in the layer k = sum of all degrees ($2 * \text{edges}$ for undirected)

Properties:

$H_c = 0$ when all nodes have the same degree (uniform distribution) H_c is maximized when degree distribution is highly uneven

Parameters

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier

Returns

Entropy value in bits

Examples

```
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1]
... ], input_type='list')
>>> entropy = layer_connectivity_entropy(network, 'L1')
>>> print(f"Connectivity entropy: {entropy:.3f}")
```

Reference:

Solé-Ribalta et al. (2013), “Spectral properties of complex networks” Shannon (1948), “A Mathematical Theory of Communication”

`py3plex.algorithms.statistics.multilayer_statistics.layer_density`(*network*: Any, *layer*: str) → float

Calculate layer density ().

Formula: $= (2E) / (N(N - 1))$ [undirected]
 $= E / (N(N - 1))$ [directed]

Measures the fraction of possible edges present in a specific layer, indicating how densely connected that layer is.

Variables:

E = number of edges in layer N = number of nodes in layer

Parameters

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier

Returns

Density value between 0 and 1

Examples

```
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network(directed=False)
>>> network.add_edges([
...     ['A', 'L1', 'B', 'L1', 1],
...     ['B', 'L1', 'C', 'L1', 1]
... ], input_type='list')
>>> density = layer_density(network, 'L1')
>>> print(f"Layer L1 density: {density:.3f}")
```

Reference:

Kivela et al. (2014), J. Complex Networks 2(3), 203-271

Contracts:

- Precondition: network must not be None
- Precondition: layer must be a non-empty string
- Postcondition: result is in [0, 1] (fundamental property of density)
- Postcondition: result is not NaN

py3plex.algorithms.statistics.multilayer_statistics.**layer_influence_centrality**(network: Any, layer: str, method: str = 'coupling', sample_size: int = 100) → float

Calculate layer influence centrality (I).

Formula (coupling): $I = C^{\wedge} / (L-1)$ Formula (flow): $I = F^{\wedge} / (L-1)$

Quantifies how much a layer influences other layers through inter-layer connections (coupling method) or information flow (flow method).

Variables:

$C^{\wedge}$ = inter-layer coupling strength between layers and $F^{\wedge}$ = flow from layer to layer (random walk transition probability) L = total number of layers

Properties:

Higher values indicate layers that strongly influence others Useful for identifying critical layers in the multilayer structure

Parameters

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier

- **method** – ‘coupling’ for structural influence, ‘flow’ for dynamic influence
- **sample_size** – Number of random walk steps for flow simulation

Returns

Influence centrality value

Examples

```
>>> influence = layer_influence_centrality(network, 'L1', method='coupling')
>>> print(f"Layer L1 influence: {influence:.3f}")
```

Reference:

Cozzo et al. (2013), “Mathematical formulation of multilayer networks” De Domenico et al. (2014), “Identifying modular flows”

`py3plex.algorithms.statistics.multilayer_statistics.layer_redundancy_coefficient`(*network*: Any, *layer_i*: str, *layer_j*: str) → float

Calculate layer redundancy coefficient.

Measures the proportion of edges in one layer that are redundant (also present) in another layer. Values close to 1 indicate high redundancy, while values close to 0 indicate complementary layers.

Formula: $R = |E_i \cap E_j| / |E_i|$

where E_i and E_j are edge sets of layers i and j .

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

Returns

Redundancy coefficient between 0 and 1

Examples

```
>>> redundancy = layer_redundancy_coefficient(network, 'social', 'work')
>>> print(f"Redundancy: {redundancy:.2%}")
```

Reference:

Nicosia & Latora (2015), “Measuring and modeling correlations in multiplex networks”

`py3plex.algorithms.statistics.multilayer_statistics.layer_similarity`(*network*: Any, *layer_i*: str, *layer_j*: str, *method*: str = ‘cosine’) → float

Calculate layer similarity (S^A).

Formula: $S^A = \langle A, A \rangle / (\|A\| \|A\|) = AA / ((A)^2) ((A)^2)$

Cosine or Jaccard similarity between adjacency matrices of two layers.

Variables:

A, A = adjacency matrices for layers and $\langle \cdot, \cdot \rangle$ = Frobenius inner product $\|\cdot\|$ = Frobenius norm

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier ()
- **layer_j** – Second layer identifier ()
- **method** – ‘cosine’ or ‘jaccard’

Returns

Similarity value between 0 and 1

Examples

```
>>> similarity = layer_similarity(network, 'L1', 'L2', method='cosine')
```

Reference:

De Domenico et al. (2013), Physical Review X 3(4), 041022

```
py3plex.algorithms.statistics.multilayer_statistics.multilayer_betweenness_surface(network:
Any,
normal-
ized:
bool =
True,
weight:
str |
None =
None)
→
ndarray
```

Calculate multilayer betweenness surface (tensor representation).

Computes betweenness centrality for each node-layer pair and organizes the results as a 2D array (nodes $\times$ layers) that can be visualized as a heatmap or surface plot.

Formula: $\text{Surface}[i,] = B$

where B is the betweenness centrality of node i in layer .

Parameters

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize betweenness values
- **weight** – Edge weight attribute name (None for unweighted)

Returns

2D numpy array of shape (num_nodes, num_layers) containing betweenness values. Also returns tuple of (node_labels, layer_labels) for axis labeling.

Examples

```
>>> surface, (nodes, layers) = multilayer_betweenness_surface(network)
>>> import matplotlib.pyplot as plt
>>> plt.imshow(surface, aspect='auto', cmap='viridis')
>>> plt.xlabel('Layers')
>>> plt.ylabel('Nodes')
>>> plt.xticks(range(len(layers)), layers)
>>> plt.yticks(range(len(nodes)), nodes)
>>> plt.colorbar(label='Betweenness Centrality')
>>> plt.title('Multilayer Betweenness Surface')
>>> plt.show()
```

Reference:

De Domenico et al. (2015), “Structural reducibility of multilayer networks”

`py3plex.algorithms.statistics.multilayer_statistics.multilayer_clustering_coefficient` (*network*: Any, *node*: Any | None = None) → float | Dict[Any, float]

Calculate multilayer clustering coefficient (C).

Formula: $C = T / T$

Extends transitivity to account for triangles that span multiple layers.

Variables:

T = number of closed triplets (triangles) involving node i across all layers T = maximum possible triplets = $k(k - 1)/2$ for undirected networks Average over all nodes: $C = (1/N) \sum C_i$

Parameters

- **network** – py3plex multi_layer_network object
- **node** – If specified, compute for single node; otherwise compute for all

Returns

Clustering coefficient value or dict of values per node

Examples

```
>>> clustering = multilayer_clustering_coefficient(network)
>>> node_clustering = multilayer_clustering_coefficient(network, node='A')
```

Reference:

Battiston et al. (2014), Section III.C


```
py3plex.algorithms.statistics.multilayer_statistics.multilayer_modularity(network: Any,
                                                                    communities:
                                                                    Dict[Tuple[Any,
                                                                    Any], int], gamma:
                                                                    float | Dict[Any,
                                                                    float] = 1.0, omega:
                                                                    float | ndarray =
                                                                    1.0, weight: str =
                                                                    'weight') → float
```

Calculate multilayer modularity (Q).

This is a wrapper for the existing `multilayer_modularity` implementation in `py3plex.algorithms.community_detection.multilayer_modularity`.

Formula: $Q = (1/2) [(A - P) +] (g, g)$

Extension of Newman-Girvan modularity to multiplex networks (Mucha et al., 2010). Measures community quality across layers.

Variables:

w_{ij} = total edge weight in supra-network A = adjacency matrix element for layer $P = \text{kk}/(2m)$ is the null model (configuration model) γ = resolution parameter for layer ω = inter-layer coupling strength δ_{ij} = Kronecker delta (1 if $i=j$, 0 otherwise) $\delta_{ij}^{(l)}$ = Kronecker delta (1 if $i=j$, 0 otherwise) $(g, g) = 1$ if node i in layer l and node j in layer l are in same community

Parameters

- **network** – py3plex multi_layer_network object
- **communities** – Dictionary mapping (node, layer) tuples to community IDs
- **gamma** – Resolution parameter(s)
- **omega** – Inter-layer coupling strength
- **weight** – Edge weight attribute

Returns

Modularity value Q

Examples

```
>>> communities = {('A', 'L1'): 0, ('B', 'L1'): 0, ('C', 'L1'): 1}
>>> Q = multilayer_modularity(network, communities)
```

Reference:

Mucha et al. (2010), Science 328(5980), 876-878

```
py3plex.algorithms.statistics.multilayer_statistics.multilayer_motif_frequency(network:
                                                                    Any,
                                                                    motif_size:
                                                                    int = 3) →
                                                                    Dict[str,
                                                                    float]
```

Calculate multilayer motif frequency (f).

Formula: $f = n / n$

Frequency of recurring subgraph patterns across layers.

Variables:

n = count of motif type m n = total count of all motifs

Note: This is a simplified implementation counting basic patterns (intra-layer vs. inter-layer triangles). Complete multilayer motif enumeration includes many more configurations and is computationally expensive.

Parameters

- **network** – py3plex multi_layer_network object
- **motif_size** – Size of motifs to count (default: 3 for triangles)

Returns

Dictionary of motif type frequencies

Examples

```
>>> motifs = multilayer_motif_frequency(network, motif_size=3)
```

Reference:

Battiston et al. (2014), Section IV

```
py3plex.algorithms.statistics.multilayer_statistics.multiplex_betweenness centrality(network:
Any,
normalized:
bool
=
True,
weight:
str |
None
=
None)
→
Dict[Tuple[Any,
Any],
float]
```

Calculate multiplex betweenness centrality.

Computes betweenness centrality on the supra-graph, accounting for paths that traverse inter-layer couplings. This extends the standard betweenness definition to multiplex networks where paths can cross layers.

Formula: $B = ((i) /)$

where i is the total number of shortest paths from s to t , and (i) is the number of those paths passing through node i in layer $.$

Parameters

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize by the number of node pairs
- **weight** – Edge weight attribute name (None for unweighted)

Returns

Dictionary mapping (node, layer) tuples to betweenness centrality values

Examples

```
>>> betweenness = multiplex_betweenness centrality(network)
>>> top_nodes = sorted(betweenness.items(), key=lambda x: x[1], reverse=True)[:5]
```

Reference:

De Domenico et al. (2015), “Structural reducibility of multilayer networks”

`py3plex.algorithms.statistics.multilayer_statistics.multiplex_closeness centrality`(*network*: Any, *normalized*: bool = True, *weight*: str | None = None) → Dict[Tuple[Any, Any], float]

Calculate multiplex closeness centrality.

Computes closeness centrality on the supra-graph, where shortest paths can traverse inter-layer edges. This captures how quickly a node-layer can reach all other node-layers in the multiplex network.

Formula: $C = (N * L - 1) / d(i, j)$

where $d(i, j)$ is the shortest path distance from node i in layer to node j in layer , and $N * L$ is the total number of node-layer pairs.

Parameters

- **network** – py3plex multi_layer_network object
- **normalized** – Whether to normalize by network size
- **weight** – Edge weight attribute name (None for unweighted)

Returns

Dictionary mapping (node, layer) tuples to closeness centrality values

Examples

```
>>> closeness = multiplex_closeness centrality(network)
>>> central_nodes = {k: v for k, v in closeness.items() if v > 0.5}
```

Reference:

De Domenico et al. (2015), “Structural reducibility of multilayer networks”

`py3plex.algorithms.statistics.multilayer_statistics.multiplex_rich_club_coefficient`(*network*: Any, *k*: int, *normalized*: bool = True) → float

Calculate multiplex rich-club coefficient.

Measures the tendency of high-degree nodes to be more densely connected to each other than expected by chance, accounting for the multiplex structure.

Formula: $\langle k \rangle = E(>k) / (N(>k) * (N(>k)-1) / 2)$

where $E(>k)$ is the number of edges among nodes with overlapping degree $> k$, and $N(>k)$ is the number of such nodes.

Parameters

- **network** – py3plex multi_layer_network object
- **k** – Degree threshold
- **normalized** – Whether to normalize by random expectation

Returns

Rich-club coefficient value

Examples

```
>>> rich_club = multiplex_rich_club_coefficient(network, k=10)
>>> print(f"Rich-club coefficient: {rich_club:.3f}")
```

Reference:

Alstott et al. (2014), “powerlaw: A Python Package for Analysis of Heavy-Tailed Distributions” Extended to multiplex networks

`py3plex.algorithms.statistics.multilayer_statistics.node_activity`(*network*: Any, *node*: Any) → float

Calculate node activity (a).

Formula: $a = (1/L) \sum_{v \in V} V$

Fraction of layers in which node i is active (has at least one connection).

Variables:

L = total number of layers ($\sum_{v \in V} V$) = indicator function (1 if node i is active in layer, 0 otherwise) V = set of active nodes in layer

Parameters

- **network** – py3plex multi_layer_network object
- **node** – Node identifier

Returns

Activity value between 0 and 1

Examples

```
>>> activity = node_activity(network, 'A')
```

Reference:

Kivelä et al. (2014), J. Complex Networks 2(3), 203-271

Contracts:

- Precondition: network must not be None
- Precondition: node must not be None
- Postcondition: result is in [0, 1] (fraction of layers)
- Postcondition: result is not NaN

`py3plex.algorithms.statistics.multilayer_statistics.percolation_threshold`(*network: Any*,
removal_strategy: str = 'random',
trials: int = 10) → float

Estimate percolation threshold for the multiplex network.

Determines the fraction of nodes that must be removed before the network fragments into disconnected components. Uses sampling to estimate threshold.

Parameters

- **network** – py3plex multi_layer_network object
- **removal_strategy** – ‘random’, ‘degree’, or ‘betweenness’
- **trials** – Number of trials for averaging

Returns

Estimated percolation threshold (fraction of nodes)

Examples

```
>>> threshold = percolation_threshold(network, removal_strategy='degree')
>>> print(f"Percolation threshold: {threshold:.2%}")
```

Reference:

Buldyrev et al. (2010), “Catastrophic cascade of failures in interdependent networks”

`py3plex.algorithms.statistics.multilayer_statistics.resilience`(*network: Any*, *perturbation_type: str = 'layer_removal'*,
perturbation_param: str | float | None = None) → float

Calculate resilience (R).

Formula: $R = S' / S_0$

Ratio of largest connected component after perturbation to original size.

Variables:

S_0 = size of largest connected component in original network S' = size of largest connected component after perturbation

Perturbation types:

1. Layer removal: Remove all nodes/edges in a specific layer
2. Coupling removal: Remove a fraction of inter-layer edges

Properties:

$R = 1$ indicates full resilience (no impact from perturbation) $R = 0$ indicates complete fragmentation $0 < R < 1$ indicates partial resilience

Parameters

- **network** – py3plex multi_layer_network object
- **perturbation_type** – ‘layer_removal’ or ‘coupling_removal’
- **perturbation_param** – Layer to remove or fraction of inter-layer edges

Returns

Resilience ratio between 0 and 1

Examples

```
>>> r = resilience(network, 'layer_removal', perturbation_param='L1')
>>> r = resilience(network, 'coupling_removal', perturbation_param=0.5)
```

Reference:

Buldyrev et al. (2010), Nature 464, 1025-1028

py3plex.algorithms.statistics.multilayer_statistics.**supra_laplacian_spectrum**(network: Any, k: int = 10) → ndarray

Calculate supra-Laplacian spectrum ().

Formula: = -

Eigenvalue spectrum of the supra-Laplacian matrix; captures diffusion properties. Uses sparse eigenvalue computation when beneficial.

Variables:

= supra-adjacency matrix ($NL \times NL$ block matrix containing all layers and inter-layer couplings) = supra-degree matrix (diagonal matrix with row sums of) = supra-Laplacian matrix = $\{0, 1, \dots, 1\}$ with $0 = 0 \ 1 \dots \ 1$

Parameters

- **network** – py3plex multi_layer_network object
- **k** – Number of smallest eigenvalues to compute

Returns

Array of k smallest eigenvalues

Examples

```
>>> spectrum = supra_laplacian_spectrum(network, k=10)
```

Reference:

De Domenico et al. (2013), Gomez et al. (2013)

Notes

- Uses sparse eigsh() for sparse matrices (more efficient)
- Falls back to dense computation for small matrices ($n < 100$) or when k is large relative to n
- Laplacian is always symmetric for undirected graphs (PSD with smallest eigenvalue = 0)

`py3plex.algorithms.statistics.multilayer_statistics.targeted_layer_removal`(*network: Any*,
layer: str,
return_resilience: bool = False) →
Any | Tuple[Any, float]

Simulate targeted removal of an entire layer.

Removes all edges in a specified layer and returns the modified network or resilience score.

Parameters

- **network** – py3plex multi_layer_network object
- **layer** – Layer identifier to remove
- **return_resilience** – If True, return resilience score instead of network

Returns

Modified network or resilience score

Examples

```
>>> resilience = targeted_layer_removal(network, 'social', return_resilience=True)
>>> print(f"Resilience after removing social layer: {resilience:.3f}")
```

Reference:

Buldyrev et al. (2010), “Catastrophic cascade of failures”

`py3plex.algorithms.statistics.multilayer_statistics.unique_redundant_edges`(*network: Any*,
layer_i: str,
layer_j: str) →
Tuple[int, int]

Count unique and redundant edges between two layers.

Returns the number of edges unique to the first layer and the number of edges present in both layers (redundant).

Parameters

- **network** – py3plex multi_layer_network object
- **layer_i** – First layer identifier
- **layer_j** – Second layer identifier

Returns

Tuple of (unique_edges, redundant_edges)

Examples

```
>>> unique, redundant = unique_redundant_edges(network, 'social', 'work')
>>> print(f"Unique: {unique}, Redundant: {redundant}")
```

`py3plex.algorithms.statistics.multilayer_statistics.versatility_centrality`(*network: Any*,
centrality_type: str = 'degree',
alpha: Dict[str, float] | None = None) →
Dict[Any, float]

Calculate versatility centrality (V).

Formula: $V = w C$

Weighted combination of node centrality values across layers; measures overall influence.

Variables:

w = weight for layer (typically $1/L$ for uniform weighting, $w = 1$) C = centrality of node i in layer (can be degree, betweenness, closeness, etc.)

Parameters

- **network** – py3plex multi_layer_network object
- **centrality_type** – Type of centrality ('degree', 'betweenness', 'closeness')
- **alpha** – Layer weights (default: uniform weights)

Returns

Dictionary mapping nodes to versatility centrality values

Examples

```
>>> versatility = versatility_centrality(network, centrality_type='degree')
```

Reference:

De Domenico et al. (2015), Nature Communications 6, 6868

`py3plex.algorithms.statistics.topology.basic_pl_stats`(*degree_sequence: List[int]*) → *Tuple[float, float]*

Calculate basic power law statistics for a degree sequence.

Parameters

degree_sequence – Degree sequence of individual nodes

Returns

Tuple of (alpha, sigma) values

`py3plex.algorithms.statistics.topology.plot_power_law`(*degree_sequence: List[int]*, *title: str*, *xlabel: str*, *plabel: str*, *ylabel: str = 'Number of nodes'*, *formula_x: int = 70*, *formula_y: float = 0.05*, *show: bool = True*, *use_normalization: bool = False*) → *Any*

`py3plex.algorithms.statistics.correlation_networks.default_correlation_to_network(matrix: ndarray, input_type: str = 'matrix', preprocess: str = 'standard') → ndarray`

Convert correlation matrix to network using optimal thresholding.

Parameters

- **matrix** – Input data matrix
- **input_type** – Type of input (default: “matrix”)
- **preprocess** – Preprocessing method (default: “standard”)

Returns

Binary adjacency matrix

`py3plex.algorithms.statistics.correlation_networks.pick_threshold(matrix: ndarray) → float`

Pick optimal threshold for correlation network construction.

Parameters

matrix – Input data matrix

Returns

Optimal threshold value

`py3plex.algorithms.statistics.basic_statistics.core_network_statistics(G: Graph, labels: Any | None = None, name: str = 'example') → DataFrame`

Compute core statistics for a network.

Parameters

- **G** – NetworkX graph to analyze
- **labels** – Optional label matrix with shape attribute
- **name** – Name identifier for the network (default: “example”)

Returns

DataFrame containing network statistics

`py3plex.algorithms.statistics.basic_statistics.ensure(*args, **kwargs)`

`py3plex.algorithms.statistics.basic_statistics.identify_n_hubs(G: Graph, top_n: int = 100, node_type: str | None = None) → Dict[Any, int]`

Identify the top N hub nodes in a network based on degree centrality.

Parameters

- **G** – NetworkX graph to analyze
- **top_n** – Number of top hubs to return (default: 100)

- **node_type** – Optional filter for specific node type

Returns

Dictionary mapping node identifiers to their degree values

Contracts:

- Precondition: top_n must be positive
- Postcondition: result has at most top_n entries
- Postcondition: all degree values are non-negative integers

```
py3plex.algorithms.statistics.basic_statistics.require(*args, **kwargs)
```

Multilayer Algorithms

Multilayer/Multiplex Network Centrality Measures

This module implements various centrality measures for multilayer and multiplex networks, following standard definitions from multilayer network analysis literature.

Weight Handling Notes:

For path-based centralities (betweenness, closeness): - Edge weights from the supra-adjacency matrix are converted to distances (inverse of weight) - NetworkX algorithms use these distances for shortest path computation - betweenness centrality: weight parameter specifies the edge attribute for path computation - closeness centrality: distance parameter specifies the edge attribute for path computation

For disconnected graphs: - closeness centrality uses wf_improved parameter (Wasserman-Faust scaling) by default - When wf_improved=True, scores are normalized by reachable nodes only - When wf_improved=False, unreachable nodes contribute infinite distance

Weight Constraints: - Weights should be positive (> 0) for shortest path algorithms - Zero or negative weights will cause undefined behavior - For unweighted analysis, use weighted=False parameters

Authors: py3plex contributors Date: 2025

```
class py3plex.algorithms.multilayer_algorithms.centrality.MultilayerCentrality(network:  
Any)
```

Bases: object

Class for computing centrality measures on multilayer networks.

This class provides implementations of various centrality measures specifically designed for multilayer/multiplex networks, including degree-based, eigenvector-based, and path-based measures.

accessibility centrality(h=2)

Compute accessibility centrality (entropy-based reach within h steps).

Accessibility measures the diversity of nodes reachable within h steps using entropy of the probability distribution.

Access_r = $\exp(H_r)$ where H_r is the entropy of the h-step distribution

Parameters

h – Number of steps (default: 2)

Returns

{(node, layer): accessibility}

Return type
dict

Note

Accessibility is measured as the effective number of h-step destinations, using the entropy of the random walk distribution.

aggregate_to_node_level(*node_layer_centralities*, *method*='sum', *weights*=None)

Aggregate node-layer centralities to node level.

Parameters

- **node_layer_centralities** – dict with {(node, layer): value} entries
- **method** – 'sum', 'mean', 'max', 'weighted_sum'
- **weights** – dict with {layer: weight} for weighted_sum method

Returns

{node: aggregated_value}

Return type
dict

bridging centrality()

Compute bridging centrality for nodes in the supra-graph.

Bridging centrality combines betweenness with the bridging coefficient, which measures how much a node connects sparse regions of the network.

$\text{Bridging}(u) = B(u) * \text{BCoeff}(u)$ where $\text{BCoeff}(u) = (1 / k_u) * \sum_{v \in N(u)} [1 / k_v]$

Returns

{(node, layer): bridging centrality}

Return type
dict

Note

Nodes with higher bridging centrality act as important bridges connecting different parts of the network.

collective_influence(*radius*=2)

Compute collective influence (CI_) for multiplex networks.

Collective influence identifies influential spreaders by considering not just immediate neighbors but also nodes at distance .

$\text{CI}_-(u) = (k_u - 1) * \sum_{v \in \text{Ball}_-(u)} (k_v - 1)$

Parameters

radius – Radius for the ball boundary (default: 2)

Returns

{(node, layer): collective_influence}

Return type
dict

Note

Uses overlapping degree across all layers for each physical node.

communicability_betweenness centrality(*normalized=True*)

Compute communicability betweenness centrality (Estrada-style).

This measure quantifies how much a node contributes to the communicability between other pairs of nodes. It uses the matrix exponential to account for all walks between nodes.

Returns

{(node, layer): communicability_betweenness}

Return type

dict

Note

This is computationally expensive as it requires computing the matrix exponential multiple times. For large networks, this may take significant time.

current_flow_betweenness centrality()

Compute current-flow betweenness centrality via supra Laplacian pseudoinverse.

This measure is based on the electrical current flow through each node.

Returns

{(node, layer): current_flow_betweenness}

Return type

dict

current_flow_closeness centrality()

Compute current-flow closeness centrality via supra Laplacian pseudoinverse.

This measure is based on the resistance distance in electrical networks.

Returns

{(node, layer): current_flow_closeness}

Return type

dict

edge_betweenness centrality(*normalized=True*)

Compute edge betweenness centrality on the supra-graph.

Edge betweenness measures the fraction of shortest paths that pass through each edge.

Returns

{(source, target): edge_betweenness}

Return type

dict

Note

This returns edge-level centrality, not node-level.

flow_betweenness centrality(*samples=100*)

Compute flow betweenness based on maximum flow.

Flow betweenness measures how much flow passes through a node when maximizing flow between random source-target pairs.

Parameters

samples – Number of source-target pairs to sample (default: 100)

Returns

{(node, layer): flow_betweenness}

Return type

dict

Note

This uses NetworkX's maximum flow algorithms. For large networks, this can be computationally expensive.

harmonic_closeness centrality()

Compute harmonic closeness centrality on the supra-graph.

Harmonic closeness handles disconnected graphs better than standard closeness by summing the reciprocals of distances instead of taking reciprocal of sum.

$HC(u) = \sum_v (1 / d(u,v))$ for finite distances

Returns

{(node, layer): harmonic_closeness}

Return type

dict

Note

This measure naturally handles disconnected components as unreachable nodes contribute 0 (instead of infinity) to the sum.

hits centrality(*max_iter=1000, tol=1e-06*)

Compute HITS (hubs and authorities) centrality on the supra-graph.

For undirected networks, this equals eigenvector centrality. For directed networks, computes separate hub and authority scores.

Parameters

- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

Returns

If directed network: {'hubs': {(node, layer): score}, 'authorities': {(node, layer): score}}

If undirected network: {(node, layer): score} (equivalent to eigenvector centrality)

Return type

dict

information centrality()

Compute Information Centrality (Stephenson-Zelen style) on the supra-graph.

Information centrality measures the importance of a node based on the information flow through the network. It uses the inverse of a modified Laplacian matrix.

For each physical node u :

$$I(u) = 1 / \text{mean}_{\{a \text{ in } U\}} [G[a,a] - (2/(N*L)) * \sum_b G[a,b] + (1/(N*L)^2) * \sum_{\{b,c\}} G[b,c]]$$

where G is the inverse of $B = L + (1/(N*L)) * 1 * 1^T$

Returns

$\{(node, layer): \text{information_centrality}\}$

Return type

dict

Note

This implementation uses NetworkX's `information centrality` for computation. Falls back to harmonic closeness if information centrality computation fails.

katz_bonacich centrality(alpha=0.1, beta=None)

Compute Katz-Bonacich centrality on the supra-graph.

$$z = \lim_{t \rightarrow \infty} M^t b = (I - M)^{-1} b$$

Parameters

- **alpha** – Attenuation parameter (should be $< 1/(M)$). If None, automatically computes a safe value as $0.85/(M)$ where (M) is the spectral radius.
- **beta** – Exogenous preference vector. If None, uses vector of ones.

Returns

$\{(node, layer): \text{centrality_value}\}$

Return type

dict

layer_degree centrality(layer: str | None = None, weighted: bool = False, direction: str = 'out') → Dict[str | Tuple[str, str], float]

Compute layer-specific degree (or strength) centrality.

For undirected networks:

$$k^u_i = \sum_j 1(A^u_{ij} > 0) \text{ [unweighted]} \quad s^u_i = \sum_j A^u_{ij} \text{ [weighted]}$$

For directed networks:

$$k^{[out]}_i = \sum_j 1(A^{[out]}_{ij} > 0) \text{ [out-degree]} \quad k^{[in]}_i = \sum_j 1(A^{[in]}_{ji} > 0) \text{ [in-degree]}$$

Parameters

- **layer** – Layer to compute centrality for. If None, compute for all layers.
- **weighted** – If True, compute strength instead of degree.
- **direction** – 'out', 'in', or 'both' for directed networks.

Returns

$\{(node, layer): \text{centrality_value}\}$ if **layer** is None,
 $\{node: \text{centrality_value}\}$ if layer is specified.

Return type
dict

load_centrality()

Compute load centrality (shortest-path load).

Load centrality measures the fraction of shortest paths that pass through each node, counting all paths (not just unique pairs).

Load[k] = sum over all (s,t) pairs of [number of shortest paths through k / total shortest paths]

Returns
{(node, layer): load_centrality}

Return type
dict

Note

This is similar to betweenness but counts all paths rather than normalizing by the number of node pairs.

local_efficiency_centrality()

Compute local efficiency centrality.

Local efficiency measures how efficiently information is exchanged among a node's neighbors when the node is removed. It quantifies the fault tolerance of the network.

$LE(u) = (1 / ((N_u) * (N_u - 1))) * \sum_{ij \in N_u} [1 / d(i,j)]$

Returns
{(node, layer): local_efficiency}

Return type
dict

Note

For nodes with less than 2 neighbors, local efficiency is 0.

lp_aggregated_centrality(layer_centralities, p=2, weights=None)

Compute Lp-aggregated per-layer centrality.

Aggregates per-layer centrality values using Lp norm.

$C_i = (\sum_{l=1..L} w_l * |c^l[i]|^p)^{1/p}$ for $p < \infty$ $C_i = \max_l \{ w_l * |c^l[i]| \}$ for $p = \infty$

Parameters

- **layer_centralities** – dict of {layer: {node: centrality}} or {(node, layer): centrality}
- **p** – Lp norm parameter (default: 2). Use float('inf') for L-infinity norm.
- **weights** – dict of {layer: weight}. If None, uniform weights are used.

Returns
{node: aggregated_centrality}

Return type
dict

Note

This is a framework for aggregating any per-layer centrality measure. Input can be degree, PageRank, eigenvector, etc.

multilayer_betweenness centrality(*normalized=True, endpoints=False*)

Compute betweenness centrality on the supra-graph.

For each node-layer pair (i,), computes the fraction of shortest paths between all pairs of nodes that pass through (i,).

Parameters

- **normalized** – Whether to normalize the betweenness values.
- **endpoints** – Whether to include endpoints in path counts.

Returns

{(node, layer): betweenness centrality}

Return type

dict

Note

This is computationally expensive for large networks as it requires computing shortest paths between all pairs of nodes.

Weight handling: Edge weights from the supra-adjacency matrix are converted to distances (1/weight) for shortest path computation. Weights must be positive (> 0). Zero or negative weights will cause undefined behavior.

multilayer_closeness centrality(*normalized=True, wf_improved=True*)

Compute closeness centrality on the supra-graph.

For each node-layer pair (i,), computes: $C_c(i,) = (n-1) / \sum_{j,} d((i,), (j,))$

where $d((i,), (j,))$ is the shortest path distance in the supra-graph.

Parameters

- **normalized** – Whether to normalize by (n-1).
- **wf_improved** – If True, use Wasserman-Faust improved closeness scaling for disconnected graphs. Default is True. This affects the magnitude and ordering of scores in graphs with multiple components (e.g., low interlayer coupling). See NetworkX documentation for details.

Returns

{(node, layer): closeness centrality}

Return type

dict

Note

This implementation uses NetworkX's shortest path algorithms on the supra-graph representation. For large networks, this can be computationally expensive.

The `wf_improved` parameter controls how closeness is computed for nodes that cannot reach all other nodes. When True (default), uses the Wasserman-Faust formula that considers reachable nodes only, making scores comparable across disconnected components.

multiplex_coreness()

Alias for `multiplex_k_core` for compatibility.

Returns

{(node, layer): core_number}

Return type

dict

multiplex_eigenvector_centrality(max_iter: int = 1000, tol: float = 1e-06) → Dict

Compute multiplex eigenvector centrality (node-layer level).

$x = (1/_\text{max}) * M * x$ where $x_{\{i\}}$ is the centrality of node i in layer , and $_\text{max}$ is the spectral radius of the supra-adjacency matrix M .

Parameters

- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

Returns

{(node, layer): centrality_value}

Return type

dict

multiplex_eigenvector_versatility(max_iter: int = 1000, tol: float = 1e-06) → Dict

Compute node-level eigenvector versatility.

$x_i = _x_{\{i\}}$

Parameters

- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

Returns

{node: versatility_value}

Return type

dict

multiplex_k_core()

Compute multiplex k-core decomposition.

A node belongs to the k-core if it has at least k neighbors in the multilayer network. This implementation computes the core number for each node-layer pair.

Returns

{(node, layer): core_number}

Return type

dict

overlapping_degree centrality(*weighted: bool = False*) → Dict

Compute overlapping degree/strength centrality (node level).

$k^{\{\text{over}\}}_i = \sum k_i$ [unweighted] $s^{\{\text{over}\}}_i = \sum s_i$ [weighted]

Parameters

weighted – If True, compute overlapping strength.

Returns

{node: centrality_value}

Return type

dict

pagerank centrality(*damping=0.85, max_iter=1000, tol=1e-06*)

Compute PageRank centrality on the supra-graph.

Uses the standard PageRank algorithm on the supra-adjacency matrix representing the multilayer network. Properly handles dangling nodes (nodes with no outgoing edges) via teleportation.

This implementation preserves sparsity when possible for memory efficiency.

Parameters

- **damping** – Damping parameter (typically 0.85).
- **max_iter** – Maximum number of iterations.
- **tol** – Tolerance for convergence.

Returns

{(node, layer): centrality_value}

Return type

dict

Mathematical Invariants:

- PageRank values sum to 1.0 (within tol=1e-6)
- All values are non-negative
- Converges for strongly connected components or with teleportation

participation coefficient(*weighted: bool = False*) → Dict

Compute participation coefficient across layers.

Measures how evenly a node's degree is distributed across layers: $P_i = 1 - \sum (k_i / k^{\{\text{over}\}}_i)^2$

Set $P_i = 0$ if $k^{\{\text{over}\}}_i = 0$.

Parameters

weighted – If True, use strength instead of degree.

Returns

{node: participation_coefficient}

Return type

dict

percolation centrality(*edge_activation_prob=0.5, trials=100*)

Compute percolation centrality using bond percolation Monte Carlo simulation.

Percolation centrality measures the importance of a node based on its role in maintaining network connectivity under random edge failures.

Parameters

- **edge_activation_prob** – Probability that an edge is active (default: 0.5)
- **trials** – Number of Monte Carlo trials (default: 100)

Returns

{(node, layer): percolation_centrality}

Return type

dict

Note

Higher values indicate nodes that belong to larger connected components across different percolation realizations.

spreading_centrality(*beta=0.2, mu=0.1, trials=50, steps=100*)

Compute spreading (epidemic) centrality using SIR model.

Spreading centrality measures how influential a node is in spreading information or disease through the network.

Parameters

- **beta** – Infection rate (default: 0.2)
- **mu** – Recovery rate (default: 0.1)
- **trials** – Number of simulation trials per node (default: 50)
- **steps** – Maximum simulation steps (default: 100)

Returns

{(node, layer): spreading_centrality}

Return type

dict

Note

This measures the average outbreak size when seeding from each node. Requires discrete-time SIR simulation on the supra-graph.

subgraph_centrality()

Compute subgraph centrality via matrix exponential of the supra-adjacency matrix.

Subgraph centrality counts closed walks of all lengths starting and ending at each node. $SC_i = (e^A)_{ii}$ where A is the adjacency matrix.

Returns

{(node, layer): subgraph_centrality}

Return type

dict

supra_degree_centrality(*weighted: bool = False*) → Dict

Compute supra degree/strength centrality (node-layer level).

$k_{\{i\}} = \sum_j \mathbb{1}(M_{\{(i),(j)\}} > 0)$ [unweighted] $s_{\{i\}} = \sum_j M_{\{(i),(j)\}}$ [weighted]

Parameters

weighted – If True, compute strength instead of degree.

Returns

{(node, layer): centrality_value}

Return type

dict

total_communicability()

Compute total communicability via matrix exponential.

Total communicability is the row sum of the matrix exponential: $TC_i = \sum_j (e^A)_{ij}$

Returns

{(node, layer): total_communicability}

Return type

dict

```
py3plex.algorithms.multilayer_algorithms.centrality.communicability_centrality(supra_matrix:  
    spmatrix,  
    normalize:  
    bool = True,  
    use_sparse:  
    bool = True,  
    max_iter: int  
    = 100, tol:  
    float =  
    1e-06) →  
    ndarray
```

Compute communicability centrality for each node-layer pair.

Communicability centrality measures the weighted sum of all walks between nodes, with exponentially decaying weights for longer walks. It is computed as the row sum of the matrix exponential:

$$c_i = \sum_j (\exp(A))_{ij}$$

This implementation uses `scipy.sparse.linalg.expm_multiply` for efficient sparse matrix exponential computation.

Parameters

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **normalize** – If True, normalize output to sum to 1.
- **use_sparse** – If True, use sparse matrix operations. Falls back to dense for matrices smaller than 10000 elements.
- **max_iter** – Maximum number of iterations for sparse approximation (currently unused).
- **tol** – Tolerance for convergence (currently unused).

Returns

Communicability centrality scores for each node-layer pair (n,).

Return type

np.ndarray

Raises

Py3plexMatrixError – If matrix is invalid (non-square, empty, etc.).

Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> comm = communicability centrality(A)
>>> print(f"Communicability centrality computed for {len(comm)} node-layer pairs")
```

```
py3plex.algorithms.multilayer_algorithms.centrality.compute_all_centralities(network, in-
include_path_based=False, in-
include_advanced=False, in-
include_extended=False,
preset=None,
wf_improved=True)
```

Compute all available centrality measures for a multilayer network.

Parameters

- **network** – py3plex multi_layer_network object
- **include_path_based** – Whether to include computationally expensive path-based measures (betweenness, closeness). Default: False
- **include_advanced** – Whether to include advanced measures (HITS, current-flow, communicability, k-core). Default: False
- **include_extended** – Whether to include extended measures (information, accessibility, percolation, spreading, collective influence, load, flow betweenness, harmonic, bridging, local efficiency). Default: False
- **preset** – Convenience parameter to set all inclusion flags at once. Options: - ‘basic’: Only degree and eigenvector-based measures (default behavior) - ‘standard’: Includes path-based measures - ‘advanced’: Includes path-based and advanced measures - ‘all’: Includes all measures (path-based, advanced, and extended) - None: Use individual flags (default)
- **wf_improved** – If True, use Wasserman-Faust improved scaling for closeness centrality in disconnected graphs. Default: True

Returns

Dictionary containing all computed centrality measures with keys:

- Degree-based: “layer_degree”, “layer_strength”, “supra_degree”, “supra_strength”, “overlapping_degree”, “overlapping_strength”, “participation_coefficient”, “participation_coefficient_strength”
- Eigenvector-based: “multiplex_eigenvector”, “eigenvector_versatility”, “katz_bonacich”, “pagerank”
- Path-based (if include_path_based=True): “closeness”, “betweenness”
- Advanced (if include_advanced=True): “hits”, “current_flow_closeness”, “current_flow_betweenness”, “subgraph_centrality”, “total_communicability”, “multiplex_k_core”
- Extended (if include_extended=True): “information”, “communicability_betweenness”, “accessibility”, “harmonic_closeness”, “local_efficiency”, “edge_betweenness”, “bridging”, “percolation”, “spreading”, “collective_influence”, “load”, “flow_betweenness”

Return type
dict

Note

Path-based, advanced, and extended measures are computationally expensive for large networks. Use flags or presets to control which measures are computed.

Examples

```
>>> # Compute only basic measures (fast)
>>> results = compute_all_centralities(network)
```

```
>>> # Use preset for standard analysis
>>> results = compute_all_centralities(network, preset='standard')
```

```
>>> # Compute everything
>>> results = compute_all_centralities(network, preset='all')
```

```
>>> # Fine-grained control
>>> results = compute_all_centralities(
...     network,
...     include_path_based=True,
...     include_extended=True
... )
```

`py3plex.algorithms.multilayer_algorithms centrality.katz_centrality`(*supra_matrix*: *spmatrix*,
alpha: *float* | *None* = *None*,
beta: *float* = 1.0, *tol*: *float*
= 1e-06) → *ndarray*

Compute Katz centrality for each node-layer pair.

Katz centrality measures node influence by accounting for all paths with exponentially decaying weights. It is computed as:

$$x = (I - \alpha * A)^{-1} * \beta * 1$$

where $\alpha < 1/\lambda_{\max}(A)$ to ensure convergence. If α is not provided, it defaults to $0.85 / \lambda_{\max}(A)$.

Parameters

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **alpha** – Attenuation parameter. Must be less than $1/\text{spectral_radius}(A)$. If *None*, defaults to $0.85 / \lambda_{\max}(A)$.
- **beta** – Weight of exogenous influence (typically 1.0).
- **tol** – Tolerance for eigenvalue computation.

Returns

Katz centrality scores for each node-layer pair (n,).
Normalized to sum to 1.

Return type

np.ndarray

Raises*Py3plexMatrixError* – If matrix is invalid or alpha is out of valid range.**Example**

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> katz = katz_centrality(A)
>>> print(f"Katz centrality computed for {len(katz)} node-layer pairs")
>>> # With custom alpha
>>> katz_custom = katz_centrality(A, alpha=0.05)
```

MultiXRank: Random Walk with Restart on Universal Multilayer Networks

This module implements the MultiXRank algorithm described in: Baptista et al. (2022), “Universal multilayer network exploration by random walk with restart”, Communications Physics, 5, 170.

MultiXRank performs random walk with restart (RWR) on a supra-heterogeneous adjacency matrix built from multiple multiplexes connected by bipartite blocks.

References

- Paper: <https://doi.org/10.1038/s42005-022-00937-9>
- arXiv: <https://arxiv.org/abs/2106.07869>
- Package: <https://github.com/anthbapt/multixrank>
- Docs: <https://multixrank-doc.readthedocs.io/>

```
class py3plex.algorithms.multilayer_algorithms.multixrank.MultiXRank(restart_prob: float = 0.4,
                                                                    epsilon: float = 1e-06,
                                                                    max_iter: int = 100000,
                                                                    verbose: bool = True)
```

Bases: object

MultiXRank: Universal multilayer network exploration by random walk with restart.

This class implements the MultiXRank algorithm for node prioritization and ranking in universal multilayer networks. It builds a supra-heterogeneous adjacency matrix from multiple multiplexes and bipartite inter-multiplex connections, then performs random walk with restart.

multiplexes

Dictionary of multiplex supra-adjacency matrices

Type

Dict[str, sp.spmatrix]

bipartite_blocks

Inter-multiplex connection matrices

Type

Dict[Tuple[str, str], sp.spmatrix]

restart_prob

Restart probability (r) for RWR

Type

float

epsilon

Convergence threshold

Type

float

max_iter

Maximum number of iterations

Type

int

node_order

Node ordering for each multiplex

Type

Dict[str, List]

supra_matrix

Built supra-heterogeneous adjacency matrix

Type

sp.spmatrix

transition_matrix

Column-stochastic transition matrix

Type

sp.spmatrix

add_bipartite_block(*multiplex_from*: str, *multiplex_to*: str, *bipartite_matrix*: spmatrix | ndarray, *weight*: float = 1.0)

Add a bipartite block connecting two multiplexes.

Parameters

- **multiplex_from** – Name of source multiplex
- **multiplex_to** – Name of target multiplex
- **bipartite_matrix** – Matrix of connections (rows: from, cols: to)
- **weight** – Optional weight to scale this bipartite block

add_multiplex(*name*: str, *supra_adjacency*: spmatrix | ndarray, *node_order*: List | None = None)

Add a multiplex to the universal multilayer network.

Parameters

- **name** – Unique identifier for this multiplex
- **supra_adjacency** – Supra-adjacency matrix for the multiplex (can be from `multi_layer_network.get_supra_adjacency_matrix()`)
- **node_order** – Optional list of node IDs in the order they appear in the matrix. If None, uses integer indices.

aggregate_scores(*scores*: ndarray, *aggregation*: str = 'sum') → Dict[str, Dict]

Aggregate scores per multiplex and optionally per physical node.

Parameters

- **scores** – Probability vector from RWR (length = total supra-matrix dimension)
- **aggregation** – How to aggregate scores ('sum', 'mean', 'max')

Returns

Dictionary mapping multiplex names to dictionaries of node scores

build_supra_heterogeneous_matrix(*block_weights*: Dict[str | Tuple[str, str], float] | None = None)

Build the supra-heterogeneous adjacency matrix S.

This constructs the universal multilayer network matrix by: 1. Placing each multiplex supra-adjacency on the block diagonal 2. Adding bipartite blocks for inter-multiplex connections

Parameters

block_weights – Optional dictionary to weight blocks. Keys can be: - Multiplex names (to weight within-multiplex edges) - Tuple (multiplex_from, multiplex_to) for bipartite blocks

Returns

The supra-heterogeneous adjacency matrix

column_normalize(*handle_dangling*: str = 'uniform') → spmatrix

Column-normalize the supra-heterogeneous matrix to create a stochastic transition matrix.

This ensures each column sums to 1, making the matrix suitable for RWR.

Parameters

handle_dangling – How to handle dangling nodes (columns with zero sum): - 'uniform': Distribute mass uniformly across all nodes - 'self': Add self-loop (mass stays at the node) - 'ignore': Leave as zero (not recommended for RWR)

Returns

Column-stochastic transition matrix

get_top_ranked(*scores*: ndarray, *k*: int = 10, *multiplex*: str | None = None, *exclude_seeds*: bool = True, *seed_nodes*: List[int] | Dict[str, List] | None = None) → List[Tuple]

Get top-k ranked nodes from RWR scores.

Parameters

- **scores** – Probability vector from RWR
- **k** – Number of top nodes to return
- **multiplex** – If specified, only return nodes from this multiplex
- **exclude_seeds** – Whether to exclude seed nodes from results
- **seed_nodes** – Seed nodes to exclude (if exclude_seeds=True)

Returns

List of (global_index, score) tuples, sorted by score descending

random_walk_with_restart(*seed_nodes*: List[int] | Dict[str, List] | ndarray, *seed_weights*: ndarray | None = None, *multiplex_name*: str | None = None) → ndarray

Perform Random Walk with Restart (RWR) from seed nodes.

Parameters

- **seed_nodes** – Seed node specification. Can be: - List of global indices in the supra-matrix - Dict mapping multiplex names to lists of local node indices - NumPy array of global indices
- **seed_weights** – Optional weights for seed nodes (must match length of seed_nodes). If None, uniform weights are used.
- **multiplex_name** – If seed_nodes is a list/array of local indices, specify which multiplex they belong to

Returns

Steady-state probability vector (length = total nodes across all multiplexes)

```

py3plex.algorithms.multilayer_algorithms.multixrank.multixrank_from_py3plex_networks(networks:
    Dict[str,
    multi-
    net.multi_layer_network
    bi-
    partite_connections:
    Dict[Tuple[str,
    str],
    sp-
    ma-
    trix |
    ndarray] |
    None
    =
    None,
    seed_nodes:
    Dict[str,
    List]
    |
    None
    =
    None,
    restart_prob:
    float
    =
    0.4,
    epsilon:
    float
    = 1e-
    06,
    max_iter:
    int =
    100000,
    verbose:
    bool
    =
    True)
    →
    Tuple[MultiXRank,
    ndarray]

```

Convenience function to run MultiXRank on py3plex multi_layer_network objects.

Parameters

- **networks** – Dictionary mapping names to multi_layer_network objects
- **bipartite_connections** – Optional dict of inter-network connection matrices
- **seed_nodes** – Dict mapping network names to lists of seed node IDs
- **restart_prob** – Restart probability for RWR

- **epsilon** – Convergence threshold
- **max_iter** – Maximum iterations
- **verbose** – Whether to log progress

Returns

Tuple of (MultiXRank object, scores array)

Example

```
>>> from py3plex.core import multinet
>>> net1 = multinet.multi_layer_network()
>>> net1.load_network('network1.edgelist', ...)
>>> net2 = multinet.multi_layer_network()
>>> net2.load_network('network2.edgelist', ...)
>>>
>>> networks = {'net1': net1, 'net2': net2}
>>> seed_nodes = {'net1': ['node1', 'node2']}
>>>
>>> mxr, scores = multixrank_from_py3plex_networks(
...     networks, seed_nodes=seed_nodes
... )
>>>
>>> # Get aggregated scores per network
>>> aggregated = mxr.aggregate_scores(scores)
```

Authors: Benjamin Renoust (github.com/renoust) Date: 2018/02/13 Description: Loads a Detangler JSON format graph and compute unweighted entanglement analysis with Py3Plex

`py3plex.algorithms.multilayer_algorithms.entanglement.build_occurrence_matrix(network: Any)`
→
Tuple[ndarray,
List[Any]]

Build occurrence matrix from multilayer network.

Parameters

network – Multilayer network object

Returns

Tuple of (c_matrix, layers) where c_matrix is the normalized occurrence matrix and layers is the list of layer names

`py3plex.algorithms.multilayer_algorithms.entanglement.compute_blocks(c_matrix: ndarray)` →
Tuple[List[List[int]],
List[ndarray]]

Compute block decomposition of occurrence matrix.

Parameters

c_matrix – Occurrence matrix

Returns

Tuple of (indices, blocks) where indices are the layer indices in each block and blocks are the submatrices for each block

`py3plex.algorithms.multilayer_algorithms.entanglement.compute_entanglement(block_matrix: ndarray)` →
Tuple[List[float],
List[float]]

Compute entanglement metrics for a block.

Parameters

block_matrix – Block submatrix

Returns

Tuple of ([intensity, homogeneity, normalized_homogeneity], gamma_layers)

```
py3plex.algorithms.multilayer_algorithms.entanglement.compute_entanglement_analysis(network:
                                                                                          Any)
                                                                                          →
                                                                                          List[Dict[str,
                                                                                          Any]]
```

Compute full entanglement analysis for a multilayer network.

Parameters

network – Multilayer network object

Returns

List of block analysis dictionaries with entanglement metrics

Supra Matrix Function Centralities for Multilayer Networks.

This module implements centrality measures based on matrix functions of the supra-adjacency matrix, including: - Communicability Centrality (Estrada & Hatano, 2008) - Katz Centrality (Katz, 1953)

These measures operate directly on the sparse supra-adjacency matrix obtained from `multi_layer_network.get_supra_adjacency_matrix()`.

References

- Estrada, E., & Hatano, N. (2008). Communicability in complex networks. *Physical Review E*, 77(3), 036111.
- Katz, L. (1953). A new status index derived from sociometric analysis. *Psychometrika*, 18(1), 39-43.

Authors: py3plex contributors Date: October 2025 (Phase II)

`py3plex.algorithms.multilayer_algorithms.supra_matrix_function_centrality.communicability_centrality(supra_matrix, normalize, use_sparse, max_iter, tol, float_precision)`

Compute communicability centrality for each node-layer pair.

Communicability centrality measures the weighted sum of all walks between nodes, with exponentially decaying weights for longer walks. It is computed as the row sum of the matrix exponential:

$$c_i = \sum_j (\exp(A))_{ij}$$

This implementation uses `scipy.sparse.linalg.expm_multiply` for efficient sparse matrix exponential computation.

Parameters

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **normalize** – If True, normalize output to sum to 1.
- **use_sparse** – If True, use sparse matrix operations. Falls back to dense for matrices smaller than 10000 elements.
- **max_iter** – Maximum number of iterations for sparse approximation (currently unused).
- **tol** – Tolerance for convergence (currently unused).

Returns

Communicability centrality scores for each node-layer pair (n,).

Return type

`np.ndarray`

Raises

`Py3plexMatrixError` – If matrix is invalid (non-square, empty, etc.).

Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> comm = communicability centrality(A)
>>> print(f"Communicability centrality computed for {len(comm)} node-layer pairs")
```

```
py3plex.algorithms.multilayer_algorithms.supra_matrix_function_centrality.katz_centrality(supra_matrix:
    sp-
    ma-
    trix,
    al-
    pha:
    float
    |
    None
    =
    None,
    beta:
    float
    =
    1.0,
    tol:
    float
    =
    1e-
    06)
    →
    ndarray
```

Compute Katz centrality for each node-layer pair.

Katz centrality measures node influence by accounting for all paths with exponentially decaying weights. It is computed as:

$$x = (I - \alpha * A)^{-1} * \beta * 1$$

where $\alpha < 1/\lambda_{\max}(A)$ to ensure convergence. If α is not provided, it defaults to $0.85 / \lambda_{\max}(A)$.

Parameters

- **supra_matrix** – Sparse supra-adjacency matrix (n x n).
- **alpha** – Attenuation parameter. Must be less than $1/\text{spectral_radius}(A)$. If None, defaults to $0.85 / \lambda_{\max}(A)$.
- **beta** – Weight of exogenous influence (typically 1.0).
- **tol** – Tolerance for eigenvalue computation.

Returns

Katz centrality scores for each node-layer pair (n,).

Normalized to sum to 1.

Return type

np.ndarray

Raises

Py3plexMatrixError – If matrix is invalid or alpha is out of valid range.

Example

```
>>> from py3plex.core import random_generators
>>> net = random_generators.random_multiplex_ER(50, 3, 0.1)
>>> A = net.get_supra_adjacency_matrix()
>>> katz = katz_centrality(A)
>>> print(f"Katz centrality computed for {len(katz)} node-layer pairs")
>>> # With custom alpha
>>> katz_custom = katz_centrality(A, alpha=0.05)
```

Multiplex Participation Coefficient (MPC) for multiplex networks.

This module implements the Multiplex Participation Coefficient metric for multiplex networks (networks with identical node sets across all layers).

Authors: py3plex contributors Date: 2025

`py3plex.algorithms.multicentrality.ensure(*args, **kwargs)`

`py3plex.algorithms.multicentrality.multiplex_participation_coefficient`(*multinet*: Any,
normalized: bool =
True, check_multiplex:
bool = True) →
Dict[Any, float]

Compute the Multiplex Participation Coefficient (MPC) for multiplex networks. MPC measures how evenly a node participates across layers.

Parameters

- **multinet** (`py3plex.core.multinet.multi_layer_network`) – Multiplex network object (same node set across layers).
- **normalized** (*bool*, *optional*) – Whether to normalize MPC to [0,1]. Default: True.
- **check_multiplex** (*bool*, *optional*) – Validate that all layers share the same node set.

Returns

Node → MPC value mapping.

Return type

dict

Notes**The MPC is computed as:**

$$\text{MPC}(i) = 1 - \sum_{} (k_i^{\text{layer}} / k_i^{\text{total}})^2$$

Where: k_i^{layer} is the degree of node i in layer $-$ k_i^{total} is the total degree of node i across all layers

When `normalized=True`, the result is multiplied by $L/(L-1)$ to normalize to [0,1] range, where L is the number of layers.

References

- Battiston, F., et al. (2014). “Structural measures for multiplex networks.” *Physical Review E*, 89(3), 032804.
- De Domenico, M., et al. (2015). “Identifying modular flows on multilayer networks reveals highly overlapping organization in interconnected systems.” *Physical Review X*, 5(1), 011027.
- Harooni, M., et al. (2025). “Centrality in Multilayer Networks: Accurate Measurements with MultiNetPy.” *The Journal of Supercomputing*, 81(1), 92. DOI: 10.1007/s11227-025-07197-8

Contracts:

- Precondition: `multinet` must not be `None`
- Precondition: `normalized` and `check_multiplex` must be booleans
- Postcondition: returns a dictionary with numeric values

```
py3plex.algorithms.multicentrality.require(*args, **kwargs)
```

General Algorithms

Random walk primitives for graph-based algorithms.

This module provides foundation for higher-level algorithms like `Node2Vec`, `DeepWalk`, and diffusion processes. Implements both basic and second-order (biased) random walks with proper edge weight handling and multilayer support.

Key Features:

- Basic random walks with weighted edge sampling
- Second-order (`Node2Vec`-style) biased random walks with `p/q` parameters
- Multiple simultaneous walks with deterministic reproducibility
- Support for directed, weighted, and multilayer networks
- Efficient sparse adjacency handling

References

- Grover, A., & Leskovec, J. (2016). `node2vec`: Scalable feature learning for networks. KDD ‘16. <https://doi.org/10.1145/2939672.2939754>
- Perozzi, B., Al-Rfou, R., & Skiena, S. (2014). `DeepWalk`: Online learning of social representations. KDD ‘14. <https://doi.org/10.1145/2623330.2623732>

```
py3plex.algorithms.general.walkers.basic_random_walk(G: Graph, start_node: int | str, walk_length:
                                                    int, weighted: bool = True, seed: int | None =
                                                    None) → List[str | int]
```

Perform a basic random walk on a graph with proper edge weight handling.

The next step is sampled proportionally to the normalized edge weights of the current node. For unweighted graphs, transitions are uniform.

Parameters

- **G** – NetworkX graph (directed or undirected, weighted or unweighted)
- **start_node** – Node to start the walk from
- **walk_length** – Number of steps in the walk

- **weighted** – Whether to use edge weights (default: True)
- **seed** – Random seed for reproducibility (default: None)

Returns

List of nodes representing the walk path (includes start_node)

Raises

ValueError – If start_node not in graph or walk_length < 1

Examples

```
>>> G = nx.Graph()
>>> G.add_weighted_edges_from([(0, 1, 1.0), (1, 2, 2.0), (1, 3, 1.0)])
>>> walk = basic_random_walk(G, 0, walk_length=3, seed=42)
>>> len(walk)
4
>>> walk[0]
0
```

Note

- Handles disconnected nodes by terminating walk early
- Edge weights must be positive for weighted walks
- Sum of transition probabilities from any node equals 1.0

`py3plex.algorithms.general.walkers.general_random_walk(G, start_node, iterations=1000, teleportation_prob=0)`

Legacy random walk with teleportation (for backward compatibility).

Deprecated since version 0.95a: Use `basic_random_walk()` or `node2vec_walk()` instead. This function will be removed in version 1.0.

Parameters

- **G** – NetworkX graph
- **start_node** – Starting node
- **iterations** – Number of steps
- **teleportation_prob** – Probability of teleporting to random visited node

Returns

List of visited nodes (excluding start_node)

`py3plex.algorithms.general.walkers.generate_walks(G: Graph, num_walks: int, walk_length: int, start_nodes: List[str | int] | None = None, p: float = 1.0, q: float = 1.0, weighted: bool = True, return_edges: bool = False, seed: int | None = None) → List[List[str | int]] | List[List[Tuple[int | str, int | str]]]`

Generate multiple random walks from specified or all nodes.

This interface supports multiple simultaneous walks with deterministic reproducibility under fixed RNG seed. Can return either node sequences or edge sequences.

Parameters

- **G** – NetworkX graph
- **num_walks** – Number of walks to generate per start node
- **walk_length** – Number of steps in each walk
- **start_nodes** – Nodes to start walks from (if None, uses all nodes)
- **p** – Return parameter for Node2Vec (1.0 = no bias)
- **q** – In-out parameter for Node2Vec (1.0 = no bias)
- **weighted** – Whether to use edge weights
- **return_edges** – Return edge sequences instead of node sequences
- **seed** – Random seed for reproducibility

Returns

- List of nodes (if return_edges=False)
- List of edges as tuples (if return_edges=True)

Return type

List of walks, where each walk is either

Examples

```
>>> G = nx.karate_club_graph()
>>> # Generate 10 walks from each node
>>> walks = generate_walks(G, num_walks=10, walk_length=5, seed=42)
>>> len(walks)
340
```

```
>>> # Generate walks with Node2Vec bias
>>> walks = generate_walks(G, num_walks=5, walk_length=10, p=0.5, q=2.0, seed=42)
```

```
>>> # Get edge sequences
>>> edge_walks = generate_walks(G, num_walks=3, walk_length=5, return_edges=True,
↪seed=42)
```

Note

- With same seed, generates identical walks across runs
- If $p == q == 1.0$, uses basic random walk (faster)
- Empty walks are included if node has no neighbors

```
py3plex.algorithms.general.walkers.layer_specific_random_walk(G: Graph, start_node: int | str,
                                                             walk_length: int, layer: str | None
                                                             = None, cross_layer_prob: float =
                                                             0.0, weighted: bool = True, seed:
                                                             int | None = None) → List[str | int]
```

Perform random walk with layer constraints for multilayer networks.

In a multilayer network represented in py3plex format (where node names include layer information), this function can constrain walks to specific layers with occasional inter-layer transitions.

Parameters

- **G** – NetworkX graph (multilayer network in py3plex format)
- **start_node** – Node to start walk from (may include layer info)
- **walk_length** – Number of steps in the walk
- **layer** – Target layer to constrain walk to (None = no constraint)
- **cross_layer_prob** – Probability of crossing to different layer (0-1)
- **weighted** – Whether to use edge weights
- **seed** – Random seed for reproducibility

Returns

List of nodes representing the walk path

Examples

```
>>> # Multilayer network with layer-specific walks
>>> from py3plex.core import multinet
>>> network = multinet.multi_layer_network()
>>> network.add_layer("social")
>>> network.add_layer("biological")
>>> # ... add nodes and edges ...
>>> walk = layer_specific_random_walk(
...     network.core_network,
...     "nodeA---social",
...     walk_length=10,
...     layer="social",
...     cross_layer_prob=0.1
... )
```

Note

- If layer is None, behaves like basic_random_walk
- cross_layer_prob controls inter-layer transitions
- Node format should follow py3plex convention: “nodeID—layerID”

`py3plex.algorithms.general.walkers.node2vec_walk(G: Graph, start_node: int | str, walk_length: int, p: float = 1.0, q: float = 1.0, weighted: bool = True, seed: int | None = None) → List[str | int]`

Perform a second-order (biased) random walk following Node2Vec logic.

When transitioning from node $t \rightarrow v \rightarrow x$, the probability of choosing x is biased by parameters p (return) and q (in-out): - If $x == t$ (return to previous): weight / p - If x is neighbor of t (stay close): $\text{weight} / 1$ - If x is not neighbor of t (explore): weight / q

Parameters

- **G** – NetworkX graph (directed or undirected, weighted or unweighted)

- **start_node** – Node to start the walk from
- **walk_length** – Number of steps in the walk
- **p** – Return parameter (higher p = less likely to return to previous node)
- **q** – In-out parameter (higher q = less likely to explore further)
- **weighted** – Whether to use edge weights (default: True)
- **seed** – Random seed for reproducibility (default: None)

Returns

List of nodes representing the walk path (includes start_node)

Raises

ValueError – If $p \leq 0$ or $q \leq 0$ or start_node not in graph

Examples

```
>>> G = nx.Graph()
>>> G.add_edges_from([(0, 1), (1, 2), (1, 3), (0, 2)])
>>> # Low p, high q: tends to backtrack
>>> walk = node2vec_walk(G, 0, walk_length=5, p=0.1, q=10.0, seed=42)
>>> # High p, low q: tends to explore outward
>>> walk2 = node2vec_walk(G, 0, walk_length=5, p=10.0, q=0.1, seed=42)
```

Note

- First step is always a basic random walk (no previous node)
- Properly normalizes probabilities at each step
- Handles disconnected nodes by terminating early

References

Grover & Leskovec (2016), node2vec: Scalable feature learning for networks

Benchmark algorithms for node classification performance evaluation.

This module provides algorithms for benchmarking node classification performance, including oracle-based F1 score evaluation.

`py3plex.algorithms.general.benchmark_classification.evaluate_oracle_F1`(probs: ndarray, Y_real: ndarray) → tuple[float, float]

Evaluate oracle F1 scores for multi-label classification.

This function computes micro and macro F1 scores by selecting the top-k predictions for each sample, where k is determined by the ground truth number of labels.

Parameters

- **probs** – Predicted probability matrix of shape (n_samples, n_labels).
- **Y_real** – Ground truth binary label matrix of shape (n_samples, n_labels).

Returns

- micro: Micro-averaged F1 score

- **macro**: Macro-averaged F1 score

Return type

A tuple containing

Example

```
>>> probs = np.array([[0.9, 0.1, 0.2], [0.1, 0.8, 0.7]])
>>> Y_real = np.array([[1, 0, 0], [0, 1, 1]])
>>> micro, macro = evaluate_oracle_F1(probs, Y_real)
```

Node Ranking

`py3plex.algorithms.node_ranking.node_ranking.authority_matrix(graph: Graph) → spmatrix`

Get the authority matrix of a graph.

Parameters

graph – NetworkX graph

Returns

Authority matrix

`py3plex.algorithms.node_ranking.node_ranking.hub_matrix(graph: Graph) → spmatrix`

Get the hub matrix of a graph.

Parameters

graph – NetworkX graph

Returns

Hub matrix

`py3plex.algorithms.node_ranking.node_ranking.hubs_and_authorities(graph: Graph) → Tuple[dict, dict]`

Compute hubs and authorities scores using HITS algorithm.

Parameters

graph – NetworkX graph

Returns

Tuple of (hubs dictionary, authorities dictionary)

`py3plex.algorithms.node_ranking.node_ranking.modularity(G: Graph, communities: List[List[Any]], weight: str = 'weight') → float`

Calculate modularity of a graph partition.

Parameters

- **G** – NetworkX graph
- **communities** – List of communities (each community is a list of nodes)
- **weight** – Edge weight attribute name

Returns

Modularity value

`py3plex.algorithms.node_ranking.node_ranking.page_rank_kernel(index_row: int) → Tuple[int, ndarray]`

PageRank kernel for parallel computation.

Note: This function expects global variables `G`, `damping_hyper`, `spread_step_hyper`, `spread_percent_hyper`, and `graph` to be defined. It's designed for use with `multiprocessing.Pool.map()`.

Parameters

index_row – Row index to compute PageRank for

Returns

Tuple of (index, PageRank vector)

```
py3plex.algorithms.node_ranking.node_ranking.sparse_page_rank(matrix: spmatrix, start_nodes:
                                                                List[int] | range | None, epsilon:
                                                                float = 1e-06, max_steps: int =
                                                                100000, damping: float = 0.5,
                                                                spread_step: int = 10,
                                                                spread_percent: float = 0.3,
                                                                try_shrink: bool = False) →
                                                                ndarray
```

Compute sparse PageRank with personalization.

Parameters

- **matrix** – Sparse adjacency matrix (column-stochastic)
- **start_nodes** – List of starting node indices for personalization (can be range or None)
- **epsilon** – Convergence threshold
- **max_steps** – Maximum number of iterations
- **damping** – Damping factor (teleportation probability)
- **spread_step** – Maximum steps for spread calculation
- **spread_percent** – Percentage threshold for spread
- **try_shrink** – Whether to try matrix shrinking optimization

Returns

PageRank vector

```
py3plex.algorithms.node_ranking.node_ranking.stochastic_normalization(matrix: spmatrix) →
                                                                spmatrix
```

Normalize a sparse matrix stochastically.

Parameters

matrix – Sparse matrix to normalize

Returns

Stochastically normalized sparse matrix

```
py3plex.algorithms.node_ranking.node_ranking.stochastic_normalization_hin(matrix: spmatrix)
                                                                → spmatrix
```

Normalize a heterogeneous information network matrix stochastically.

Parameters

matrix – Sparse matrix to normalize

Returns

Stochastically normalized sparse matrix

Network Classification

`py3plex.algorithms.network_classification.label_propagation.label_propagation`(*graph_matrix*:
spmatrix,
class_matrix:
ndarray,
alpha: *float* =
0.001, *epsilon*:
float = *1e-12*,
max_steps: *int*
= *100000*,
normalization:
str | *List[str]* =
'freq') →
ndarray

Propagate labels through a graph.

Parameters

- **graph_matrix** – Sparse graph adjacency matrix
- **class_matrix** – Initial class label matrix
- **alpha** – Propagation weight parameter
- **epsilon** – Convergence threshold
- **max_steps** – Maximum number of iterations
- **normalization** – Normalization scheme(s) to apply

Returns

Propagated label matrix

`py3plex.algorithms.network_classification.label_propagation.label_propagation_normalization`(*matrix*:
sp-
ma-
trix)
→
spmatrix

Normalize a matrix for label propagation.

Parameters

matrix – Sparse matrix to normalize

Returns

Normalized sparse matrix

`py3plex.algorithms.network_classification.label_propagation.label_propagation_tf`() → None
TensorFlow-based label propagation (TODO: implement).

Placeholder for future TensorFlow implementation.

`py3plex.algorithms.network_classification.label_propagation.normalize_amplify_freq`(*mat*:
ndar-
ray) →
ndarray

Normalize and amplify matrix by frequency.

Parameters

mat – Matrix to normalize

Returns

Normalized and amplified matrix

`py3plex.algorithms.network_classification.label_propagation.normalize_exp(mat: ndarray) → ndarray`

Apply exponential normalization.

Parameters

mat – Matrix to normalize

Returns

Exponentially normalized matrix

`py3plex.algorithms.network_classification.label_propagation.normalize_initial_matrix_freq(mat: ndarray) → ndarray`

Normalize matrix by frequency.

Parameters

mat – Matrix to normalize

Returns

Normalized matrix

`py3plex.algorithms.network_classification.label_propagation.normalize_none(mat: ndarray) → ndarray`

No normalization (identity function).

Parameters

mat – Matrix to return unchanged

Returns

Original matrix

```
py3plex.algorithms.network_classification.label_propagation.validate_label_propagation(core_network:
                                                                 sp-
                                                                 ma-
                                                                 trix,
                                                                 la-
                                                                 bels:
                                                                 ndar-
                                                                 ray
                                                                 |
                                                                 sp-
                                                                 ma-
                                                                 trix,
                                                                 dataset_name:
                                                                 str
                                                                 =
                                                                 'test',
                                                                 rep-
                                                                 e-
                                                                 ti-
                                                                 tions:
                                                                 int
                                                                 =
                                                                 5,
                                                                 nor-
                                                                 mal-
                                                                 iza-
                                                                 tion_scheme:
                                                                 str
                                                                 |
                                                                 List[str]
                                                                 =
                                                                 'ba-
                                                                 sic',
                                                                 al-
                                                                 pha_value:
                                                                 float
                                                                 =
                                                                 0.001,
                                                                 ran-
                                                                 dom_seed:
                                                                 int
                                                                 =
                                                                 123,
                                                                 ver-
                                                                 bose:
                                                                 bool
                                                                 =
                                                                 False)
                                                                 →
                                                                 DataFrame
```

Validate label propagation with cross-validation.

Parameters

- **core_network** – Sparse network adjacency matrix

- **labels** – Label matrix
- **dataset_name** – Name of the dataset
- **repetitions** – Number of repetitions
- **normalization_scheme** – Normalization scheme to use
- **alpha_value** – Alpha parameter for propagation
- **random_seed** – Random seed for reproducibility
- **verbose** – Whether to print progress

Returns

DataFrame with validation results

7.29.4 Visualization

```
py3plex.visualization.multilayer.draw_multiedges(network_list: List[Graph] | Dict[Any, Graph],
                                                    multi_edge_tuple: List[Any], input_type: str =
                                                    'nodes', linepoints: str = '-.', alphachannel: float =
                                                    0.3, linecolor: str = 'black', curve_height: float = 1,
                                                    style: str = 'curve2_bezier', linewidth: float = 1,
                                                    invert: bool = False, linmod: str = 'both', resolution:
                                                    float = 0.001) → None
```

Draw edges connecting multiple layers.

Parameters

- **network_list** – List of NetworkX graphs (layers) or dict of layer_name -> graph
- **multi_edge_tuple** – Tuple specifying edges to draw
- **input_type** – Type of input (“nodes” or other)
- **linepoints** – Line style
- **alphachannel** – Transparency level
- **linecolor** – Color of the lines
- **curve_height** – Height of curved edges
- **style** – Style of edges (“curve2_bezier”, “line”, etc.)
- **linewidth** – Width of lines
- **invert** – Whether to invert drawing direction
- **linmod** – Line modification mode
- **resolution** – Resolution for curve drawing

```
py3plex.visualization.multilayer.draw_multilayer_default(network_list: List[Graph] | Dict[Any, Graph], display: bool = True, node_size: int = 10, alphalevel: float = 0.13, rectanglex: float = 1, rectangley: float = 1, background_shape: str = 'circle', background_color: str = 'rainbow', networks_color: str = 'rainbow', labels: bool = False, arrowsize: float = 0.5, label_position: int = 1, verbose: bool = False, remove_isolated_nodes: bool = False, axis: Any | None = None, edge_size: float = 1, node_labels: bool = False, node_font_size: int = 5, scale_by_size: bool = False) → None
```

Core multilayer drawing method.

Parameters

- **network_list** – List of NetworkX graphs to visualize (or dict of layer_name -> graph)
- **display** – Whether to display the plot directly
- **node_size** – Base size of nodes
- **alphalevel** – Transparency level for background shapes
- **rectanglex** – Width of rectangular backgrounds
- **rectangley** – Height of rectangular backgrounds
- **background_shape** – Background shape type (“circle” or “rectangle”)
- **background_color** – Background color scheme (“default”, “rainbow”, or None)
- **networks_color** – Network color scheme (“rainbow” or “black”)
- **labels** – Layer labels to display
- **arrowsize** – Size of edge arrows
- **label_position** – Position offset for layer labels
- **verbose** – Whether to log network information
- **remove_isolated_nodes** – Whether to remove isolated nodes
- **axis** – Matplotlib axis to draw on (None for current axis)
- **edge_size** – Width of edges
- **node_labels** – Whether to display node labels
- **node_font_size** – Font size for node labels
- **scale_by_size** – Whether to scale node size by degree

Returns

None

```
py3plex.visualization.multilayer.draw_multilayer_flow(graphs: List[Graph], multilinks: Dict[str,
List[Tuple]], labels: List[str] | None = None,
node_activity: Dict[Any, float] | None = None,
ax: Any | None = None, display: bool = True,
layer_gap: float = 3.0, node_size: float = 30,
node_cmap: str = 'viridis', flow_alpha: float
= 0.3, flow_min_width: float = 0.2,
flow_max_width: float = 4.0, aggregate_by:
Tuple[str, ...] = ('u', 'v', 'layer_u', 'layer_v'),
**kwargs) → Any
```

Draw multilayer network as layered flow visualization (alluvial-style).

Shows each layer as a horizontal band with nodes positioned along the x-axis. Intra-layer activity is encoded as node color/size, and inter-layer edges are shown as thick flow ribbons (Bezier curves) where width encodes edge weight.

Parameters

- **graphs** – List of NetworkX graphs, one per layer (from `multi_layer_network.get_layers()`)
- **multilinks** – Dictionary mapping `edge_type` -> list of multi-layer edges
- **labels** – Optional list of layer labels. If `None`, uses layer indices
- **node_activity** – Optional dict mapping `node_id` -> activity value. If `None`, computes intra-layer degree
- **ax** – Matplotlib axes to draw on. If `None`, creates new figure
- **display** – If `True`, calls `plt.show()` at the end
- **layer_gap** – Vertical distance between layer bands
- **node_size** – Base marker size for nodes
- **node_cmap** – Matplotlib colormap name for node activity coloring
- **flow_alpha** – Base transparency for flow ribbons
- **flow_min_width** – Minimum line width for flows
- **flow_max_width** – Maximum line width for flows
- **aggregate_by** – Tuple of keys for aggregating flows (currently not used, for future extension)
- ****kwargs** – Reserved for future extensions

Returns

Matplotlib axes object

Examples

```
>>> network = multi_layer_network()
>>> network.load_network("data.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = network.get_layers()
>>> draw_multilayer_flow(graphs, multilinks, labels=labels)
```

```
py3plex.visualization.multilayer.generate_random_multiedges(network_list: List[Graph],  
                                                           random_edges: int, style: str = 'line',  
                                                           linepoints: str = '-.', upper_first: int =  
2, lower_first: int = 0, lower_second:  
int = 2, inverse_tag: bool = False,  
                                                           pheight: float = 1) → None
```

Generate and draw random multi-layer edges.

Parameters

- **network_list** – List of NetworkX graphs (layers)
- **random_edges** – Number of random edges to generate
- **style** – Style of edges to draw
- **linepoints** – Line style
- **upper_first** – Upper bound for first layer
- **lower_first** – Lower bound for first layer
- **lower_second** – Lower bound for second layer
- **inverse_tag** – Whether to invert drawing
- **pheight** – Height parameter for curves

```
py3plex.visualization.multilayer.generate_random_networks(number_of_networks: int) →  
List[Graph]
```

Generate random networks for testing.

Parameters

- **number_of_networks** – Number of random networks to generate

Returns

List of NetworkX graphs with random layouts

```
py3plex.visualization.multilayer.hairball_plot(g: Graph | Any, color_list: List[str] | List[int] | None =  
None, display: bool = False, node_size: float = 1,  
                                               text_color: str = 'black', node_sizes: List[float] | None  
= None, layout_parameters: dict | None = None,  
                                               legend: Any | None = None, scale_by_size: bool =  
True, layout_algorithm: str = 'force', edge_width: float  
= 0.01, alpha_channel: float = 0.5, labels: List[str] |  
None = None, draw: bool = True, label_font_size: int  
= 2) → Any | None
```

A method for drawing force-directed plots
Args: network (networkx): A network to be visualized
color_list (list): A list of colors for nodes
node_size (float): Size of nodes
layout_parameters (dict): A dictionary of label parameters
legend (bool): Display legend?
scale_by_size (bool): Rescale nodes?
layout_algorithm (string): What type of layout algorithm is to be used?
edge_width (float): Width of edges
alpha_channel (float): Transparency level.
labels (bool): Display labels?
label_font_size (int): Sizes of labels
:returns: None

```
py3plex.visualization.multilayer.interactive_diagonal_plot(network_list: List[Graph] | Dict[Any,  
Graph], layer_labels: List[str] | None  
= None, layout_algorithm: str = 'force',  
               layer_gap: float = 4.0, node_size_base:  
int = 8, layer_colors: List[str] | None =  
None, show_interlayer_edges: bool =  
True, interlayer_edges: List[Tuple[Any,  
Any]] | None = None) → bool | Any
```

Create an interactive 2.5D diagonal multilayer plot using Plotly.

This function creates an interactive version of the diagonal multilayer visualization, mimicking the traditional 2D diagonal layout but in an interactive 3D environment. Each layer is positioned diagonally with clear visual separation, similar to the static diagonal visualization.

Parameters

- **network_list** – List of NetworkX graphs (layers) or dict of layer_name -> graph
- **layer_labels** – Optional labels for each layer
- **layout_algorithm** – Layout algorithm for nodes (“force”, “circular”, “random”)
- **layer_gap** – Distance between layers in diagonal direction (default: 2.5)
- **node_size_base** – Base size for nodes (default: 8)
- **layer_colors** – Optional list of colors for each layer (HTML color names or hex)
- **show_interlayer_edges** – Whether to show inter-layer edges
- **interlayer_edges** – List of tuples (node1, node2) for inter-layer connections

Returns

False if plotly not available, otherwise plotly figure object

Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.load_network("network.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = net.get_layers("diagonal")
>>> fig = interactive_diagonal_plot(graphs, layer_labels=labels)
```

`py3plex.visualization.multilayer.interactive_hairball_plot(G: Graph, nsizes: List[float], final_color_mapping: dict, pos: dict, colorscale: str = 'Rainbow') → bool | Any`

Create an interactive 3D hairball plot using Plotly.

Parameters

- **G** – NetworkX graph to visualize
- **nsizes** – Node sizes
- **final_color_mapping** – Mapping of nodes to colors
- **pos** – Node positions
- **colorscale** – Color scale to use

Returns

False if plotly not available, otherwise plotly figure object

`py3plex.visualization.multilayer.onclick(event: Any) → None`

Handle mouse click events on plots.

Parameters

event – Matplotlib event object

```
py3plex.visualization.multilayer.plot_edge_colored_projection(multilayer_network, layout: str =  
    'spring', node_size: int = 50,  
    layer_colors: Dict[Any, str] | None  
    = None,  
    aggregate_multilayer_edges: bool  
    = True, figsize: Tuple[float, float] =  
    (12, 9), edge_alpha: float = 0.7,  
    **kwargs)
```

Create an aggregated projection where edge colors indicate layer membership.

This visualization projects all layers onto a single 2D graph, using edge colors to distinguish which layer each edge belongs to. Useful for seeing the overall structure while maintaining layer information.

Parameters

- **multilayer_network** – A MultiLayerNetwork instance
- **layout** – Layout algorithm to use (“spring”, “circular”, “random”, “kamada_kawai”)
- **node_size** – Size of nodes
- **layer_colors** – Optional dict mapping layer names to colors; if None, auto-generated
- **aggregate_multilayer_edges** – If True, show edges from all layers with distinct colors
- **figsize** – Figure size as (width, height) tuple
- **edge_alpha** – Transparency level for edges (0-1)
- ****kwargs** – Additional arguments for customization

Returns

The created figure

Return type

matplotlib.figure.Figure

```
py3plex.visualization.multilayer.plot_ego_multilayer(multilayer_network, ego, layers: List[Any] |  
    None = None, max_depth: int = 1, layout: str =  
    'spring', figsize: Tuple[float, float] | None =  
    None, max_cols: int = 3, node_size: int = 500,  
    ego_node_size: int = 1200, **kwargs)
```

Create an ego-centric multilayer visualization.

This visualization focuses on a single node (ego) and shows its neighborhood across different layers, highlighting the ego node’s position in each layer.

Parameters

- **multilayer_network** – A MultiLayerNetwork instance
- **ego** – The ego node to focus on
- **layers** – Optional list of specific layers to visualize; if None, uses all layers
- **max_depth** – Maximum depth of neighborhood to include (number of hops)
- **layout** – Layout algorithm for each ego graph
- **figsize** – Optional figure size; if None, auto-calculated
- **max_cols** – Maximum columns in subplot grid
- **node_size** – Size of regular nodes

- **ego_node_size** – Size of the ego node (highlighted)
- ****kwargs** – Additional drawing parameters

Returns

The created figure

Return type

matplotlib.figure.Figure

```
py3plex.visualization.multilayer.plot_radial_layers(multilayer_network, base_radius: float = 1.0,
                                                    radius_step: float = 1.0, node_size: int = 500,
                                                    draw_inter_layer_edges: bool = True, figsize:
                                                    Tuple[float, float] = (12, 12), edge_alpha: float
                                                    = 0.5, draw_layer_bands: bool = True,
                                                    band_alpha: float = 0.25, **kwargs)
```

Create a radial/concentric visualization with layers as rings.

This visualization arranges layers as concentric circles, with nodes positioned on rings based on their layer. Inter-layer edges appear as radial connections.

Parameters

- **multilayer_network** – A MultiLayerNetwork instance
- **base_radius** – Radius of the innermost layer
- **radius_step** – Distance between consecutive layer rings
- **node_size** – Size of nodes (default: 500 for better visibility)
- **draw_inter_layer_edges** – If True, draw edges between layers
- **figsize** – Figure size as (width, height) tuple
- **edge_alpha** – Transparency for edges
- **draw_layer_bands** – If True, draw semi-transparent circular bands around layers
- **band_alpha** – Transparency for layer bands (default: 0.25)
- ****kwargs** – Additional drawing parameters

Returns

The created figure

Return type

matplotlib.figure.Figure

```
py3plex.visualization.multilayer.plot_small_multiples(multilayer_network, layout: str = 'spring',
                                                       max_cols: int = 3, node_size: int = 50,
                                                       shared_layout: bool = True,
                                                       show_layer_titles: bool = True, figsize:
                                                       Tuple[float, float] | None = None, **kwargs)
```

Create a small multiples visualization with one subplot per layer.

This visualization shows each layer as a separate subplot in a grid layout, making it easy to compare the structure of different layers side-by-side.

Parameters

- **multilayer_network** – A MultiLayerNetwork instance
- **layout** – Layout algorithm to use (“spring”, “circular”, “random”, “kamada_kawai”)

- **max_cols** – Maximum number of columns in the subplot grid
- **node_size** – Size of nodes in each subplot
- **shared_layout** – If True, compute one layout and reuse for all layers; if False, compute independent layouts per layer
- **show_layer_titles** – If True, show layer names as subplot titles
- **figsize** – Optional figure size as (width, height) tuple
- ****kwargs** – Additional arguments passed to `nx.draw_networkx`

Returns

The created figure

Return type

`matplotlib.figure.Figure`

```
py3plex.visualization.multilayer.plot_supra_adjacency_heatmap(multilayer_network,  
                                                             include_inter_layer: bool = False,  
                                                             inter_layer_weight: float = 1.0,  
                                                             node_order: List[Any] | None =  
                                                             None, cmap: str = 'viridis', figsize:  
                                                             Tuple[float, float] = (10, 10),  
                                                             **kwargs)
```

Create a supra-adjacency matrix heatmap visualization.

This visualization shows the multilayer network as a block matrix where each block represents the adjacency matrix of one layer. Optionally includes inter-layer connections.

Parameters

- **multilayer_network** – A `MultiLayerNetwork` instance
- **include_inter_layer** – If True, include inter-layer edges/couplings
- **inter_layer_weight** – Default weight for inter-layer connections
- **node_order** – Optional list specifying node ordering; if None, uses sorted order
- **cmap** – Colormap name for the heatmap
- **figsize** – Figure size as (width, height) tuple
- ****kwargs** – Additional arguments for `imshow`

Returns

The created figure

Return type

`matplotlib.figure.Figure`

```
py3plex.visualization.multilayer.supra_adjacency_matrix_plot(matrix: ndarray, display: bool =  
                                                             False) → None
```

Plot a supra-adjacency matrix.

Parameters

- **matrix** – Supra-adjacency matrix to plot
- **display** – Whether to display the plot immediately

```
py3plex.visualization.multilayer.visualize_multilayer_network(multilayer_network,
                                                             visualization_type: str = 'diagonal',
                                                             **kwargs)
```

High-level function to visualize multilayer networks with multiple visualization modes.

This function provides a unified interface for various multilayer network visualization techniques, making it easy to switch between different visual representations.

Parameters

- **multilayer_network** – A MultiLayerNetwork instance from `py3plex.core.multinet`
- **visualization_type** – Type of visualization to use. Options: - “diagonal”: Default layer-centric diagonal layout (existing behavior) - “small_multiples”: One subplot per layer with shared or independent layouts - “edge_colored_projection”: Aggregate projection with edge colors by layer - “supra_adjacency_heatmap”: Matrix representation of multilayer structure - “radial_layers”: Concentric circles for layers with radial inter-layer edges - “ego_multilayer”: Ego-centric view focused on a specific node
- ****kwargs** – Additional keyword arguments specific to each visualization type. See individual plot functions for details.

Returns

The created figure object

Return type

`matplotlib.figure.Figure`

Raises

ValueError – If `visualization_type` is not recognized

Examples

```
>>> from py3plex.core import multinet
>>> net = multinet.multi_layer_network()
>>> net.load_network("network.txt", input_type="multiedgelist")
>>>
>>> # Use default diagonal visualization
>>> fig = visualize_multilayer_network(net)
>>>
>>> # Use small multiples view
>>> fig = visualize_multilayer_network(net, visualization_type="small_multiples")
>>>
>>> # Use edge-colored projection
>>> fig = visualize_multilayer_network(net, visualization_type="edge_colored_
↪projection")
```

```
py3plex.visualization.drawing_machinery.draw(G, pos=None, ax=None, **kws)
```

Draw the graph *G* with Matplotlib.

Draw the graph as a simple representation with no node labels or edge labels and using the full Matplotlib figure area and no axis labels by default. See `draw_networkx()` for more full-featured drawing that allows title, axis labels etc.

Parameters

- **G** (*graph*) – A `networkx` graph

- **pos** (*dictionary, optional*) – A dictionary with nodes as keys and positions as values. If not specified a spring layout positioning will be computed. See `networkx.drawing.layout` for functions that compute node positions.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in specified Matplotlib axes.
- **kwds** (*optional keywords*) – See `networkx.draw_networkx()` for a description of optional keywords.

Examples

```
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G)
>>> nx.draw(G, pos=nx.spring_layout(G)) # use spring layout
```

See also

`draw_networkx`, `draw_networkx_nodes`, `draw_networkx_edges`, `draw_networkx_labels`, `draw_networkx_edge_labels`

Notes

This function has the same name as `pylab.draw` and `pyplot.draw` so beware when using

```
>>> from networkx import *
```

since you might overwrite the `pylab.draw` function.

With `pyplot` use

```
>>> import matplotlib.pyplot as plt
>>> import networkx as nx
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G) # networkx draw()
>>> plt.draw() # pyplot draw()
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

`py3plex.visualization.drawing_machinery.draw_circular(G, **kwargs)`

Draw the graph `G` with a circular layout.

Parameters

- **G** (*graph*) – A `networkx` graph
- **kwargs** (*optional keywords*) – See `networkx.draw_networkx()` for a description of optional keywords, with the exception of the `pos` parameter which is not used by this function.

`py3plex.visualization.drawing_machinery.draw_kamada_kawai(G, **kwargs)`

Draw the graph `G` with a Kamada-Kawai force-directed layout.

Parameters

- **G** (*graph*) – A `networkx` graph

- **kwargs** (*optional keywords*) – See `networkx.draw_networkx()` for a description of optional keywords, with the exception of the `pos` parameter which is not used by this function.

`py3plex.visualization.drawing_machinery.draw_networkx(G, pos=None, arrows=True, with_labels=True, **kws)`

Draw the graph `G` using Matplotlib.

Draw the graph with Matplotlib with options for node positions, labeling, titles, and many other drawing features. See `draw()` for simple drawing without labels or axes.

Parameters

- **G** (*graph*) – A `networkx` graph
- **pos** (*dictionary, optional*) – A dictionary with nodes as keys and positions as values. If not specified a spring layout positioning will be computed. See `networkx.drawing.layout` for functions that compute node positions.
- **arrows** (*bool, optional (default=True)*) – For directed graphs, if `True` draw arrowheads. Note: Arrows will be the same color as edges.
- **arrowstyle** (*str, optional (default='->')*) – For directed graphs, choose the style of the arrowheads. See `py:class: matplotlib.patches.ArrowStyle` for more options.
- **arrowsize** (*int, optional (default=10)*) – For directed graphs, choose the size of the arrow head's length and width. See `py:class: matplotlib.patches.FancyArrowPatch` for attribute `mutation_scale` for more info.
- **with_labels** (*bool, optional (default=True)*) – Set to `True` to draw labels on the nodes.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **odelist** (*list, optional (default G.nodes())*) – Draw only specified nodes
- **edgelist** (*list, optional (default=G.edges())*) – Draw only specified edges
- **node_size** (*scalar or array, optional (default=300)*) – Size of nodes. If an array is specified it must be the same length as `odelist`.
- **node_color** (*color string, or array of floats, (default='r')*) – Node color. Can be a single color format string, or a sequence of colors with the same length as `odelist`. If numeric values are specified they will be mapped to colors using the `cmap` and `vmin,vmax` parameters. See `matplotlib.scatter` for more details.
- **node_shape** (*string, optional (default='o')*) – The shape of the node. Specification is as `matplotlib.scatter` marker, one of `'so^>v<dph8'`.
- **alpha** (*float, optional (default=1.0)*) – The node and edge transparency
- **cmap** (*Matplotlib colormap, optional (default=None)*) – Colormap for mapping intensities of nodes
- **vmin** (*float, optional (default=None)*) – Minimum and maximum for node colormap scaling
- **vmax** (*float, optional (default=None)*) – Minimum and maximum for node colormap scaling
- **linewidths** (*[None | scalar | sequence]*) – Line width of symbol border (default = 1.0)
- **width** (*float, optional (default=1.0)*) – Line width of edges

- **edge_color** (*color string, or array of floats (default='r')*) – Edge color. Can be a single color format string, or a sequence of colors with the same length as edge-list. If numeric values are specified they will be mapped to colors using the `edge_cmap` and `edge_vmin`, `edge_vmax` parameters.
- **edge_cmap** (*Matplotlib colormap, optional (default=None)*) – Colormap for mapping intensities of edges
- **edge_vmin** (*floats, optional (default=None)*) – Minimum and maximum for edge colormap scaling
- **edge_vmax** (*floats, optional (default=None)*) – Minimum and maximum for edge colormap scaling
- **style** (*string, optional (default='solid')*) – Edge line style (solid|dashed|dotted,dashdot)
- **labels** (*dictionary, optional (default=None)*) – Node labels in a dictionary keyed by node of text labels
- **font_size** (*int, optional (default=12)*) – Font size for text labels
- **font_color** (*string, optional (default='k' black)*) – Font color string
- **font_weight** (*string, optional (default='normal')*) – Font weight
- **font_family** (*string, optional (default='sans-serif')*) – Font family
- **label** (*string, optional*) – Label for graph legend

Notes

For directed graphs, arrows are drawn at the head end. Arrows can be turned off with keyword `arrows=False`.

Examples

```
>>> G = nx.dodecahedral_graph()
>>> nx.draw(G)
>>> nx.draw(G, pos=nx.spring_layout(G)) # use spring layout
```

```
>>> import matplotlib.pyplot as plt
>>> limits = plt.axis('off') # turn of axis
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

See also

`draw`, `draw_networkx_nodes`, `draw_networkx_edges`, `draw_networkx_labels`,
`draw_networkx_edge_labels`

```
py3plex.visualization.drawing_machinery.draw_networkx_edge_labels(G, pos, edge_labels=None,
                                                                    label_pos=0.5, font_size=10,
                                                                    font_color='k',
                                                                    font_family='sans-serif',
                                                                    font_weight='normal',
                                                                    alpha=1.0, bbox=None,
                                                                    ax=None, rotate=True,
                                                                    **kwds)
```

Draw edge labels.

Parameters

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **alpha** (*float*) – The text transparency (default=1.0)
- **edge_labels** (*dictionary*) – Edge labels in a dictionary keyed by edge two-tuple of text labels (default=None). Only labels for the keys in the dictionary are drawn.
- **label_pos** (*float*) – Position of edge label along edge (0=head, 0.5=center, 1=tail)
- **font_size** (*int*) – Font size for text labels (default=12)
- **font_color** (*string*) – Font color string (default='k' black)
- **font_weight** (*string*) – Font weight (default='normal')
- **font_family** (*string*) – Font family (default='sans-serif')
- **bbox** (*Matplotlib bbox*) – Specify text box shape and colors.
- **clip_on** (*bool*) – Turn on clipping at axis boundaries (default=True)

Returns

dict of labels keyed on the edges

Return type

dict

Examples

```
>>> G = nx.dodecahedral_graph()
>>> edge_labels = nx.draw_networkx_edge_labels(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

See also

[draw](#), [draw_networkx](#), [draw_networkx_nodes](#), [draw_networkx_edges](#), [draw_networkx_labels](#)

```
py3plex.visualization.drawing_machinery.draw_networkx_edges(G, pos, edgelist=None, width=1.0,
    edge_color='k', style='solid',
    alpha=1.0, arrowstyle='->',
    arrowsize=10, edge_cmap=None,
    edge_vmin=None, edge_vmax=None,
    ax=None, arrows=True, label=None,
    node_size=300, nodelist=None,
    node_shape='o', **kwds)
```

Draw the edges of the graph G.

This draws only the edges of the graph G.

Parameters

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **edgelist** (*collection of edge tuples*) – Draw only specified edges (default=G.edges())
- **width** (*float, or array of floats*) – Line width of edges (default=1.0)
- **edge_color** (*color string, or array of floats*) – Edge color. Can be a single color format string (default='r'), or a sequence of colors with the same length as edgelist. If numeric values are specified they will be mapped to colors using the edge_cmap and edge_vmin, edge_vmax parameters.
- **style** (*string*) – Edge line style (default='solid') (solid|dashed|dotted,dashdot)
- **alpha** (*float*) – The edge transparency (default=1.0)
- **cmap** (*edge*) – Colormap for mapping intensities of edges (default=None)
- **edge_vmin** (*floats*) – Minimum and maximum for edge colormap scaling (default=None)
- **edge_vmax** (*floats*) – Minimum and maximum for edge colormap scaling (default=None)
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **arrows** (*bool, optional (default=True)*) – For directed graphs, if True draw arrow-heads. Note: Arrows will be the same color as edges.
- **arrowstyle** (*str, optional (default='->')*) – For directed graphs, choose the style of the arrow heads. See :py:class: *matplotlib.patches.ArrowStyle* for more options.
- **arrowsize** (*int, optional (default=10)*) – For directed graphs, choose the size of the arrow head head's length and width. See :py:class: *matplotlib.patches.FancyArrowPatch* for attribute *mutation_scale* for more info.
- **label** (*[None | string]*) – Label for legend

Returns

- *matplotlib.collection.LineCollection* – *LineCollection* of the edges
- *list of matplotlib.patches.FancyArrowPatch* – *FancyArrowPatch* instances of the directed edges
- *Depending whether the drawing includes arrows or not.*

Notes

For directed graphs, arrows are drawn at the head end. Arrows can be turned off with keyword `arrows=False`. Be sure to include `node_size` as a keyword argument; arrows are drawn considering the size of nodes.

Examples

```
>>> G = nx.dodecahedral_graph()
>>> edges = nx.draw_networkx_edges(G, pos=nx.spring_layout(G))
```



```
>>> G = nx.DiGraph()
>>> G.add_edges_from([(1, 2), (1, 3), (2, 3)])
>>> arcs = nx.draw_networkx_edges(G, pos=nx.spring_layout(G))
>>> alphas = [0.3, 0.4, 0.5]
>>> for i, arc in enumerate(arcs): # change alpha values of arcs
...     arc.set_alpha(alphas[i])
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

See also

[*draw*](#), [*draw_networkx*](#), [*draw_networkx_nodes*](#), [*draw_networkx_labels*](#),
[*draw_networkx_edge_labels*](#)

```
py3plex.visualization.drawing_machinery.draw_networkx_labels(G, pos, labels=None, font_size=1,
                                                             font_color='k',
                                                             font_family='sans-serif',
                                                             font_weight='normal', alpha=1.0,
                                                             bbox=None, ax=None, **kws)
```

Draw node labels on the graph G.

Parameters

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **labels** (*dictionary, optional (default=None)*) – Node labels in a dictionary keyed by node of text labels Node-keys in labels should appear as keys in *pos*. If needed use: *{n:lab for n,lab in labels.items() if n in pos}*
- **font_size** (*int*) – Font size for text labels (default=12)
- **font_color** (*string*) – Font color string (default='k' black)
- **font_family** (*string*) – Font family (default='sans-serif')
- **font_weight** (*string*) – Font weight (default='normal')
- **alpha** (*float*) – The text transparency (default=1.0)
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.

Returns

dict of labels keyed on the nodes

Return type

dict

Examples

```
>>> G = nx.dodecahedral_graph()
>>> labels = nx.draw_networkx_labels(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

See also

[*draw*](#), [*draw_networkx*](#), [*draw_networkx_nodes*](#), [*draw_networkx_edges*](#),
[*draw_networkx_edge_labels*](#)

```
py3plex.visualization.drawing_machinery.draw_networkx_nodes(G, pos, nodelist=None,  
                                                             node_size=300, node_color='r',  
                                                             node_shape='o', alpha=1.0,  
                                                             cmap=None, vmin=None,  
                                                             vmax=None, ax=None,  
                                                             linewidths=None, edgecolors=None,  
                                                             label=None, **kwds)
```

Draw the nodes of the graph G.

This draws only the nodes of the graph G.

Parameters

- **G** (*graph*) – A networkx graph
- **pos** (*dictionary*) – A dictionary with nodes as keys and positions as values. Positions should be sequences of length 2.
- **ax** (*Matplotlib Axes object, optional*) – Draw the graph in the specified Matplotlib axes.
- **nodelist** (*list, optional*) – Draw only specified nodes (default G.nodes())
- **node_size** (*scalar or array*) – Size of nodes (default=300). If an array is specified it must be the same length as nodelist.
- **node_color** (*color string, or array of floats*) – Node color. Can be a single color format string (default='r'), or a sequence of colors with the same length as nodelist. If numeric values are specified they will be mapped to colors using the cmap and vmin,vmax parameters. See matplotlib.scatter for more details.
- **node_shape** (*string*) – The shape of the node. Specification is as matplotlib.scatter marker, one of 'so^>v<dph8' (default='o').
- **alpha** (*float or array of floats*) – The node transparency. This can be a single alpha value (default=1.0), in which case it will be applied to all the nodes of color. Otherwise, if it is an array, the elements of alpha will be applied to the colors in order (cycling through alpha multiple times if necessary).
- **cmap** (*Matplotlib colormap*) – Colormap for mapping intensities of nodes (default=None)
- **vmin** (*floats*) – Minimum and maximum for node colormap scaling (default=None)
- **vmax** (*floats*) – Minimum and maximum for node colormap scaling (default=None)
- **linewidths** (*[None | scalar | sequence]*) – Line width of symbol border (default =1.0)
- **edgecolors** (*[None | scalar | sequence]*) – Colors of node borders (default = node_color)
- **label** (*[None | string]*) – Label for legend

Returns*PathCollection* of the nodes.**Return type**

matplotlib.collections.PathCollection

Examples

```
>>> G = nx.dodecahedral_graph()
>>> nodes = nx.draw_networkx_nodes(G, pos=nx.spring_layout(G))
```

Also see the NetworkX drawing examples at https://networkx.github.io/documentation/latest/auto_examples/index.html

See also

draw, *draw_networkx*, *draw_networkx_edges*, *draw_networkx_labels*,
draw_networkx_edge_labels

py3plex.visualization.drawing_machinery.**draw_random**(G, **kwargs)

Draw the graph G with a random layout.

Parameters

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_shell**(G, **kwargs)

Draw networkx graph with shell layout.

Parameters

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_spectral**(G, **kwargs)

Draw the graph G with a spectral layout.

Parameters

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

py3plex.visualization.drawing_machinery.**draw_spring**(G, **kwargs)

Draw the graph G with a spring layout.

Parameters

- **G** (*graph*) – A networkx graph
- **kwargs** (*optional keywords*) – See networkx.draw_networkx() for a description of optional keywords, with the exception of the pos parameter which is not used by this function.

Color utilities for py3plex visualization.

`py3plex.visualization.colors.RGB_to_hex( RGB: List[int] ) → str`

Convert RGB list to hex color.

Parameters

RGB – RGB values as [R, G, B] list

Returns

Hex color string like “#FFFFFF”

`py3plex.visualization.colors.color_dict( gradient: List[List[int]] ) → Dict[str, List]`

Takes in a list of RGB sub-lists and returns dictionary of colors in RGB and hex form for use in a graphing function defined later on.

Parameters

gradient – List of RGB color values

Returns

Dictionary with ‘hex’, ‘r’, ‘g’, ‘b’ keys

`py3plex.visualization.colors.hex_to_RGB( hex: str ) → List[int]`

Convert hex color to RGB list.

Parameters

hex – Hex color string like “#FFFFFF”

Returns

RGB values as [R, G, B] list

`py3plex.visualization.colors.linear_gradient(start_hex: str, finish_hex: str = '#FFFFFF', n: int = 10)
→ Dict[str, List]`

Returns a gradient list of (n) colors between two hex colors.

Parameters

- **start_hex** – Starting color as six-digit hex string (e.g., “#FFFFFF”)
- **finish_hex** – Ending color as six-digit hex string (default: “#FFFFFF”)
- **n** – Number of colors in gradient (default: 10)

Returns

Dictionary with ‘hex’, ‘r’, ‘g’, ‘b’ keys containing gradient colors

`py3plex.visualization.layout_algorithms.compute_force_directed_layout(g: Graph,
layout_parameters:
Dict[str, Any] | None =
None, verbose: bool =
True, gravity: float = 0.2,
strongGravityMode:
bool = False,
barnesHutTheta: float =
1.2,
edgeWeightInfluence:
float = 1, scalingRatio:
float = 2.0, forceImport:
bool = True, seed: int |
None = None) →
Dict[Any, ndarray]`

Compute force-directed layout for a graph using ForceAtlas2 or NetworkX spring layout.

Parameters

- **g** – NetworkX graph to layout
- **layout_parameters** – Optional parameters to pass to layout algorithm
- **verbose** – Whether to print progress information
- **gravity** – Attraction force towards the center (must be non-negative)
- **strongGravityMode** – Use strong gravity mode
- **barnesHutTheta** – Barnes-Hut approximation parameter
- **edgeWeightInfluence** – Influence of edge weights on layout
- **scalingRatio** – Scaling factor for the layout (must be positive)
- **forceImport** – Whether to use ForceAtlas2 (if available)
- **seed** – Random seed for reproducibility in fallback spring layout

Returns

Dictionary mapping nodes to 2D position arrays

Note

For large networks (>1000 nodes), this may be slow. Consider using faster layouts (circular, random, spectral) or matrix visualization.

Contracts:

- Precondition: graph must not be None and be a NetworkX graph
- Precondition: graph must have at least one node
- Precondition: gravity must be non-negative
- Precondition: scalingRatio must be positive
- Postcondition: result is a dictionary
- Postcondition: result has positions for all nodes

`py3plex.visualization.layout_algorithms.compute_random_layout(g: Graph, seed: int | None = None) → Dict[Any, ndarray]`

Compute a random layout for the graph.

Parameters

- **g** – NetworkX graph
- **seed** – Random seed for reproducibility

Returns

Dictionary mapping nodes to 2D positions

Contracts:

- Precondition: graph must not be None and be a NetworkX graph
- Precondition: graph must have at least one node
- Postcondition: result is a dictionary
- Postcondition: result has positions for all nodes

```
py3plex.visualization.layout_algorithms.ensure(*args, **kwargs)
```

```
py3plex.visualization.layout_algorithms.require(*args, **kwargs)
```

```
py3plex.visualization.bezier.bezier_calculate_dfy(mp_y: float, path_height: float, x0: float,  
                                                  midpoint_x: float, x1: float, y0: float, y1: float,  
                                                  dfx: ndarray, mode: str = 'upper') → ndarray
```

Calculate y-coordinates for bezier curve.

Parameters

- **mp_y** – Midpoint y-coordinate
- **path_height** – Height of the path
- **x0** – Start x-coordinate
- **midpoint_x** – Midpoint x-coordinate
- **x1** – End x-coordinate
- **y0** – Start y-coordinate
- **y1** – End y-coordinate
- **dfx** – Array of x-coordinates
- **mode** – Mode for curve calculation (“upper” or “bottom”)

Returns

Array of y-coordinates

```
py3plex.visualization.bezier.draw_bezier(total_size: int, p1: Tuple[float, float], p2: Tuple[float, float],  
                                         mode: str = 'quadratic', inversion: bool = False, path_height:  
                                         float = 2, linemode: str = 'both', resolution: float = 0.1) →  
                                         Tuple[ndarray, ndarray]
```

Draw bezier curve between two points.

Parameters

- **total_size** – Total size of the drawing area
- **p1** – First point coordinates (x0, x1)
- **p2** – Second point coordinates (y0, y1)
- **mode** – Drawing mode (default: “quadratic”)
- **inversion** – Whether to invert the curve
- **path_height** – Height of the path
- **linemode** – Line drawing mode (“upper”, “bottom”, or “both”)
- **resolution** – Resolution for curve sampling

Returns

Tuple of (x-coordinates, y-coordinates) arrays

```
py3plex.visualization.benchmark_visualizations.generic_grouping(fname: DataFrame, score_name:  
                                                                str, threshold: float = 1.0,  
                                                                percentages: bool = True) →  
                                                                DataFrame
```

```

py3plex.visualization.benchmark_visualizations.plot_core_macro(fname: DataFrame) → int
    A very simple visualization of the results..
py3plex.visualization.benchmark_visualizations.plot_core_macro_box(fname: str) → int
py3plex.visualization.benchmark_visualizations.plot_core_macro_gg(fname: DataFrame) → None
py3plex.visualization.benchmark_visualizations.plot_core_micro(fname: DataFrame) → int
    A very simple visualization of the results..
py3plex.visualization.benchmark_visualizations.plot_core_micro_gg(fname: DataFrame) → None
py3plex.visualization.benchmark_visualizations.plot_core_micro_grid(fname: str) → None
py3plex.visualization.benchmark_visualizations.plot_core_time(fname: DataFrame) → int
py3plex.visualization.benchmark_visualizations.plot_core_time_gg(fname: str) → None
py3plex.visualization.benchmark_visualizations.plot_core_variability(fname: str) → None
py3plex.visualization.benchmark_visualizations.plot_critical_distance(fname: str, num_algo:
                                                                    int = 14) → None
py3plex.visualization.benchmark_visualizations.plot_mean_times(fn: DataFrame) → None
py3plex.visualization.benchmark_visualizations.plot_robustness(infile: DataFrame) → None
py3plex.visualization.benchmark_visualizations.table_to_latex(fname,
                                                                outfolder='./final_results/tables/',
                                                                threshold=1)

```

7.29.5 Wrappers

class py3plex.wrappers.benchmark_nodes.TopKRanker(estimator, *, n_jobs=None, verbose=0)

Bases: OneVsRestClassifier

predict(X: ndarray, top_k_list: List[int]) → List[List]

Predict top K labels for each sample.

Parameters

- **X** – Feature matrix
- **top_k_list** – List of K values for each sample

Returns

List of predicted labels for each sample

set_partial_fit_request(*, classes: bool | None | str = '\$UNCHANGED\$') → TopKRanker

Configure whether metadata should be requested to be passed to the `partial_fit` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config()`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- **True**: metadata is requested, and passed to `partial_fit` if provided. The request is ignored if metadata is not provided.
- **False**: metadata is not requested and the meta-estimator will not pass it to `partial_fit`.

- `None`: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str`: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

Parameters

classes (`str`, `True`, `False`, or `None`, `default=sklearn.utils.metadata_routing.UNCHANGED`) – Metadata routing for `classes` parameter in `partial_fit`.

Returns

self – The updated object.

Return type

object

set_predict_request(*, *top_k_list*: `bool` | `None` | `str` = `'$UNCHANGED$'`) → *TopKRanker*

Configure whether metadata should be requested to be passed to the `predict` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config()`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- `True`: metadata is requested, and passed to `predict` if provided. The request is ignored if metadata is not provided.
- `False`: metadata is not requested and the meta-estimator will not pass it to `predict`.
- `None`: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- `str`: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

Parameters

top_k_list (`str`, `True`, `False`, or `None`, `default=sklearn.utils.metadata_routing.UNCHANGED`) – Metadata routing for `top_k_list` parameter in `predict`.

Returns

self – The updated object.

Return type

object

set_score_request(*, *sample_weight*: `bool` | `None` | `str` = `'$UNCHANGED$'`) → *TopKRanker*

Configure whether metadata should be requested to be passed to the `score` method.

Note that this method is only relevant when this estimator is used as a sub-estimator within a meta-estimator and metadata routing is enabled with `enable_metadata_routing=True` (see `sklearn.set_config()`). Please check the User Guide on how the routing mechanism works.

The options for each parameter are:

- **True**: metadata is requested, and passed to `score` if provided. The request is ignored if metadata is not provided.
- **False**: metadata is not requested and the meta-estimator will not pass it to `score`.
- **None**: metadata is not requested, and the meta-estimator will raise an error if the user provides it.
- **str**: metadata should be passed to the meta-estimator with this given alias instead of the original name.

The default (`sklearn.utils.metadata_routing.UNCHANGED`) retains the existing request. This allows you to change the request for some parameters and not others.

Added in version 1.3.

Parameters

sample_weight (*str, True, False, or None, default=sklearn.utils.metadata_routing.UNCHANGED*) – Metadata routing for `sample_weight` parameter in `score`.

Returns

self – The updated object.

Return type

object

`py3plex.wrappers.benchmark_nodes.benchmark_node_classification`(*path: str, core_network: Any, labels_matrix: Any, percent: Any = 'all'*) → Dict[Any, Any]

Benchmark node classification using embeddings.

Parameters

- **path** – Path to embeddings file
- **core_network** – Network adjacency matrix
- **labels_matrix** – Labels for nodes
- **percent** – Training percentage or “all” for multiple percentages

Returns

Dictionary of classification results

`py3plex.wrappers.benchmark_nodes.main()`

`py3plex.wrappers.benchmark_nodes.sparse2graph`(*x: spmatrix*) → Dict[str, List[str]]

Convert sparse matrix to graph dictionary.

Parameters

x – Sparse matrix representation of graph

Returns

Dictionary mapping node IDs to lists of neighbor IDs

7.29.6 I/O Operations

For the complete auto-generated API documentation, see the `AUTOGEN_results` directory after building the docs. Aggregation and Network Operations ————— Vectorized multiplex aggregation for multilayer networks.

This module provides optimized implementations for aggregating edges across multiple layers using vectorized NumPy and SciPy sparse operations, replacing slower Python loops with efficient matrix operations.

Performance targets: - $3\times$ speedup for 1M edges across 4 layers compared to legacy loop-based methods - Memory-efficient sparse matrix output by default - Float tolerance of $1e-6$ for numerical equivalence with legacy methods

Author: py3plex development team License: MIT

`py3plex.multinet.aggregation.aggregate_layers`(*edges: ndarray | list, weight_col: str | int = 'w', reducer: Literal['sum', 'mean', 'max'] = 'sum', to_sparse: bool = True*) $\rightarrow$ *csr_matrix | ndarray*

Aggregate edge weights across multiple layers using vectorized operations.

This function replaces Python loops with efficient NumPy and SciPy sparse matrix operations for superior performance on large multilayer networks.

Complexity: $O(E)$ where E is the number of edges **Memory:** $O(E)$ for sparse output, $O(N^2)$ for dense output

Parameters

- **edges** – Edge data as ndarray with shape $(E, \geq 3)$ containing (layer, src, dst, [weight]) columns. If no weight column, assumes $\text{weight}=1.0$ for all edges. Can also accept list of lists.
- **weight_col** – Column name or index for weights (default “w”). If edges is ndarray, must be int index (0-based). Column 3 is used if it exists, else weights default to 1.
- **reducer** – Aggregation method - one of: - “sum”: Sum weights for edges appearing in multiple layers (default) - “mean”: Average weights across layers - “max”: Take maximum weight across layers
- **to_sparse** – If True, return `scipy.sparse.csr_matrix` (default, memory-efficient). If False, return dense `numpy.ndarray`.

Returns

Aggregated adjacency matrix in requested format (sparse CSR or dense). Shape is (N, N) where N is the maximum node ID + 1.

Raises

- **ValueError** – If edges array has wrong shape or reducer is invalid.
- **TypeError** – If edges is not ndarray or list-like.

Examples

```
>>> import numpy as np
>>> # Create edge list: (layer, src, dst, weight)
>>> edges = np.array([
...     [0, 0, 1, 1.0],
...     [0, 1, 2, 2.0],
...     [1, 0, 1, 0.5], # Same edge in layer 1
...     [1, 2, 3, 1.5],
... ])
>>> mat = aggregate_layers(edges, reducer="sum")
>>> mat.shape
(4, 4)
>>> mat[0, 1] # Sum of weights from both layers
1.5
```

```
>>> # With mean aggregation
>>> mat_mean = aggregate_layers(edges, reducer="mean", to_sparse=False)
```

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```
>>> mat_mean[0, 1]  # Average of 1.0 and 0.5
0.75
```

Notes

- Node IDs are assumed to be integers starting from 0
- Self-loops are supported
- For directed graphs, (i,j) and (j,i) are different edges
- Sparse output recommended for large networks (N > 1000)
- Deterministic output for fixed input order

Performance:

- Achieves 3× speedup vs loop-based aggregation on 1M edges: - Legacy loop: ~2.5s for 1M edges, 4 layers
- Vectorized: ~0.8s for same dataset (measured on standard hardware)

```
py3plex.multinet.aggregation.ensure(*args, **kwargs)
```

```
py3plex.multinet.aggregation.require(*args, **kwargs)
```

7.29.7 Profiling Utilities

Performance profiling utilities for py3plex.

This module provides decorators and utilities for tracking function execution time, memory usage, and performance metrics. These tools enable performance regression detection and optimization efforts.

Example

```
>>> from py3plex.profiling import profile_performance
>>>
>>> @profile_performance
>>> def slow_function():
...     # ... computation
...     pass
>>>
>>> slow_function()  # Logs execution time automatically
```

```
class py3plex.profiling.PerformanceMonitor
```

Bases: object

Global performance monitoring registry.

Tracks execution times and call counts for profiled functions. Can be used to generate performance reports and detect regressions.

enabled

Whether profiling is enabled globally

stats

Dictionary mapping function names to performance statistics

clear()

Clear all collected statistics.

get_report() → str

Generate a performance report.

Returns

String containing formatted performance statistics

record(*func_name: str, elapsed: float, memory_delta: float | None = None*)

Record performance metrics for a function call.

Parameters

- **func_name** – Name of the function
- **elapsed** – Execution time in seconds
- **memory_delta** – Memory usage change in MB (optional)

`py3plex.profiling.benchmark`(*func: Callable, iterations: int = 100, warmup: int = 10, args: tuple = (), kwargs: dict | None = None*) → Dict[str, float]

Benchmark a function with multiple iterations.

Runs the function multiple times and collects statistics about execution time. Includes a warmup phase to allow JIT compilation and caching.

Parameters

- **func** – Function to benchmark
- **iterations** – Number of iterations to run (default: 100)
- **warmup** – Number of warmup iterations (default: 10)
- **args** – Positional arguments for the function
- **kwargs** – Keyword arguments for the function

Returns

- mean: Average execution time (seconds)
- median: Median execution time (seconds)
- min: Minimum execution time (seconds)
- max: Maximum execution time (seconds)
- std: Standard deviation (seconds)
- total: Total time for all iterations (seconds)

Return type

Dictionary containing benchmark statistics

Example

```
>>> from py3plex.profiling import benchmark
>>>
>>> def my_function(n):
...     return sum(range(n))
>>>
```

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```
>>> stats = benchmark(my_function, iterations=1000, args=(1000,))
>>> print(f"Average time: {stats['mean']*1000:.3f}ms")
```

`py3plex.profiling.get_monitor()` → *PerformanceMonitor*

Get the global performance monitor instance.

Returns

Global performance monitoring instance

Return type

PerformanceMonitor

Example

```
>>> from py3plex.profiling import get_monitor
>>> monitor = get_monitor()
>>> print(monitor.get_report())
```

`py3plex.profiling.profile_performance(func: Callable | None = None, *, log_args: bool = False, track_memory: bool = False) → Callable`

Decorator to track function execution time and optionally memory usage.

This decorator measures the wall-clock time taken by a function and logs it. It can also track memory usage changes if requested. Performance metrics are stored in the global performance monitor for later analysis.

Parameters

- **func** – Function to decorate (when used without arguments)
- **log_args** – If True, log function arguments (default: False)
- **track_memory** – If True, track memory usage (default: False)

Returns

Decorated function that logs execution time

Examples

Basic usage: `>>> @profile_performance ... def my_function(x, y): ... return x + y`

With options: `>>> @profile_performance(log_args=True, track_memory=True) ... def expensive_function(data): ... # ... expensive computation ... return result`

Manual wrapping: `>>> def my_function(): ... pass >>> profiled_func = profile_performance(my_function)`

Note

Memory tracking requires the `tracemalloc` module and adds overhead. Use it only when investigating memory issues.

`py3plex.profiling.timed_section(name: str, log_level: str = 'info')`

Context manager for timing code blocks.

Useful for timing specific sections of code without wrapping entire functions.

Parameters

- **name** – Name of the code section being timed

- **log_level** – Logging level ('debug', 'info', 'warning', 'error')

Example

```
>>> from py3plex.profilng import timed_section
>>>
>>> with timed_section("data loading"):
...     data = load_large_dataset()
>>>
>>> with timed_section("computation", log_level="debug"):
...     result = complex_calculation()
```

Yields

None

7.29.8 Logging Configuration

Logging configuration for py3plex.

This module provides centralized logging configuration for the py3plex library.

`py3plex.logging_config.get_logger(name: str | None = None, level: int = 20) → Logger`

Get or create a logger for py3plex modules.

Parameters

- **name** – Logger name. If None, returns the root py3plex logger.
- **level** – Logging level (default: INFO)

Returns

Configured logger instance

Example

```
>>> from py3plex.logging_config import get_logger
>>> logger = get_logger(__name__)
>>> logger.info("Processing network...")
```

`py3plex.logging_config.setup_logging(level: int = 20, format_string: str | None = None) → Logger`

Configure logging for py3plex with custom settings.

Parameters

- **level** – Logging level (default: INFO)
- **format_string** – Custom format string (optional)

Returns

Root py3plex logger

Example

```
>>> from py3plex.logging_config import setup_logging
>>> logger = setup_logging(level=logging.DEBUG)
```

7.29.9 I/O Schema and Validation

Schema definitions for multilayer graphs using dataclasses.

This module provides dataclass representations of multilayer graph components with built-in validation and serialization support.

```
class py3plex.io.schema.Edge(src: ~typing.Hashable, dst: ~typing.Hashable, src_layer: ~typing.Hashable,
                             dst_layer: ~typing.Hashable, key: int = 0, attributes: ~typing.Dict[str,
                             ~typing.Any] = <factory>)
```

Bases: object

Represents an edge in a multilayer network.

src

Source node ID

Type

Hashable

dst

Destination node ID

Type

Hashable

src_layer

Source layer ID

Type

Hashable

dst_layer

Destination layer ID

Type

Hashable

key

Optional edge key for multigraphs (default: 0)

Type

int

attributes

Dictionary of edge attributes (must be JSON-serializable)

Type

Dict[str, Any]

attributes: Dict[str, Any]

dst: Hashable

dst_layer: Hashable

edge_tuple() → Tuple[Hashable, Hashable, Hashable, Hashable, int]

Return edge as a tuple for uniqueness checking.

Returns

Tuple of (src, dst, src_layer, dst_layer, key)

classmethod **from_dict**(*data: Dict[str, Any]*) → Edge

Create edge from dictionary.

Parameters

data – Dictionary containing edge data

Returns

Edge instance

key: int = 0

src: Hashable

src_layer: Hashable

to_dict() → Dict[str, Any]

Convert edge to dictionary.

Returns

Dictionary representation of the edge

class py3plex.io.schema.**Layer**(*id: ~typing.Hashable, attributes: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Represents a layer in a multilayer network.

id

Unique identifier for the layer

Type

Hashable

attributes

Dictionary of layer attributes (must be JSON-serializable)

Type

Dict[str, Any]

attributes: Dict[str, Any]

classmethod **from_dict**(*data: Dict[str, Any]*) → Layer

Create layer from dictionary.

Parameters

data – Dictionary containing layer data

Returns

Layer instance

id: Hashable

to_dict() → Dict[str, Any]

Convert layer to dictionary.

Returns

Dictionary representation of the layer


```
class py3plex.io.schema.MultiLayerGraph(nodes: ~typing.Dict[~typing.Hashable,
~py3plex.io.schema.Node] = <factory>, layers:
~typing.Dict[~typing.Hashable, ~py3plex.io.schema.Layer] =
<factory>, edges: ~typing.List[~py3plex.io.schema.Edge] =
<factory>, directed: bool = True, attributes: ~typing.Dict[str,
~typing.Any] = <factory>)
```

Bases: object

Represents a complete multilayer graph.

nodes

Dictionary mapping node IDs to Node objects

Type

Dict[Hashable, py3plex.io.schema.Node]

layers

Dictionary mapping layer IDs to Layer objects

Type

Dict[Hashable, py3plex.io.schema.Layer]

edges

List of Edge objects

Type

List[py3plex.io.schema.Edge]

directed

Whether the graph is directed

Type

bool

attributes

Dictionary of graph-level attributes

Type

Dict[str, Any]

add_edge(edge: Edge)

Add an edge to the graph.

Parameters

edge – Edge to add

Raises

- **ReferentialIntegrityError** – If edge references non-existent nodes or layers
- **SchemaValidationError** – If edge is duplicate

add_layer(layer: Layer)

Add a layer to the graph.

Parameters

layer – Layer to add

Raises

- **SchemaValidationError** – If layer ID already exists

add_node(*node: Node*)

Add a node to the graph.

Parameters

node – Node to add

Raises

SchemaValidationError – If node ID already exists

attributes: Dict[str, Any]

directed: bool = True

edges: List[Edge]

classmethod from_dict(*data: Dict[str, Any]*) → MultiLayerGraph

Create graph from dictionary.

Parameters

data – Dictionary containing graph data

Returns

MultiLayerGraph instance

layers: Dict[Hashable, Layer]

nodes: Dict[Hashable, Node]

to_dict() → Dict[str, Any]

Convert graph to dictionary.

Returns

Dictionary representation of the graph

class py3plex.io.schema.**Node**(*id: ~typing.Hashable, attributes: ~typing.Dict[str, ~typing.Any] = <factory>*)

Bases: object

Represents a node in a multilayer network.

id

Unique identifier for the node

Type

Hashable

attributes

Dictionary of node attributes (must be JSON-serializable)

Type

Dict[str, Any]

attributes: Dict[str, Any]

classmethod from_dict(*data: Dict[str, Any]*) → Node

Create node from dictionary.

Parameters

data – Dictionary containing node data

Returns

Node instance

id: Hashable

to_dict() → Dict[str, Any]

Convert node to dictionary.

Returns

Dictionary representation of the node

Public API for reading and writing multilayer graphs.

This module provides the main entry points for I/O operations with format detection and a registry system for extensibility.

`py3plex.io.api.ensure(*args, **kwargs)`

`py3plex.io.api.read(filepath: str | Path, format: str | None = None, **kwargs) → MultiLayerGraph`

Read a multilayer graph from a file.

Parameters

- **filepath** – Path to the input file
- **format** – Format name (e.g., 'json', 'csv'). If None, auto-detected from extension
- ****kwargs** – Additional arguments passed to the format-specific reader

Returns

MultiLayerGraph instance

Raises

- **FormatUnsupportedError** – If format is not supported or cannot be detected
- **FileNotFoundError** – If file does not exist

Example

```
>>> graph = read('network.json')
>>> graph = read('network.csv', format='csv')
```

`py3plex.io.api.register_reader(format_name: str, reader_func: Callable[[...], MultiLayerGraph]) → None`

Register a reader function for a specific format.

Parameters

- **format_name** – Name of the format (e.g., 'json', 'csv', 'graphml')
- **reader_func** – Function that takes (filepath, **kwargs) and returns MultiLayerGraph

Example

```
>>> def my_reader(filepath, **kwargs):
...     # Custom reading logic
...     return MultiLayerGraph(...)
>>> register_reader('myformat', my_reader)
```

Contracts:

- Precondition: format_name must be a non-empty string
- Precondition: reader_func must be callable

- Postcondition: reader is registered in `_READERS`

`py3plex.io.api.register_writer(format_name: str, writer_func: Callable[[...], None]) → None`

Register a writer function for a specific format.

Parameters

- **format_name** – Name of the format (e.g., 'json', 'csv', 'graphml')
- **writer_func** – Function that takes (graph, filepath, ****kwargs**) and writes to file

Example

```
>>> def my_writer(graph, filepath, **kwargs):
...     # Custom writing logic
...     pass
>>> register_writer('myformat', my_writer)
```

Contracts:

- Precondition: format_name must be a non-empty string
- Precondition: writer_func must be callable
- Postcondition: writer is registered in `_WRITERS`

`py3plex.io.api.require(*args, **kwargs)`

`py3plex.io.api.supported_formats(read: bool = True, write: bool = True) → Dict[str, List[str]]`

Get list of supported formats for read and/or write operations.

Parameters

- **read** – Include formats that support reading
- **write** – Include formats that support writing

Returns

Dictionary with 'read' and/or 'write' keys containing lists of format names

Example

```
>>> formats = supported_formats()
>>> print(formats)
{'read': ['json', 'jsonl', 'csv'], 'write': ['json', 'jsonl', 'csv']}
```

Contracts:

- Postcondition: result is a dictionary
- Postcondition: result contains 'read' key when read=True
- Postcondition: result contains 'write' key when write=True

`py3plex.io.api.write(graph: MultiLayerGraph, filepath: str | Path, format: str | None = None, **kwargs) → None`

Write a multilayer graph to a file.

Parameters

- **graph** – MultiLayerGraph to write
- **filepath** – Path to the output file
- **format** – Format name (e.g., 'json', 'csv'). If None, auto-detected from extension
- ****kwargs** – Additional arguments passed to the format-specific writer

Raises

FormatUnsupportedError – If format is not supported or cannot be detected

Example

```
>>> write(graph, 'network.json')
>>> write(graph, 'network.csv', format='csv', deterministic=True)
```

7.29.10 Hedwig Rule Learning

`py3plex.algorithms.hedwig.build_graph(kwarg: Dict[str, Any]) → Any`

`py3plex.algorithms.hedwig.generate_rules_report(kwarg: ~typing.Dict[str, ~typing.Any],
rules_per_target:
~typing.List[~typing.Tuple[~typing.Any,
~typing.List[~typing.Any]]], human:
~typing.Callable[~typing.Any, ~typing.Any],
~typing.Any] = <function <lambda>>) → str`

`py3plex.algorithms.hedwig.rule_kernel(target: Any) → Tuple[Any, List[Any]]`

`py3plex.algorithms.hedwig.run(kwarg: Dict[str, Any], cli: bool = True, generator_tag: bool = False,
num_threads: str | int = 'all') → List[Tuple[Any, List[Any]]]`

`py3plex.algorithms.hedwig.run_learner(kwarg: Dict[str, Any], kb: ExperimentKB, validator: Validate,
generator: bool = False, num_threads: str | int = 'all') →
List[Tuple[Any, List[Any]]]`

Example-related classes.

@author: anze.vavpetic@ijs.si

class `py3plex.algorithms.hedwig.core.example.Example(id, label, score, annotations=None,
weights=None)`

Bases: `object`

Represents an example with its score, label, id and annotations.

ClassLabeled = 'class'

Ranked = 'ranked'

Predicate-related classes.

@author: anze.vavpetic@ijs.si

class `py3plex.algorithms.hedwig.core.predicate.BinaryPredicate(label, pairs, kb,
producer_pred=None)`

Bases: `Predicate`

A binary predicate.

```
class py3plex.algorithms.hedwig.core.predicate.Predicate(label, kb, producer_pred)
```

Bases: object

Represents a predicate as a member of a certain rule.

i = -1

```
class py3plex.algorithms.hedwig.core.predicate.UnaryPredicate(label, members, kb,
                                                              producer_pred=None,
                                                              custom_var_name=None,
                                                              negated=False)
```

Bases: *Predicate*

A unary predicate.

The rule class.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.core.rule.Rule(kb, predicates=None, target=None)
```

Bases: object

Represents a rule, along with its description, examples and statistics.

clone()

Returns a clone of this rule. The predicates themselves are NOT cloned.

clone_append(predicate_label, producer_pred, bin=False)

Returns a copy of this rule where 'predicate_label' is appended to the rule.

clone_negate(target_pred)

Returns a copy of this rule where 'target_pred' is negated.

clone_swap_with_subclass(target_pred, child_pred_label)

Returns a copy of this rule where 'target_pred' is swapped for 'child_pred_label'.

examples(positive_only=False)

Returns the covered examples.

property positives

precision()

rule_report(show_uris=False, latex=False)

Rule as string with some statistics.

static ruleset_examples_json(rules_per_target, show_uris=False)

static ruleset_report(rules, show_uris=False, latex=False, human=<function Rule.<lambda>>)

similarity(rule)

Calculates the similarity between this rule and 'rule'.

size()

Returns the number of conjuncts.

static to_json(rules_per_target, show_uris=False)

7.29.11 Force Atlas 2 Visualization

```
class py3plex.visualization.fa2.forceatlas2.ForceAtlas2(outboundAttractionDistribution=False,
                                                    linLogMode=False, adjustSizes=False,
                                                    edgeWeightInfluence=1.0,
                                                    jitterTolerance=1.0,
                                                    barnesHutOptimize=True,
                                                    barnesHutTheta=1.2,
                                                    multiThreaded=False, scalingRatio=2.0,
                                                    strongGravityMode=False, gravity=1.0,
                                                    verbose=True)
```

Bases: object

```
forceatlas2(G, pos=None, iterations=30)
```

```
forceatlas2_igraph_layout(G, pos=None, iterations=100, weight_attr=None)
```

```
forceatlas2_networkx_layout(G, pos=None, iterations=100)
```

```
init(G, pos=None)
```

```
class py3plex.visualization.fa2.forceatlas2.Timer(name="Timer")
```

Bases: object

```
display()
```

```
start()
```

```
stop()
```

```
class py3plex.visualization.fa2.fa2util.Edge
```

Bases: object

```
class py3plex.visualization.fa2.fa2util.Node
```

Bases: object

```
class py3plex.visualization.fa2.fa2util.Region(nodes)
```

Bases: object

```
applyForce(n, theta, coefficient=0)
```

```
applyForceOnNodes(nodes, theta, coefficient=0)
```

```
buildSubRegions()
```

```
updateMassAndGeometry()
```

```
py3plex.visualization.fa2.fa2util.adjustSpeedAndApplyForces(nodes, speed, speedEfficiency,
                                                            jitterTolerance)
```

```
py3plex.visualization.fa2.fa2util.apply_attraction(nodes, edges, distributedAttraction, coefficient,
                                                    edgeWeightInfluence)
```

```
py3plex.visualization.fa2.fa2util.apply_gravity(nodes, gravity, useStrongGravity=False)
```

```
py3plex.visualization.fa2.fa2util.apply_repulsion(nodes, coefficient)
```

```
py3plex.visualization.fa2.fa2util.linAttraction(n1, n2, e, distributedAttraction, coefficient=0)
```

```
py3plex.visualization.fa2.fa2util.linGravity(n, g)
py3plex.visualization.fa2.fa2util.linRepulsion(n1, n2, coefficient=0)
py3plex.visualization.fa2.fa2util.linRepulsion_region(n, r, coefficient=0)
py3plex.visualization.fa2.fa2util.strongGravity(n, g, coefficient=0)
```

7.29.12 Embedding Visualization

```
py3plex.visualization.embedding_visualization.embedding_visualization.visualize_embedding(multinet,
                                                                                          labels=None,
                                                                                          verbose=True)
```

7.29.13 Network Generation and Benchmarking

7.29.14 Additional Statistics and Analysis

```
py3plex.algorithms.statistics.bayesiantests.correlated_ttest(x, rope, runs=1, verbose=False,
                                                             names=('C1', 'C2'))

py3plex.algorithms.statistics.bayesiantests.correlated_ttest_MC(x, rope, runs=1,
                                                                nsamples=50000)
```

See correlated_ttest module for explanations

```
py3plex.algorithms.statistics.bayesiantests.heaviside(X)
```

Compute the Heaviside step function.

The Heaviside function returns 1 for positive values, 0.5 for zero, and 0 for negative values. This is used in signed-rank tests for Bayesian comparisons.

Parameters

X – Input array or scalar

Returns

Array with same shape as X containing:

- 1.0 where $X > 0$
- 0.5 where $X == 0$
- 0.0 where $X < 0$

Return type

np.ndarray

Examples

```
>>> heaviside(np.array([-1, 0, 1, 2]))
array([0. , 0.5, 1. , 1. ])
```

```
py3plex.algorithms.statistics.bayesiantests.hierarchical(diff, rope, rho, upperAlpha=2,
                                                         lowerAlpha=1, lowerBeta=0.01,
                                                         upperBeta=0.1, std_upper_bound=1000,
                                                         verbose=False, names=('C1', 'C2'))
```


Perform hierarchical Bayesian test for comparing algorithms across multiple datasets.

This test accounts for the hierarchical structure of the data (multiple datasets, each with multiple folds) and correlations due to overlapping training sets.

Parameters

- **diff** – Array of differences between classifier scores
- **rope** – Width of the region of practical equivalence (ROPE)
- **rho** – Correlation between folds (typically around $1/n_folds$)
- **upperAlpha** – Upper bound for alpha parameter of Gamma prior (default: 2)
- **lowerAlpha** – Lower bound for alpha parameter of Gamma prior (default: 1)
- **lowerBeta** – Lower bound for beta parameter of Gamma prior (default: 0.01)
- **upperBeta** – Upper bound for beta parameter of Gamma prior (default: 0.1)
- **std_upper_bound** – Upper bound multiplier for standard deviation prior (default: 1000)
Posterior is insensitive to this if large enough (>100)
- **verbose** – Whether to print probability results (default: False)
- **names** – Tuple of classifier names for verbose output (default: ("C1", "C2"))

Returns

(p_left, p_rope, p_right)

- p_left: Probability that first classifier is worse
- p_rope: Probability that classifiers are practically equivalent
- p_right: Probability that first classifier is better

Return type

Tuple[float, float, float]

Notes

- The Gamma distribution parameters control the prior on degrees of freedom
- The hierarchical structure models between-dataset and within-dataset variance
- Use when comparing algorithms across multiple datasets with cross-validation

References

- Benavoli, A., Corani, G., & Mangili, F. (2016). Should we really use post-hoc tests based on mean-ranks? The Journal of Machine Learning Research.

See also

hierarchical_MC: Monte Carlo sampling version correlated_ttest: Simpler test for single dataset comparisons

```
py3plex.algorithms.statistics.bayesian_tests.hierarchical_MC(diff, rope, rho, upperAlpha=2,
                                                            lowerAlpha=1, lowerBeta=0.01,
                                                            upperBeta=0.1,
                                                            std_upper_bound=1000,
                                                            names=('C1', 'C2'))
```

Monte Carlo sampling for hierarchical Bayesian test.

Generates Monte Carlo samples from the posterior distribution for hierarchical comparison of algorithms across multiple datasets with cross-validation.

Parameters

- **diff** – Array of differences between classifier scores (shape: `n_datasets x n_folds`)
- **rope** – Width of the region of practical equivalence (ROPE)
- **rho** – Correlation between folds due to overlapping training sets
- **upperAlpha** – Upper bound for alpha parameter of Gamma prior (default: 2)
- **lowerAlpha** – Lower bound for alpha parameter of Gamma prior (default: 1)
- **lowerBeta** – Lower bound for beta parameter of Gamma prior (default: 0.01)
- **upperBeta** – Upper bound for beta parameter of Gamma prior (default: 0.1)
- **std_upper_bound** – Upper bound multiplier for standard deviation prior (default: 1000)
- **names** – Tuple of classifier names for identification (default: (“C1”, “C2”))

Returns

Monte Carlo samples with shape (`n_samples`, 3)

Each row contains [`p_left`, `p_rope`, `p_right`] for one MC sample

Return type

`np.ndarray`

Notes

- Uses PyStan for Bayesian inference with hierarchical model
- Data is rescaled by mean standard deviation for numerical stability
- Hierarchical structure captures both within-dataset and between-dataset variance
- Requires pystan package to be installed

Implementation:

- Uses Stan’s NUTS sampler with 4 chains
- Each chain runs 100 iterations (including warmup)
- Total posterior samples: ~200 after warmup

See also

`hierarchical`: Main function that processes MC samples
`correlated_ttest_MC`: Simpler MC version for single dataset

Raises

ImportError – If pystan is not installed

```
py3plex.algorithms.statistics.bayesiantests.plot_posterior(samples, names=('C1', 'C2'),
                                                         proba_triplet=None)
```

Parameters

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **names** (*pair of str*) – the names of the two classifiers

Returns

matplotlib.pyplot.figure

```
py3plex.algorithms.statistics.bayesiantests.plot_simplex(points, names=('C1', 'C2'),
                                                         proba_triplet=None)
```

```
py3plex.algorithms.statistics.bayesiantests.signrank(x, rope, prior_strength=0.6, prior_place=1,
                                                      nsamples=50000, verbose=False,
                                                      names=('C1', 'C2'))
```

Parameters

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 0.6)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples
- **verbose** (*bool*) – report the computed probabilities
- **names** (*pair of str*) – the names of the two classifiers

Returns

p_left, p_rope, p_right

```
py3plex.algorithms.statistics.bayesiantests.signrank_MC(x, rope, prior_strength=0.6,
                                                         prior_place=1, nsamples=50000)
```

Parameters

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 0.6)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples

Returns

2-d array with rows corresponding to samples and columns to probabilities [*p_left, p_rope, p_right*]

```
py3plex.algorithms.statistics.bayesiantests.signtest(x, rope, prior_strength=1, prior_place=1,
                                                      nsamples=50000, verbose=False,
                                                      names=('C1', 'C2'))
```

Parameters

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 1)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples
- **verbose** (*bool*) – report the computed probabilities
- **names** (*pair of str*) – the names of the two classifiers

Returns

p_left, p_rope, p_right

```
py3plex.algorithms.statistics.bayesiantests.signtest_MC(x, rope, prior_strength=1, prior_place=1,
                                                         nsamples=50000)
```

Parameters

- **x** (*array*) – a vector of differences or a 2d array with pairs of scores.
- **rope** (*float*) – the width of the rope
- **prior_strength** (*float*) – prior strength (default: 1)
- **prior_place** (*LEFT, ROPE or RIGHT*) – the region to which the prior is assigned (default: ROPE)
- **nsamples** (*int*) – the number of Monte Carlo samples

Returns

2-d array with rows corresponding to samples and columns to probabilities [p_left, p_rope, p_right]

7.29.15 Community Detection Advanced

Node Ranking and Clustering (NoRC) module for community detection.

This module implements algorithms for node ranking and hierarchical clustering in networks, including parallel PageRank computation and hierarchical merging.

```
py3plex.algorithms.community_detection.NoRC.NoRC_communities_main(input_graph, clustering_scheme='hierarchical',
                                                                    max_com_num=100,
                                                                    verbose=False,
                                                                    parallel_step=None,
                                                                    prob_threshold=0.0005,
                                                                    community_range=None,
                                                                    fine_range=3,
                                                                    lag_threshold=10)
```

```
py3plex.algorithms.community_detection.NoRC.page_rank_kernel(index_row)
```

Compute normalized PageRank vector for a given node.

Parameters

index_row – Node index for which to compute PageRank

Returns

Tuple of (node_index, normalized_pagerank_vector)

Community-based node ranking framework.

This module implements a framework for ranking nodes within and across communities using PageRank-based metrics and hierarchical clustering.

```
py3plex.algorithms.community_detection.community_ranking.create_tree(centers: ndarray) →
                                                                    Dict[int, Dict[str, List]]
```

```
py3plex.algorithms.community_detection.community_ranking.page_rank_kernel(index_row: int) →
                                                                    Tuple[int, ndarray]
```

```
py3plex.algorithms.community_detection.community_ranking.return_infomap_communities(network:
                                                                                      Any)
                                                                                      →
                                                                                      List[List[int]]
```

7.29.16 Node Ranking and Classification

Node ranking algorithms for multilayer networks.

This module provides various node ranking algorithms including PageRank variants, HITS (Hubs and Authorities), and personalized PageRank (PPR) for network analysis.

Key Functions:

- `sparse_page_rank`: Compute PageRank scores using sparse matrix operations
- `run_PPR`: Run Personalized PageRank in parallel across multiple cores
- `hubs_and_authorities`: Compute HITS scores for nodes
- `stochastic_normalization`: Normalize adjacency matrix to stochastic form

Notes

The PageRank implementations use sparse matrices for memory efficiency and support parallel computation for large-scale networks.

```
py3plex.algorithms.node_ranking.authority_matrix(graph: Graph) → ndarray
```

Get the authority matrix representation of a graph.

Computes the matrix $A = A.T @ A$ where A is the adjacency matrix. The authority matrix is used in HITS algorithm computation.

Parameters

graph – NetworkX graph

Returns

Authority matrix (N x N) where N is number of nodes

Return type

np.ndarray

Notes

- For directed graphs: $A[i,j]$ = number of nodes that point to both i and j
- Used internally by HITS algorithm to compute authority scores

See also

hub_matrix: Complementary hub matrix hubs_and_authorities: Compute actual hub/authority scores

py3plex.algorithms.node_ranking.damping_hyper: float

py3plex.algorithms.node_ranking.hub_matrix(*graph: Graph*) → ndarray

Get the hub matrix representation of a graph.

Computes the matrix $H = A @ A.T$ where A is the adjacency matrix. The hub matrix is used in HITS algorithm computation.

Parameters

graph – NetworkX graph

Returns

Hub matrix ($N \times N$) where N is number of nodes

Return type

np.ndarray

Notes

- For directed graphs: $H[i,j]$ = number of nodes pointed to by both i and j
- Used internally by HITS algorithm to compute hub scores

See also

authority_matrix: Complementary authority matrix hubs_and_authorities: Compute actual hub/authority scores

py3plex.algorithms.node_ranking.hubs_and_authorities(*graph: Graph*) → Tuple[dict, dict]

Compute HITS (Hubs and Authorities) scores for all nodes in a graph.

Implements the Hyperlink-Induced Topic Search (HITS) algorithm to identify hub nodes (nodes that point to many authorities) and authority nodes (nodes pointed to by many hubs) in a network.

Parameters

graph – NetworkX graph (directed or undirected)

Returns

(hub_scores, authority_scores)

- hub_scores: Dictionary mapping node -> hub score
- authority_scores: Dictionary mapping node -> authority score

Return type

Tuple[dict, dict]

Notes

- Uses scipy-based implementation from NetworkX (nx.hits_scipy)
- Scores are normalized so that the sum of squares equals 1
- For undirected graphs, hub and authority scores are identical

- Converges using power iteration method

Examples

```
>>> import networkx as nx
>>> G = nx.DiGraph([(0, 1), (0, 2), (1, 2)])
>>> hubs, authorities = hubs_and_authorities(G)
>>> # Node 0 has high hub score (points to others)
>>> # Node 2 has high authority score (pointed to by others)
```

See also

hub_matrix: Get the hub matrix representation authority_matrix: Get the authority matrix representation

`py3plex.algorithms.node_ranking.page_rank_kernel(index_row: int) → Tuple[int, ndarray]`

Compute PageRank vector for a single starting node (multiprocessing kernel).

This function is designed to be called in parallel via `multiprocessing.Pool.map()`. It computes the personalized PageRank vector starting from a single node.

Parameters

index_row – Index of the starting node for personalized PageRank

Returns

(node_index, normalized_pagerank_vector)

- node_index: The input index (for tracking results)
- pagerank_vector: L2-normalized PageRank scores for all nodes

Return type

Tuple[int, np.ndarray]

Notes

- Accesses global variables: `__graph_matrix`, `damping_hyper`, `spread_step_hyper`, `spread_percent_hyper` (set by `run_PPR` before parallel execution)
- Returns zero vector if normalization fails
- L2 normalization ensures comparable magnitudes across different starting nodes

See also

`run_PPR`: Main function that sets up parallel execution `sparse_page_rank`: Core PageRank computation

`py3plex.algorithms.node_ranking.run_PPR(network: spmatrix, cores: int | None = None, jobs: List[range] | None = None, damping: float = 0.85, spread_step: int = 10, spread_percent: float = 0.3, targets: List[int] | None = None, parallel: bool = True) → Generator[Tuple[int, ndarray] | List[Tuple[int, ndarray]], None, None]`

Run Personalized PageRank (PPR) in parallel for multiple starting nodes.

Computes personalized PageRank vectors for multiple nodes using parallel processing. This is useful for creating node embeddings or analyzing node importance from different perspectives in the network.

Parameters

- **network** – Sparse adjacency matrix (will be automatically normalized to stochastic form)
- **cores** – Number of CPU cores to use (default: all available cores)
- **jobs** – Custom job batches as list of ranges (default: auto-generated)
- **damping** – Damping factor for PageRank (default: 0.85) Higher values (0.85-0.99) emphasize network structure
- **spread_step** – Steps to check spread pattern for optimization (default: 10)
- **spread_percent** – Max node fraction for shrinkage optimization (default: 0.3)
- **targets** – Specific node indices to compute PPR for (default: all nodes)
- **parallel** – Enable parallel processing (default: True) Set to False for debugging or single-core execution

Yields

Union[Tuple[int, np.ndarray], List[Tuple[int, np.ndarray]]] – - If parallel=True: Lists of (node_index, pagerank_vector) tuples (batched) - If parallel=False: Individual (node_index, pagerank_vector) tuples

Notes

- Automatically normalizes input matrix to column-stochastic form
- Uses multiprocessing.Pool for parallel execution
- Global variables are used to share the graph matrix across processes
- Results are yielded incrementally (generator pattern) to save memory
- Each pagerank_vector is L2-normalized for comparability

Examples

```
>>> import scipy.sparse as sp
>>> # Create a small network
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>>
>>> # Compute PPR for all nodes in parallel
>>> for batch in run_PPR(adj, cores=2, parallel=True):
...     for node_idx, pr_vector in batch:
...         print(f"Node {node_idx}: {pr_vector}")
>>>
>>> # Compute PPR for specific nodes without parallelism
>>> for node_idx, pr_vector in run_PPR(adj, targets=[0, 1], parallel=False):
...     print(f"Node {node_idx}: {pr_vector}")
```

Performance:

- Parallel speedup scales with number of cores (up to ~0.8 * cores efficiency)
- Memory usage: $O(N * N_{\text{targets}})$ for storing results
- For large networks (>100K nodes), consider processing targets in batches

See also

`sparse_page_rank`: Core PageRank computation
`page_rank_kernel`: Worker function for parallel execution
`stochastic_normalization`: Matrix normalization (called internally)

```
py3plex.algorithms.node_ranking.sparse_page_rank(matrix: spmatrix, start_nodes: List[int] | range,
                                                  epsilon: float = 1e-06, max_steps: int = 100000,
                                                  damping: float = 0.5, spread_step: int = 10,
                                                  spread_percent: float = 0.3, try_shrink: bool =
                                                  True) → ndarray
```

Compute personalized PageRank using sparse matrix operations.

Implements an efficient personalized PageRank algorithm with adaptive sparsification to reduce memory usage and computation time. The algorithm uses a power iteration method with early stopping based on convergence criteria.

Parameters

- **matrix** – Column-stochastic sparse adjacency matrix (use `stochastic_normalization` first)
- **start_nodes** – List or range of starting nodes for personalized PageRank
- **epsilon** – Convergence threshold for L1 norm difference (default: 1e-6)
- **max_steps** – Maximum number of iterations (default: 100000)
- **damping** – Damping factor / teleportation probability (default: 0.5) Higher values (e.g., 0.85) favor network structure over random jumps
- **spread_step** – Number of steps to check for sparsity pattern (default: 10)
- **spread_percent** – Maximum fraction of nodes to consider for shrinkage (default: 0.3)
- **try_shrink** – Enable adaptive shrinkage to reduce computation (default: True)

Returns

PageRank scores for all nodes, with `start_nodes` set to 0

Return type

`np.ndarray`

Notes

- Assumes matrix is column-stochastic (use `stochastic_normalization` first)
- Adaptive shrinkage identifies nodes unreachable from `start_nodes` and excludes them from computation for efficiency
- Convergence is measured by L1 norm of rank vector difference
- Start nodes are zeroed out in the final result to avoid self-importance

Complexity:

- Time: $O(k * E)$ where k is iterations and E is edges
- Space: $O(N)$ for rank vectors, plus matrix storage

Examples

```
>>> import scipy.sparse as sp
>>> # Create and normalize adjacency matrix
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> adj_norm = stochastic_normalization(adj)
>>> # Compute PageRank from node 0
>>> pr = sparse_page_rank(adj_norm, [0], damping=0.85)
>>> pr # PageRank scores (node 0 will be 0)
array([0. , 0.5, 0.5])
```

Raises

AssertionError – If start_nodes is empty

See also

stochastic_normalization: Required preprocessing step
run_PPR: Parallel wrapper for computing multiple PageRank vectors

py3plex.algorithms.node_ranking.spread_percent_hyper: float

py3plex.algorithms.node_ranking.spread_step_hyper: int

py3plex.algorithms.node_ranking.stochastic_normalization(matrix: spmatrix) → spmatrix

Normalize a sparse matrix to stochastic form (column-stochastic).

Converts an adjacency matrix to a stochastic matrix where each column sums to 1. This normalization is required for PageRank-style random walk algorithms.

Parameters

matrix – Sparse adjacency matrix to normalize

Returns

Column-stochastic sparse matrix where each column sums to 1

Return type

sp.spmatrix

Notes

- Removes self-loops (sets diagonal to 0) before normalization
- Handles zero-degree nodes by leaving corresponding columns as zeros
- Preserves sparsity structure for memory efficiency

Examples

```
>>> import scipy.sparse as sp
>>> adj = sp.csr_matrix([[0, 1, 1], [1, 0, 1], [1, 1, 0]])
>>> stoch_adj = stochastic_normalization(adj)
>>> stoch_adj.sum(axis=1).A1 # Column sums should be 1
array([1., 1., 1.])
```

`py3plex.algorithms.network_classification.benchmark_classification`(*matrix: spmatrix, labels: ndarray, alpha_value: float = 0.85, iterations: int = 30, normalization_scheme: str = 'freq', dataset_name: str = 'example', verbose: bool = False, test_size: float | None = None*) → DataFrame

`py3plex.algorithms.network_classification.label_propagation`(*graph_matrix: spmatrix, class_matrix: ndarray, alpha: float = 0.001, epsilon: float = 1e-12, max_steps: int = 100000, normalization: str | List[str] = 'freq'*) → ndarray

Propagate labels through a graph.

Parameters

- **graph_matrix** – Sparse graph adjacency matrix
- **class_matrix** – Initial class label matrix
- **alpha** – Propagation weight parameter
- **epsilon** – Convergence threshold
- **max_steps** – Maximum number of iterations
- **normalization** – Normalization scheme(s) to apply

Returns

Propagated label matrix

`py3plex.algorithms.network_classification.label_propagation_normalization`(*matrix: spmatrix*) → spmatrix

Normalize a matrix for label propagation.

Parameters

matrix – Sparse matrix to normalize

Returns

Normalized sparse matrix

`py3plex.algorithms.network_classification.label_propagation_tf`() → None

`py3plex.algorithms.network_classification.normalize_amplify_freq`(*mat: ndarray*) → ndarray

Normalize and amplify matrix by frequency.

Parameters

mat – Matrix to normalize

Returns

Normalized and amplified matrix

`py3plex.algorithms.network_classification.normalize_exp`(*mat: ndarray*) → ndarray

Apply exponential normalization.

Parameters

mat – Matrix to normalize

Returns

Exponentially normalized matrix

`py3plex.algorithms.network_classification.normalize_initial_matrix_freq(mat: ndarray) → ndarray`

Normalize matrix by frequency.

Parameters

mat – Matrix to normalize

Returns

Normalized matrix

`py3plex.algorithms.network_classification.validate_label_propagation(core_network: spmatrix, labels: ndarray | spmatrix, dataset_name: str = 'test', repetitions: int = 5, normalization_scheme: str | List[str] = 'basic', alpha_value: float = 0.001, random_seed: int = 123, verbose: bool = False) → DataFrame`

Validate label propagation with cross-validation.

Parameters

- **core_network** – Sparse network adjacency matrix
- **labels** – Label matrix
- **dataset_name** – Name of the dataset
- **repetitions** – Number of repetitions
- **normalization_scheme** – Normalization scheme to use
- **alpha_value** – Alpha parameter for propagation
- **random_seed** – Random seed for reproducibility
- **verbose** – Whether to print progress

Returns

DataFrame with validation results

7.29.17 Embeddings and Wrappers

`py3plex.wrappers.train_node2vec_embedding.call_node2vec_binary(input_graph: str, output_graph: str, p: float = 1, q: float = 1, dimension: int = 128, directed: bool = False, weighted: bool = True, binary: str | None = None, timeout: int = 300) → None`

Call the Node2Vec C++ binary with specified parameters.

Parameters

- **input_graph** – Path to input graph file
- **output_graph** – Path to output embedding file

- **p** – Return parameter
- **q** – In-out parameter
- **dimension** – Embedding dimension
- **directed** – Whether graph is directed
- **weighted** – Whether graph is weighted
- **binary** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or “./node2vec”)
- **timeout** – Maximum execution time in seconds

Raises

ExternalToolError – If binary is not found or execution fails

```
py3plex.wrappers.train_node2vec_embedding.learn_embedding(core_network: Any, labels: List[Any] |
                                                         None = None, ssize: float = 0.5,
                                                         embedding_outfile: str = 'out.emb', p:
                                                         float = 0.1, q: float = 0.1, binary_path:
                                                         str | None = None, parameter_range: str
                                                         = '[0.25,0.50,1,2,4]', timeout: int =
                                                         300) → Tuple[str, float]
```

Learn node embeddings for a network.

Parameters

- **core_network** – NetworkX graph
- **labels** – Node labels
- **ssize** – Sample size
- **embedding_outfile** – Output file for embeddings
- **p** – Return parameter
- **q** – In-out parameter
- **binary_path** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or “./node2vec”)
- **parameter_range** – String representation of parameter range list
- **timeout** – Maximum execution time in seconds per call

Returns

Tuple of (method_name, elapsed_time)

```
py3plex.wrappers.train_node2vec_embedding.n2v_embedding(G: Any, targets: Any, verbose: bool =
                                                         False, sample_size: float = 0.5,
                                                         outfile_name: str = 'test.emb', p: float |
                                                         None = None, q: float | None = None,
                                                         binary_path: str | None = None,
                                                         parameter_range: List[float] | None =
                                                         None, embedding_dimension: int = 128,
                                                         timeout: int = 300) → None
```

Train Node2Vec embeddings with parameter optimization.

Parameters

- **G** – NetworkX graph

- **targets** – Target labels for nodes
- **verbose** – Whether to print verbose output
- **sample_size** – Sample size for training
- **outfile_name** – Output embedding file name
- **p** – Return parameter (None triggers grid search)
- **q** – In-out parameter (None triggers grid search)
- **binary_path** – Path to node2vec binary (defaults to PY3PLEX_NODE2VEC_BINARY env var or “./node2vec”)
- **parameter_range** – Range of parameters to search
- **embedding_dimension** – Dimension of embeddings
- **timeout** – Maximum execution time in seconds per call

7.29.18 Network Motifs and Patterns

7.29.19 HINMINE Data Structures

```
class py3plex.core.HINMINE.dataStructures.Class(lab_id, name, members)
```

Bases: object

```
class py3plex.core.HINMINE.dataStructures.HeterogeneousInformationNetwork(network,  
                                                                           label_delimiter,  
                                                                           weight_tag=False,  
                                                                           target_tag=True)
```

Bases: object

```
add_label(node, label_id, label_name=None)
```

```
calculate_decomposition_candidates(max_decomposition_length=10)
```

```
calculate_schema()
```

```
create_label_matrix(weights=None)
```

```
decompose_from_iterator(name, weighing, summing, generator=None, degrees=None, parallel=False,  
                        pool=None)
```

```
midpoint_generator(node_sequence, edge_sequence)
```

```
process_network(label_delimiter)
```

```
split_to_indices(train_indices=(), validate_indices=(), test_indices=())
```

```
split_to_parts(lst, n)
```

7.29.20 Command-Line Interface

Command-line interface for py3plex.

This module provides a comprehensive CLI tool for multilayer network analysis with full coverage of main algorithms.

`py3plex.cli.cmd_aggregate(args: Namespace) → int`

Aggregate multilayer network into single layer.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_centrality(args: Namespace) → int`

Compute node centrality measures.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_check(args: Namespace) → int`

Lint and validate a graph data file.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success, non-zero for errors)

`py3plex.cli.cmd_community(args: Namespace) → int`

Detect communities in the network.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_convert(args: Namespace) → int`

Convert network between different formats.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_create(args: Namespace) → int`

Create a new multilayer network.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_help(args: Namespace) → int`

Show detailed help information.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_load(args: Namespace) → int`

Load and inspect a network.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_quickstart(args: Namespace) → int`

Run quickstart demo - creates a tiny demo graph and demonstrates basic operations.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_run_config(args: Namespace) → int`

Run workflow from configuration file.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_selftest(args: Namespace) → int`

Run self-test to verify installation and core functionality.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_stats(args: Namespace) → int`

Compute multilayer network statistics.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.cmd_visualize(args: Namespace) → int`

Visualize the multilayer network.

Parameters

args – Parsed command-line arguments

Returns

Exit code (0 for success)

`py3plex.cli.create_parser() → ArgumentParser`

Create and configure the argument parser for py3plex CLI.

Returns

Configured ArgumentParser instance

`py3plex.cli.main(argv: List[str] | None = None) → int`

Main entry point for the CLI.

Parameters

argv – Command-line arguments (defaults to `sys.argv`)

Returns

Exit code (0 for success, non-zero for errors)

7.29.21 Validation Utilities

Input validation utilities for Py3plex.

This module provides pre-validation functions to catch common input errors early and provide clear, actionable error messages to users.

`py3plex.validation.validate_csv_columns(file_path: str, required_columns: List[str], optional_columns: List[str] | None = None) → None`

Validate that a CSV file has required columns.

Parameters

- **file_path** – Path to CSV file
- **required_columns** – List of column names that must be present
- **optional_columns** – List of column names that are optional

Raises

ParsingError – If required columns are missing

Contracts:

- Precondition: `file_path` must be a non-empty string
- Precondition: `required_columns` must be a non-empty list of strings

`py3plex.validation.validate_edgelist_format(file_path: str, delimiter: str | None = None) → None`

Validate simple edgelist file format (source target weight).

Parameters

- **file_path** – Path to edgelist file
- **delimiter** – Optional delimiter (default: whitespace)

Raises

ParsingError – If file format is invalid

Contracts:

- Precondition: `file_path` must be a non-empty string
- Precondition: `delimiter` must be `None` or a string

`py3plex.validation.validate_file_exists(file_path: str) → None`

Validate that a file exists and is readable.

Parameters

file_path – Path to file to validate

Raises

ParsingError – If file doesn't exist or isn't readable

Contracts:

- Precondition: `file_path` must be a non-empty string

`py3plex.validation.validate_input_type(input_type: str, valid_types: Set[str] | None = None) → None`

Validate that `input_type` is recognized.

Parameters

- **`input_type`** – The input type string to validate
- **`valid_types`** – Optional set of valid types (uses default if `None`)

Raises

[ParsingError](#) – If `input_type` is not valid

Contracts:

- Precondition: `input_type` must be a non-empty string
- Precondition: `valid_types` must be `None` or a set

`py3plex.validation.validate_multiedgelist_format(file_path: str, delimiter: str | None = None) → None`

Validate multiedgelist file format (source target layer weight).

Parameters

- **`file_path`** – Path to multiedgelist file
- **`delimiter`** – Optional delimiter (default: whitespace)

Raises

[ParsingError](#) – If file format is invalid

Contracts:

- Precondition: `file_path` must be a non-empty string
- Precondition: `delimiter` must be `None` or a string

`py3plex.validation.validate_network_data(file_path: str, input_type: str) → None`

Validate network data before parsing.

This is the main entry point for validation. It performs appropriate validation based on the input type.

Parameters

- **`file_path`** – Path to network file
- **`input_type`** – Type of input file

Raises

[ParsingError](#) – If validation fails

Contracts:

- Precondition: `file_path` must be a string
- Precondition: `input_type` must be a non-empty string

7.29.22 Network Comparison and Testing

7.29.23 Hedwig Learning Algorithms

Main learner class.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.learners.learner.Learner(kb, n=None, min_sup=1, sim=1, depth=4,  
                                                         target=None, use_negations=False,  
                                                         optimal_subclass=False)
```

Bases: object

Learner class, supporting various types of induction from the knowledge base.

Default = 'default'

Improvement = 'improvement'

Similarity = 'similarity'

can_specialize(*rule*)

Is the rule good enough to be further refined?

extend(*rules, specializations*)

Extends the ruleset in the given way.

extend_replace_worst(*rules, specializations*)

Extends the list by replacing the worst rules.

extend_with_similarity(*rules, specializations*)

Extends the list based on how similar is 'new_rule' to the rules contained in 'rules'.

get_subclasses(*pred*)

get_superclasses(*pred*)

group_score(*rules*)

Calculates the score of the whole list of rules.

induce()

Induces rules for the given knowledge base.

is_implicit_root(*pred*)

non_redundant(*rule, new_rule*)

Is the rule non-redundant compared to its immediate generalization?

specialize(*rule*)

Returns a list of all specializations of 'rule'.

specialize_add_relation(*rule*)

Specialize with new binary relation.

Main learner class.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.learners.bottomup.BottomUpLearner(kb, n=None, min_sup=1,
                                                                    sim=1, depth=4, target=None,
                                                                    use_negations=False)
```

Bases: object

Bottom-up learner.

Default = 'default'

Improvement = 'improvement'

Similarity = 'similarity'

bottom_clause()

get_subclasses(pred)

get_superclasses(pred)

induce()

Induces rules for the given knowledge base.

is_implicit_root(pred)

Main learner class.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.learners.optimal.OptimalLearner(kb, n=None, min_sup=1, sim=1,
                                                                    depth=4, target=None,
                                                                    use_negations=False,
                                                                    optimal_subclass=True)
```

Bases: [Learner](#)

Finds the optimal top-k rules.

induce()

Induces rules for the given knowledge base.

7.29.24 Hedwig Statistics and Scoring

Score function definitions.

@author: anze.vavpetic@ijs.si

py3plex.algorithms.hedwig.stats.scorefunctions.**chisq**(rule)

py3plex.algorithms.hedwig.stats.scorefunctions.**enrichment_score**(rule)

py3plex.algorithms.hedwig.stats.scorefunctions.**interesting**(rule)

Checks if a given rule is interesting for the given score function

py3plex.algorithms.hedwig.stats.scorefunctions.**kaplan_meier_AUC**(rule)

py3plex.algorithms.hedwig.stats.scorefunctions.**leverage**(rule)

py3plex.algorithms.hedwig.stats.scorefunctions.**lift**(rule)

py3plex.algorithms.hedwig.stats.scorefunctions.**precision**(rule)

`py3plex.algorithms.hedwig.stats.scorefunctions.t_score(rule)`

`py3plex.algorithms.hedwig.stats.scorefunctions.wracc(rule)`

`py3plex.algorithms.hedwig.stats.scorefunctions.z_score(rule)`

Module for ruleset validation.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.stats.validate.Validate(kb, significance_test=<function
    apply_fisher>, adjustment=<function
    fdr>)
```

Bases: object

test(ruleset, alpha=0.05, q=0.01)

Tests the given ruleset and returns the significant rules.

7.29.25 Hedwig Core Components

```
py3plex.algorithms.hedwig.core.converters.convert_mapping_to_rdf(input_mapping_file,
    extract_subnode_info=False,
    split_node_by=':',
    keep_index=1,
    layer_type='uniprotkb', annotation_mapping_file='test.gaf',
    go_identifier='GO:',
    prepend_string=None)
```

`py3plex.algorithms.hedwig.core.converters.obo2n3(obofile, n3out, gaf_file)`

Knowledge-base class.

@author: anze.vavpetic@ijs.si

```
class py3plex.algorithms.hedwig.core.kb.ExperimentKB(triplets, score_fun, instances_as_leaves=True)
```

Bases: object

The knowledge base for one specific experiment.

add_sub_class(sub, obj)

Adds the resource 'sub' as a subclass of 'obj'.

bits_to_indices(bits)

Converts the bitset to a set of indices.

get_domains(predicate)

Returns the bitsets for input and output examples of the binary predicate 'predicate'.

get_empty_domain()

Returns a bitset covering no examples.

get_examples()

Returns all examples for this experiment.

get_full_domain()

Returns a bitset covering all examples.

get_members(*predicate*, *bit=True*)

Returns the examples for this predicate, either as a bitset or a set of ids.

get_reverse_members(*predicate*, *bit=True*)

Returns the examples for this predicate, either as a bitset or a set of ids.

get_root()

Root predicate, which covers all examples.

get_score(*ex_idx*)

Returns the score for example id 'ex_idx'.

get_subclasses(*predicate*, *producer_pred=None*)

Returns a list of subclasses (as predicate objects) for 'predicate'.

indices_to_bits(*indices*)

Converts the indices to a bitset.

is_discrete_target()

n_examples()

Returns the number of examples.

n_members(*predicate*)

super_classes(*pred*)

Returns all super classes of pred (with transitivity).

Global settings file.

@author: anze.vavpetic@ijs.si

class py3plex.algorithms.hedwig.core.settings.Defaults

Bases: object

ADJUST = 'fwer'

ALPHA = 0.05

BEAM_SIZE = 20

COVERED = None

DEPTH = 5

FDR_Q = 0.05

FORMAT = 'n3'

LEARNER = 'heuristic'

LEAVES = False

MODE = 'subgroups'

NEGATIONS = False

NO_CACHE = False

OPTIMAL_SUBCLASS = False

```

OUTPUT = None

SCORE = 'lift'

SUPPORT = 0.1

TARGET = None

URIS = False

VERBOSE = False

```

7.29.26 Time Series and Temporal Analysis

7.29.27 Advanced Visualization

Sankey diagram visualization for multilayer networks.

This module provides inter-layer flow visualization to show connection strength between layers in multilayer networks. The visualization displays flows as arrows with widths proportional to the number of inter-layer connections.

Note: This uses a simplified flow diagram approach rather than matplotlib's Sankey class, as the Sankey class is designed for more complex flow networks and doesn't map directly to multilayer network inter-layer connections.

```

py3plex.visualization.sankey.draw_multilayer_sankey(graphs: List[Graph], multilinks: Dict[str, List[Tuple]], labels: List[str] | None = None, ax: Any | None = None, display: bool = True, **kwargs) → Any

```

Draw inter-layer flow diagram showing connection strength in multilayer networks.

Creates a flow visualization where: - Each layer is represented in the diagram - Flows between layers show the strength (number) of inter-layer connections - Flow width/text indicates the number of inter-layer edges

Parameters

- **graphs** – List of NetworkX graphs, one per layer
- **multilinks** – Dictionary mapping edge_type -> list of multi-layer edges
- **labels** – Optional list of layer labels. If None, uses layer indices
- **ax** – Matplotlib axes to draw on. If None, creates new figure
- **display** – If True, calls plt.show() at the end
- ****kwargs** – Reserved for future extensions

Returns

Matplotlib axes object

Examples

```

>>> network = multi_layer_network()
>>> network.load_network("data.txt", input_type="multiedgelist")
>>> labels, graphs, multilinks = network.get_layers()
>>> draw_multilayer_sankey(graphs, multilinks, labels=labels)

```

Note

This visualization is most effective for networks with 2-5 layers. For networks with many layers, the diagram may become cluttered. The implementation uses a simplified flow visualization approach.

7.29.28 Link Prediction

7.30 Citation and References

If you use py3plex in your research, please **cite our work**.

7.30.1 Primary Citation

Recommended citation for py3plex:

```
@Article{Škrlj2019,
  author    = {{\v{S}}krlj, Bla{\v{z}} and Kralj, Jan and Lavra{\v{c}}, Nada},
  title     = {Py3plex toolkit for visualization and analysis of multilayer networks},
  journal   = {Applied Network Science},
  year      = {2019},
  volume    = {4},
  number    = {1},
  pages     = {94},
  doi       = {10.1007/s41109-019-0203-7},
  url       = {https://doi.org/10.1007/s41109-019-0203-7}
}
```

Plain text:

Škrlj, B., Kralj, J., & Lavrač, N. (2019). Py3plex toolkit for visualization and analysis of multilayer networks. *Applied Network Science*, 4(1), 94. <https://doi.org/10.1007/s41109-019-0203-7>

7.30.2 Conference Paper

For the **initial algorithmic work**:

```
@InProceedings{Škrlj2019Conference,
  author    = {{\v{S}}krlj, Bla{\v{z}} and Kralj, Jan and Lavra{\v{c}}, Nada},
  editor    = {Aiello, Luca Maria and Cherifi, Chantal and Cherifi, Hocine
              and Lambiotte, Renaud and Li{\v{o}}, Pietro and Rocha, Luis M.},
  title     = {Py3plex: A Library for Scalable Multilayer Network Analysis and
  ↪Visualization},
  booktitle = {Complex Networks and Their Applications VII},
  year      = {2019},
  publisher = {Springer International Publishing},
  address   = {Cham},
  pages     = {757--768},
  isbn      = {978-3-030-05411-3},
  doi       = {10.1007/978-3-030-05411-3_60}
}
```


7.30.3 Algorithm-Specific Citations

When using **specific algorithms**, please also cite the **original papers**:

Multilayer Modularity

```
@Article{Mucha2010,
  author = {Mucha, Peter J. and Richardson, Thomas and Macon, Kevin
            and Porter, Mason A. and Onnela, Jukka-Pekka},
  title  = {Community structure in time-dependent, multiscale, and multiplex networks},
  journal = {Science},
  year   = {2010},
  volume = {328},
  number = {5980},
  pages  = {876--878},
  doi    = {10.1126/science.1184819}
}
```

Node2Vec

```
@InProceedings{Grover2016,
  author = {Grover, Aditya and Leskovec, Jure},
  title  = {node2vec: Scalable Feature Learning for Networks},
  booktitle = {Proceedings of the 22nd ACM SIGKDD International Conference
               on Knowledge Discovery and Data Mining},
  year    = {2016},
  pages   = {855--864},
  doi     = {10.1145/2939672.2939754}
}
```

Louvain Algorithm

```
@Article{Blondel2008,
  author = {Blondel, Vincent D. and Guillaume, Jean-Loup and Lambiotte, Renaud
            and Lefebvre, Etienne},
  title  = {Fast unfolding of communities in large networks},
  journal = {Journal of Statistical Mechanics: Theory and Experiment},
  year   = {2008},
  volume = {2008},
  number = {10},
  pages  = {P10008},
  doi    = {10.1088/1742-5468/2008/10/P10008}
}
```

Infomap

```
@Article{Rosvall2008,
  author = {Rosvall, Martin and Bergstrom, Carl T.},
  title  = {Maps of random walks on complex networks reveal community structure},
  journal = {Proceedings of the National Academy of Sciences},
  year   = {2008},
  volume = {105},
```

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```
number = {4},
pages  = {1118--1123},
doi     = {10.1073/pnas.0706851105}
}
```

MultiXRank

```
@Article{Baptista2022,
  author = {Baptista, Anthony and Gonzalez, Aur{\e}lien and Baudot, Ana{\"}{i}s},
  title  = {Universal multilayer network exploration by random walk with restart},
  journal = {Communications Physics},
  year   = {2022},
  volume = {5},
  number = {1},
  pages  = {170},
  doi    = {10.1038/s42005-022-00937-9}
}
```

DeepWalk

```
@InProceedings{Perozzi2014,
  author = {Perozzi, Bryan and Al-Rfou, Rami and Skiena, Steven},
  title  = {DeepWalk: Online Learning of Social Representations},
  booktitle = {Proceedings of the 20th ACM SIGKDD International Conference
    on Knowledge Discovery and Data Mining},
  year    = {2014},
  pages   = {701--710},
  doi     = {10.1145/2623330.2623732}
}
```

Multilayer Network Theory

Foundational papers on multilayer networks:

```
@Article{Kivela2014,
  author = {Kivel{\"}{a}, Mikko and Arenas, Alex and Barthelemy, Marc
    and Gleeson, James P. and Moreno, Yamir and Porter, Mason A.},
  title  = {Multilayer networks},
  journal = {Journal of Complex Networks},
  year   = {2014},
  volume = {2},
  number = {3},
  pages  = {203--271},
  doi    = {10.1093/comnet/cnu016}
}
```

```
@Article{Boccaletti2014,
  author = {Boccaletti, Stefano and Bianconi, Ginestra and Criado, Regino
    and del Genio, Charo I. and Gomez-Garde{\~{n}}es, Jes{\'}{u}s
    and Romance, Miguel and Sendi{\~{n}}a-Nadal, Irene and Wang, Zhen
    and Zanin, Massimiliano},
}
```

(continues on next page)

(continued from previous page)

```

title   = {The structure and dynamics of multilayer networks},
journal = {Physics Reports},
year    = {2014},
volume  = {544},
number  = {1},
pages   = {1--122},
doi     = {10.1016/j.physrep.2014.07.001}
}

```

```

@Article{DeDomenico2013,
  author = {De Domenico, Manlio and Sol{\`e}-Ribalta, Albert and Cozzo, Emanuele
            and Kivel{\`a}, Mikko and Moreno, Yamir and Porter, Mason A.
            and G{\`o}mez, Sergio and Arenas, Alex},
  title   = {Mathematical formulation of multilayer networks},
  journal = {Physical Review X},
  year    = {2013},
  volume  = {3},
  number  = {4},
  pages   = {041022},
  doi     = {10.1103/PhysRevX.3.041022}
}

```

7.30.4 Complete Citation List

For algorithm citations, see the `algorithm_guide` which includes all major algorithms with their original references and DOIs.

7.30.5 Acknowledgments

Development and Contributors

py3plex is developed and maintained by:

- **Blaž Škrlič** (lead developer) - Jožef Stefan Institute, Ljubljana, Slovenia
- **Jan Kralj** (contributor) - Jožef Stefan Institute
- **Nada Lavrač** (contributor) - Jožef Stefan Institute

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This work has been **supported by**:

- **Jožef Stefan Institute**, Ljubljana, Slovenia
- **Slovenian Research Agency** (research program P2-0103)

External Libraries

py3plex builds upon **excellent open-source libraries**:

- **NetworkX** - Graph data structures and algorithms
- **NumPy** - Numerical computing
- **SciPy** - Scientific computing
- **Matplotlib** - Visualization

- **scikit-learn** - Machine learning utilities

Community Acknowledgments

We thank all contributors, users, and the broader network science community for their support, feedback, and contributions.

Special thanks to contributors who have submitted pull requests, reported issues, or provided feedback.

7.30.6 License Information

py3plex is released under the **MIT License**.

MIT License

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Note: Some bundled algorithms may have **different licenses**:

- Infomap community detection code: **AGPLv3**
- Louvain community detection: **BSD-3-Clause**

See LICENSE file in the repository for detailed license compatibility information.

7.30.7 Contact

For questions about citing py3plex or collaboration opportunities:

- **Email:** blaz.skrj@ijs.si
- **GitHub:** <https://github.com/SkBlaz/py3plex>
- **Issues:** <https://github.com/SkBlaz/py3plex/issues>

7.30.8 Related Work

Other tools and libraries for multilayer network analysis:

- **muxViz** - Multilayer network visualization (R)

- **pymnet** - Multilayer networks in Python
- **multinet** - R package for multilayer networks
- **graphistry** - GPU-accelerated graph visualization

For a comprehensive comparison and positioning of py3plex, see the original publication.

7.30.9 Usage in Publications

If you've used py3plex in your research and would like to be listed here, please:

1. Open an issue or pull request
2. Provide citation to your paper
3. Brief description of how py3plex was used

We maintain a list of publications using py3plex to showcase applications and build community.

7.30.10 See Also

- [acknowledgements](#) - Full acknowledgments
- [algorithm_guide](#) - Complete algorithm citations with references
- [py3plex GitHub](#) - Source code and issues
- [Applied Network Science paper](#) - Primary publication

EXAMPLES

The best way to learn py3plex is through examples.

All examples are available in the [examples/](#) directory.

Key examples:

- `tutorial_10min.py` - Executable version of the 10-minute tutorial
- `example_random_walks.py` - Random walk primitives (Node2Vec, DeepWalk)
- `example_multilayer_visualization.py` - Network visualization
- `example_community_detection.py` - Community detection with Louvain and Infomap
- `example_network_decomposition.py` - Meta-path feature extraction
- `example_multilayer_statistics.py` - Computing multilayer metrics
- `example_n2v_embedding.py` - Node2Vec embeddings
- `example_label_propagation.py` - Semi-supervised learning

CITATION

If you use py3plex in your research, please cite:

```
@Article{Skrlj2019,  
  author={Škrlj, Blaž and Kralj, Jan and Lavrač, Nada},  
  title={Py3plex toolkit for visualization and analysis of multilayer networks},  
  journal={Applied Network Science},  
  year={2019},  
  volume={4},  
  number={1},  
  pages={94},  
  doi={10.1007/s41109-019-0203-7}  
}
```


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- `search`

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